

(12) **United States Patent**
Daoura et al.

(10) **Patent No.:** **US 12,412,465 B1**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **XCB TRACKING DEVICES, METHODS AND SYSTEMS**

(71) Applicant: **PB Inc.**, Issaquah, WA (US)

(72) Inventors: **Daniel J. Daoura**, Renton, WA (US);
Pawel P. Kazimierowicz, Cracow (PL);
Kal K. Lambert, Hammond, OR (US)

(73) Assignee: **PB Inc.**, Issaquah, WA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 558 days.

(21) Appl. No.: **17/870,385**

(22) Filed: **Jul. 21, 2022**

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/163,403, filed on Jan. 30, 2021, now Pat. No. 11,450,196, (Continued)

(51) **Int. Cl.**
H04W 4/029 (2018.01)
G08B 21/24 (2006.01)
H04W 4/02 (2018.01)

(52) **U.S. Cl.**
CPC **G08B 21/24** (2013.01); **H04W 4/025** (2013.01); **H04W 4/029** (2018.02)

(58) **Field of Classification Search**
CPC H04L 67/10; H04L 67/12; H04L 67/306;
H04L 67/04; H04L 67/52; H04L 67/535;
H04L 67/563

(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,402,455 B1 * 9/2019 Phanse G01C 21/3611
2014/0180711 A1 * 6/2014 Kamen G16H 70/40
705/2

(Continued)

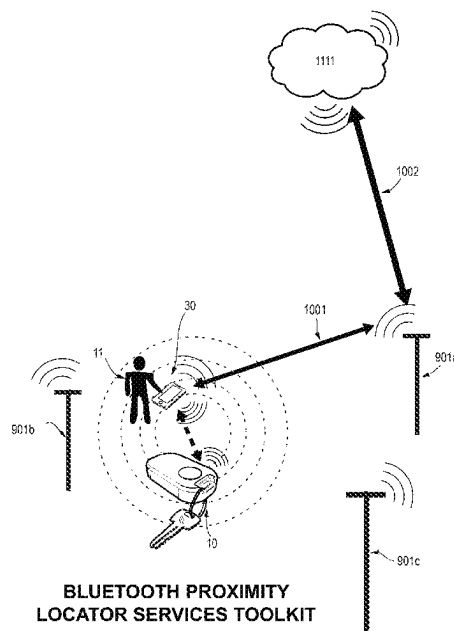
Primary Examiner — Joseph Arevalo

(74) *Attorney, Agent, or Firm* — Christensen O'Connor Johnson Kindness PLLC

(57) **ABSTRACT**

XCB poly-radio devices as finders, locators, scanners, sensors, and Bluetooth (BT) radio topology reporters as part of a global lost-and-found services system. XCB devices function as “smart objects” and combine dual transceivers operating in the telecommunications and the low power ISM radio spectra, with default power management overrides under local control of the BT modem. “Familiar zones” are defined, and a “familiarity index” versus a “strangeness index” based on meta-analysis of BT radio contact logs are used to assign location. Location, in concert with other sensor data, is coupled to conditional logic rules programmable by the user or the system administrator. Locations, determined by monitoring familiar and unfamiliar BT radio topology, trigger executable commands to smartphones, to smart devices generally, and to remote machines, and can for example, generate alerts to a user if radiotagged assets, children or pets have strayed, are lost, or are left behind. When tied to cloud predictive services or increased edge computing capacity, incipient loss of radiotag objects can be detected before an owner knows the smart object is lost. In another embodiment, XCB devices having a novel chip architecture for improved power management of cellular and MCU circuits mediated by BT radio modem that achieves “situational awareness” by monitoring the BT signal environment around the device.

25 Claims, 48 Drawing Sheets



Related U.S. Application Data

which is a continuation-in-part of application No. 16/950,666, filed on Nov. 17, 2020, now Pat. No. 11,678,141, and a continuation-in-part of application No. 16/575,315, filed on Sep. 18, 2019, now Pat. No. 11,184,858.

- (60) Provisional application No. 63/114,464, filed on Nov. 16, 2020, provisional application No. 63/108,843, filed on Nov. 2, 2020, provisional application No. 62/968,105, filed on Jan. 30, 2020, provisional application No. 62/936,588, filed on Nov. 17, 2019, provisional application No. 62/732,945, filed on Sep. 18, 2018.

(58) **Field of Classification Search**

USPC 455/3.01, 7; 370/318, 352, 252, 310,
370/329, 409, 254, 338, 216, 350

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2015/0127607 A1* 5/2015 Savage G06F 16/1873
707/693
2018/0246762 A1* 8/2018 Tarsa G06F 9/5083

* cited by examiner

FIG. 1

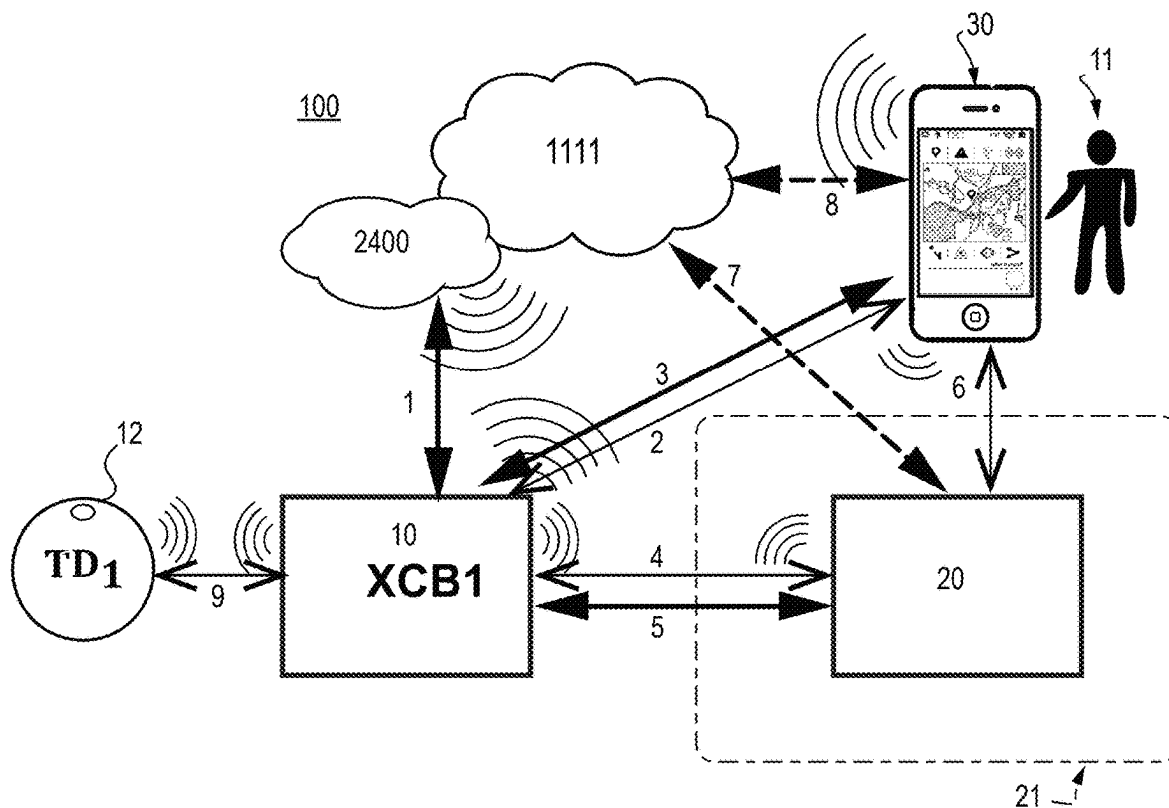


FIG. 2A

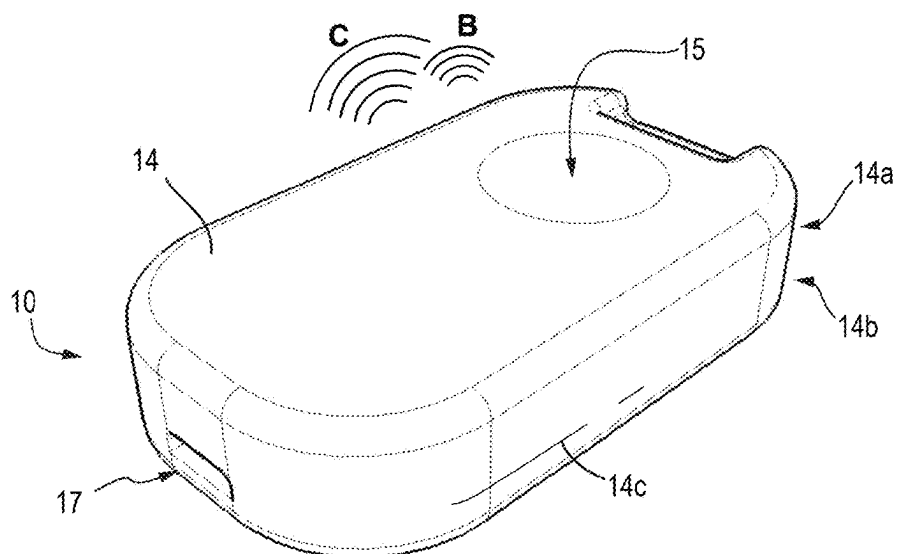


FIG. 2B

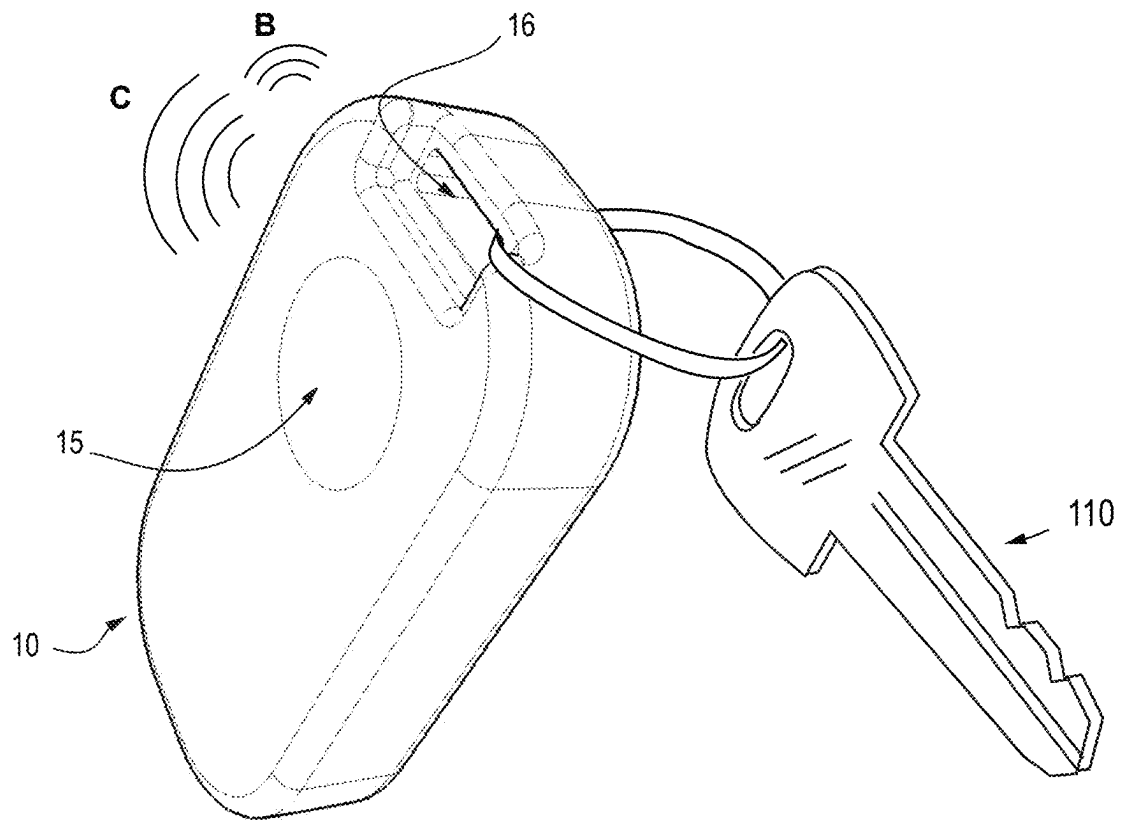
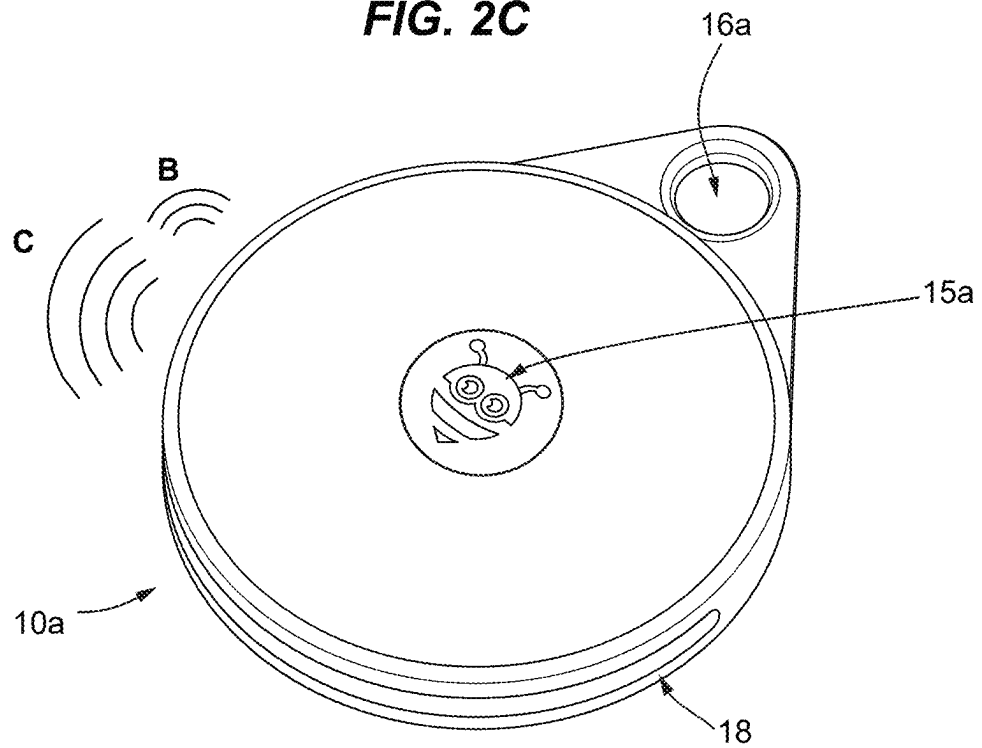
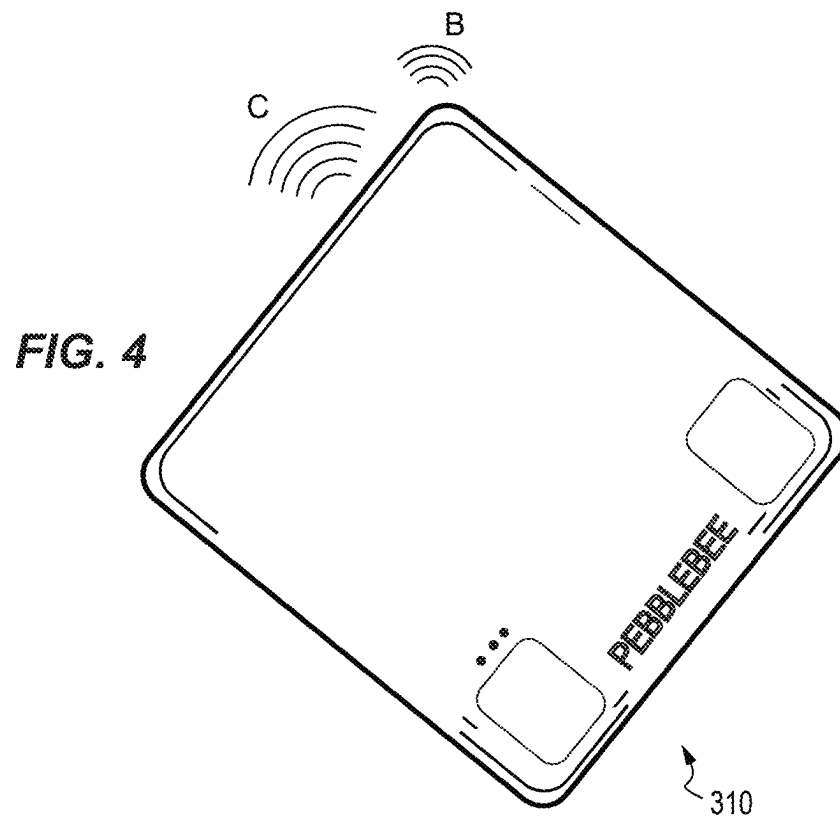
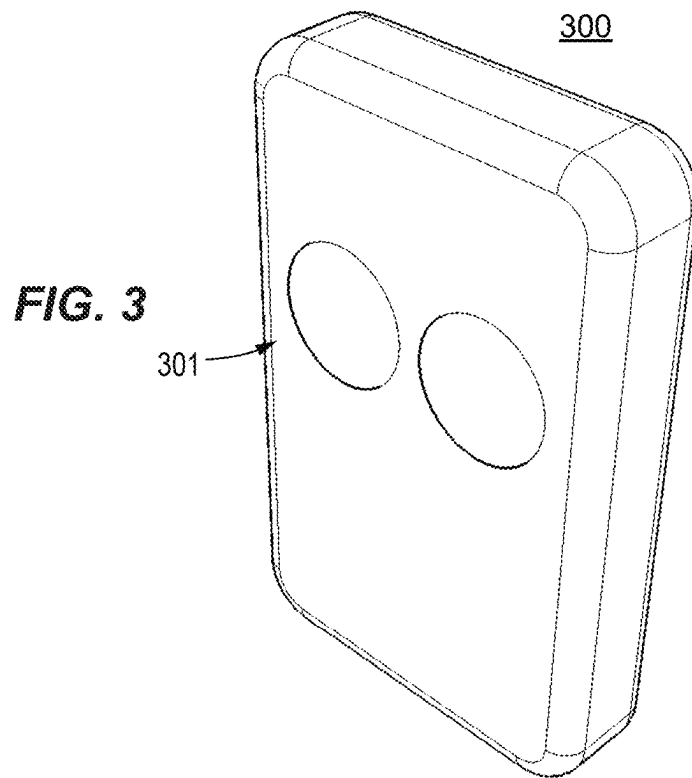


FIG. 2C





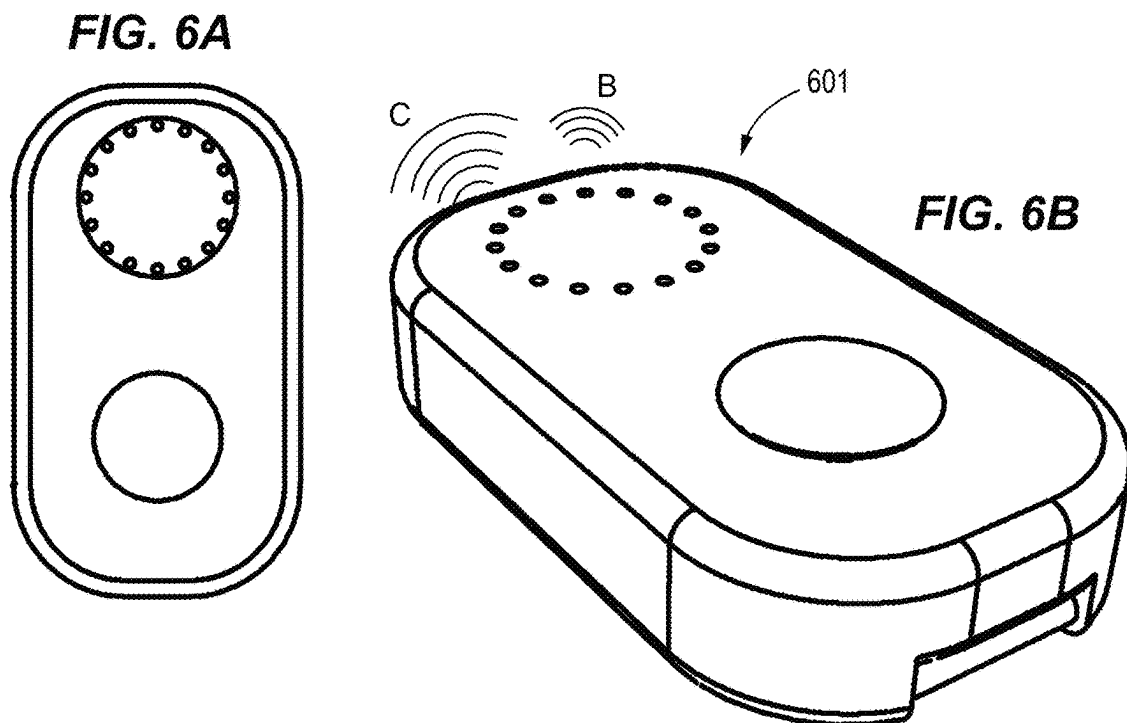
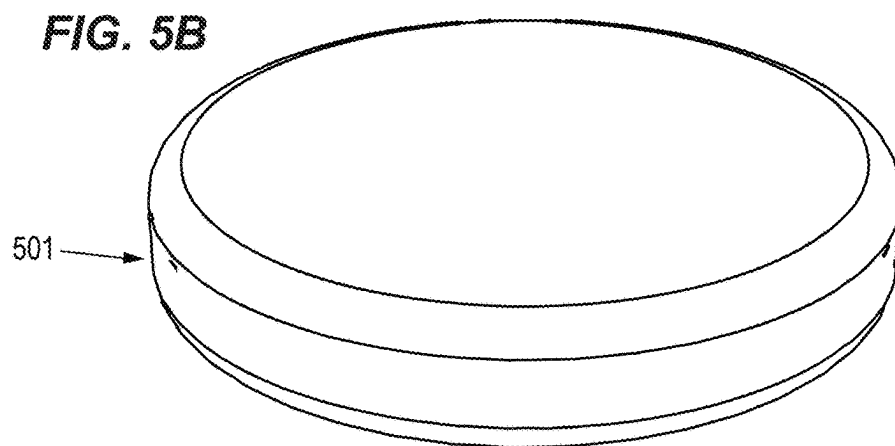
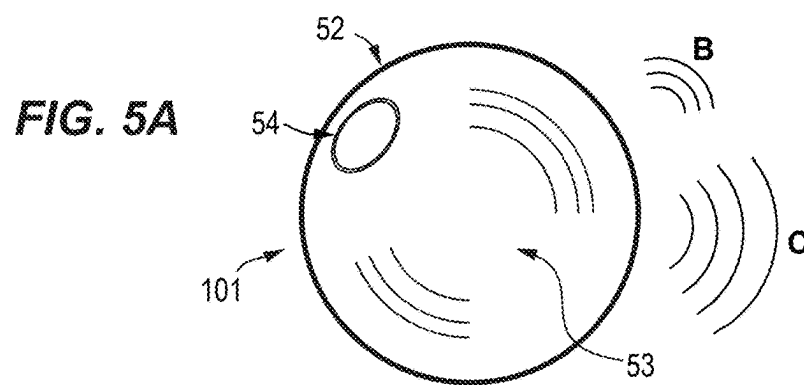


FIG. 6C

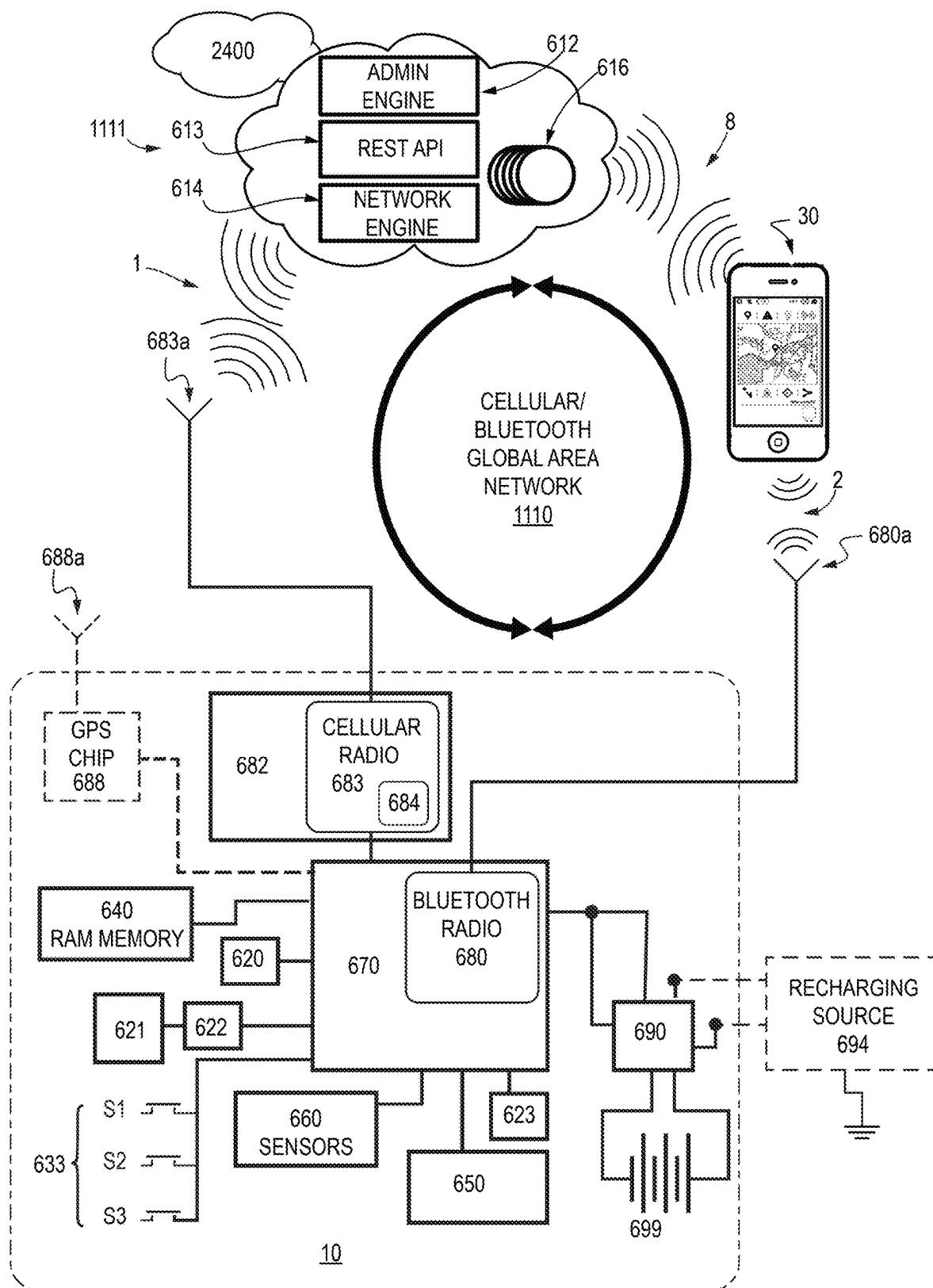


FIG. 7A

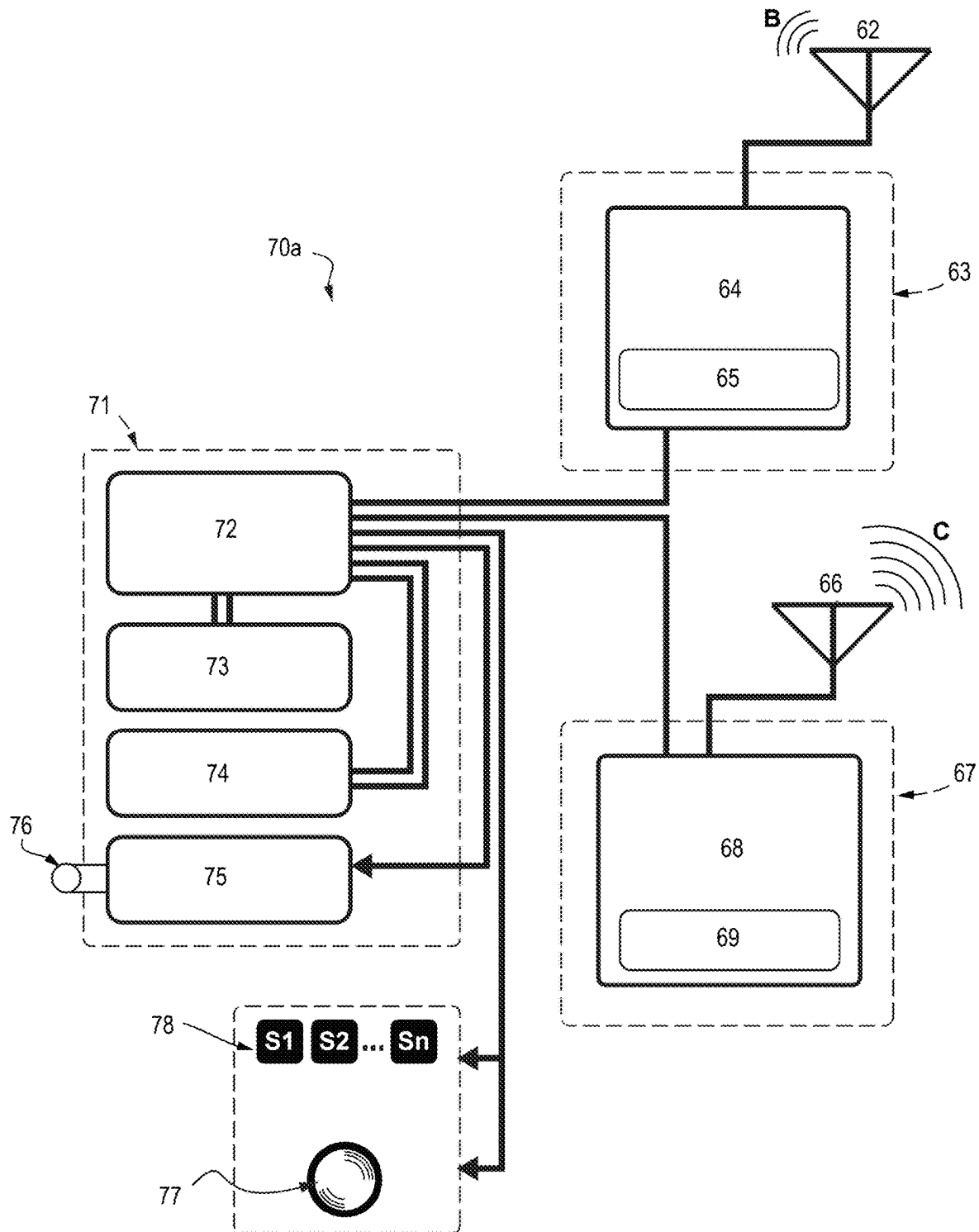


FIG. 7B

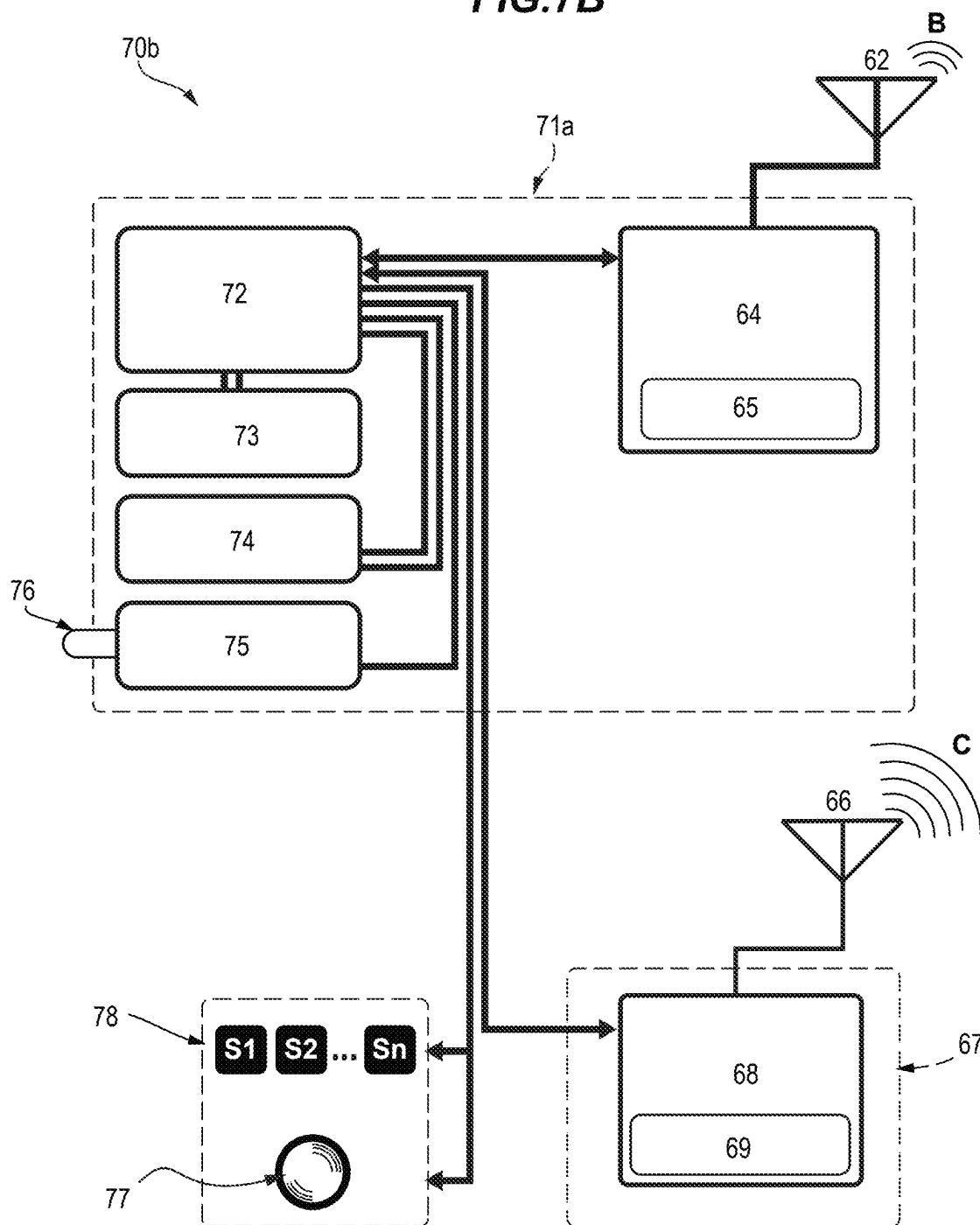


FIG. 7C

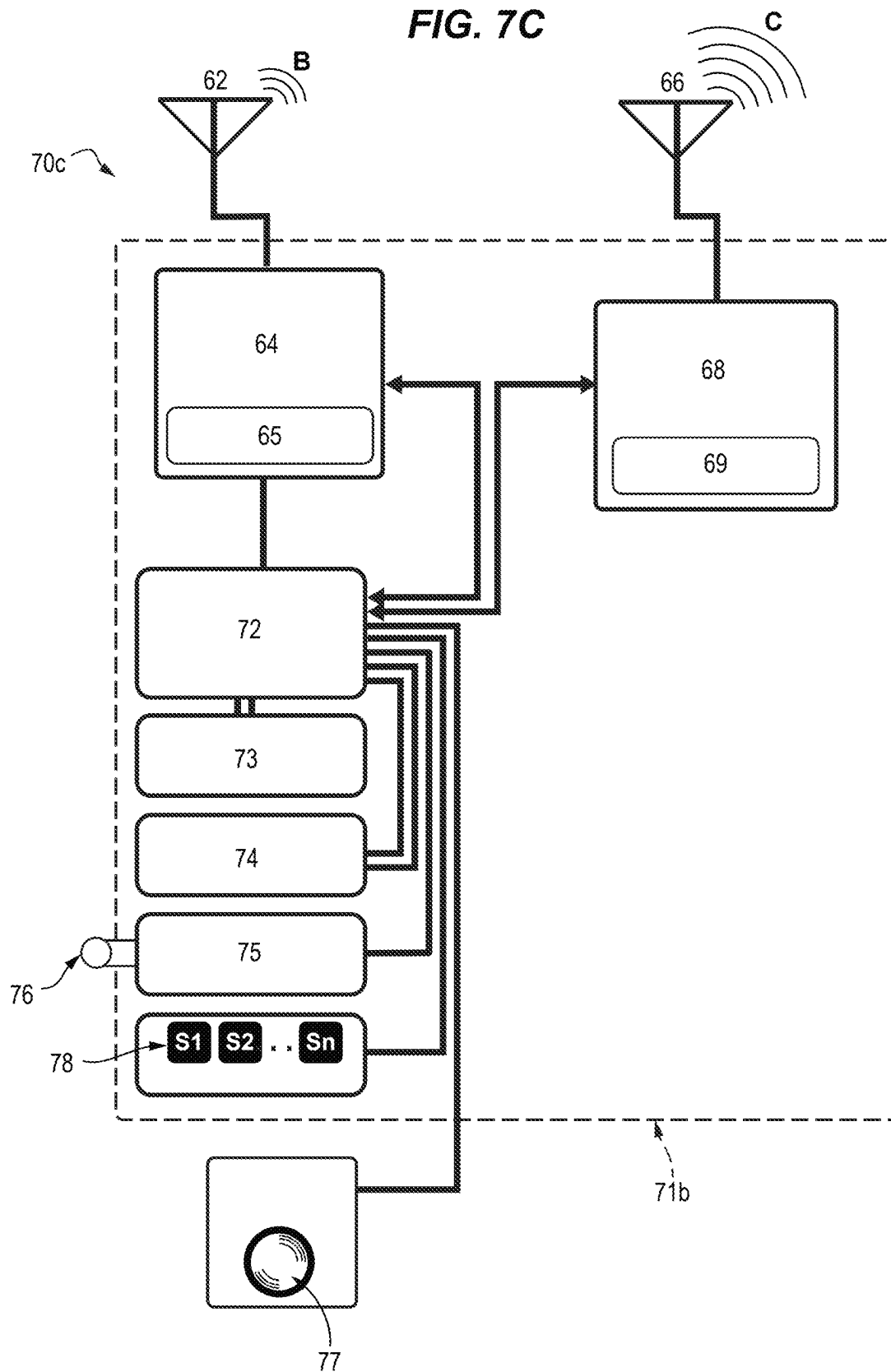


Fig. 8A

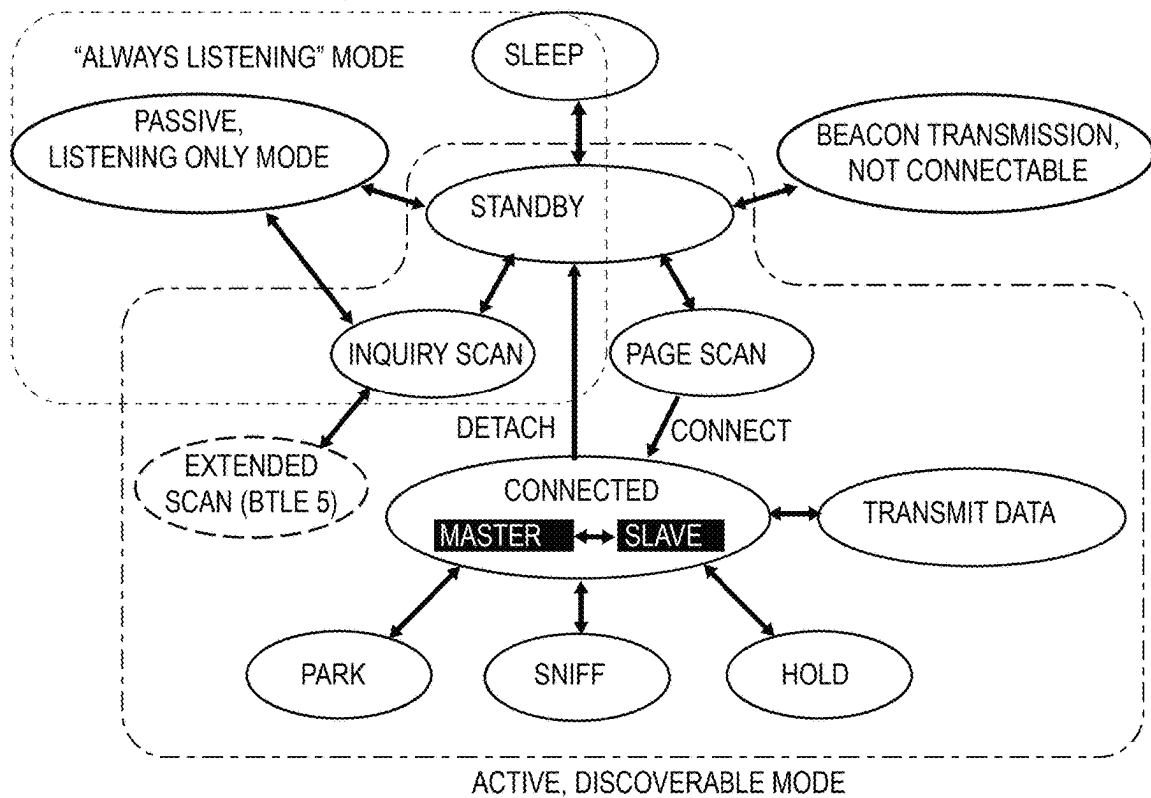


Fig. 8B

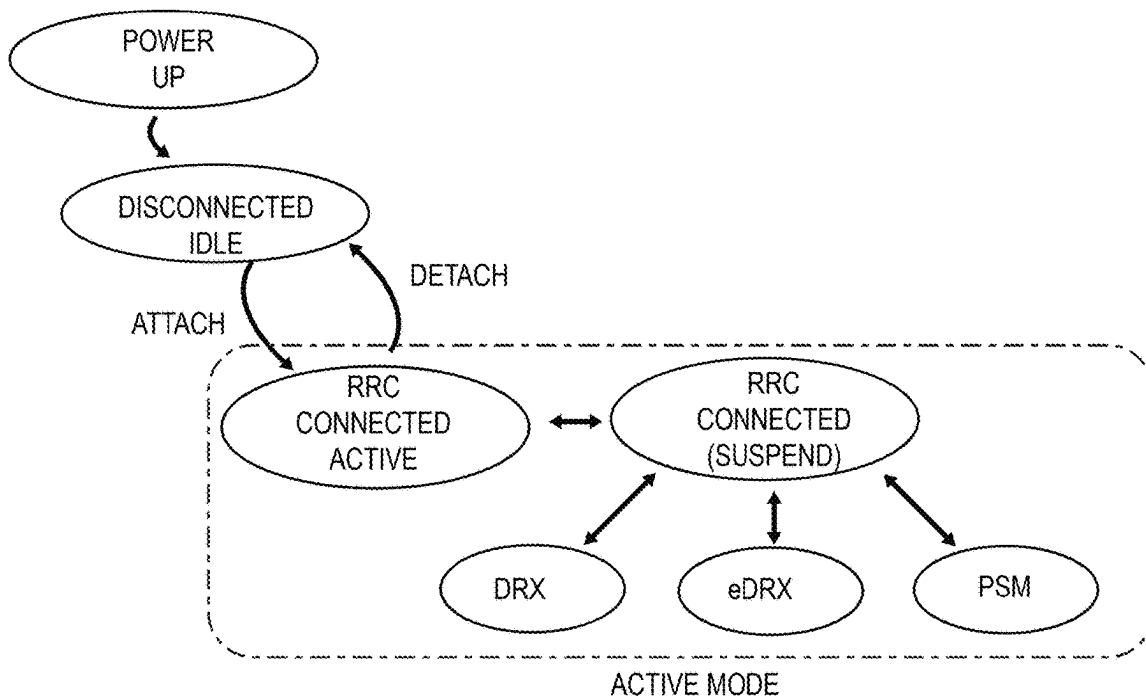


FIG. 9

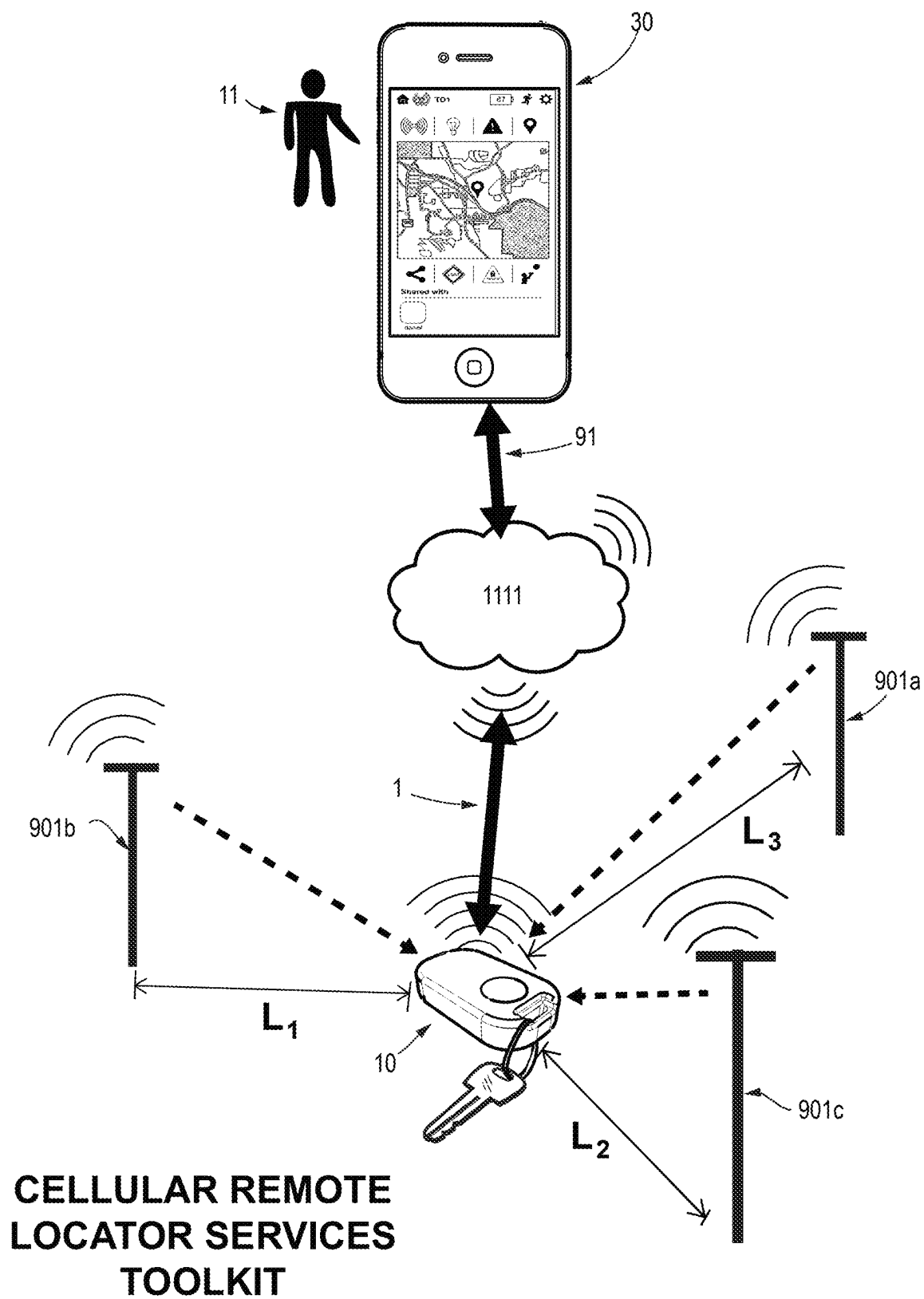


FIG. 10

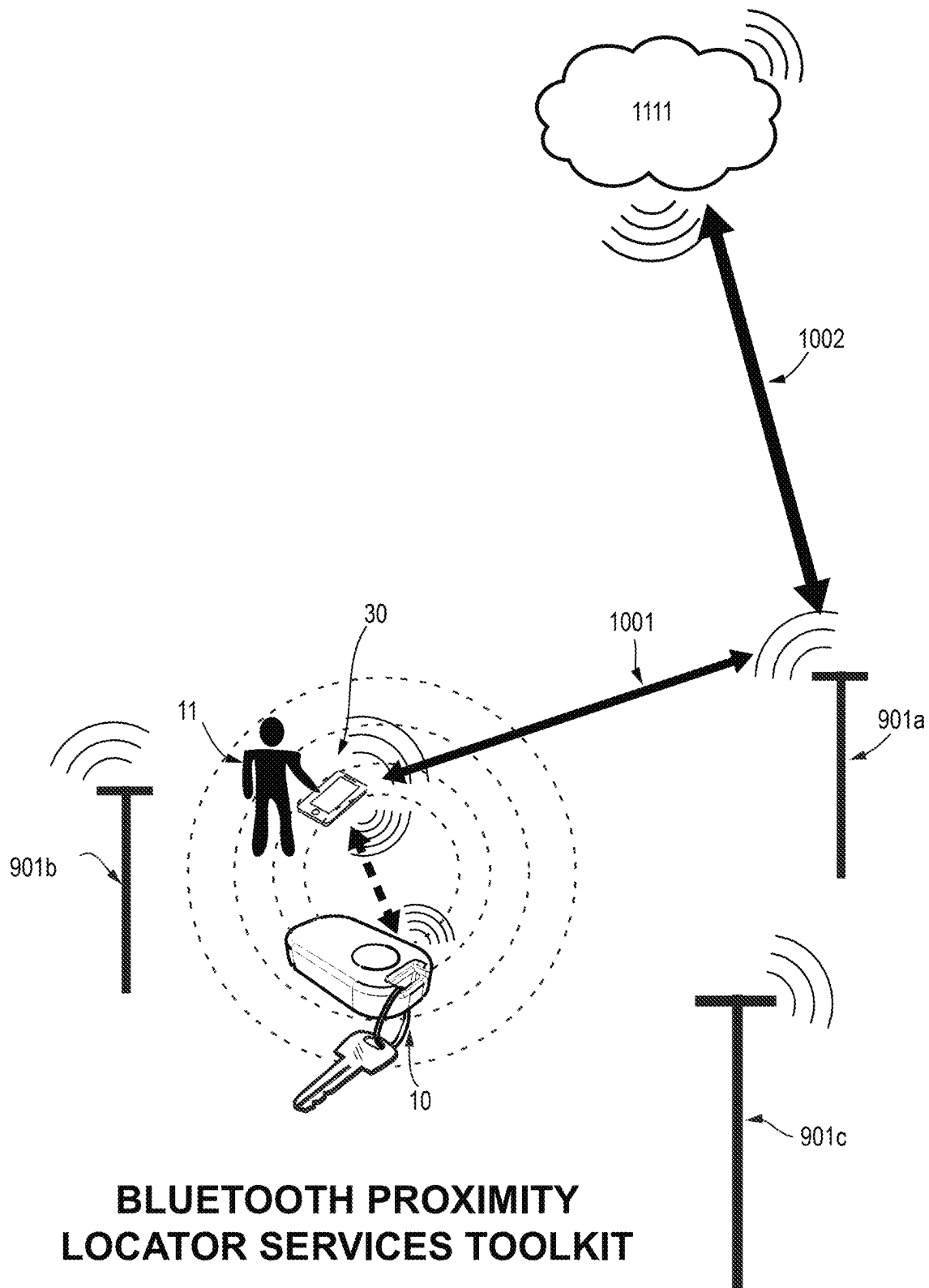


FIG. 11**FOR ANY RADIOTAG:SMARTPHONE PAIR**

TIME 0 **LOCATION KNOWN?** **YES or NO?**

IF LOCATION KNOWN "YES", THEN IN "SAFE ZONE"?

TIME 0 **IN SAFE ZONE?** **YES NO IF YES, THEN SLEEP**

**IF NOT IN SAFE ZONE OR NO LOCATION, THEN CK MOTION.
COMPARE MOTION OF RADIOTAG VERSUS SMARTPHONE:**

MOTION (Truth Table)

TAG	PHONE	
F	F	GO TO 1
F	T	GO TO 2
T	T	GO TO 3
T	F	GO TO 3

1. ENTER OR CONTINUE IN SLEEP MODE

**2. GENERATE "LEFT BEHIND" ALERT AND "CALL HOME"
IF RADIO TETHER IS BROKEN**

3. CHECK PROXIMITY CHANGE ($dRSSI/dT$)

TIME (dT) **PROXIMITY CHANGE ($dRSSI/dT$)***

~ > <

* ~: Unchanged; > Strengthening; < Fading

IF UNCHANGED, THEN SLEEP.

**IF FADING OR LOST RSSI, THEN GENERATE "LOST" ALERT
AND "CALL HOME".**

IF STRENGTHENING RSSI, LOOP $T=T+t$ AND ReCK MOTION

FIG. 12

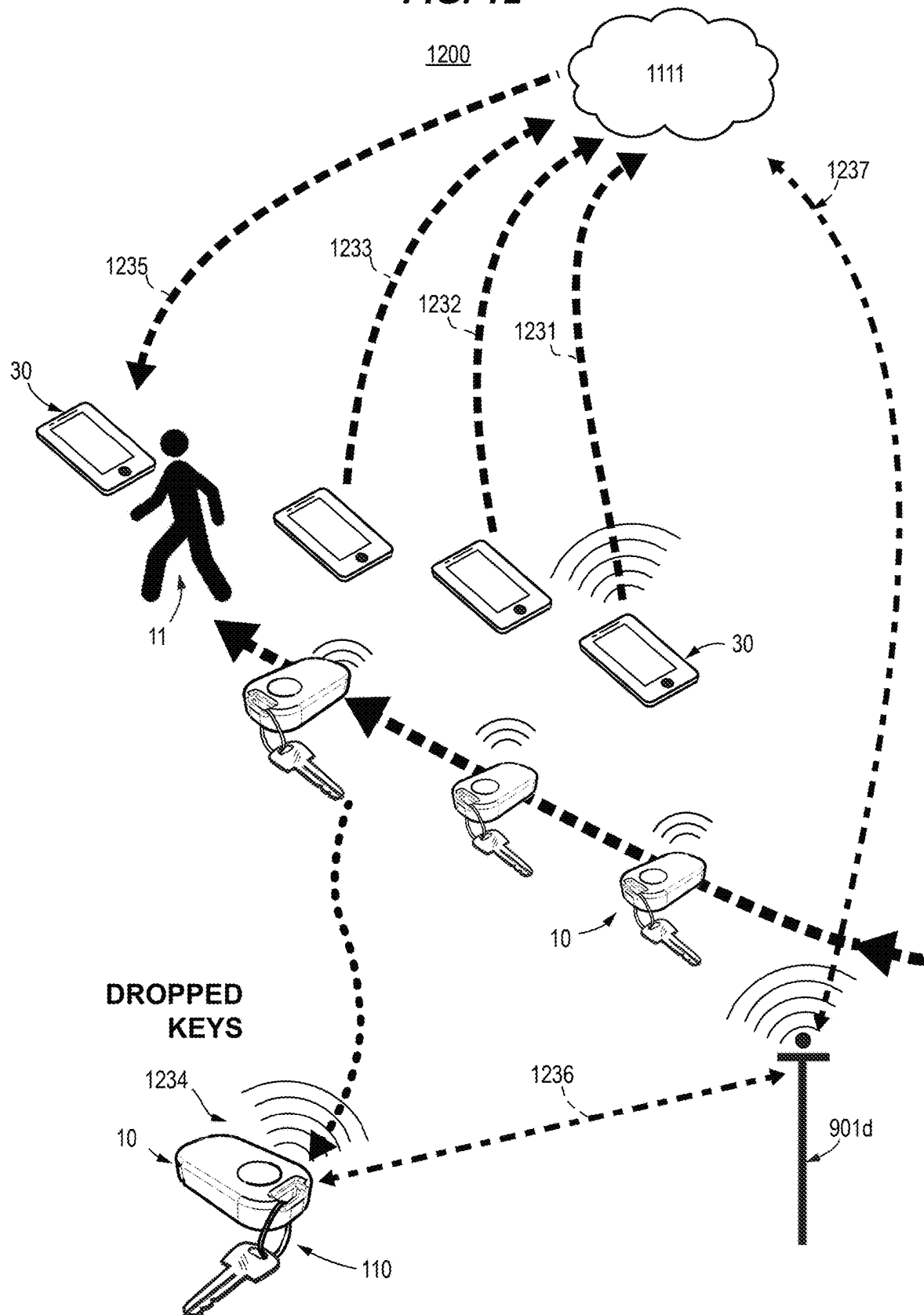


FIG. 13A

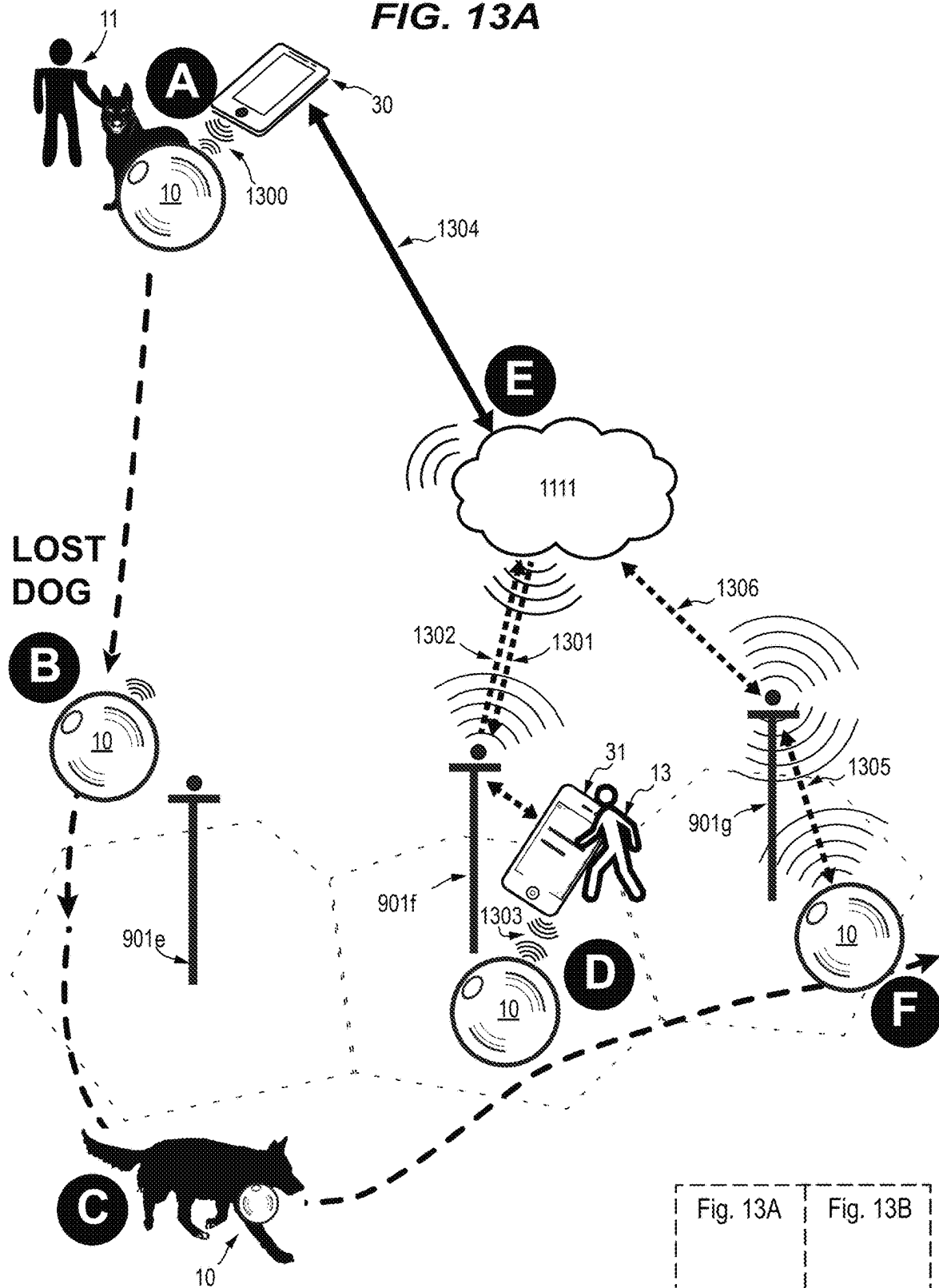


FIG. 13B

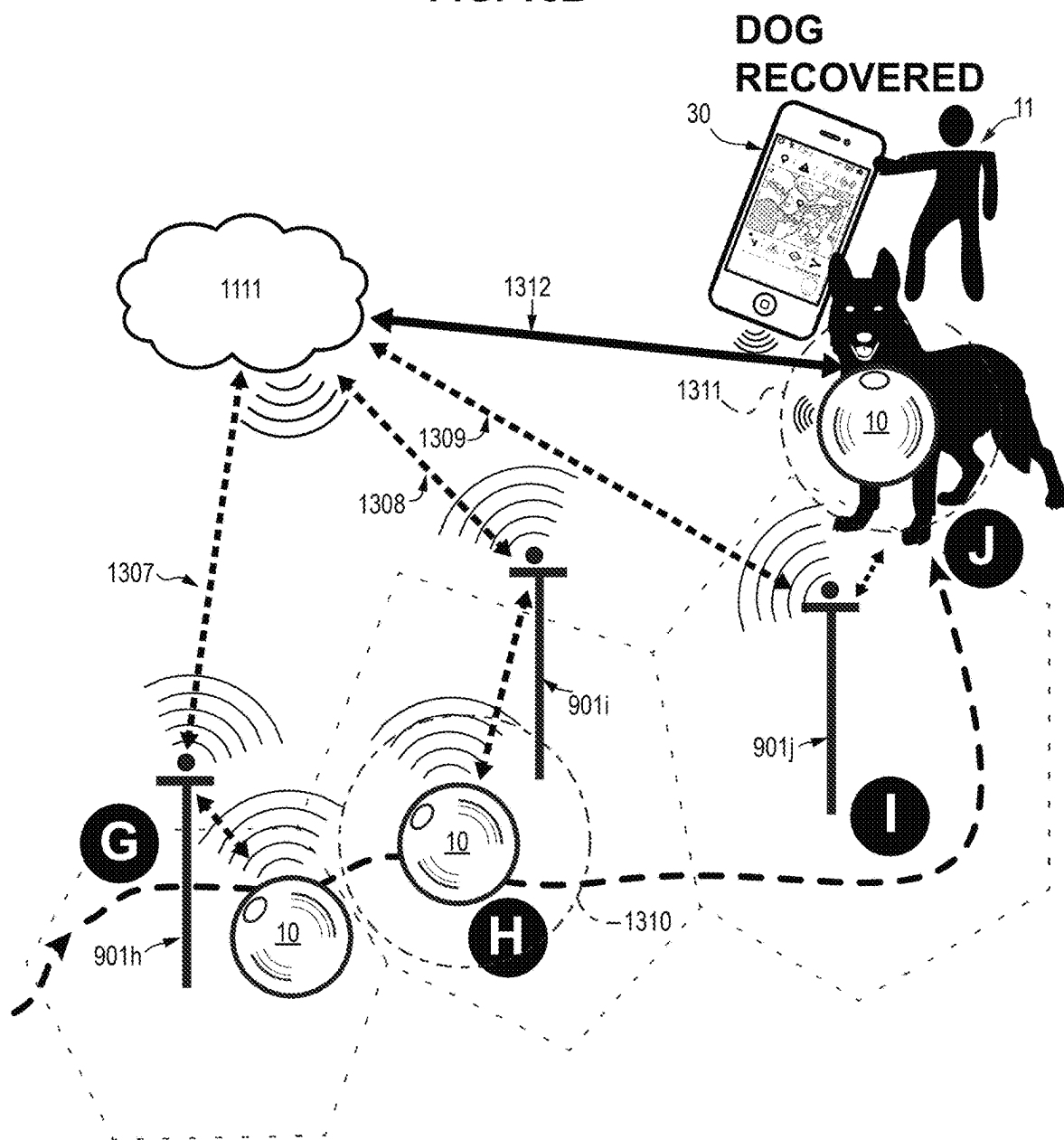


Fig. 13A	Fig. 13B
----------	----------

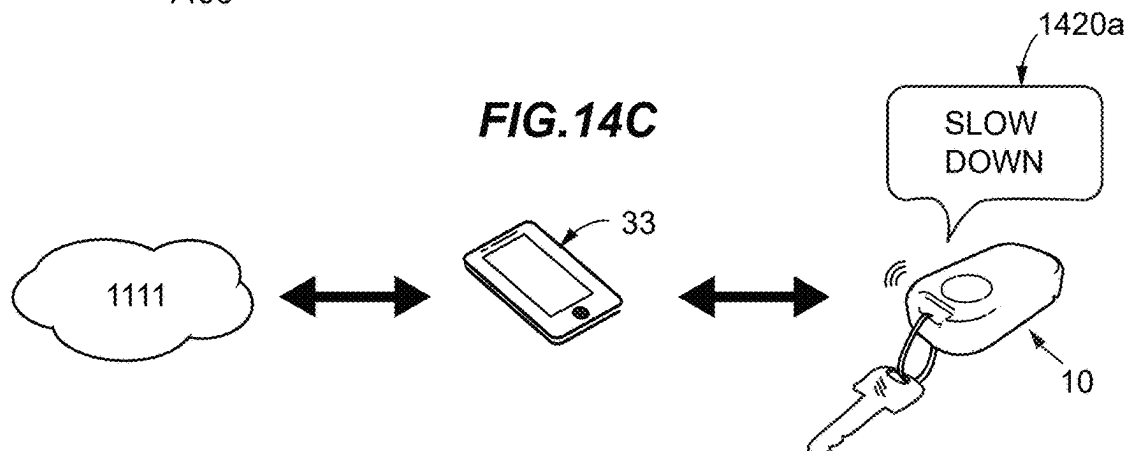
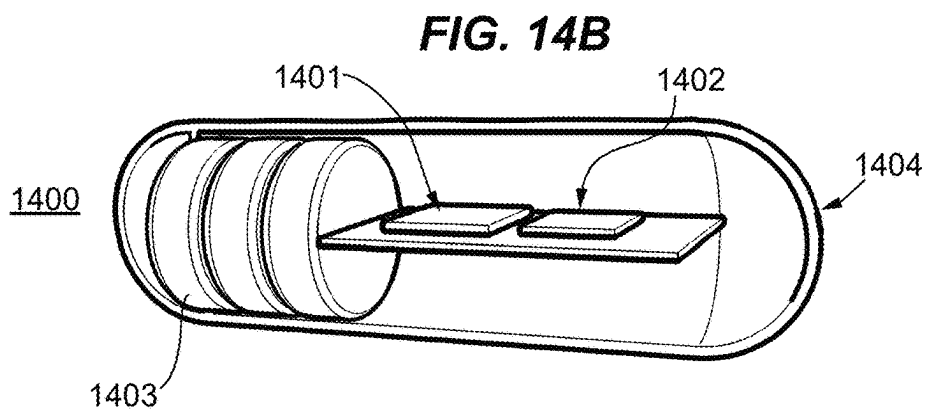
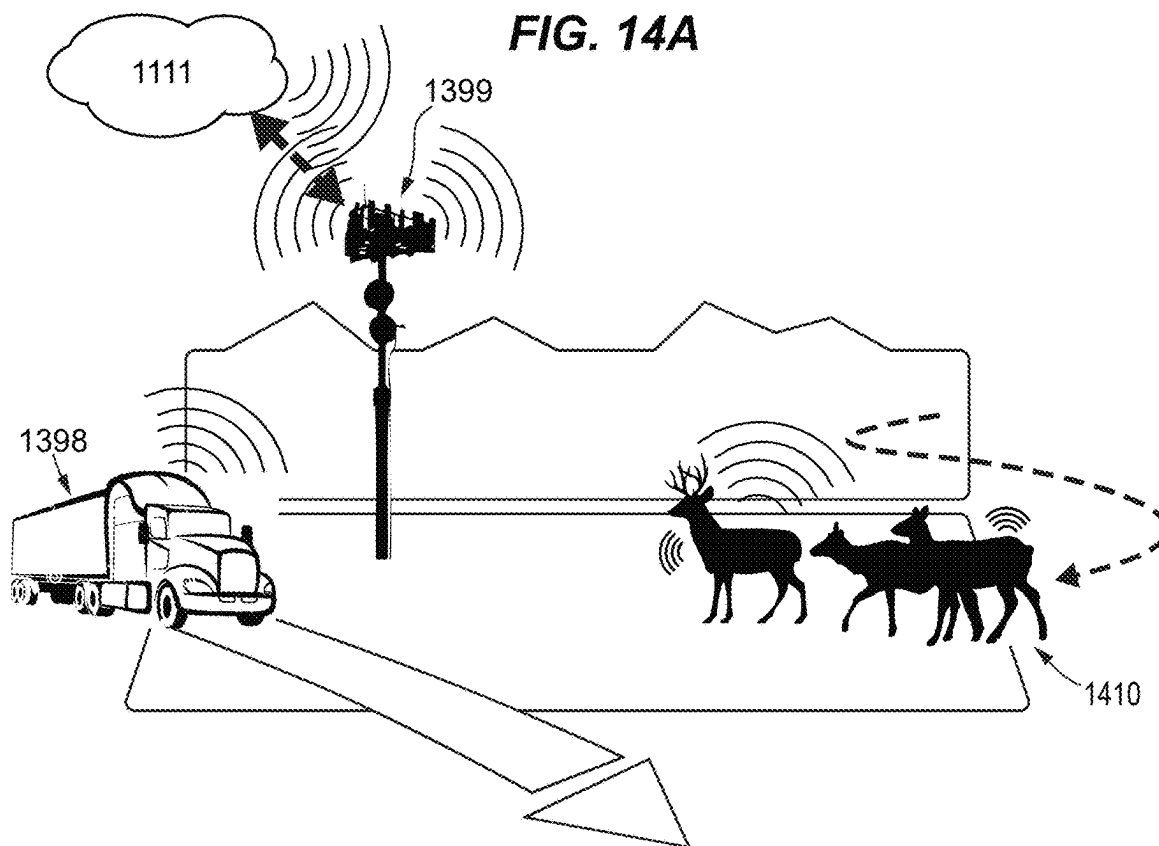


FIG. 15A

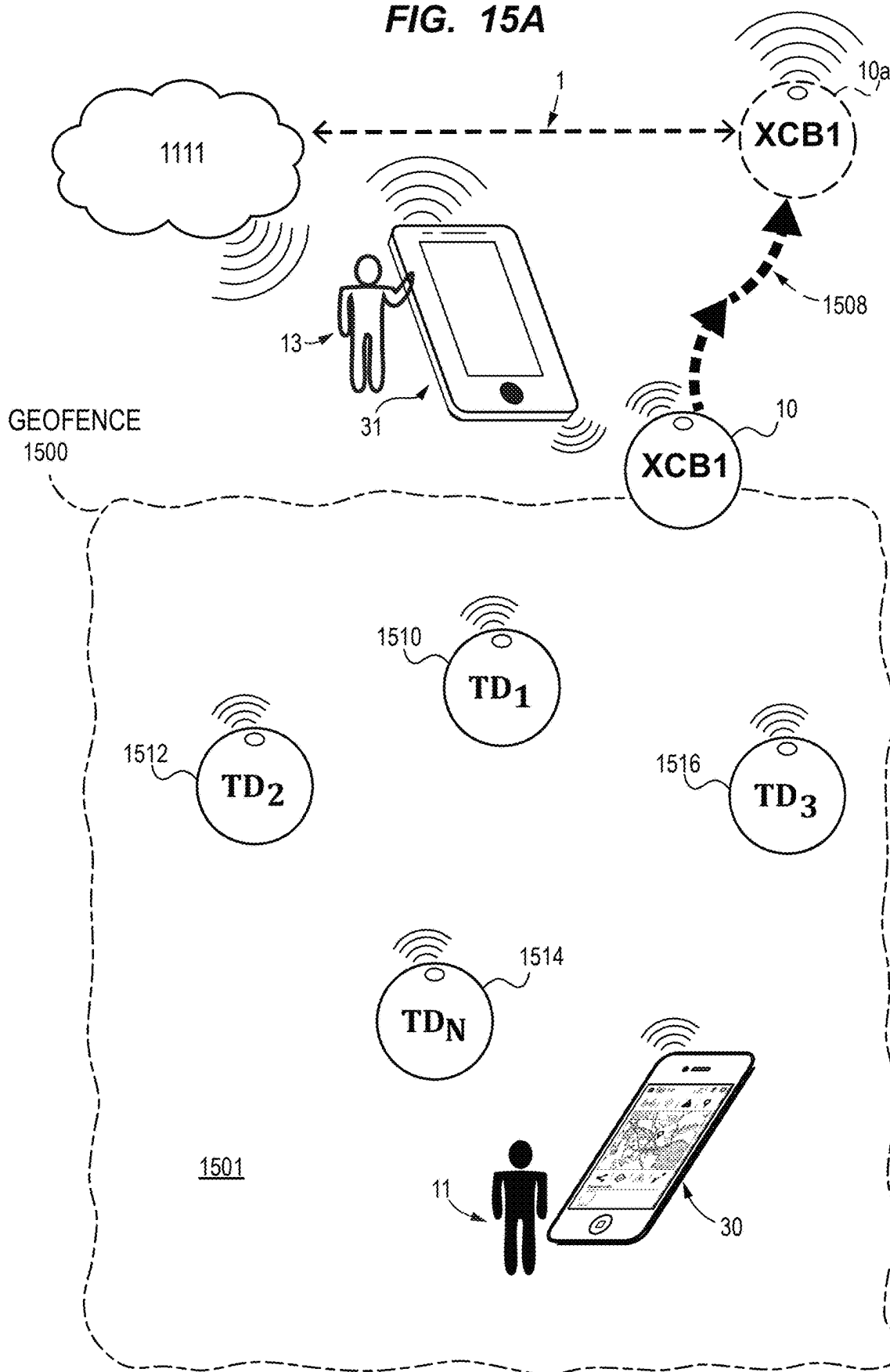


FIG. 15B

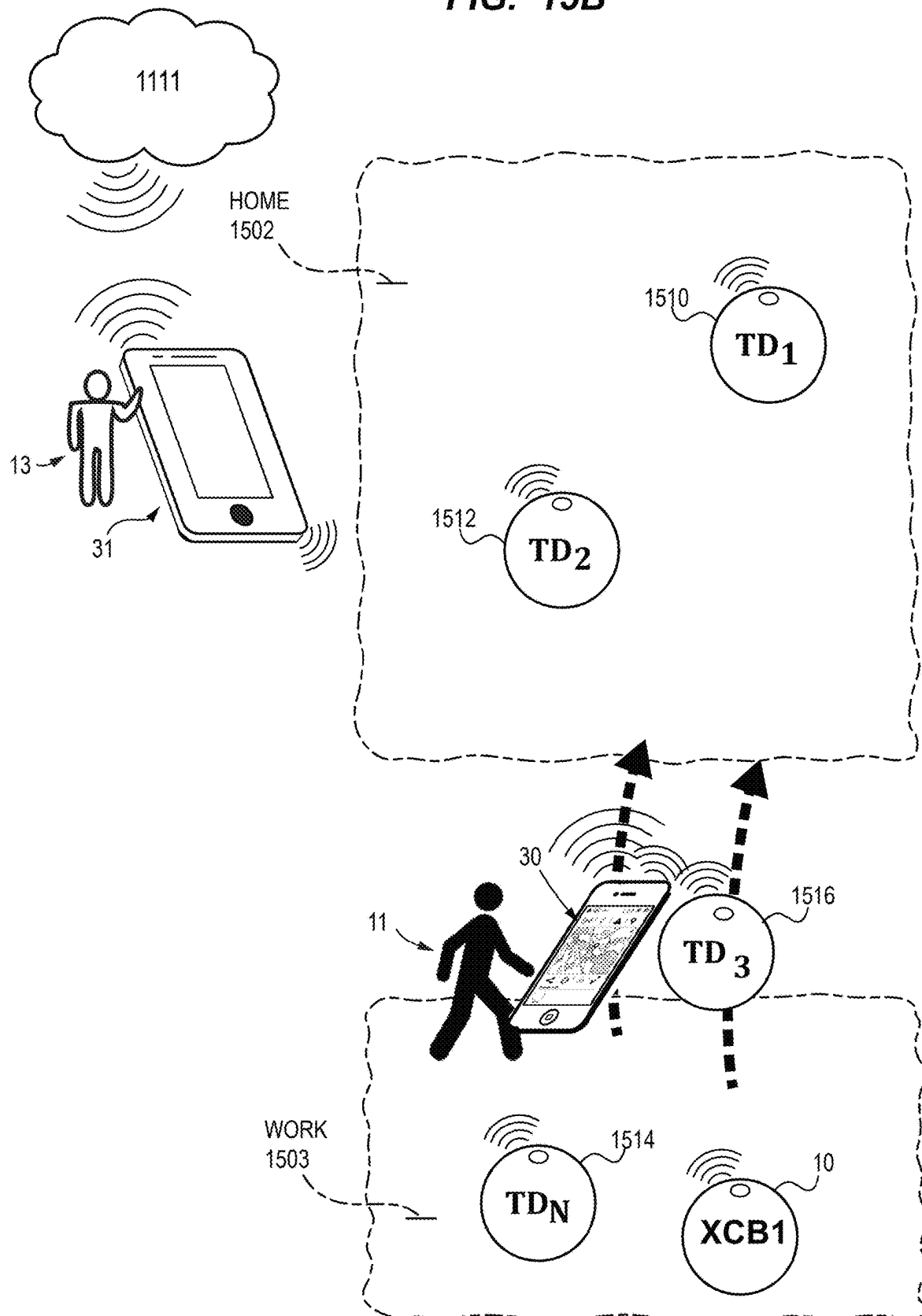


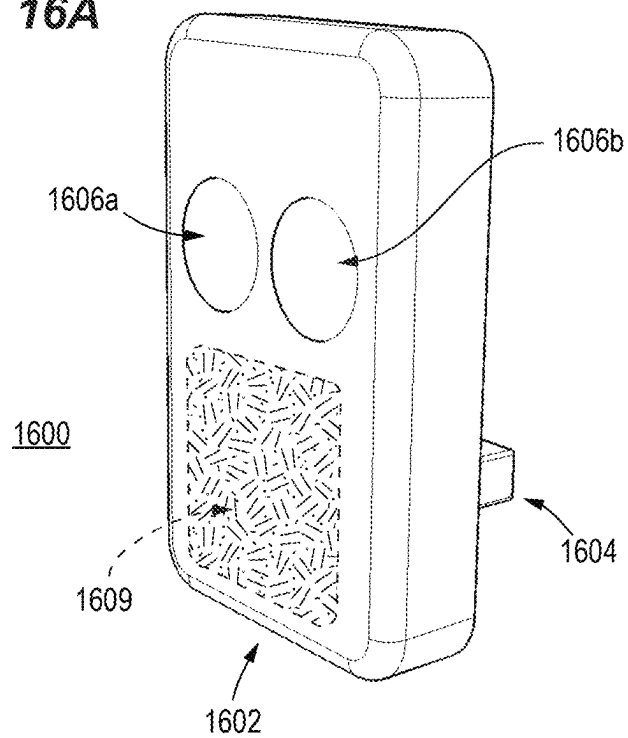
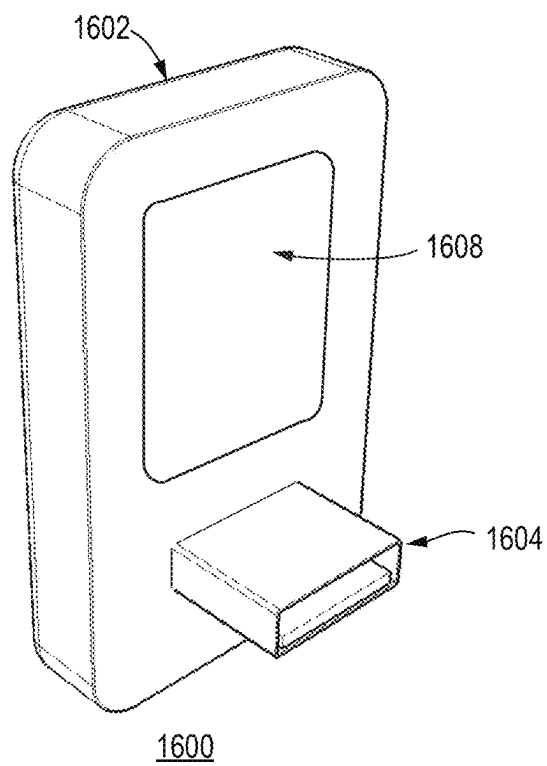
FIG. 16A**FIG. 16B**

FIG. 17

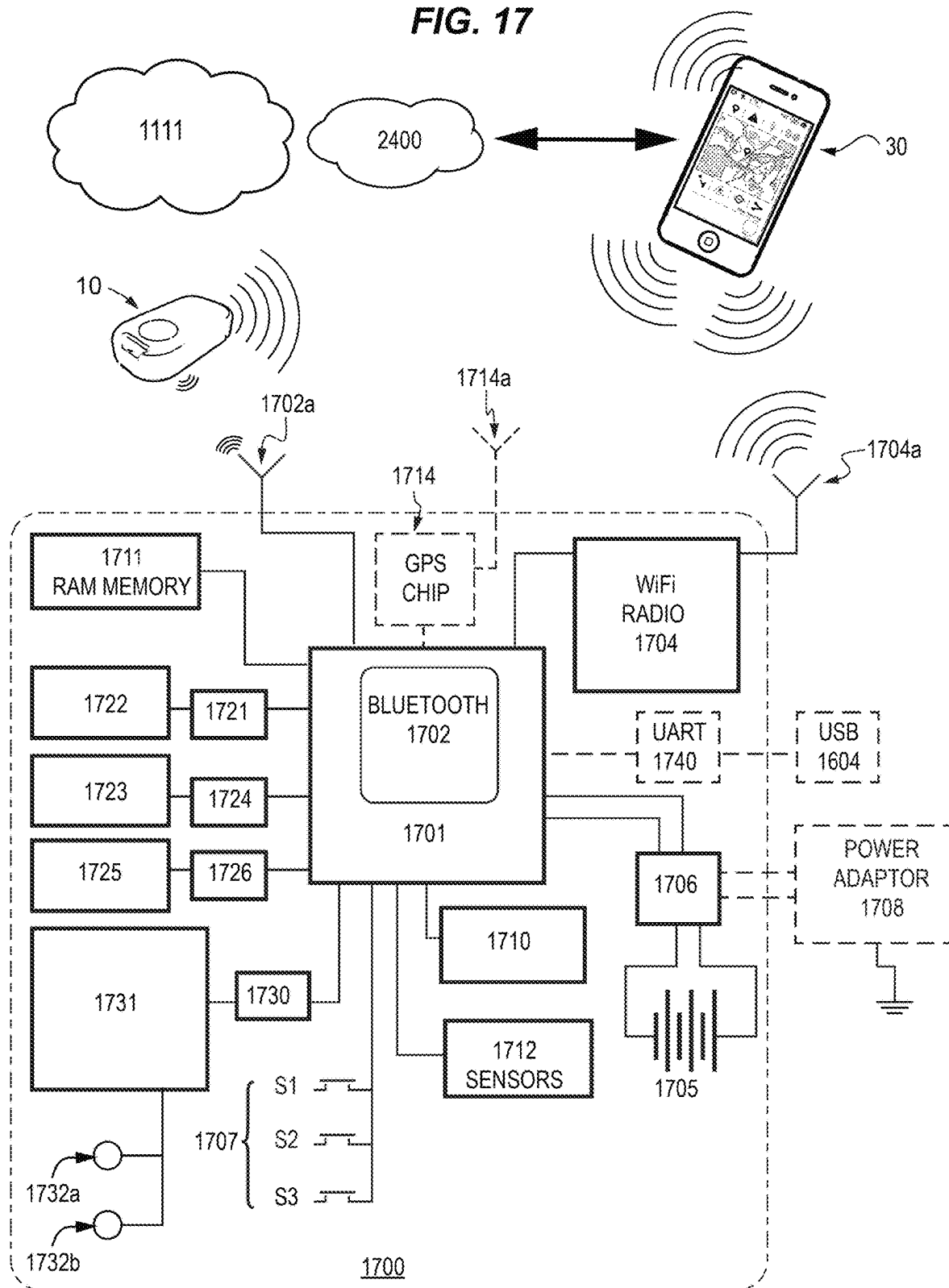


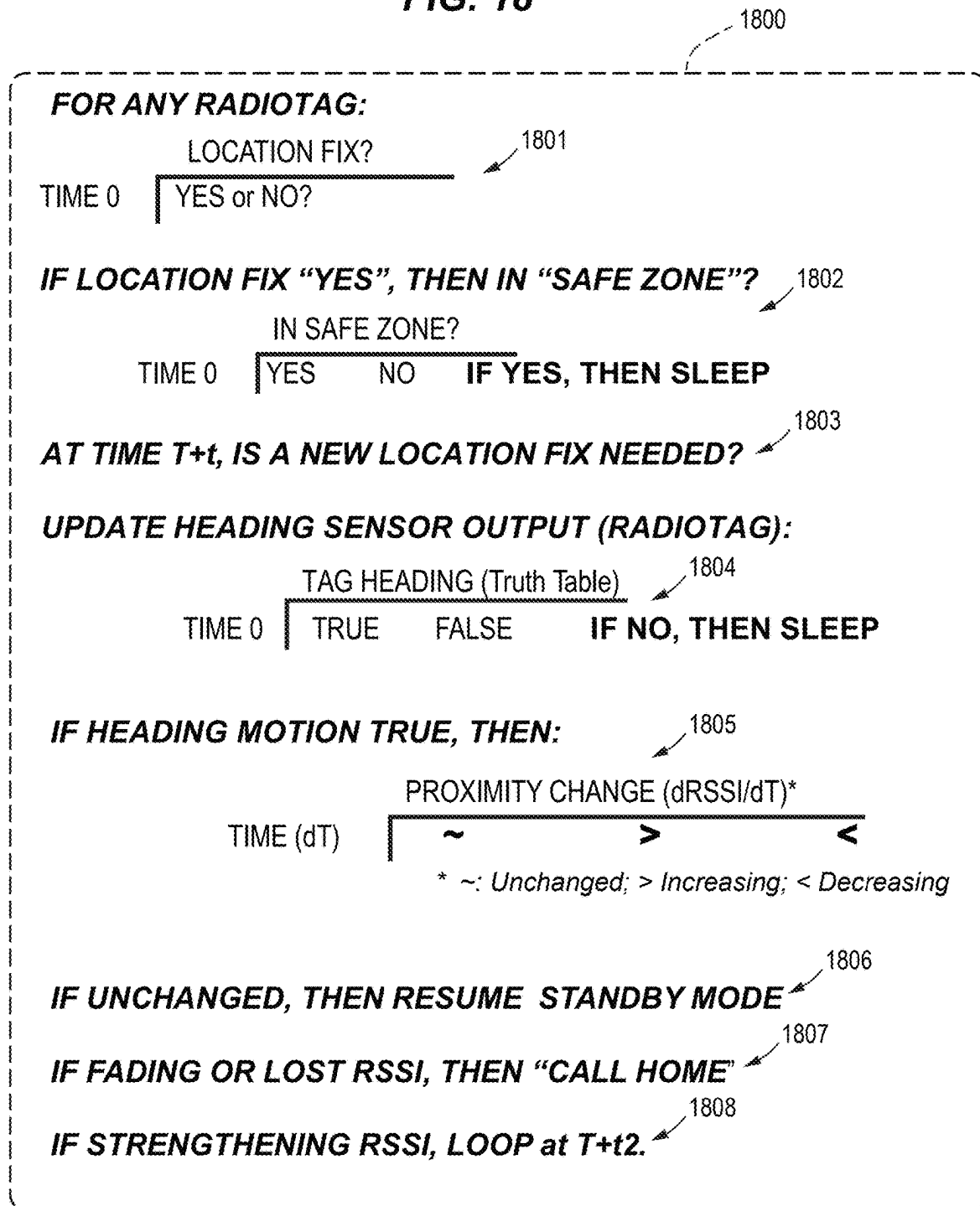
FIG. 18

FIG. 19

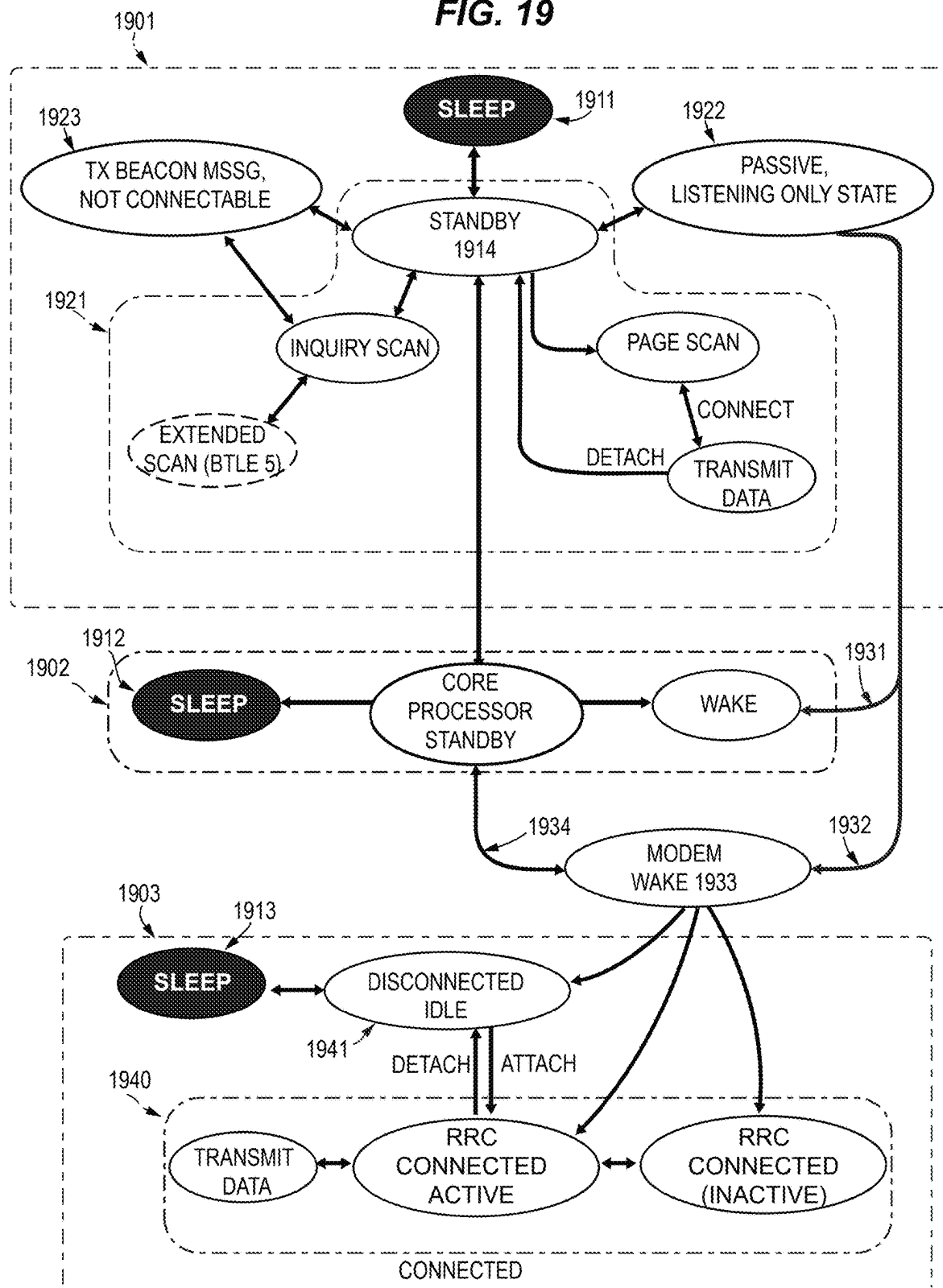


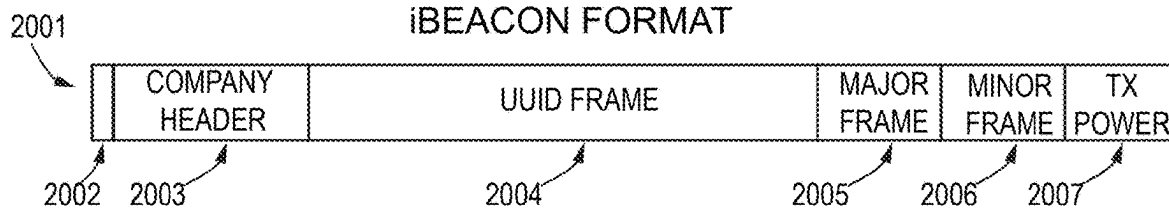
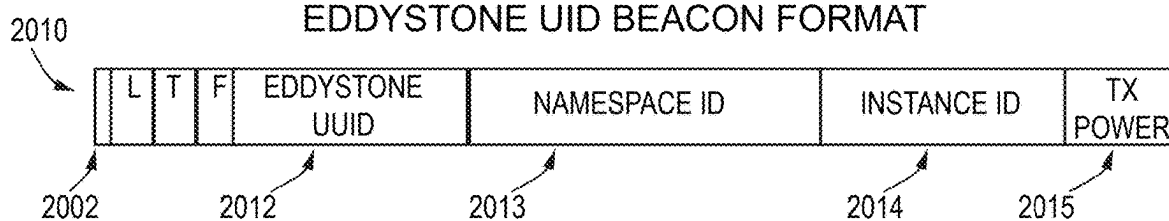
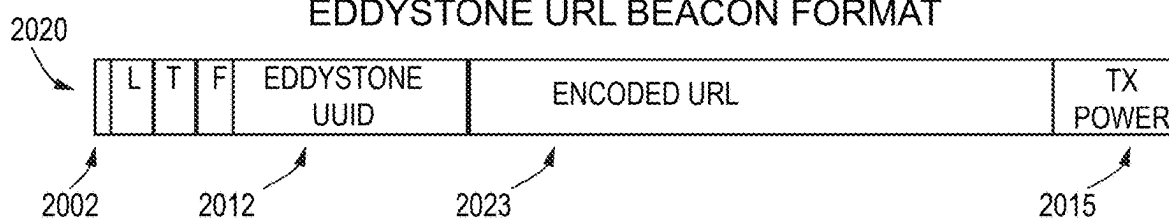
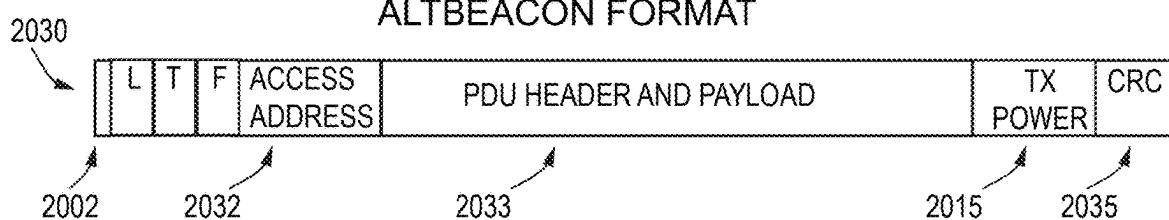
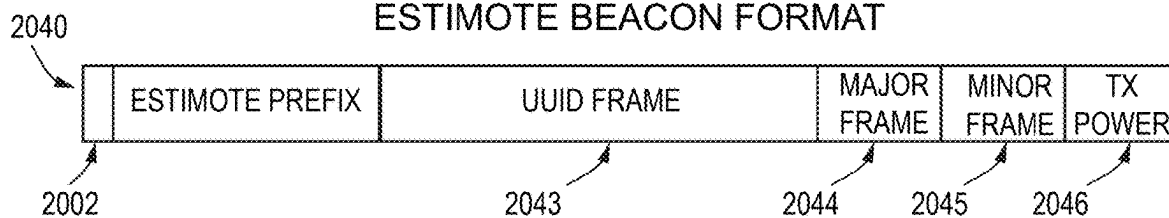
FIG. 20A**iBEACON FORMAT****FIG. 20B****EDDYSTONE UID BEACON FORMAT****FIG. 20C****EDDYSTONE URL BEACON FORMAT****FIG. 20D****ALTBEACON FORMAT****FIG. 20E****ESTIMATE BEACON FORMAT**

FIG. 21

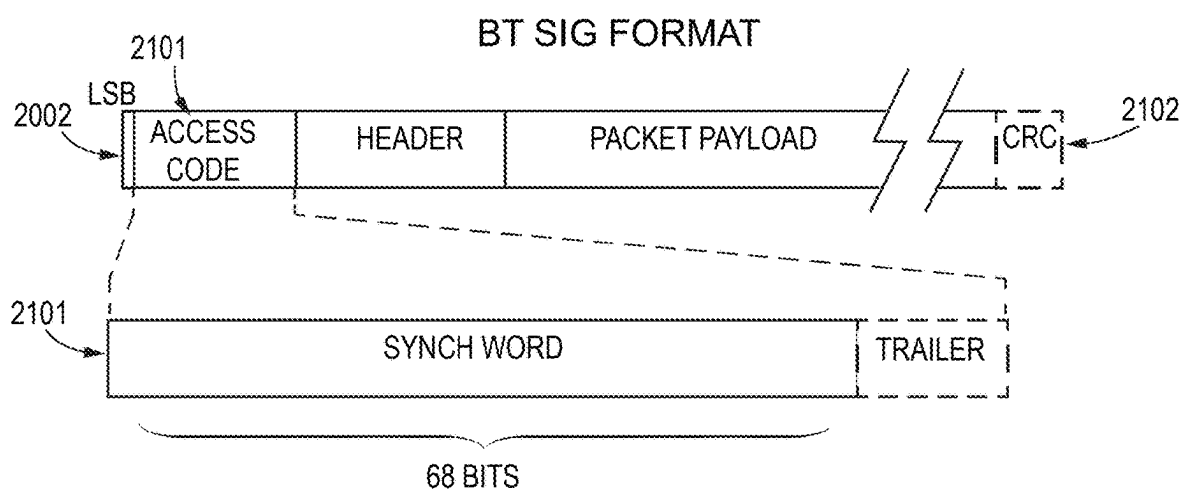


FIG. 22A

**BT INQUIRY PACKET
(ID PACKET)**

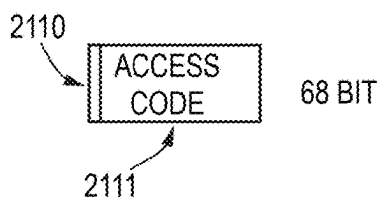


FIG. 22B

BT POLL or NULL PACKET

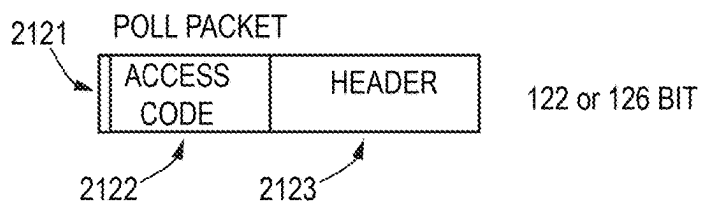


FIG. 22C

BT FSH PACKET

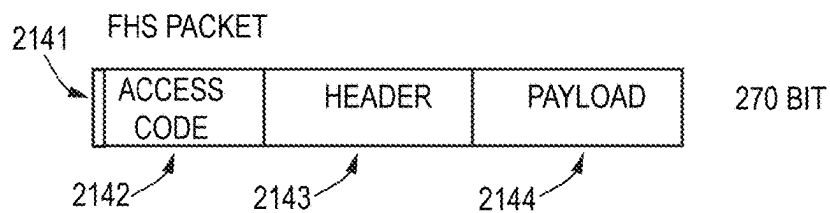


FIG. 23
BT ADVERTISING PACKET DECOMPOSITION

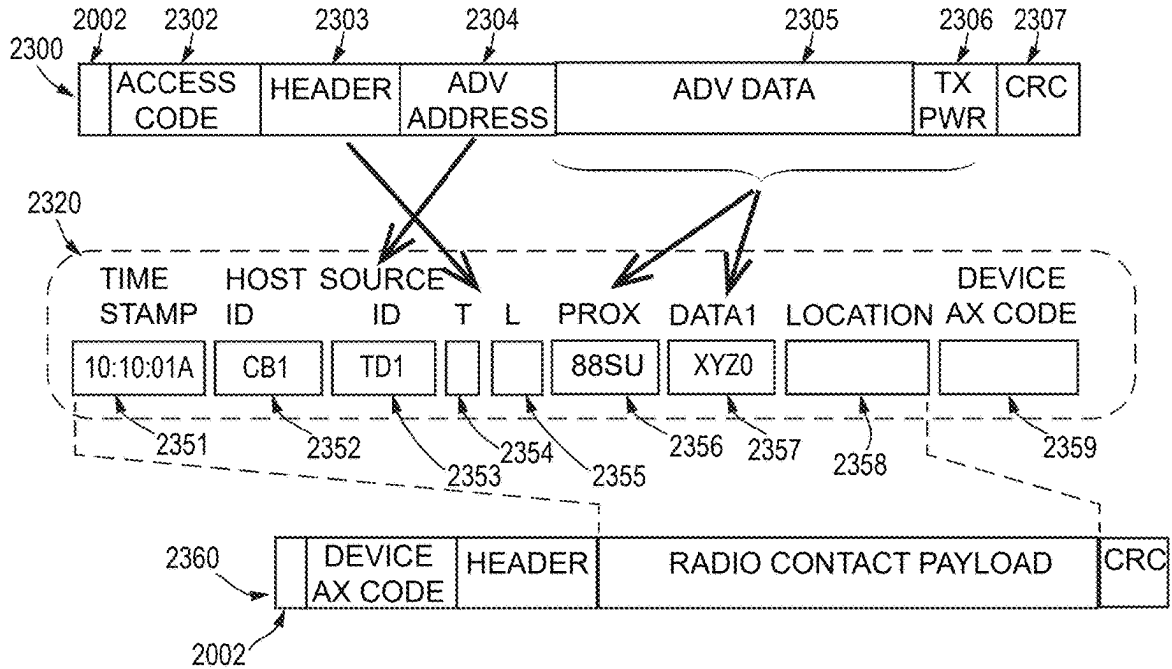


FIG. 24
BT CONNECTION REQUEST PACKET

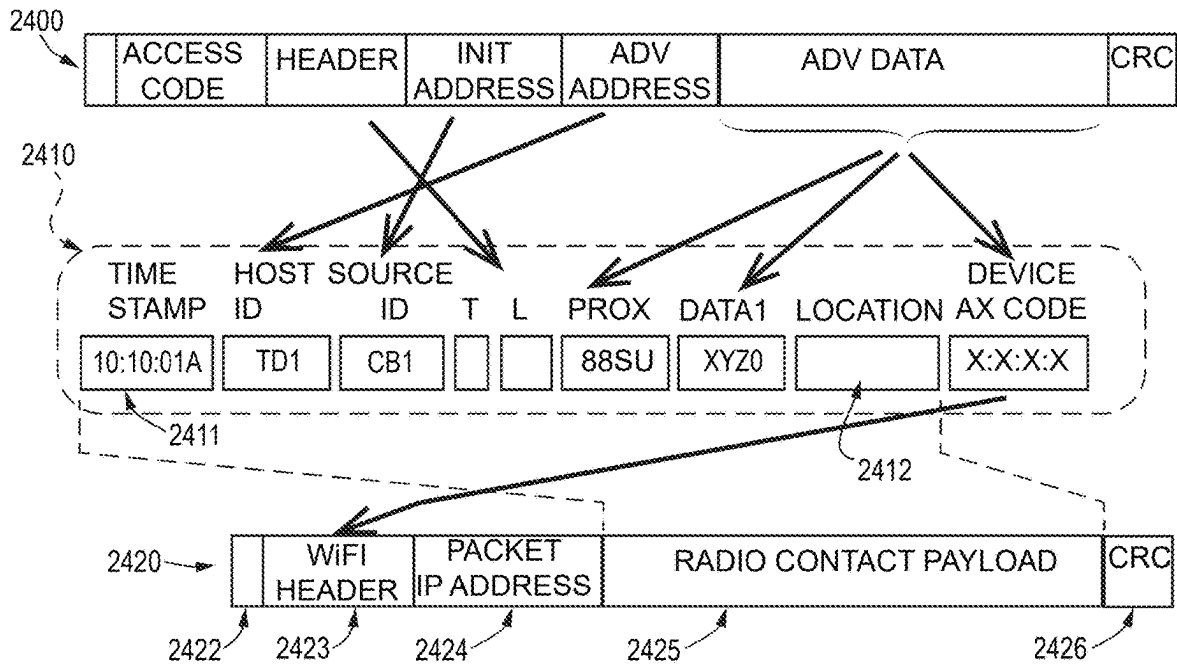


FIG. 25

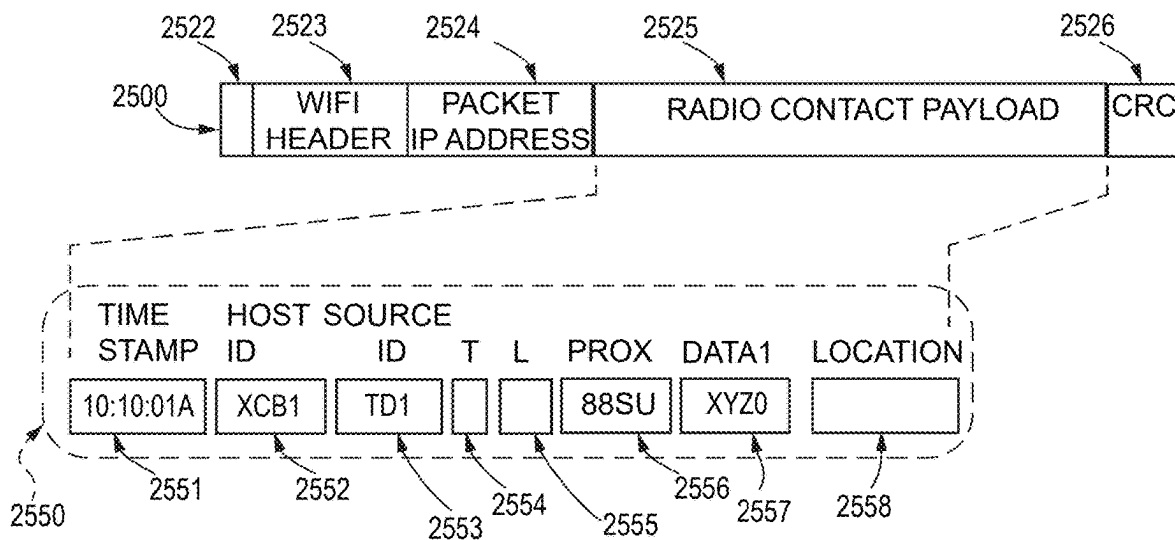


FIG. 26

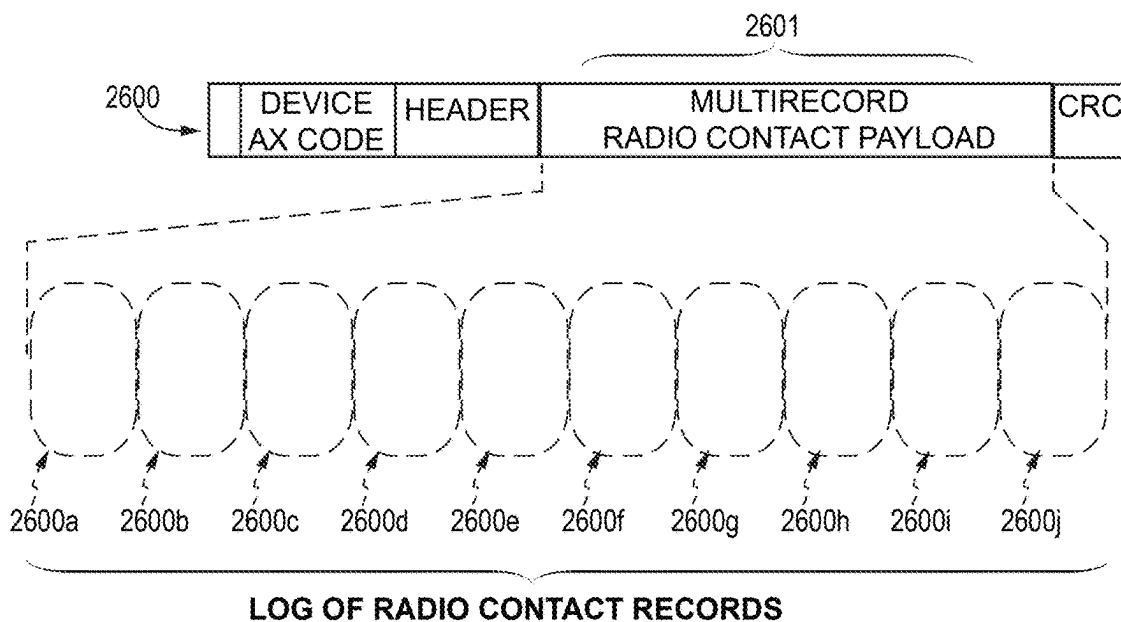


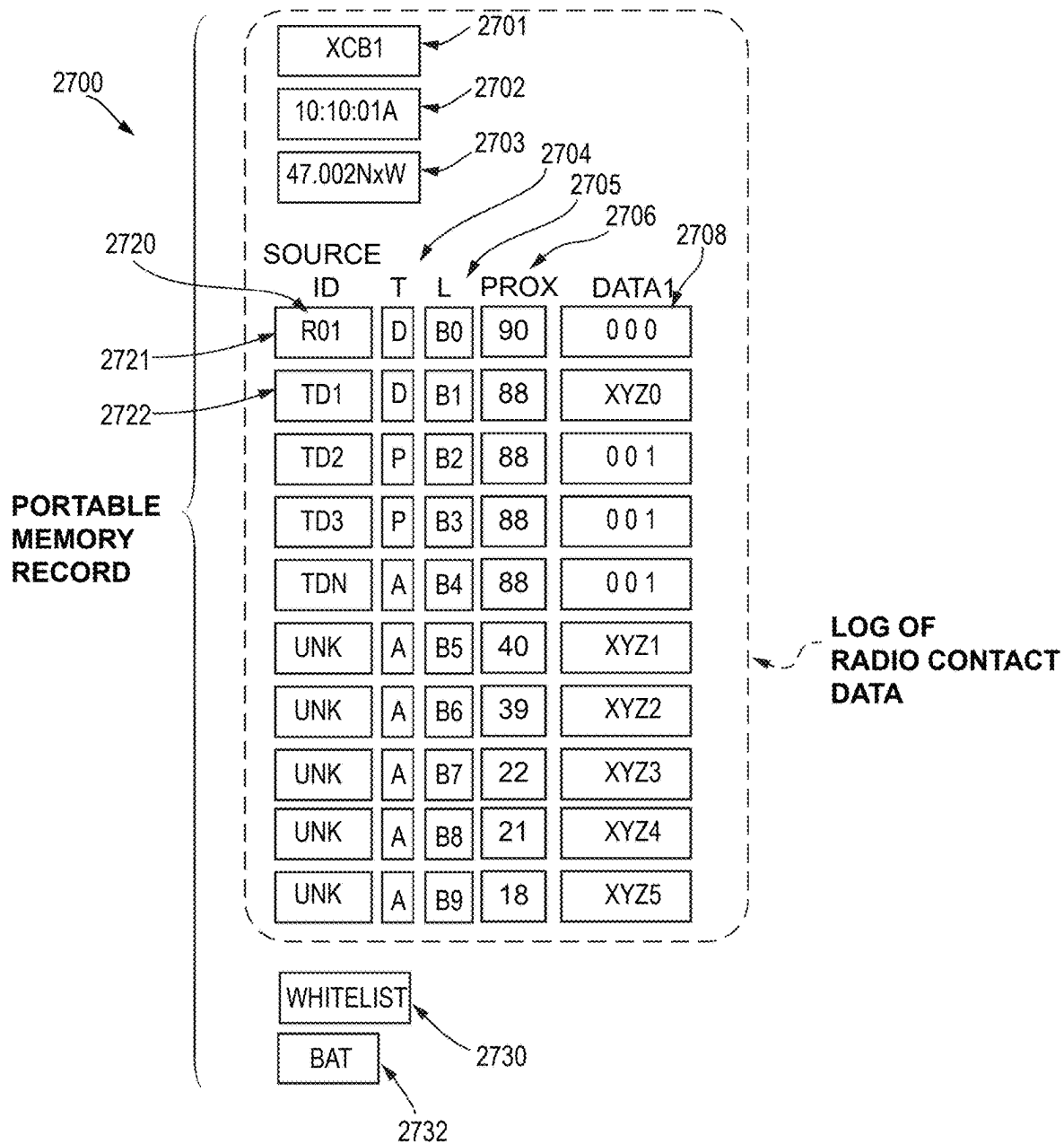
FIG. 27

FIG. 28

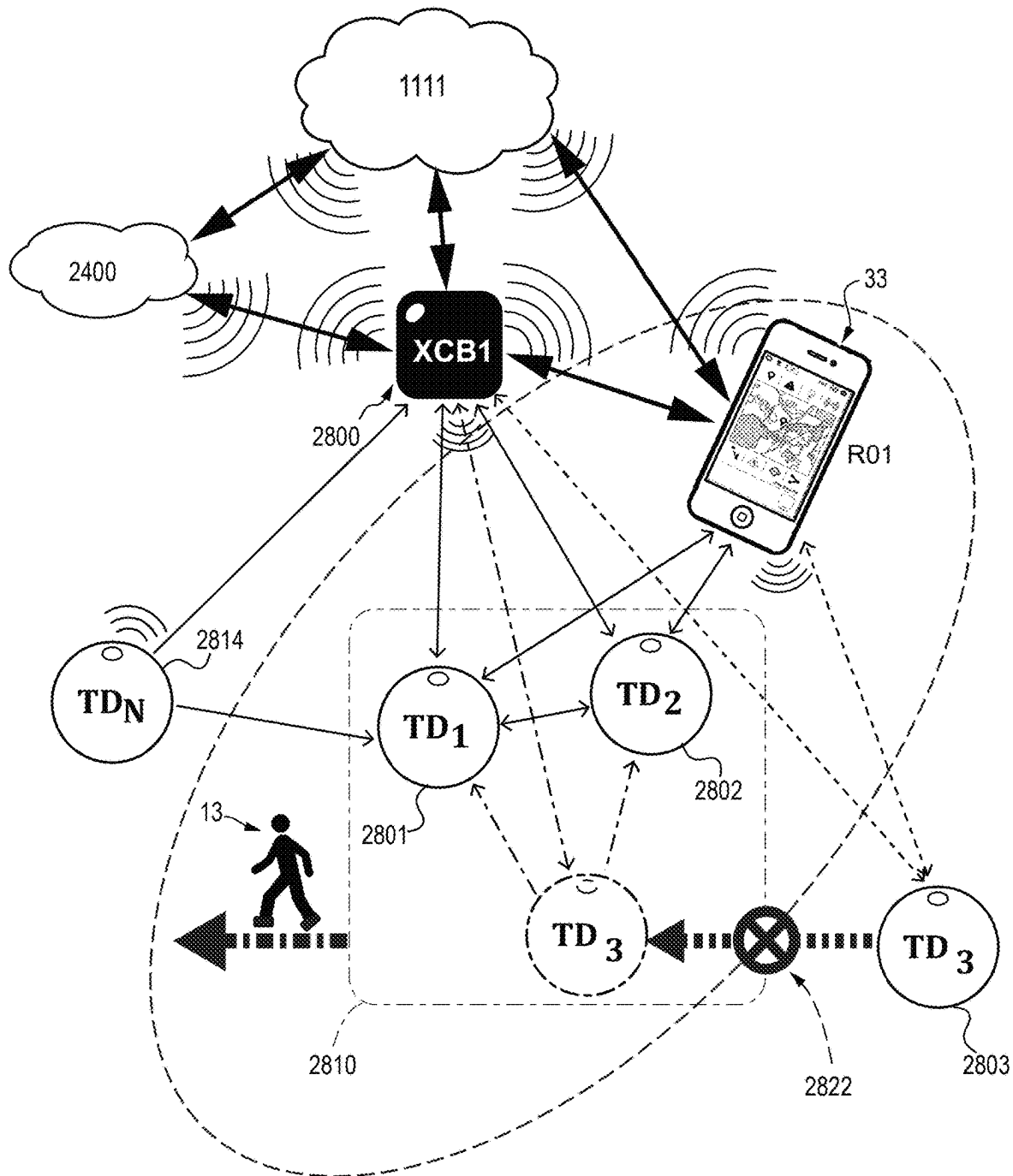


FIG. 29

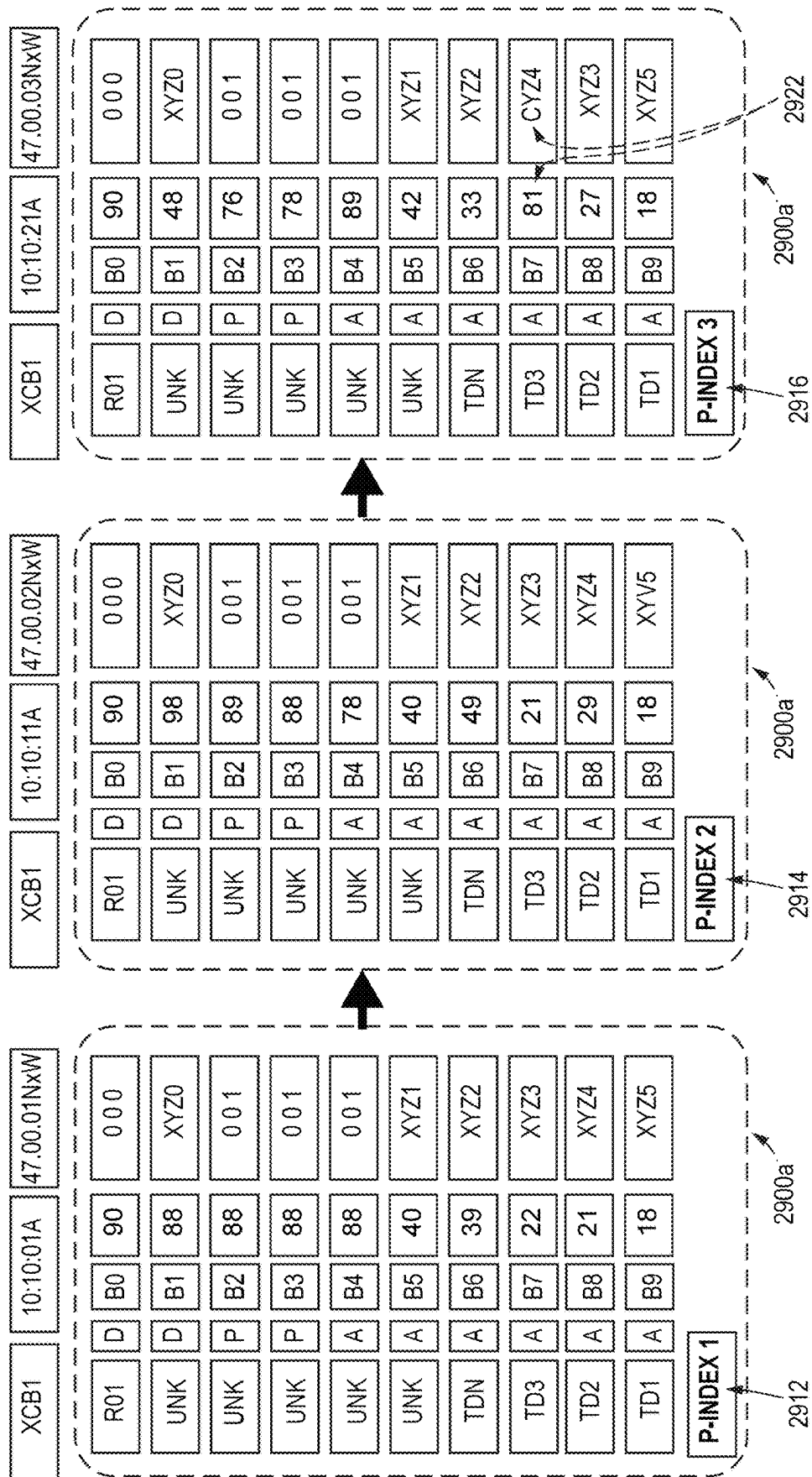


FIG. 30

NEW
IN

RADIO CONTACT RECORDS

TIME STAMP	MAC/UUID ADDRESS	RSSI PROX	LOCATION	DATA1	DATA2
100	XXX	021	90.01		
110	XXX	021	90.01		
120	001	080	90.01	AAA	XYZ
130	XXX	021	90.01		
140	001	080	90.01	BBB	XYZ
150	XXX	033	90.01	XXX	XXX
160	XXX	022	90.01	XXX	XXX
170	001	080	90.00	BBB	XYZ
180	XXX	042	89.88		
190	003	055	88.33	XXX	XYZ
200	002	045	88.33	ABC	QRT
210	XXX	044	88.18	BBB	XYZ
220	XXX	038	87.01		
230	001	080	84.93	BBB	CDE

OLD
OUT

3030

FIG. 31

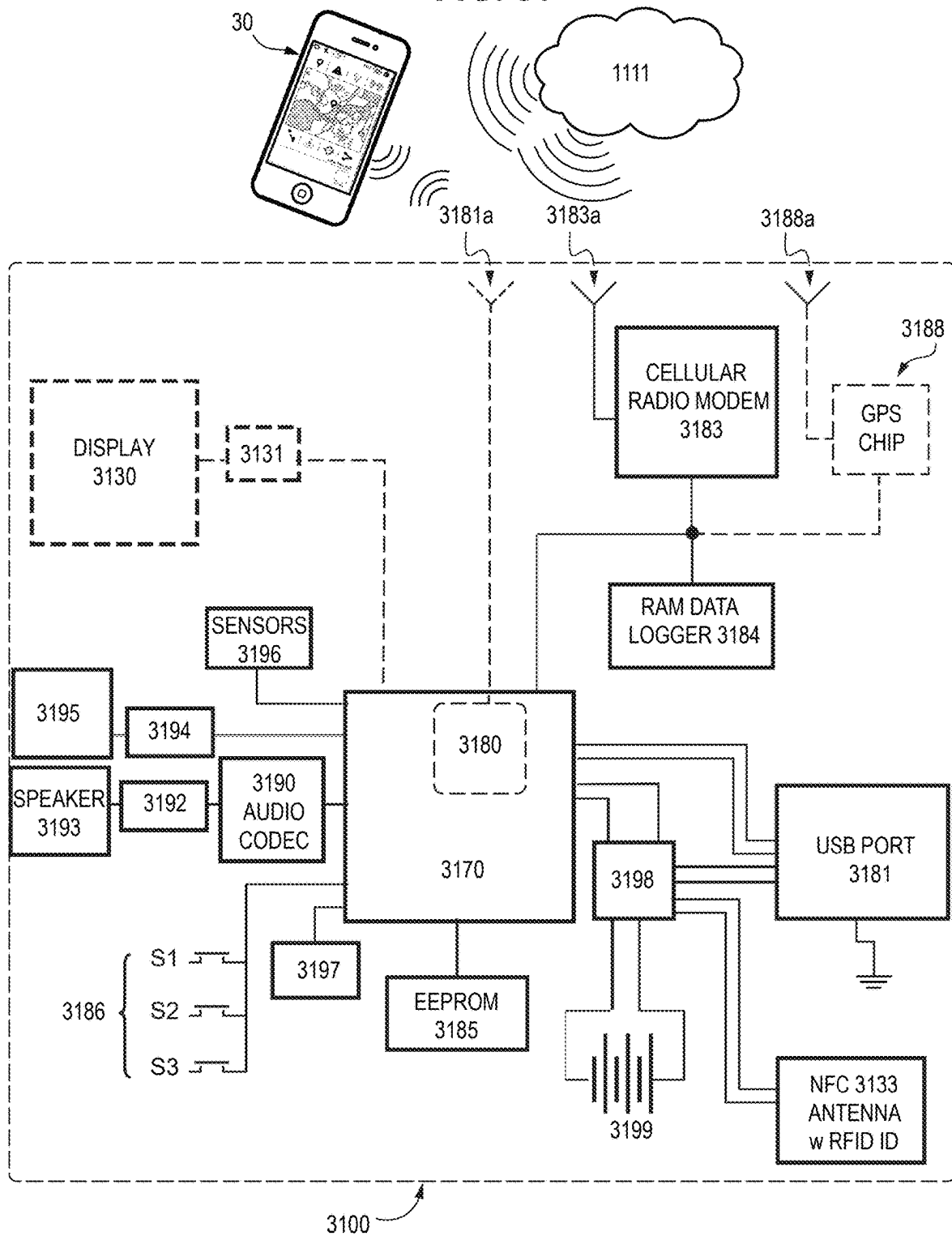


FIG. 32

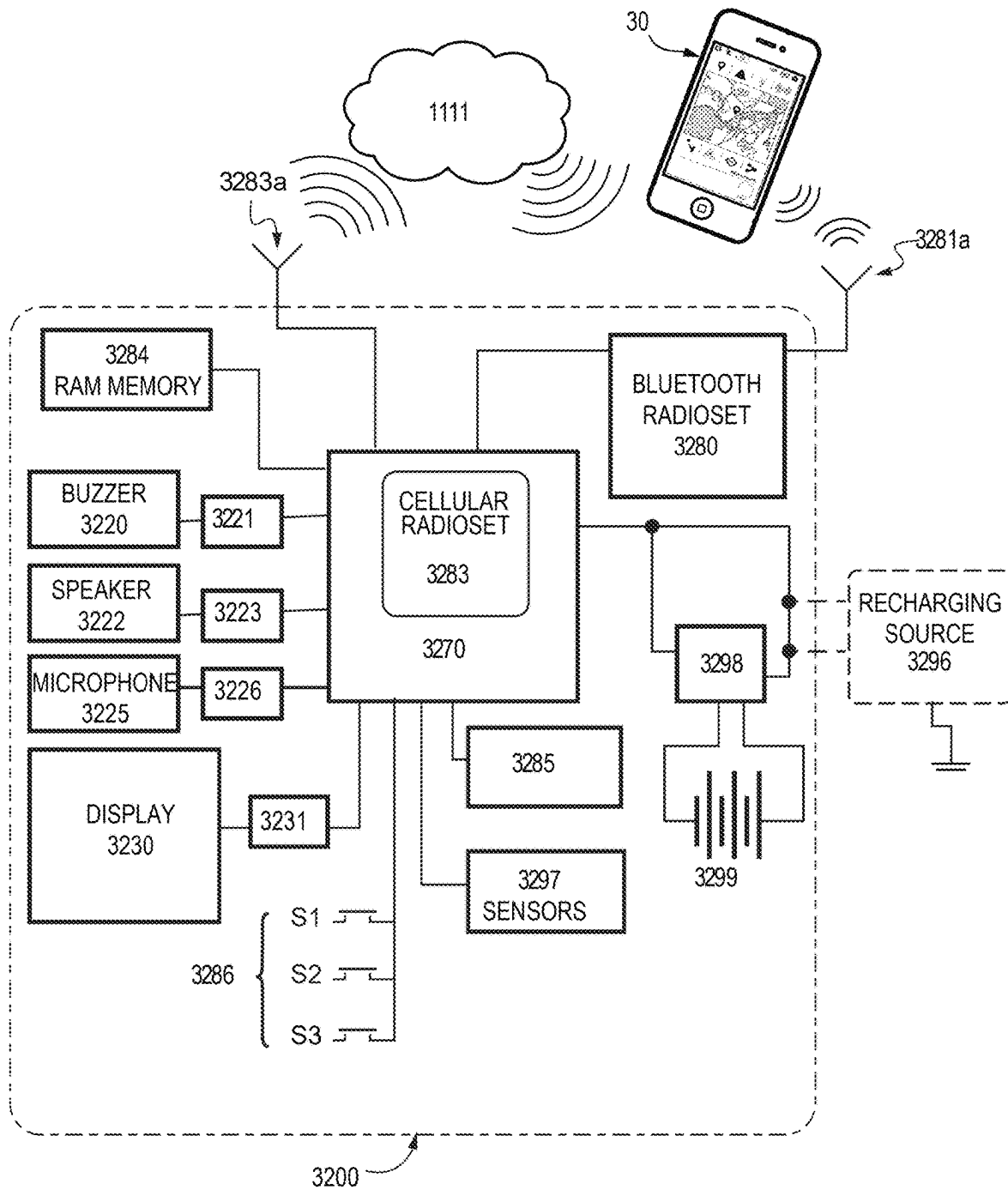
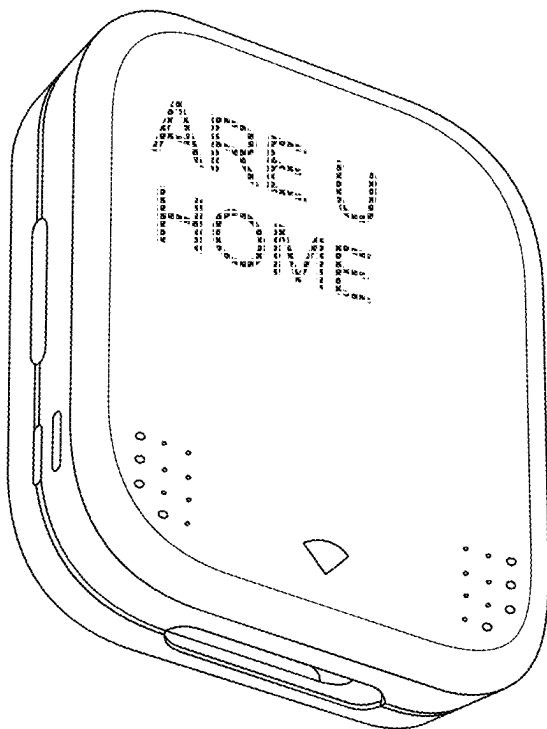


FIG. 33A

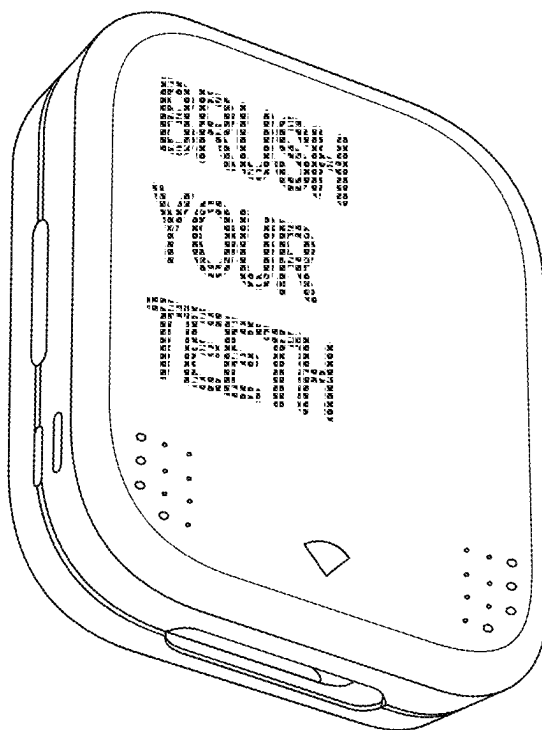
MESSAGE
LED DISPLAY



3300

FIG. 33B

MESSAGE
LED DISPLAY



3300

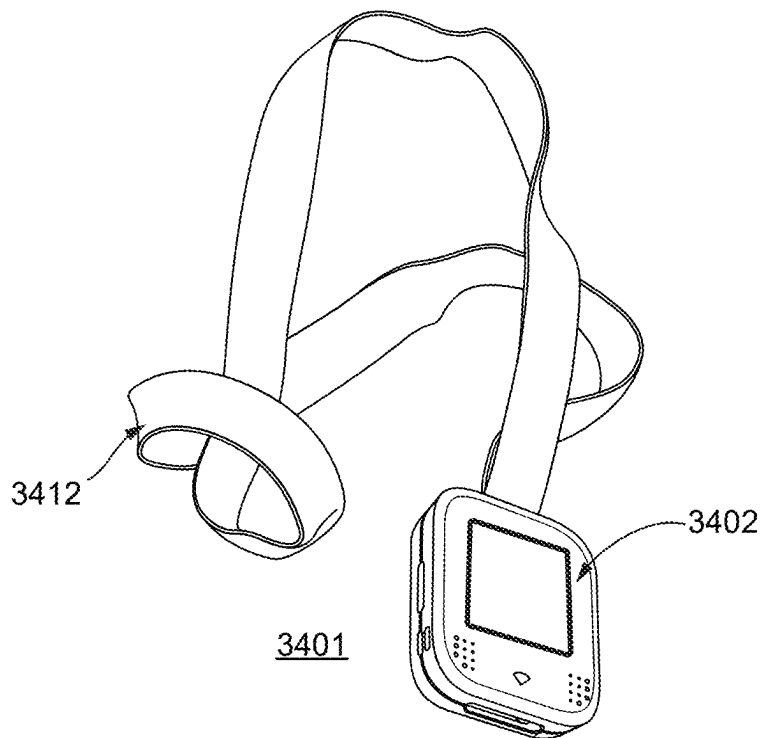
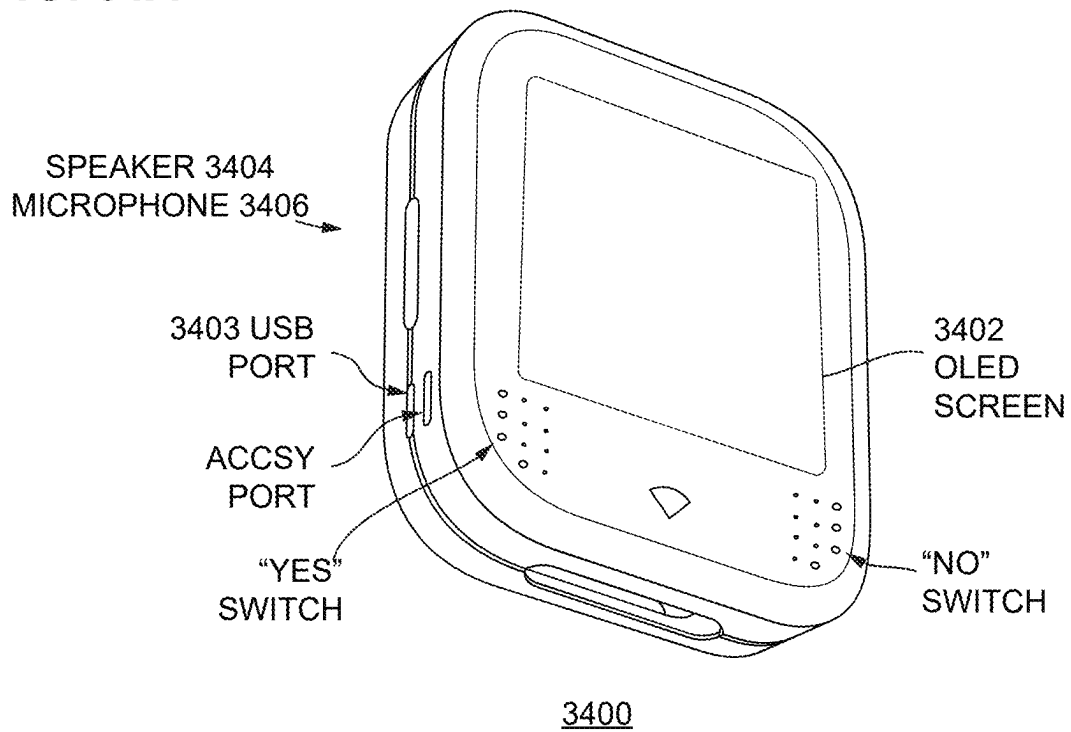
FIG. 34A**FIG. 34B**

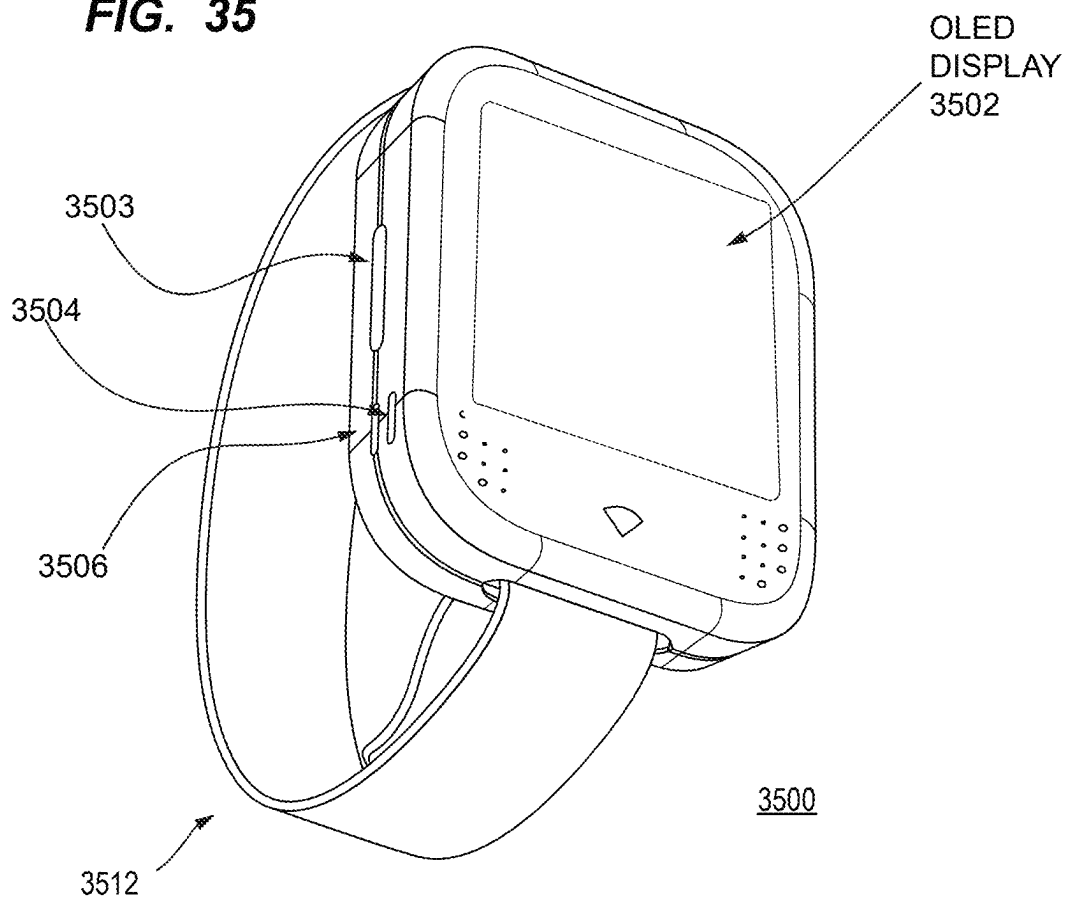
FIG. 35

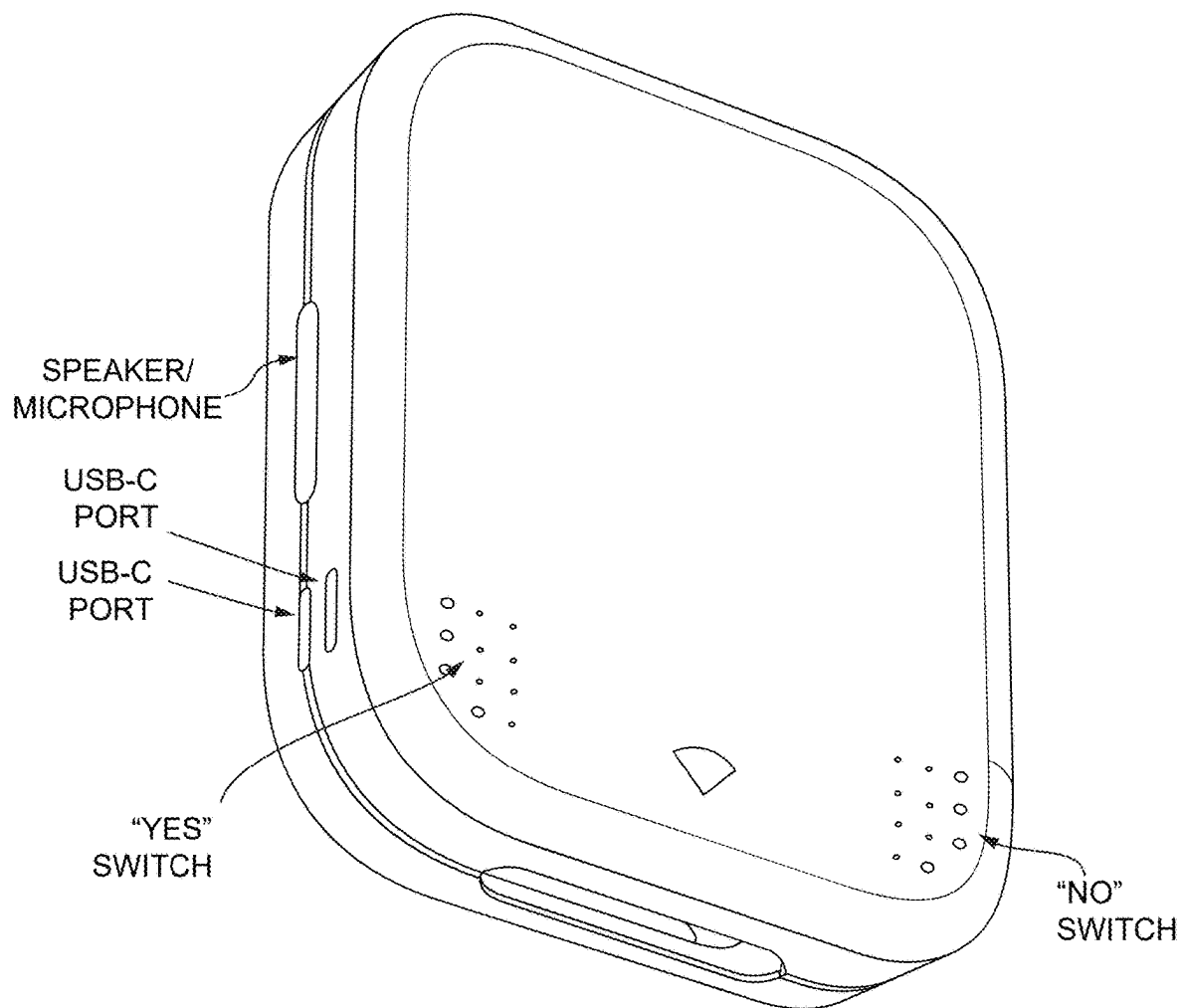
FIG. 36

FIG. 37A

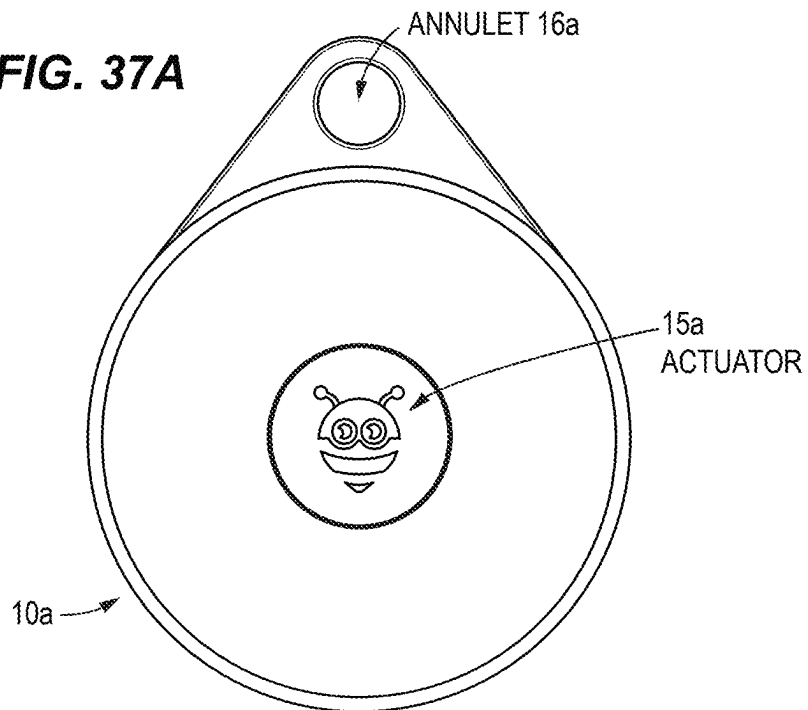


FIG. 37B

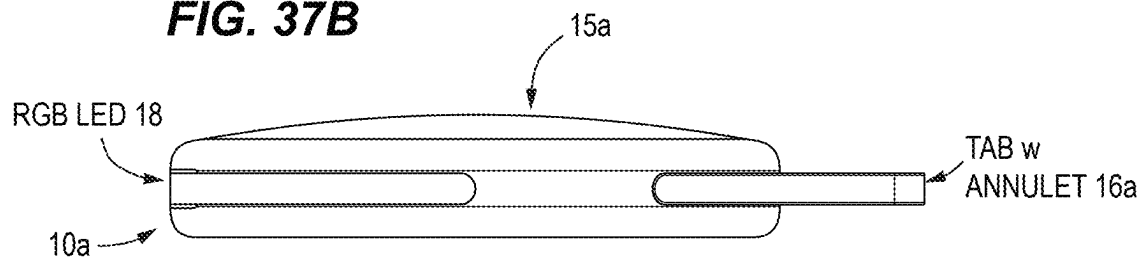


FIG. 37C

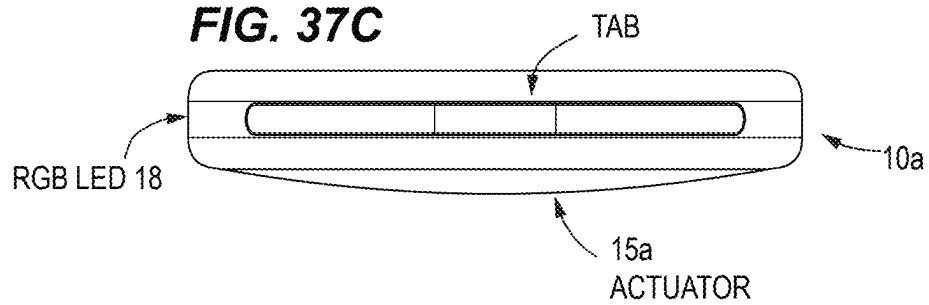


FIG. 37D

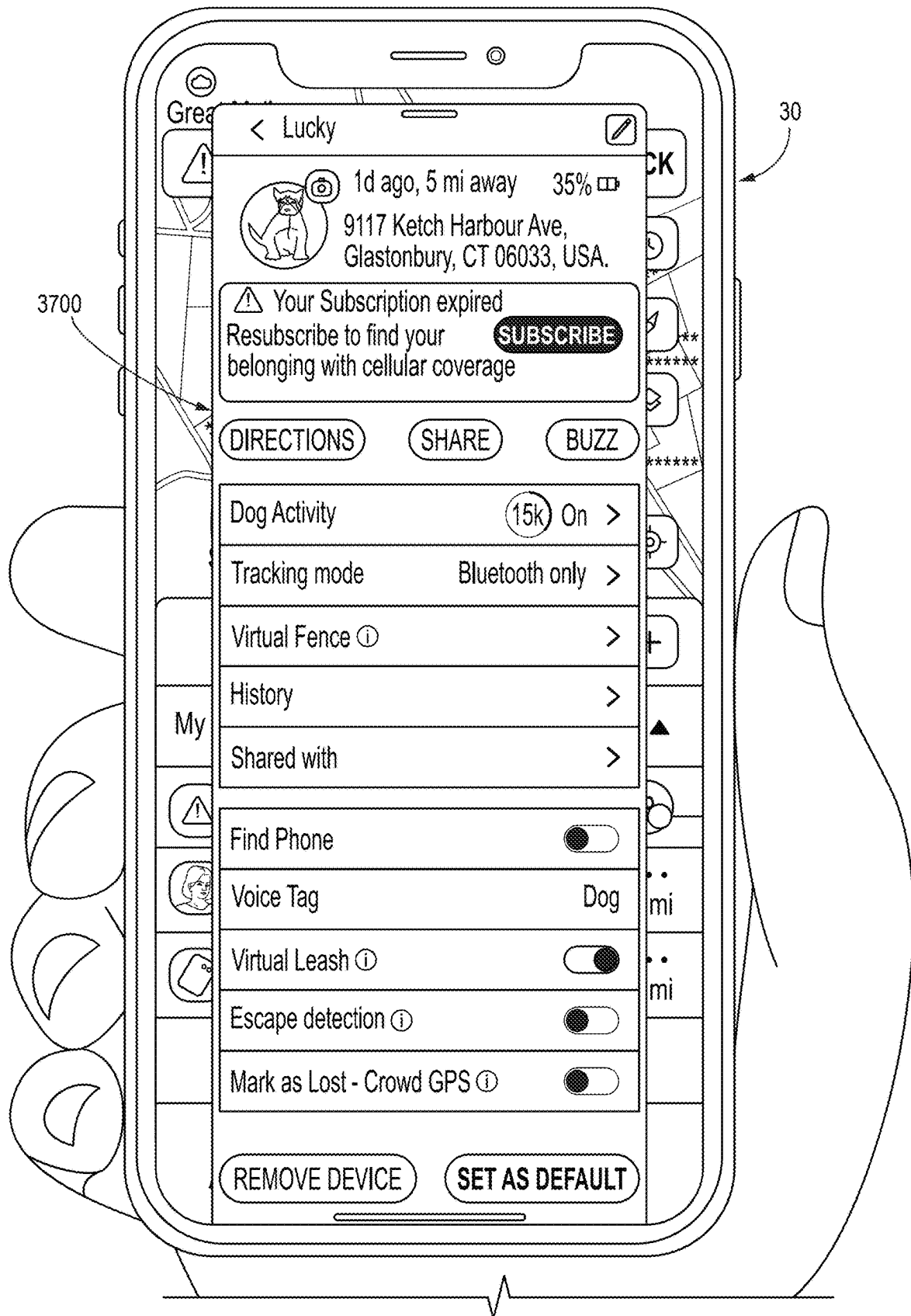


FIG. 37E

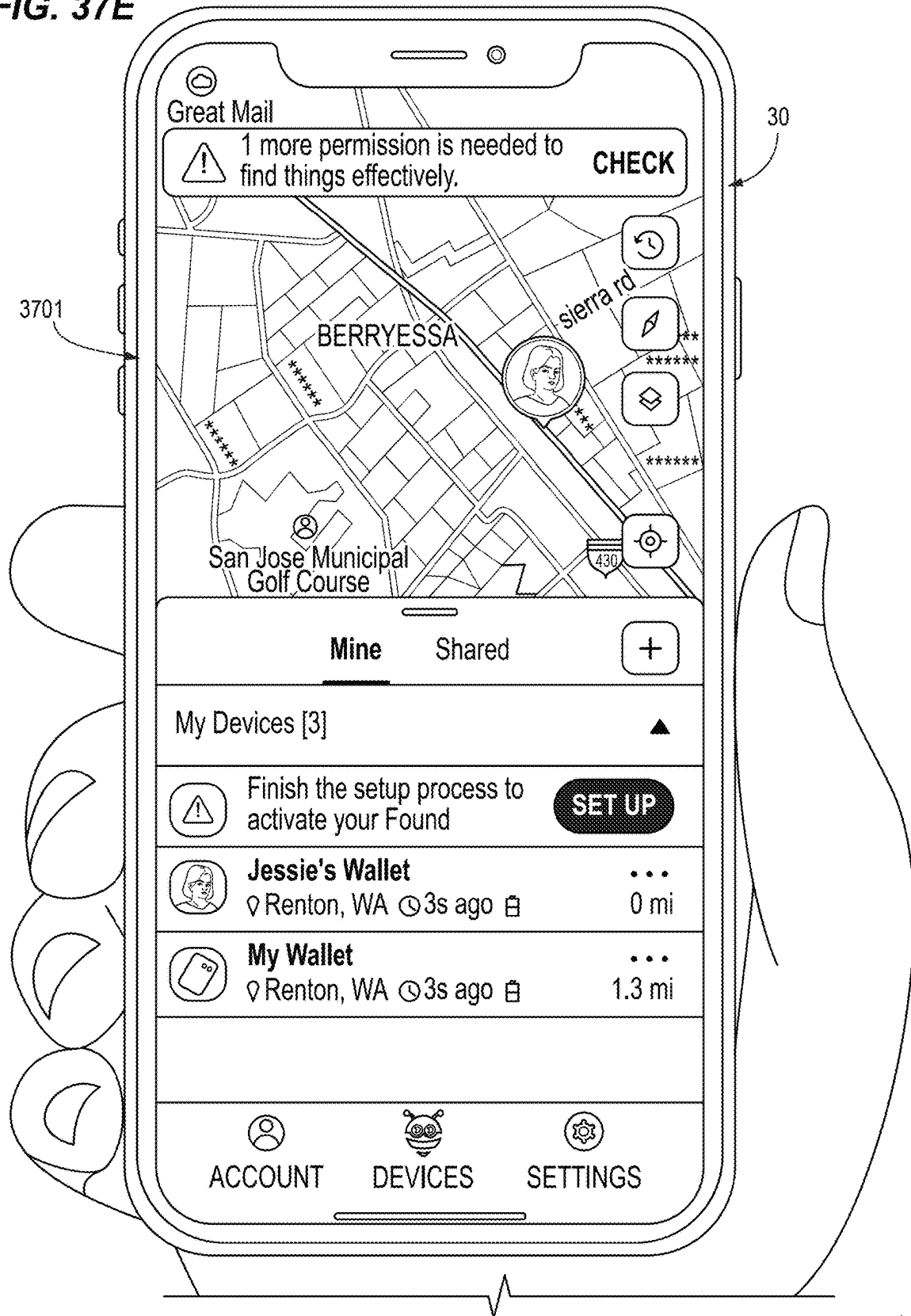


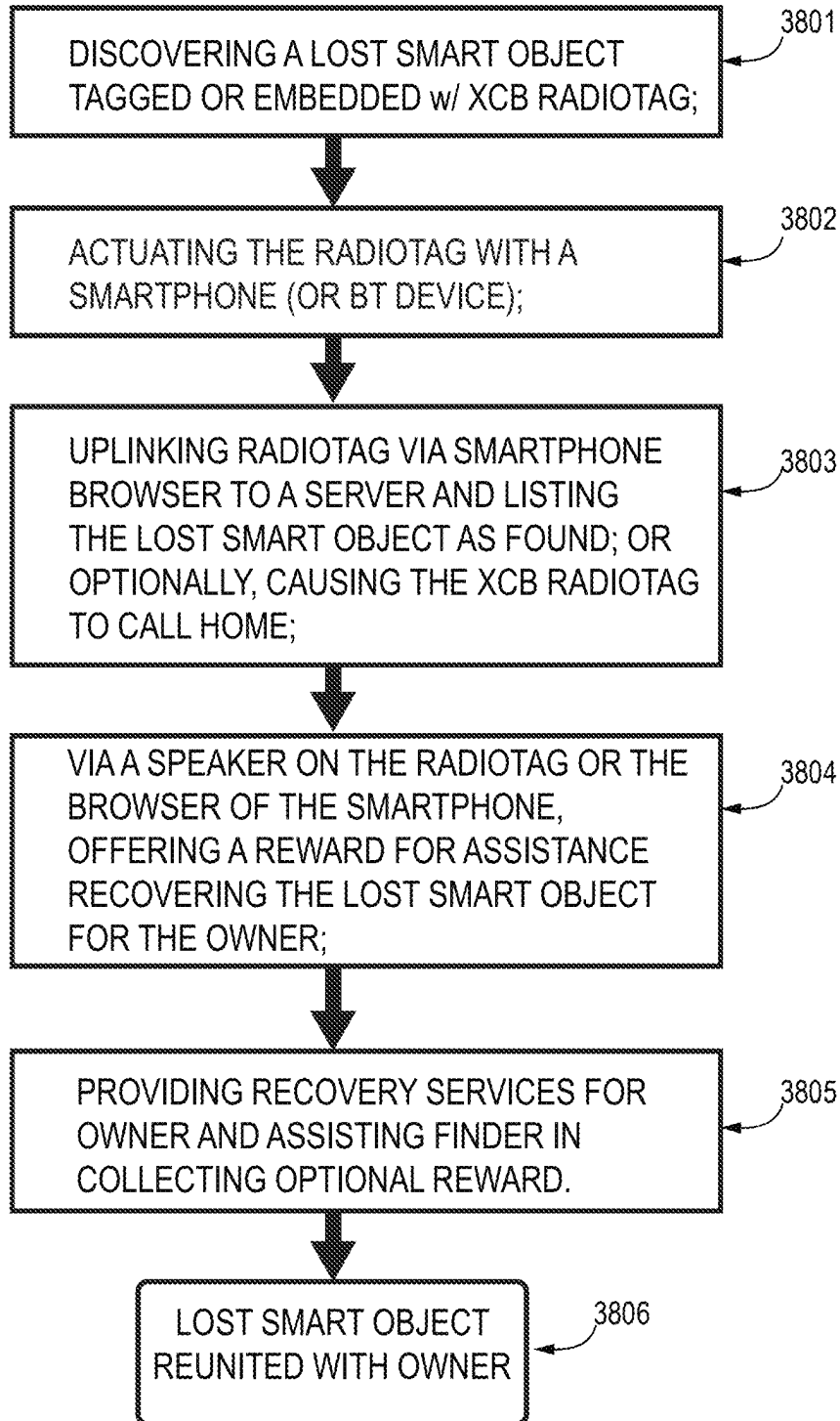
FIG. 383800

FIG. 39

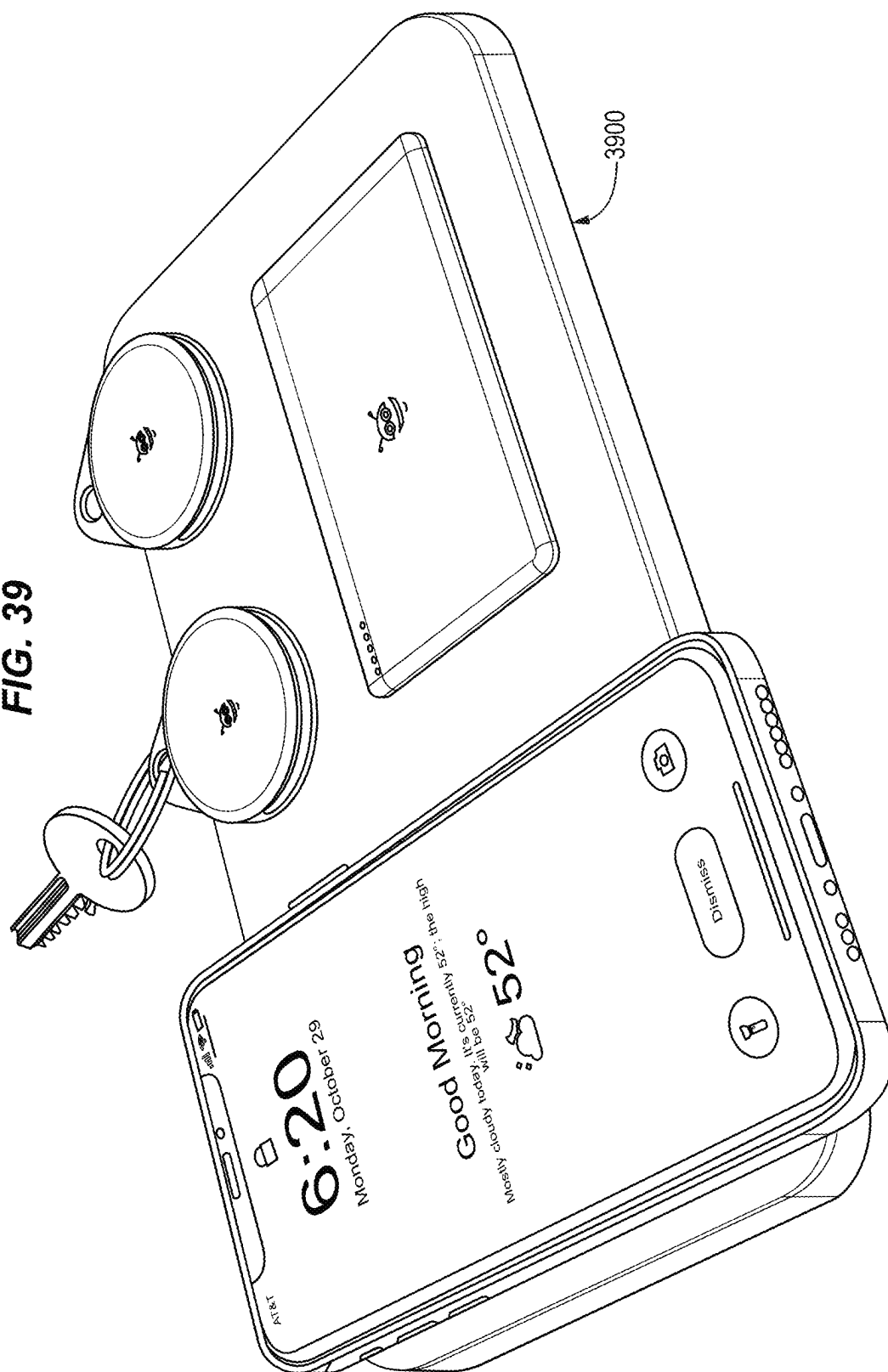


FIG. 40A
COLD CHAIN DATA LOGGING

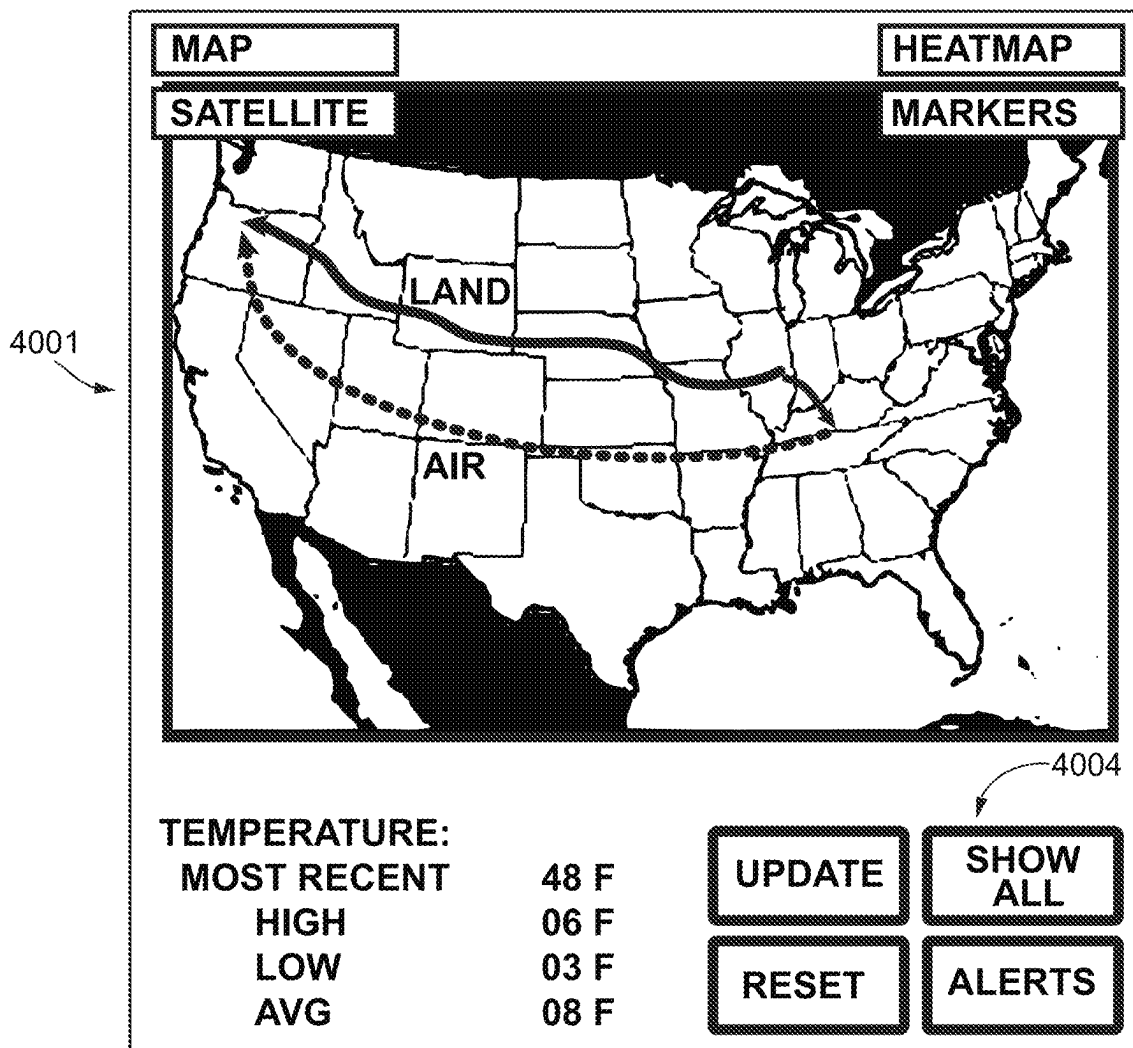


FIG. 40B

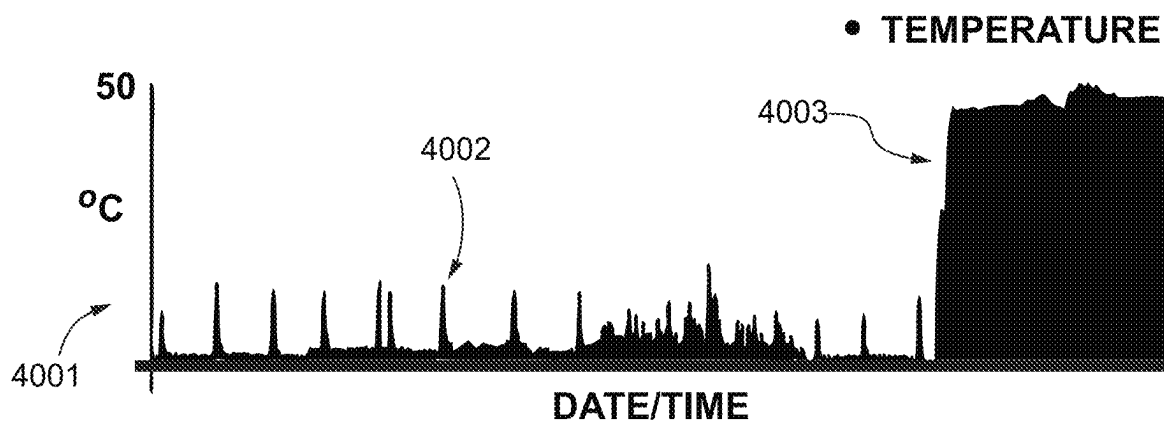


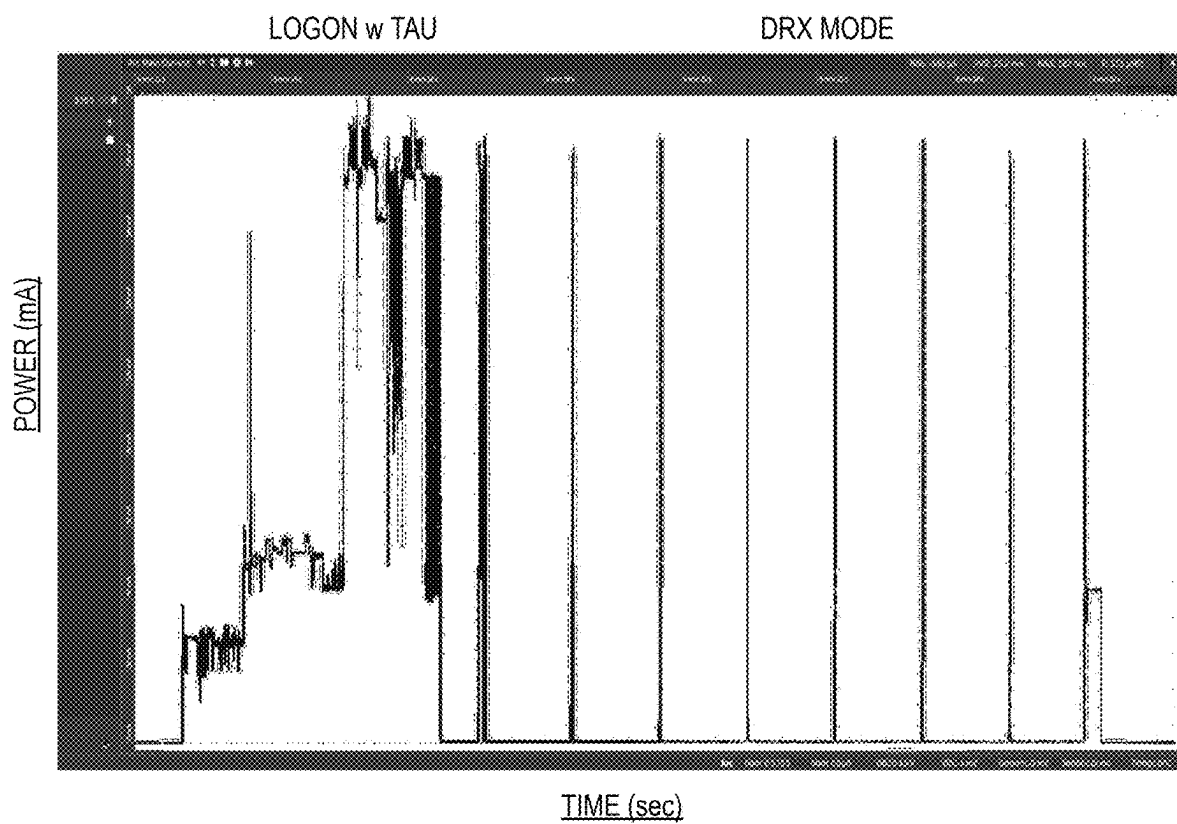
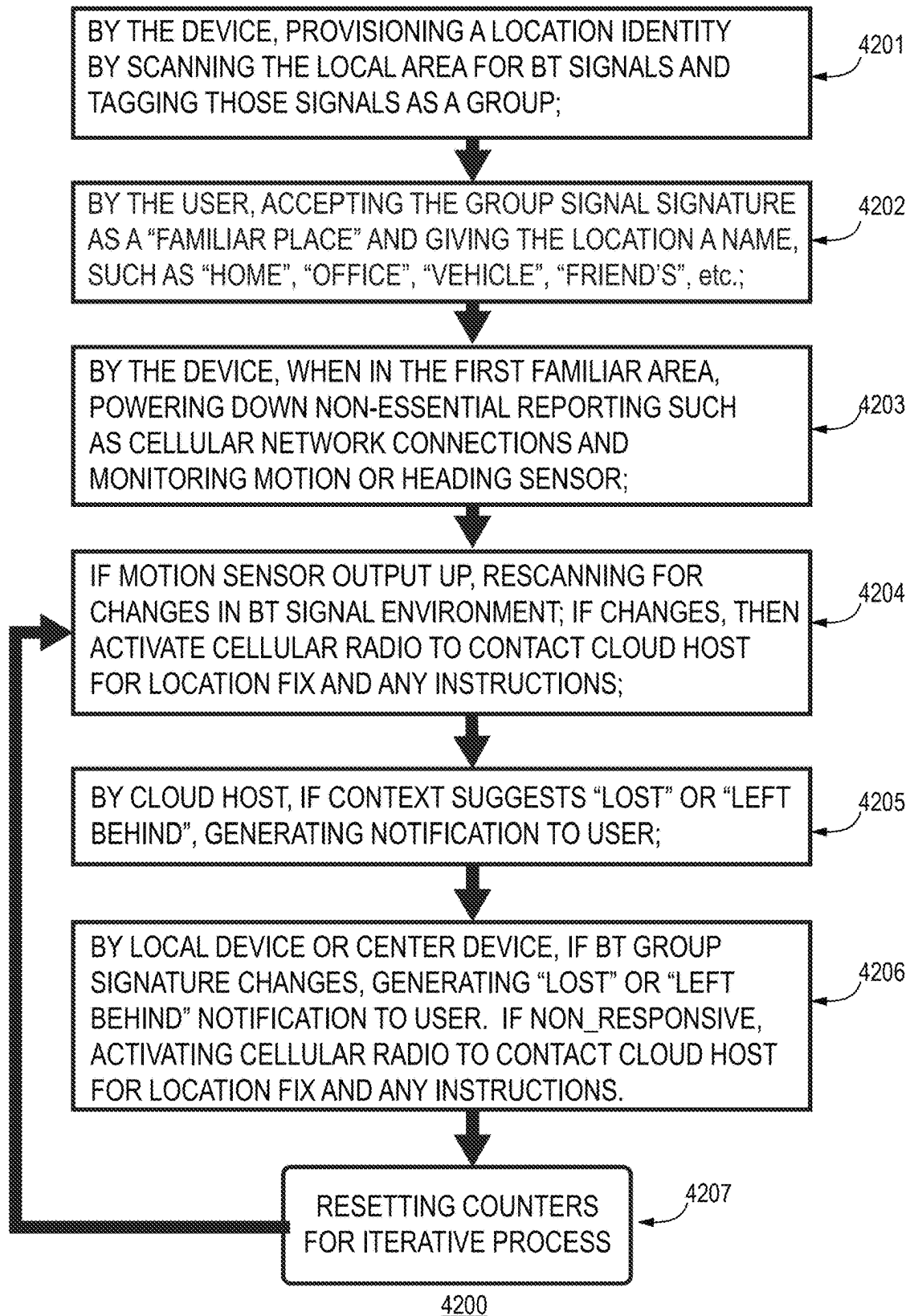
FIG. 414100

FIG. 42

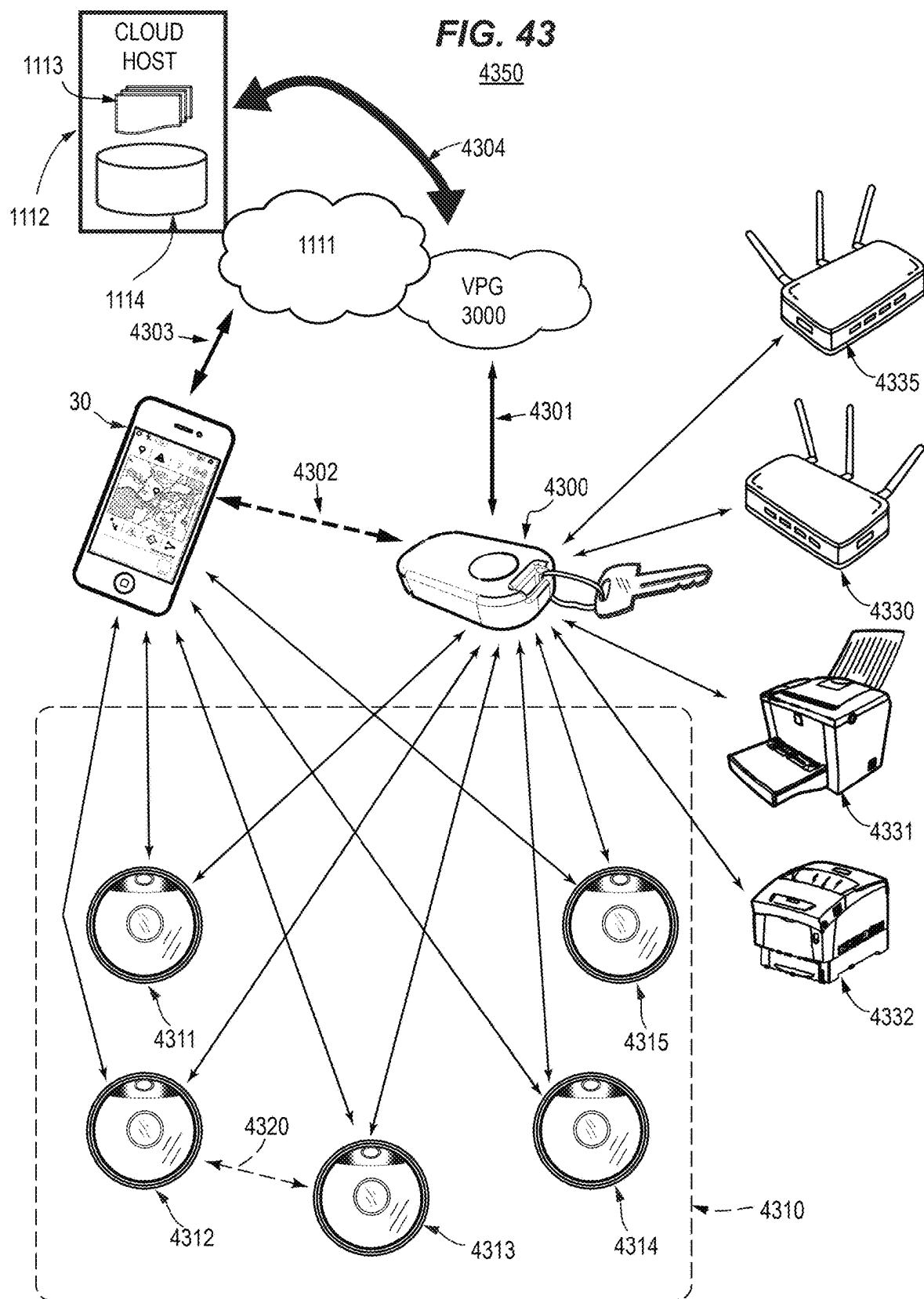


FIG. 44

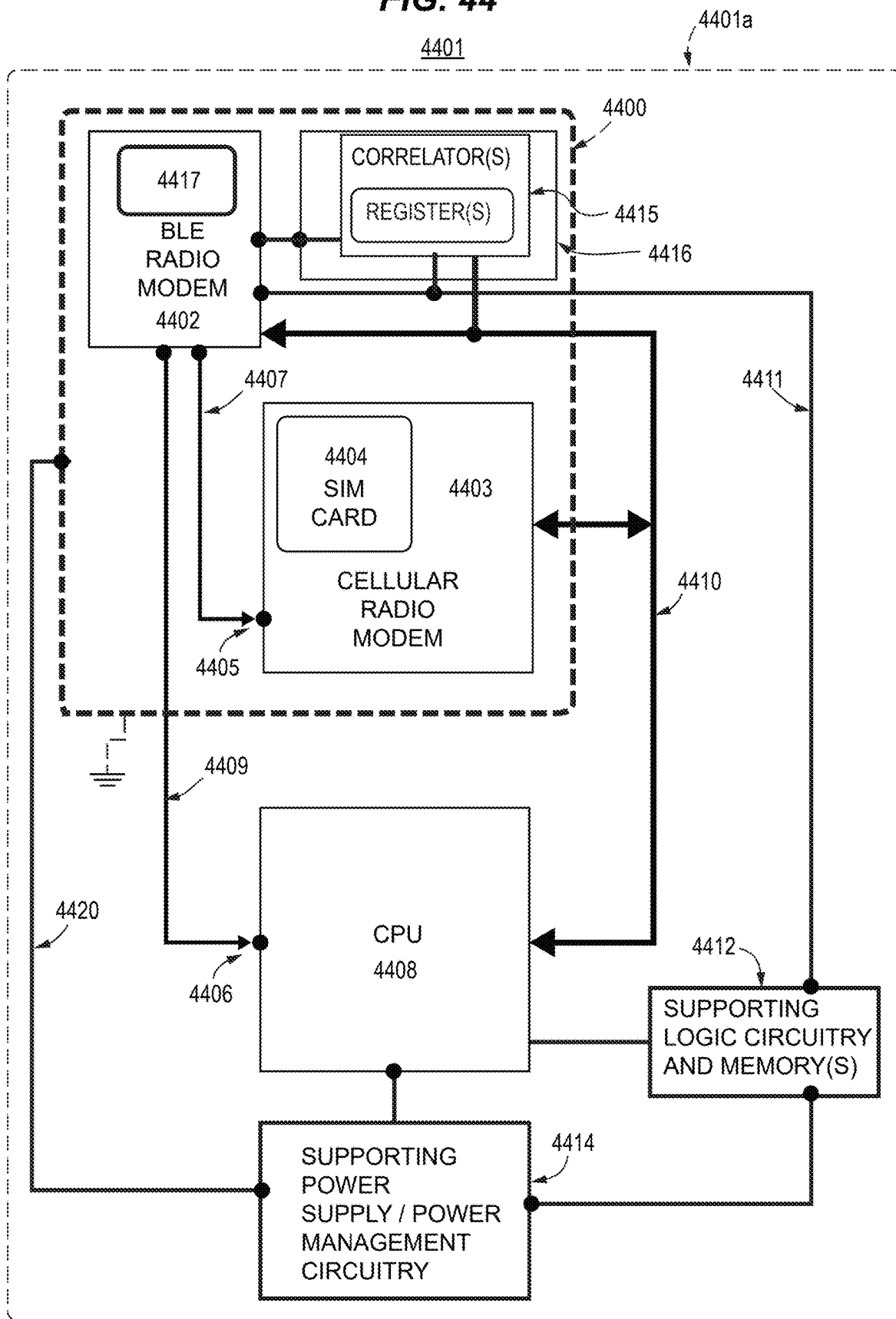
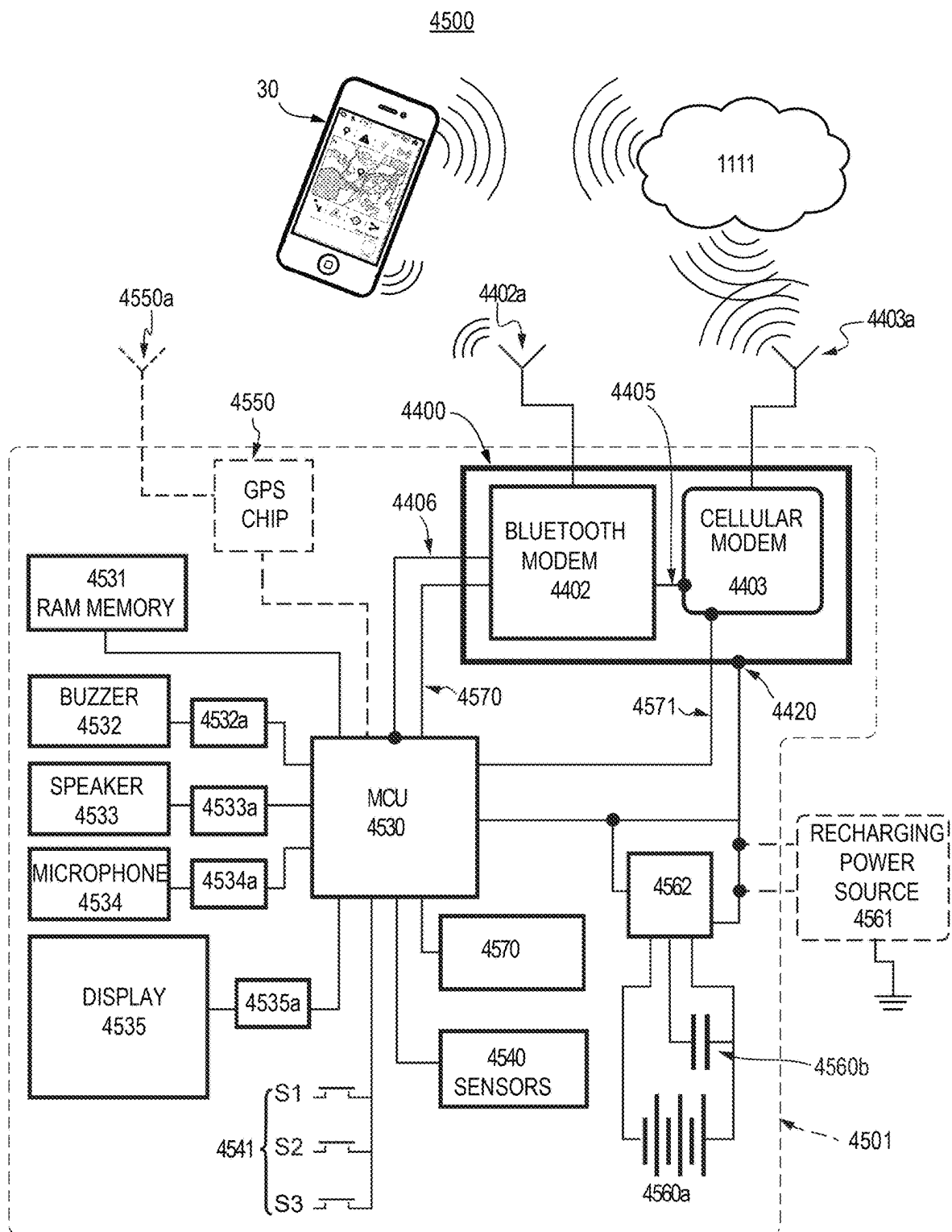


FIG. 45



1

XCB TRACKING DEVICES, METHODS AND SYSTEMS

TECHNICAL FIELD

Radiotags with poly-radio modems coupled to power management features; methods and systems for wayfinding, networking, and location-based lost-and-found services.

BACKGROUND

An ad hoc radio network and standard for data and voice sharing was long envisaged but has not yet been successfully integrated into the 5G+ radio access network (RAN) on which the Internet is operated. Long ago, in a first effort at universal connectivity, “Citizens Band” (CB) radio was created in the United States in 1945 for wireless short range messaging and remote control at 26.9 to 27.4 MHz. However, the standard was not widely adopted and today, CB radio finds only limited use; for example as “Channel 19” in vehicle-mounted walkie-talkies. The Zigbee radio standard is another example that has had a limited following. But more recently, Bluetooth radio has become an explosive success—over 40 billion devices worldwide—with more than 5B new devices shipping annually. There are many more BT radios on the planet than people.

At this time, Bluetooth (BT) radios do not provide direct access to the Internet, and despite the efforts of Internet founder Tim Berners-Lee, Personium Project Manager Akio Shimono, and Center for Human Technology co-founder Tristan Harris, among many others, the Internet remains a private conglomerate directed primarily at monetization of data with no core toolset for human-centric personal control of that data. The recently released Bluetooth Low Energy protocol (BLE) has added new limitations that teach away from personal control and favor Telecomm utility-type control, particularly cable and cell. Major manufacturers have committed to a “handset-centric model” in which the “center device” (also termed the “master”) is a telecomm-supplied smartphone with by-subscription service and restricted access except for emergency calls (eCall and CMAS). The handset is the master device that monopolizes control of satellite BT devices. This bias toward centralized control of data and connectivity has been revenue driven, and aligns with “connectivity-by-subscription” trends in social media that promote private monetization of the public airwaves. Both iOS and Android still allow developers a window within which to transmit BT advertising packets in an open mesh network, but the peer-to-peer open discovery envisaged by the Bluetooth Special Interest Group (SIG) is not supported by recent editions of iOS or Android for cellular handsets. The intent is instead to limit open connectivity to a brief window in which a new device is discovered and claimed by a handset, headset or other class of “master” device. This bias has perhaps defeated the fullest potential of the 2009 release (WiFi 4, IEEE 802.11n standard) of the 2.4 GHz and 5 GHz ISM bands for single and dual band WiFi and BT radio use, including BT 5.0 and beyond.

This bias will continue to handicap efficient use of 5G+ national critical infrastructure by limiting the mesh network connectivity beneath the SIM-authenticated radio access network (RAN) layer unless a new platform is developed. A new platform is needed that grafts devices having SIM-authenticated connectivity with the capacity to connect to or at least monitor open peer-to-peer networks such as BLE in both a credentialed and uncredentialed way, without the cellular network dominance.

2

What Bluetooth offers in sheer number of deployed devices is balanced by their limited range—typically no more than a distance of about 100 meters at 0 dBm. BLE 5+ offers 20 dBm in some countries, but increasing use of overlapping WiFi results in increasing packet error rate at greater distances despite the spread spectrum, forward error correction and other innovations. The intimacy of Bluetooth in local areas due to limited range is helpful in informing proximity-driven applications but becomes a critical limitation when location and data services must be provided over greater distances.

In some instances, even more intimacy is needed, and higher wavelengths (in the 6 to 10+ GHz range, e.g. UWB) provide a needed short range and specificity in making connections based on proximity, such as for directed wireless transfer of files. Very short range transmissions ensure a reasonable level of privacy and specificity in making connections, and can be used to bootstrap other wireless connections such as WiFi 802.11a/b/g/n/ac/ax to build sensor nets or complete larger data transfers by allowing for exchange of encryption keys without a likelihood of interception. The “tap-to-pay” system of NFC (near field communications) radio is another example of very short range radio authentication and credentials transfer, and also is used to bootstrap more robust ISM wireless connections for larger data packet transfer. Recently proposed THz bands for “body-area networks” also seem to offer a fusion of biometrics and nano-area radio networks, all of which can relieve some of the congestion in the increasingly crowded 2.45 and 5 GHz radio spectra.

The deployment of 5G and 6G radio networks is bringing increased complexity and variety to radio use. The backbone of 4G and 5G systems for most users has been the cellular broadband network that supports voice and mixed media communications as well as data. Access to this system is tightly controlled by authenticating of users based on an IMSI (International Mobile Subscriber Identity) physically or at least virtually encoded in the subscription identifier module (SIM) in every cellular device. But the need for security comes at the cost of energy consumption due to continuous network authentication and commercial exploitation, which limits the field battery life of detached cellular radio devices. The heavy cellular handsets carry large, heavy batteries and are easily dropped. More than 50M phone screens are cracked every year in the United States alone, according to a 2018 report.

An improved radiotag and system would enable owners to expect and to realize an extraordinary level of power management sophistication while enabling both long distance and short range location management tools, including finding, tracking and wayfinding. Power management must be so stringent for new IoT applications, for example, that an owner can attach a radio device to a pet or other asset, and be able to track the signal over long distances if the asset goes missing weeks or months later—while having adequate broadcast range and no unacceptable network latency or dead time. The advantage would be a single solution that allows an owner to interact with a lost radiotag, or for example to interact with any passersby who encounters the radiotag in order to enlist their help, for example a lost child at an airport in need of being reunited with a parent. A solution to this complex problem has not been achieved with conventional radiotags to date.

SUMMARY

An unmet need exists for a hybrid XCB radio device and platform that has the intimacy and ubiquity of a BT piconet

or mesh open network, but also the power to connect globally with the packet data backbone of LTE-Cat M and emerging cellular and WLAN 5G networks. A new platform is introduced that combines local BT “situational awareness” for power management and security with a capacity to report and make connections via the global IP networks and their virtual private gateway cousins. An immediate application is in the development of systems providing global lost-and-found services. In another innovation, also introduced is a new integrated circuit (IC, chip) to elevate the performance of this new platform.

The new radio devices introduce BT radiotags with a capacity to scan and report the BT radio topology associated with their current location, where “BT radio topology” includes BT radio traffic as an envelope of propagated digital radio signals that may broadly encompass the BT “spread spectrum” of multiple channels. Radio signals in the BT spread spectrum are treated as meta-data by which ambient BT signals are analyzed for patterns and signatures that are associated with location, activity, day, calendar, and any familiar place, including safe and threat environments. Familiar environments are marked by the unique radio signatures of a local ensemble of BT radios aggregated there. Threat security alerts may be issued upon detection of abnormal radio activity in a network and the same BT signatures may be used to detect session hijacking using “man-in-the-middle”, “replay”, “teardrop”, “ping-of-death”, malware, and botnet attacks. Whereas hackers are increasingly sophisticated in exploiting code weaknesses to game network resources, steal credentials or deny service, the hacker cannot readily mimic the “local color” of a known BT radio topology that surrounds a genuine user who is familiar to the system and is in a safe and known location. This is a radio version of “Recaptcha”, a very successful piggyback authentication routine designed to block bot DOS attacks. The radio version has the advantage that it can be fully engaged without user inconvenience. Two users, across a table, can exchange private personal data such as contact information in an open environment if the radio signature both radio units are sensing is copacetic. Local radio signature authentication is used to authenticate another local radio unit before divulging sensitive information, and may supplement or supplant the system of private and public keys that are vulnerabilities in 5G networks, and also leads to establishment of a decentralized system for secure storage and access to personal data, while under personal local control. In today’s world, metadata is exploited for commercialization, but here it is turned to support private users in their need to selectively share data in intimate, range-limited radio linkages such as when meeting face to face.

This can include medical data and financial data, for example lists of drug prescriptions, prior surgeries, phone numbers of relatives, and COVID vaccine history that currently must be recited out loud in the lobby of a Doctor’s office.

Because BT relies on FSK radio modulation and has a specialized packet structure, BT radio contact feeds cannot be directly transmitted across the 5G radio access network (RAN), most of which operates with PSK radio modulation. The RAN and CORE IP packeting environments define multiple unique packet types that have not been supported in BLE standards. IPv6 support over BLE (Bluetooth Low Energy) with the adaptation of 6LoWPAN and Thread protocols can result in a crossover from WLAN to BLE. IPV6 can be emulated over BLE (as defined in RFC 7668) according to the Internet Engineering Task Force. But this standard does not allow native BLE transmissions to be

routed in their native state onto the IP packet data networks. The devices described here map BT radio packets to a standardized database log entry format or “radio contact record”, and these database records are shared as a sort of “radio snapshot” of the BT signal bubble or “topology” around the originating device. This database log format is data that the 5G network environment readily transmits and accepts. This radio snapshot can be used for authentication, but also finds a myriad of uses in provisioning new devices, in erecting geofences, in wayfinding and locating, and in maximizing throughput in the ISM spread spectrum by identifying pathways around congestion. The snapshot may include 1280 octets or larger when transmitted over WiFi or using the 6LoWPAN standard, but may be assembled from smaller bitwise snippets extracted from the advertising packets and PDUs in the intercepted radio traffic.

As disclosed here, these new radio devices are termed “crossover cellular Bluetooth” radios (XCB). XCB devices include at least a Bluetooth (BT) radio modem and a cellular radio modem, with a processor, supporting circuitry and power management features that default to a “sleep” mode for extended battery life. Synergically, by engineering the BT radio as part of the device controller and attaching a cellular modem to the controller, the BT radio can be used as a “wake up receiver” or “always listening radio” (ALR) for activating the cellular modem. The devices differ from true “always listening” radios (which remain largely theoretical because the energy for their operation must be derived from the received radio signal) because there is a latency between scans. The latency can be adjusted in 10 msec windows, but the data is analogous to the snapshots of a motion picture film, like a long ribbon, and is engineered to create an illusion of “always listening” just as a motion picture, when viewed at 30 or 60 frames per second, is a convincing illusion of continuous action to the brain.

These poly-radio XCB devices may also include one or more WiFi radiosets operating in the ISM band and one or more of the WPAN radios such as for Thread, 6LoWPAN, pulsed UWB, and so forth. Synthetic radio with “chirp” capability is also compatible. The ISM bandwidth available for mesh BT and WiFi radio includes 2.4 and 5 GHz and combines the 802.11a/b/g/n/ac/ah/ax WiFi, 802.15.1-3 Bluetooth, and 802.15.4a/g WPAN radio standards. BT standards most relevant today are 4.0, 5.0 and 6.0 (and continuing innovations), each of which adds new complexity to the ISM bands.

Cellular radios operate in LTE and 5G bands that differ from country to country, but with suitable clocks, oscillators and antennae, may be included in the package. While cellular radio includes some sub-GHz bands, the components of radio spectrum relevant for cellular are the LTE/GSM/5G bands dedicated for telecommunications. Use of synthetic radio has the potential to build clock oscillators that can generate multiple frequencies with adjustable impedance, and in some embodiments, the poly-radio devices are provided with multi-band antennas that satisfy communications needs over several disjoint radio bands, such as the LTE cluster, or dual-band WiFi, or a combination antenna for LTE, Bluetooth and WiFi, for example. Fractal, PIFA, ceramic, stacked patch, diversity, and phased array antennas are also included. Electronically, impedance matched receivers are included that can jump from band to band depending on the transmission/reception feed. Radio antennas form a large segment of the patent literature, for example U.S. Pat. No. 4,381,566 to Kane, U.S. Pat. No. 6,452,553 to Cohen, U.S. Pat. No. 7,148,850 to Baliarda, U.S. Pat. No. 7,202,822 to Baliarda, U.S. Pat. No. 8,456,365

to Pros, U.S. Pat. No. 9,000,985 to Baliarda, and U.S. Pat. No. 9,755,314 to Baliarda, for example. Also relevant are papers by Hong, S S et al. 2012, Picasso, Flexible RF and Spectrum Slicing, In ACM SIGCOMM 42:37-48. Helsinki F I. [doi.org/10.1145/2377677.2377683] and Hong, W et al. 2017. Multibeam Antenna Technologies for 5G Wireless Communications. IEEE Transactions on Antennas and Propagation, 65:6231-6249 [doi.org/10.1109/TAP.2017.2712819]. This literature is public domain and readily accessible to those skilled in the art.

Inexpensive “software-defined radio” chips are already available. In one embodiment, we suggest that “synthetic radio” oscillators of XCB devices may include a “software-defined antenna” that advantageously can be upgraded via firmware rather than requiring new hardware, as will be increasingly important with the proliferation of radio technologies to be implemented in 4G, 5G, 6G and user-defined networks. Given that most XCB devices are powered to operate BT radio at less than 20 dBm, more typically 4 or even 0 dBm, regulatory restrictions and SAR (biological exposure) considerations are minimal. We introduce a software-defined antenna that uses a programmable FET gate array to achieve impedance matching and beamforming in a wrist-sized or wearable device.

While typically defaulted to sleep state for power savings, when activated, an XCB cellular modem can make a new cellular network connection or reestablish a network connection in a process generically termed here a “CALL HOME”. Any call home is at least a status update and offers an opportunity to share a location fix with the network or to receive network assistance in establishing a location fix. Location data is the key to many community services facilitated with XCB radio devices.

XCB radiotags function in locating, tracking and monitoring lost or wayward objects, children or pets for example. In use, a radiotag is attached to a child, object or pet by an owner of the object, and can wirelessly report the radiotag’s location from around the world or can make itself found by emitting an audible alarm when misplaced out of sight. Because XCB radiotags can self-initiate a call to an owner of a radiotagged asset, the owner is not strictly dependent on the goodness of strangers to recover what is lost. Advantageously, the current location can be displayed on the owner’s smartphone for example, and updated locations can be displayed as a series of waypoints that track the location of the lost child, object or pet until safely recovered. The sent data can also include biometric data.

Output from onboard sensors such as motion, electronic heading, photocell, body temperature sensors, or a combination of radio traffic sensor data and motion sensor data, for example, also can result in a CALL HOME if something is amiss. In one embodiment, the BT radio functions as a “radio traffic sensor” that reports the surrounding BT radio topology. Sensor data stored in memory may be shared in an uplink with the network.

During radio contacts mediated by the system, the system may also send new instructions to the cellular radioset of an XCB device. The instructions may modify its cellular wake cycling so that it can be tracked with more frequent cellular updates. In some instances, the system will know that a radiotag is lost before the owner does.

Importantly, battery power is not wasted on useless cellular radio activity before a relevant event occurs. In one embodiment, the lost radiotag may be activated by someone who comes across the lost child, asset or pet and presses a button or other activation switch on the radiotag. Or activates the device unknowingly just by its close proximity of

the passerby’s smartphone. After activation, the radiotag begins a location determination and a cellular broadcast. Because the radiotags may include an LTE-M cellular radio, location is readily determined by AGPS or other network-assisted location service, and the location is readily transmitted to a system administrative host, so that the system can generate a notification to the owner/subscriber, for example. The system will react promptly so that the radiotag can be reprogrammed to broadcast in an “active-tracking mode” for a period of several weeks, or even a month or more, using cellular eDRX power savings mode with frequent wake up for paging opportunities and location update, so that the system can generate a notification to the owner/subscriber and owner is able to track and recover the radiotag and attached asset.

BT radio control of power management over the processor(s) and dual radios is a design strategy that enables extended portable operation of the devices in which the BT radio acts as a virtual “wake up radio”. Carriers in the 5 to 11 GHz bands may also be employed for wake-up radio and bootstrapping of LAN interconnecting of peripherals and centrals, servers and clients. But advantageously, the digital packet structure of BT radio signals is compatible with data, voice and video broadcasts. In the XCB devices, a polyphony of radio interconnections is achieved. Synthetic radio is used to sample the digital radio envelope in the MHz and GHz bands, yielding an organic network of community monitoring devices with a high level of “situational awareness” to reduce use of spyware and malicious software and to offer services such as location finding, wayfinding, data sharing, and detection of a lost or left behind item before the owner knows the item is about to be lost, for example.

As an introductory consumer product, for recovery of a lost or left behind item, each radiotag or device can include a speaker or LED that can be activated remotely, for example a beeper that emits an audible tone, or a speaker that interacts with nearby persons, to aid in locating it when misplaced. For more privacy, a display screen operated on a copacetic smartphone can be used to guide the user to the lost time. A proximity locator toolbox (described in earlier filings) is active to 100 meters or more. When used with these poly-radio radiotags, a cloud host combines global cellular and BT networks into a highly granular wide area network for finding lost pets, children and wayward assets—generally with meter resolution on a minimal energy budget. A synthetic radio chipset with multiple antennas, diversity antennas, or “on demand” multiband antennas enables an economy of manufacture and the flexibility to be wirelessly upgraded to improved performance configurations or new features. Alternatively, an IC approach that combines the BT and cellular radios on a single chip, linked to an off-chip general purpose processor with larger flash memory unit, is an economical means for making a mass-production XCB radiotag.

In more advanced consumer products, a display or voice interface may be provided to map the lost item in a virtual topology of landmarks and vectors and to provide distances and directional cues that speed its recovery, the electronic equivalent of a “you are getting hotter/you are getting colder” for recovering moving targets.

BT radios may be configured to operate in one or more low power BT-radio states, and will transition from standby to an active BT-radio awake mode in response to a qualified BT radio signal or traffic pattern, for example, or will actively scan for and respond to inquiry scan requests and connection requests. When the BT radio is in passive “always listening” mode, the BT radio may report inter-

cepted radio traffic to its modem and even populate a flash memory database with a rolling stack of records that constitute each a “radio snapshot” of surrounding radio activity, while not necessarily responding actively to the captured radio signals. In active mode, the BT radio may solicit responses or may actively respond to inquiries from other BT radios. BT radio traffic may include actionable qualified BT radio signals and non-qualified BT radio signals. Both qualified and non-qualified BT radio signals are part of the BT radio topology or “bubble” around a BT radio that may intercepted when the BT radio is listening.

By adjustments to the duty cycle, XCB devices spend a significant amount of time in sleep mode—but because of the low latency, in practical use they appear to be “always listening radios” (ALR). A qualified radio signal addressed to the BT radio will wake the BT processing circuitry, or can wake a cellular or WLAN radio if needed. Cellular, WLAN or PLAN network connectivity may include IP packet data connections that enable full global-area network (IPv6 Internet) communications, M2M remote activation, or interaction with a cloud host, for example.

The cellular radio transceiver is configured to cycle between at least one cellular-radio sleep mode and a cellular-radio awake mode. The radio will transition from the cellular-radio sleep mode to a cellular-radio awake mode in a paging window, or when an internal connection request is generated. A qualified radio signal addressed to the cellular modem includes a cellular radio unit identifier and is transmitted in a paging window on a cellular network for which authentication has been made and access has been granted.

But once awakened, either BT or cellular radio will receive and convey radio commands to the processing circuit(s), commands that are executed while the device is in the processing-circuit awake mode before reverting to its default sleep mode. The processing circuit is enabled to cycle between a processing-circuit sleep mode and a processing-circuit awake mode in response to activity from the BT radio, the cellular radio, from an onboard sensor, a clock signal, or according to machine intelligence.

Lightweight XCB devices are rapidly achieving success as wearables and as radiotags for attachment to assets, pets and even children. Embedded embodiments have been realized. By tapping into the power of BT and cellular networks to do positioning and proximity mapping, the tracking and finding of radiotagged objects becomes simple and doable from anywhere with a basic package. Voice user interface functions bring an intimacy to the 5G networks that permeate the IoT and a universal user interface is achieved with just a microphone and speaker as part of an XCB radio device. XCB display functions need not provide the photographic impact of a smartphone, and in fact by eliminating the display screen, very significant increases in battery life are achieved.

The radiotag devices may include a user interface for voice messaging as part of a voice control package. The user interface may include a switch to activate an “on demand” location fix and communication package useful in child, asset or pet location services and recovery. For example, smart systems enable alerts, messaging, maps and mixed media support with “community open arbors” system access for location, wayfinding and proximity management services.

It is to be expressly understood, however, that the drawings and examples are for illustration and description only and are not intended as a definition of the limits of the inventions. The various elements, features, steps, and combinations thereof that characterize aspects of the inventions

are pointed out with particularity in the claims annexed to and forming part of this disclosure. The invention(s) do not necessarily reside in any one of these aspects taken alone, but rather in the invention(s) taken as a whole.

BRIEF DESCRIPTION OF THE DRAWINGS

The teachings of the present invention(s) are more readily understood by considering the drawings, in which:

FIG. 1 depicts a networked system with XCB radiotag 10, reference hub 20, user equipment 30, and cloud host(s). A CALL HOME 1 is depicted.

FIG. 2A is a sketch of a first XCB device 10 containing a Bluetooth and cellular radio pair with button switch on a keychain. In these drawings, signals from the radios are indicated by concentric arcs that connote electromagnetic waves, wider for cellular radio signals C and narrower for BT radio signals B.

FIG. 2B is a perspective view of a first XCB radiotag 10 with button and keychain.

FIG. 2C is a CAD view of a second XCB device.

FIG. 3 is a view of an alternate XCB device with pocket user interface.

FIG. 4 is a view of a card-sized wallet device with basic user interface.

FIG. 5A is a view of an XCB radiotag with annulet for mounting on a chain or link.

FIG. 5B is a view of a sealed XCB device having a disc shape.

FIGS. 6A and 6B are plan and perspective views of an XCB radiotag with user interface.

FIG. 6C is a network schematic showing an XCB radiotag and cloud host(s) in a combined BT and cellular global network 1110.

FIGS. 7A, 7B, and 7C are schematics of alternate dual-radio XCB device embodiments.

FIG. 8A is a view of power states of a BT radio showing multiple sleep and standby states with functional modularization of power consumption. An “always listening” mode or state is defined.

FIG. 8B is a view of power states of a cellular modem according to a 5G network standard.

FIG. 9 and FIG. 10 are views comparing and contrasting BT and cellular localization strategies. Depicted are a Bluetooth Proximity Locator Services Toolkit and a Cellular Remote Locator Services Toolkit.

FIG. 11 is a flow chart for driving location management logic and power consumption using motion sensor data from a radiotag and a smartphone. The concept of a “safe zone” is introduced.

FIG. 12 illustrates a common problem encountered in tracking lost assets, here a “dropped keys” scenario.

FIGS. 13A and 13B illustrate an analysis and systems approach to a “lost dog” scenario.

FIGS. 14A, 14B and 14C illustrates a radiotag solution to proximity monitoring and avoidance for vehicles and wildlife.

FIGS. 15A and 15B are views of location tracking applications using safe zones. FIG. 15C extends the concept of a mobile radio safe zone as relate to pet location monitoring services.

FIGS. 16A and 16B are views of a “reference hub”, shown here with a USB-A male power fitting for plugging into a wall adaptor, for example in a home.

FIG. 17 is a schematic describing circuitry of a reference hub with Bluetooth and WiFi radios in a network context.

FIG. 18 is a flow chart for driving location management logic and power consumption using motion or heading sensor and radio proximity data collected by a reference hub.

FIG. 19 is a view of power management in an XCB device in which the BT radio modem in listening only state controls wake pin of an attached cellular radio modem; by-passing the core processor.

FIGS. 20A, 20B, 20C, 20D, and 20E are sample views of digital signal format of advertising packets of various radio-beacon standard types. Open and proprietary standards are compared.

FIG. 21 is a generic view of a BT digital radio packet structure.

FIGS. 22A, 22B and 22C are sample packet structures that relate to the BT advertising and link layers.

FIG. 23 is a more detailed view of a BT advertising packet and demonstrates how data having characteristics of location and proximity can be extracted from an advertising packet to build a radio contact record.

FIG. 24 is a view of a BT connection request packet and demonstrates how location and proximity characteristics can be extracted from a connection request packet to build a radio contact record.

FIGS. 25 and 26 show how a multirecord payload of a log of radio contacts may be structured.

FIG. 27 is a view of a containerized portable memory record that comprises a radio snapshot or log of radio contact records.

FIG. 28 demonstrates how a radio snapshot is used in assessing situational awareness in a system context.

FIG. 29 illustrates a series of radio snapshots in a chronology.

FIG. 30 is a view of a rolling memory stack containing radio contact records in flash memory of an XCB radiotag.

FIG. 31 is a schematic of another embodiment of an XCB device.

FIG. 32 is a block diagram view of an XCB radiotag and a global area network with cellular network, Bluetooth network, and a cloud host. The system includes voice and display capability for cellular and Bluetooth voice and data networking.

FIGS. 33A and 33B are views of an XCB radiotag with LED Dot Array Display configured for “banner” message display and with capacity for simple inquiries and responses using tactile switches on the front panel.

FIGS. 34A and 34B show the XCB device that includes a OLED display screen for video graphics,

FIG. 35 is a view of an XCB device worn as a wrist strap that functions as a location and wayfinding monitor and as a messaging center.

FIG. 36 is a schematic of an XCB device having voice and display capability for cellular and BT voice and data networking.

FIGS. 37A, 37B and 37C are plan and elevation views of an XCB radiotag for subscription services.

FIG. 37D is a screenshot listing exemplary features accessible for XCB radiotag users.

FIG. 37E provides a screenshot of the tracking application on a smartphone

FIG. 38 shows a flow chart for the steps of a Tap-2-Connect found recovery operation effected by a cloud host in cooperation with an XCB radiotag and the smartphone of a passerby.

FIG. 39 is a view of an ensemble of XCB radiotags with a smartphone on a Qi charging pad.

FIGS. 40A and 40B show a map with a shipping route and a plot of temperature data reported by a data logger device during a cross-country shipment.

FIG. 41 is an oscilloscope image of instantaneous power consumption during a connection event followed by a series of paging opportunities in one of the DRX modes of the cellular modem.

FIG. 42 describes a method of identifying familiar places using the BT radio signal environment in the location.

FIG. 43 is a system view of an XCB radiotag at the center of a BT and cellular network.

FIG. 44 is a block diagram of a chip including a BT radio modem designed to control power to a processor and cellular modem.

FIG. 45 shows the BT/Cellular integrated circuit as part of a radiotag device in its system context.

The drawing figures are not necessarily to scale. Certain features or components herein may be shown in somewhat schematic form and some details of conventional elements may not be shown in the interest of clarity, explanation, and conciseness. The drawing figures are hereby made part of the specification, written description and teachings disclosed herein.

GLOSSARY

The following definitions supplement those set forth elsewhere in this specification. Certain meanings are defined here as intended by the inventors, i.e., they are intrinsic meanings. Other words and phrases used herein take their meaning as consistent with usage as would be apparent to one skilled in the relevant arts. In case of conflict, the present specification, including definitions, will control.

Certain terms are used throughout the following description to refer to particular features, steps or components, and are used as terms of description and not of limitation. As one skilled in the art will appreciate, different persons may refer to the same feature, step or component by different names. Components, steps or features that differ in name but not in structure, function or action are considered equivalent and not distinguishable, and may be substituted in the concepts disclosed herein without departure from the inventive content.

“Bluetooth Radio” or “BT radio”—includes “Bluetooth Low Energy” (BLE), “Bluetooth Classic”, and “Bluetooth Dual Mode” (BTDM) radio. BT radios share signals with other BT radios in an open, shared ISM band between 2.400 and 2.483 GHz. A spread spectrum with frequency hopping is used to reduce interference. The operational frequency band is split into channels spaced by increments of 1 MHz or 2 MHz. Digital data is transmitted by GFSK (Gaussian Frequency Shift Keying) in which a binary “one” is modulated as an increased frequency and a binary “zero” as a decreased frequency, or by DPSK (Differential Phase Shift Keying) as known in the art. Frequency modulation ensures a constant amplitude envelope around the signal that allows higher RF amplification, and achieves a satisfactory bit error rate and tolerance to interference while at low power. DPSK achieves higher data rates. Robust repeat and error correction including CRC, ARQ, or other checksum support functions is combined with whitening techniques known to improve fidelity at low power.

In short, BT is designed for short range, robust, low power transmission and reception—with a built in promiscuity that enables easy interactions between newly discovered devices with minimal setup and requiring essential no dedicated connections. The magic of BT piconets is in a system of

access codes that are part of the packet/header structure. Details are described in the Bluetooth Core Specification, originally published in 1999, with periodic updates.

However, the BT standard also allows for BT radios to “belong” to certain small networks, termed “piconets”, and for members of those network units recognize each other and quickly recover a state of more intimate CONNECTED data exchange termed “pairing”—while not losing the capacity to listen for other radio traffic and to respond selectively or non-selectively when addressed nonspecifically or specifically.

The original BT Specification also allows for BT devices to communicate in “scatternets” formed as loose and transient associations between devices operating in different piconets. Piconets may even share BT devices in different scatternets. This capacity was limited in BLE to conserve energy, but the capacity remains in the core specification and is included here in some embodiments. While less mainstream, research has also been conducted into mesh networks by which data moves from scatternet to scatternet through internodal devices as relays. The capacity to modify the BT Specification to permit this, while preserving backwards compatibility, is anticipated here, and the use of XCB devices of this disclosure as internodes between piconets, scatternets and as members of mesh networks is, by intent, not precluded in future applications.

BT radio signals are formatted as digital packets. BT devices include packet composers and decomposers. BT radios also include correlators with registers for sorting and identifying received signals based on access code or service, for example as described in US Pat. Publ. Nos. 2002/048330 and 2009/0086711, which are incorporated herein for all they teach and reference.

“Access codes” are part of a header that addresses BT radio traffic. For example, a general inquiry access code (GIAC) identifies traffic broadcast to any listening device, and indicates a discoverable device. Other inquiry access codes may be directed to individual devices, such as particular members of a piconet. The access code may be derived from, but is not the radio unit identifier of the transmitting device or the intended receiving device.

BT unique radio unit identifiers (RUI) or more broadly, “radio signatures”, as used here, may be a MAC address of a BT radio device or may be a universal unique identifier (UUID) or a part of a UUID, and may include a serial identifier assigned by the Bluetooth Special Interest Group (Bluetooth SIG) as administered through the IEEE Standardization group (accessible via a WHOIS-style lookup). The RUI may also include a part number given by the manufacturer or a nickname assigned by a device owner. The SIG standard also permits developers to encode a “group identifier” (or as used here a “community identifier”) inside an extended unique identifier (EUID) issued by the manufacturer, inside the BD_ADDR, inside a Service UUID, or inside the universal unique identifier of the radio unit (where degeneracy of the 128 bit identifier permits). Proxy identifiers such as service UUIDs are linked to services associated with a discovered BT device. Identifiers may include payload URLs and payload unique identifiers (UIDs) that identify proprietary services. The payload may include frames or “values” containing more information. Payload information may include sub-type or location, advertising data, sensor output in digital form, and as used here, records of Bluetooth radio contacts, for example. Community identifiers may also point to a designated IP address by which WAN networked communications may be directed to a cloud administrative host, for example.

Service identifiers inform the radio of protocols to be followed in sending or receiving data, and allow developers to create tools that incorporate elements of the payload as “deep intent” triggers for software applications. Advertising messages may include one or more identifiers and service UUIDs, for example. Other messages may not provide sufficient information to identify all services associated with a device, but a qualified BT receiver can respond to a signal to obtain more information without actually connecting.

For example, identifiers in a message actuate protocols in receiving BT radios that can wake smart devices, direct a smart device to a URL, push a notification to a remote device, or pull attachments from cloud library resources, for example. A smart device, hub, or qualified radiotag, can receive Bluetooth radio traffic from any Bluetooth device in radio proximity, and forward that traffic to an IP address associated with a Bluetooth group or community, after adding a timestamp or a location stamp. By doing so, the smart device serves as a “hub” to transfer Bluetooth traffic radio contact records to the broader cellular network (or vice versa), enabling a host of location-driven services that can be modified according to sensor data.

Many BT devices broadcast their RUI or MAC address in the open, or in response to a SCAN REQUEST. A class of “BT Sniffers” may detect these addresses and compile stacks of addresses and device names as BT traffic metrics. For dedicated peripheral devices, a client application can scan for devices offering services or features associated with a UUID that specifies the GATT services the BT device supports, and in full CONNECTION, data specific to a service or feature can be transmitted across the connection.

The RUI address can be an advertising address, a device address, a dedicated address of a piconet device, a virtual address, or a subscriber address, as is useful in mesh networks and for creating whitelists. Some address standards are open, others are proprietary or are obfuscated or made dynamic to prevent BT snooping.

In recent trends, BT signal payloads may include URLs that link the device to the physical web. Alternatively a community identifier is transmitted in a message as part of a header, routing address, or payload that when recognized by packet decomposer in a receiving device, causes the message to be forwarded to an IP Address and associated cloud host. This approach has enabled community lost-and-found services such as described in U.S. Pat. No. 9,774,410 to Daoura, which is co-assigned and incorporated in full by reference. These community lost and found services may be facilitated by offering a gratuity to any smartphone operators who install the needed software and enroll in a community open arbors “lost-and-found” network. Because the found notifications are made when a BT signal of a lost smart object is discovered, and the remaining uplink and reporting is done automatically in background on the smartphone, the process entails little or no effort by participating community members but serves a valuable need to reunite lost assets with their true owners.

The radio header and payload may also include resource identifiers that direct communications protocols in the link layer and activate software applications keyed to the resource identifiers. This approach is seen frequently with smartphones—installed Apps react in real time to BT transmissions. For example, a received BT transmission can wake up a sleeping device (US Pat. Appl. Publ. No. 2020/0242549), which is incorporated here by reference. More recently, data supplied in the fields or payload of a BT transmission can cause an App to be installed, or if the App is installed and the appropriate permissions are in place, the

App can be run at a particular instance in the program as most relevant to contextual clues in the received BT signal. This is termed “deep intent” to indicate that the App anticipates the user’s thought process and causes the client smartphone to display the most relevant materials from a resource or takes an appropriate action in anticipation of the need. More recently the process has been extended to wall screens, so that “walk up” computing is increasingly automated by invisible BT radio transmissions that identify the user and guess the user’s intent from radio proximity or accelerometry data. For example, if a user picks up a shoe in a shoestore, a BT radiotag attached to the shoe will send a sensor output and a wall monitor will display more information about the shoe, or push that information onto the user’s smartphone. Apps, however, may be limited in their capacity to control smartphone wake states. It appears to be the policy of Android to discourage “wakelock” software, and iBeacon wake signals are compatible only with iPhone handsets.

In connected links, BT signals transmit data. Newer BT 5.0, 5.1 and 5.2 standards permit multi-slot messages for sharing larger amounts of information, even encoding of speech. Connectionless data sharing is also supported in the newer protocols.

“System” is a category of integration in which independent functional units are connected to enable more complex and synergic capacities. Of special interest are probability matching functions. System accessible probability calculations include Bayesian algorithms and probability match functions that may be executed in the radiotag or in the cloud. In some cases, radiotag devices, smartphones, hubs, smartwatches, or smart devices generally, may be provided with software and firmware configured to solve probability match functions from matrix data, for example, as a hybrid node, a nest of coprocessors, or as a set of device subfunctions. Although not underestimating the level of AI appropriate and required to fully maintain and operate an auto-provisionable BT-layer connected to an uplinkable/downlinkable CT/IP addressable packet data network, on the fly—as described here, an intelligent global cloud-based lost-and-found service and system (FIGS. 1, 6, 12, 22, 25, 26, 32, 34, 35), that provides a higher level of rapid response to lost, missing, left behind, wayward, runaway, or stolen “items” (broadly) tagged or embedded with BT and XCB radiobeacons and radiotags—brings the systems to a new capacity, the capacity to identify a high probability risk that a left behind item is becoming lost before it is lost . . . from an adjustable radio proximity of zero to fifty meters, with 0.0 or millisecond network latency. The system may also be configured to locate tagged items with the assistance of a smartphone world community, or a smartphone and beacon world community. The beacon world community topographically follows the earth and its structures, and is connected to the IoT through the system-connected to the lower satellite windows, connected radiotag-to-radiotag by ad hoc, passive, active or selective stealth network connections, loosely defined as a unidirectional or bidirectional exchange of data over a spread-spectrum radio channel or channels, and connected via cellular and cable networks. In one embodiment, the broadcast radio data not associated with user-to-user message content is mined for location-specific unique radio unit identifiers and location-suggestive patterns in the signal characteristics, and the location matching functions are done at one or more levels of the systems (FIGS. 1, 6, 12, 22, 25, 26, 32, 34, 35). Motion or heading data from the radiotag may be used to improve the speed and accuracy of the predictions.

“Smart device” is defined by example. The most commonly recognized smart device is the ubiquitous cellphone or “smartphone” (also termed here a “handset”), having a user/subscriber interface, a powerful battery, a cellular radio, highly advanced computational capacity, an operating system, capacious memory for programs and “apps”, capacitive touch screens, typically a BT radio, and commonly tens of sensors, all in a pocket-sized device. However, laptops, PDAs, Google glasses, smart wrist watches, wearables generally, and any generally portable device with Internet connectivity and onboard processing power is commonly understood to be a “smart device” sensu lato. Smart devices are typically provided with a SIM card when used in cellular telephonic radio communications and each such device is given an IMSI identification number that points to one particular unique device and more generally is termed a “cellular radio unit identifier”. The XCB devices disclosed here may also have a BT manufacture’s EUID, or a derivative thereof, possibly one or more UUIDs in volatile or non-volatile memory, and a cellular IMSI and IMEI, in addition to any serial or lot number given by the manufacturer. The XCB radiotags may qualify as a “smart device” sensu lato when operated as a platform with memory and computational power.

IMSI is an “international mobile subscriber identifier.”

IMEI is an “international mobile equipment identifier.”

eDRX abbreviates “extended discontinuous reception.”

PSM abbreviates “power savings mode.”

VPD is a Virtual Private Gateway.

RSSI abbreviates “received signal strength indicator”, which is an indication of incoming signal strength power or power present in an incoming radio signal and is commonly specified as an up to 8-bit unitless integer. RCPI is an improved version of RSSI, and stands for “received channel power indicator” and is logarithmic in dB, measuring total power received in a defined channel bandwidth at the antenna connector. A more general characteristic is “path loss” which relates to radiation power (dBm) minus RSSI (dBm). Transmission power is specified in some beacon signals, which allows a well-defined estimation of range by use of a path loss calculation. Other variants of RSSI exist, and the definition given here is intended to broadly encompass all such indicators of received signal strength regardless of the details of the indicium. A method provided in U.S. Pat. No. 8,879,993 provides for bidirectional exchange of signal strength to better establish proximity. The BT chip includes DC compensation that is active during receiving of the preamble (first 8 bits) of a BT message, and that function outputs the RSSI.

“Timestamp” is a temporal tag given to a data record. Timestamping is an automated function performed as a background service in most smart devices. Timestamps should be standardized as per ISO 8601 using YEAR, MONTH (MM), DAY (DD), followed by HHMMSS.SSS to specify milliseconds, and optionally as Zulu time (or with a specified offset for local timezone or daylight savings) for best practice.

The capacity of a device to timestamp data is dependent on its clock function and memory organization. Conventional Bluetooth radio tags do not provide their own timestamping (instead data is time-tagged by a companion smart device or by a network host when data was received). But the XCB radiotags of the invention may actively timestamp data for storage in local memory or prior to uploading the data to a cloud host. Timestamping may be an integral process with geostamping.

“Geostamp” is a map tag given to a data record that generally indicates the location of a receiver of a transmission, but in some instances, the location may be a transmitter, as from a lighthouse radiobeacon. In other instances the location of the receiver or the location of the transmitter may be stored in a radio contact record. Each radio contact detected is assigned to a record having a timestamp and geostamp in which the geostamp can be the host device location or the source device location, for example. Related services are “location history” services offered as part of the Android platform. Location is generally stored in coordinates given as latitude and longitude.

Geostamping can be an onboard function, much as a camera associates an image in memory with a location (determined by accessing GPS signals and by making a calculation of latitude and longitude; generally on a dedicated chip included in the device for that purpose). Satellite location systems include GPS, BeiDou, the Indian Nav Satellite System (NAVIC), GLONASS, Galileo, QZSS, DORIS, transmitting in L5 and S bands (1.1, 1.5, 1.6 and 2.4 GHz) for position, navigation and timing, for example. Iridium and Starlink may degrade the quality of Satnav signals in the L1 bands. Cellular network-assisted location services include AGPS and POLTE. Cloud host servers may further refine location accuracy using aggregated data or by correlating POLTE and GPS positioning, for example. Bluetooth radiobeacons having known fixed locations can also be used to refine location, particularly in indoor environments, much as a lighthouse marks a landmark place. Google supplies Eddystone and a Proximity Beacon REST API that allows users to register a radiobeacon with location (Lat/Long) and indoor floor level, for example, and use the radiotag as a geotag marking commercial establishments and places of interest on the Google cloud.

A geostamp or “geotag” associated with a radio contact or sensor data record can be given by a XCB device if the device is provided with satnav capability or with a processor enabled to receive and store location fixes from a network. The functions of geostamping and timestamping can be coordinated, and can be synchronous or performed separately, either on a single XCB device or at different network levels during network signal processing. Because discoverable BT devices reveal information about the user’s proximate location, an explicit permission is often required in the device discovery process. But for some platforms such as Android 8.0 (API level 26) or higher, the Companion Device Manager API gives a generic permission for registered applications to perform device discovery with location information disclosures.

“Sensor”—includes any device having a measurement function, either qualitative or quantitative, parametric or non-parametric. Sensors may measure physical properties such as temperature or motion. Sensors output a digital signal to the processor indicative of the parameters of the physical properties. The XCB and BT devices of the system may function as sensor tags that monitor and report local conditions to a higher level network. Once uploaded, aggregated sensor data may be used in generating a composite map of the local environment as has found application in “crowdsourced functions” such as weather mapping, traffic mapping, and hazard anticipation. Other sensors measure and report background noise level, particular sound patterns, radio traffic level, particular radio signals such as from Bluetooth beacons, and so forth. Sensors include photocells, radiation sensors, motion sensors, velocity sensors, accelerometers, jolt sensors, gyroscopic sensors, gesture sensors, gravitational sensors, heading sensors, magnetic sensors,

compass sensors, clock sensors, switch open/closed sensors, vibration sensors, audio-pattern-detection sensors, vehicle-performance sensors, biological-agent sensors, biochemical-agent sensors, pollution sensors, chemical-agent sensors, temperature sensors, humidity sensors, windspeed sensors, pressure sensors, location sensors, proximity sensors, altitude sensors, smoke sensors, oxygen sensors, carbon-monoxide sensors, global-positioning-satellite sensors, relative-radio-signal-strength sensors, and radio-traffic sensors, for example. Sensors may be provided as packages having multiple sensors or individually. Sensors packages having audio sensors, such as a microphone or diaphragm, may include some level of acoustic-pattern-matching capability embedded in the sensor package. In some embodiments, a sensor is a combined 9-axis motion sensor and temperature sensor. In one preferred device, a sensor is an integrated package having an accelerometer, gyroscope, and magnetometer for each axis. In some instances, the sensor package is incorporated into a processor or an integrated circuit. Also contemplated are sensors for gases such as methane, CO, CO₂, NOX, CBD vehicle performance indicia, QR sensors, aerosol particulate levels, history of sub-zero temperature, history of product over-temperature, analytes such as chemical or biological substances, and the like. More generally some sensors can detect biological agents, biochemical agents, and/or chemical agents for example. Sensor data may be stored in a rolling sensor data log.

“Electronic heading sensors” are solid state devices that combine a 3-axis magnetometer with a 3-axis accelerometer and rate gyroscope that are integrated with a processor for establishing magnetic heading even when the magnetometer is not level with the horizon. The heading sensors may also report turns and tilts.

MEMS-based motion sensors include an accelerometer and a gyroscope. An accelerometer can be used to measure linear acceleration. The physical mechanisms underlying MEMS-based accelerometers include capacitive, piezoresistive, electromagnetic, piezoelectric, ferroelectric, optical and tunneling. MEMS-based accelerometers can be simple devices consisting of a cantilever beam with a predetermined test mass (also known as proof mass seismic mass). Under the influence of external accelerations, the mass deflects from its neutral position. This deflection is measured in an analog or digital manner. Commonly, the capacitance between a set of fixed beams and a set of beams attached to the proof mass is measured. MEMS-based accelerometers generally operate in-plane, that is, they are designed to be sensitive only to a direction of the plane of the die. By integrating two devices perpendicularly on a single die a two-axis accelerometer can be made. By adding an additional out-of-plane device, three axes can be measured. Accelerometers with integral electronics offer readout electronics and self-test capability.

A compass is an instrument used for determining direction relative to the earth’s magnetic pole. It consists of a magnetized pointer free to align itself with the earth’s magnetic field. Miniature compasses are usually built out of two or three magnetic field sensors, for example Hall sensors, that provide data for a microprocessor. The correct heading relative to the compass is calculated using trigonometry. Often, a miniature compass is a discrete component which outputs either a digital or analog signal proportional to its orientation. This signal is interpreted by a controller or microprocessor. The compass can use highly calibrated internal electronics to measure the response of the compass to the earth’s magnetic field. Examples of miniature compasses available in the marketplace include the HMC1051Z

single-axis and the HMC1052 two-axis magneto-resistive sensors sold by Honeywell International Inc., the AK8973 3-axis electronic compass sold by Asahi Kasei Microdevices Corporation, and the AMI 201 (2-axis) and the AMI 302 (3-axis) electronic compass modules sold by Aichi Micro Intelligent Corporation of Japan.

A gyroscope is a device used for measuring or maintaining orientation, based on the principles of conservation of angular momentum. MEMS-based gyroscopes use vibrating proof masses. Those masses typically vibrate at a high frequency. As the sensor housing rotates in inertial space a Coriolis force is induced on the proof mass. The Coriolis force causes a vibration in an orthogonal plane and the amplitude of the orthogonal motion can be measured. This type of device is also known as a Coriolis vibratory gyro because as the plane of oscillation is rotated, the response detected by the transducer results from the Coriolis term in its equations of motion ("Coriolis force"). A vibrating structure gyroscope can be implemented as a tuning fork resonator, a vibrating wheel or a wine glass resonator using MEMS technology.

Accelerometers, compasses and gyroscopes can be used to detect movement when a direction or speed of movement changes and are termed more generally as a class, "movement sensors" or "heading sensors", and are contrasted with the more limited motion sensors that rely strictly on accelerometry.

Sensors also include radio devices designed to detect radio traffic, such as a "ping" from a proximate radio device. Such sensors may detect received radio signal strength. Other sensors may be GPS sensors having a function of fixing a location in present time, and may be combined with other data such as by registering a radio contact, a sensor datum with a time stamp and a geostamp. Sensor or location data may be sent in real time without timestamp by the transmitting device, or may be recorded in a memory with a timestamp for later transmission. Sensor data may be stored in a rolling sensor data log.

Sensors may function as triggers when linked to an enabled machine having instructions for receiving and acting on a sensor output value, where the remote machine is linked to the sensor through a network having at least one node and at least one cloud host server, and where a conditional rule has been set up so that sensor data value(s) or trends may be logically evaluated, for example as greater or less than a threshold value defined by a rule and associated with a conditional executable function. Self-reporting of machine state, such as reporting a low battery level, is also included in the scope of contextual awareness supported by sensor data sensu lato. Preferred sensors are miniaturized so that they may be co-housed with the radio controller and encoder module and sensor activity is controlled to reduce power draw.

"Media" as used to store or transport information generally includes any media that can be accessed by a computing device. "Computer-readable media" may include computer storage media, wired and wireless communication media, or any combination thereof. Additionally, computer-readable media typically embodies computer-readable instructions, data structures, program modules, or other memory containing data. Data also may be stored or transmitted in a modulated data signal such as a carrier wave, data signal, or other transport mechanism and includes any information delivery media. The terms "modulated data signal," and "carrier-wave signal" include a signal that has one or more of its characteristics set or changed in such a manner as to encode information, instructions, data, and the like, in the

signal. By way of example, communication media includes wireless media such as acoustic, radio frequency, infrared, and other wireless media, and wired media such as twisted pair, coaxial cable, fiber optics, wave guides, and other wired media.

As used in this application, the terms "component," "module," "system," or the like can, but need not, refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution. For example, a component might be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a controller and the controller can be a component. One or more components can reside within a process and/or thread of execution and a component can be localized on one computer and/or distributed between two or more computers.

"Remote machine"—may be what is termed here an "effector machine", indicating that the machine executes a physical transformation, such as a garage door opener, or what is termed here an "actuation device" more generally, indicating that the device may be a display for displaying content to a user/subscriber, or may be enabled to transmit or broadcast to other machines and to recruit other machines and devices to actuate performance of designated functions.

A "server" refers to a software engine or a computing machine on which that software engine runs, and provides a service or services to a client software program running on the same computer or on other computers distributed over a network. A client software program typically provides a user/subscriber interface and performs some or all of the processing on data or files received from the server, but the server typically maintains the data and files and processes the data requests. A "client-server model" divides processing between clients and servers, and refers to an architecture of the system that can be co-localized on a single computing machine or can be distributed throughout a network or a cloud. Specialized servers are sometimes termed "Cloud Host" here and are used for operating the Internet and its services. The server may also serve the function of distributing an "Application" to devices in need thereof. Applications are intended to augment the functionalities of the smart device on which they are installed and operated and may also be required to access supplemental resources or databases of the server. Typically, a user or "subscriber" will register with the server's administrator when downloading and installing an Application from the server.

By installing what are called "applications" on a smart device, a developer can add functionality, and for example can supply an interactive display or graphical user interface (GUI) that permits the user/subscriber to navigate through and customize programmable features of a Bluetooth or XCB device that has been paired to the host smart device, for example. In practice, an installable application can support a dashboard for easy access to all the XCB and BT radiotag devices operated by a user/subscriber, for quick adjustments to setup, and for viewing maps or plots of sensor data with trendlines and threshold monitoring that will result in automated actions taken by the system according to logic rules programmed by the user. The GUI typically may also provide any needed programming tools and support for creating conditional rules, geofence definitions, and other user customizations. The user interface may access a "user profile" stored locally or in a cloud host; the user profile may include user identifiers, radio unit identifiers (RUI), condi-

tional rules programmed by a user, geofence definitions, and logs of sensor data uplinked from a radiotag.

“Computer” means a virtual or physical computing machine that accepts information in digital or similar form and manipulates it for a specific result based on a sequence of instructions. “Computing machine” is used in a broad sense, and may include logic circuitry having a processor, programmable memory or firmware, random access memory, and generally one or more ports to I/O devices such as a graphical user interface, a pointer, a keypad, a sensor, imaging circuitry, a radio or wired communications link, and so forth. One or more processors may be integrated into the display, sensor and communications modules of an apparatus of an embodiment, and may communicate with other microprocessors or with a network via wireless or wired connections known to those skilled in the art. Processors are generally supported by static (programmable) and dynamic memory, a timing clock or clocks, and digital input and outputs as well as one or more communications protocols. Computers are frequently formed into networks, and networks of computers may be referred to here by the term “computing machine.” In one instance, informal internet networks known in the art as “cloud computing” may be functionally equivalent computing machines, for example.

“Processor” refers to a digital device that accepts information in digital form and manipulates it for a specific result based on a sequence of programmed instructions. Processors are used as parts of digital circuits generally including a clock, random access memory and non-volatile memory (containing programming instructions), and may interface with other digital devices or with analog devices through I/O ports, for example.

As used herein, the terms “infer” and “inference” generally refer to the process of reasoning about or inferring states of the system, environment, and/or user from a set of observations as captured via events and/or data. Inference can be employed to identify a specific context or action, or can generate a probability distribution over states, for example. The inference can be probabilistic—that is, the computation of a probability distribution over states of interest based on a consideration of data and events. Inference can also refer to techniques employed for composing higher-level events from a set of events and/or data. Such inference results in the construction of new events or actions from a set of observed events and/or stored event data, whether or not the events are correlated in close temporal proximity, and whether the events and data come from one or several event and data sources. Probabilistic inferences lead to predictions, and are an arm of artificial intelligence (AI).

“Edge computing” is a distributed information technology architecture in which raw data is processed at the periphery of the network, as close to the originating source as possible. In basic terms, edge computing moves a portion of storage and compute resources out of a central data center or server and closer to the source of the data itself. Rather than transmitting raw data to a central server for processing and analysis, that work is instead performed where the data is actually generated.

General connection terms including, but not limited to “connected”, “attached”, “conjoined”, “secured”, and “affixed” are not meant to be limiting, such that structures so “associated” may have more than one way of being associated. “Fluidly connected” indicates a connection for conveying a fluid therethrough. “Digitally connected” indicates a connection in which digital data may be conveyed there-

through. “Electrically connected” indicates a connection in which units of electrical charge are conveyed therethrough.

Relative terms should be construed as such. For example, the term “front” is meant to be relative to the term “back,” the term “upper” is meant to be relative to the term “lower,” the term “vertical” is meant to be relative to the term “horizontal,” the term “top” is meant to be relative to the term “bottom,” and the term “inside” is meant to be relative to the term “outside,” and so forth. Unless specifically stated otherwise, the terms “first,” “second,” “third,” and “fourth” are meant solely for purposes of designation and not for order or for limitation.

Reference to “one embodiment,” “an embodiment,” or an “aspect,” means that a particular feature, structure, step, combination or characteristic described in connection with the embodiment or aspect is included in at least one realization of the present invention. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment and may apply to multiple embodiments. Furthermore, particular features, structures, or characteristics of the embodiments may be combined in any suitable manner in one or more embodiments.

“Adapted to” includes and encompasses the meanings of “capable of” and additionally, “designed to”, as applies to those uses intended by the patent. In contrast, a claim drafted with the limitation “capable of” also encompasses unintended uses and misuses of a functional element beyond those uses indicated in the disclosure, referencing *Aspex Eyewear v Marchon Eyewear* 672 F3d 1335, 1349 (Fed Circ 2012). “Configured to”, as used here, is taken to indicate is able to, is designed to, and is intended to function in support of the inventive structures, and is thus more stringent than “enabled to” or “capable of”.

As used herein, the terms “include” and “comprise” are used synonymously, the terms and variants of which are intended to be construed as non-limiting.

The words “exemplary” and “representative” are used here to mean serving as an example, instance, or an illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Rather, illustrations and examples given are intended to present concepts in a concrete fashion.

“Conventional” refers to a term or method designating that which is known and commonly understood in the technology to which this invention relates.

It should be noted that the terms “may,” “can,” and “might” are used to indicate alternatives and optional features and only should be construed as a limitation if specifically included in the claims. The various components, features, steps, or embodiments thereof are all “preferred” whether or not specifically so indicated. Claims not including a specific limitation should not be construed to include that limitation. For example, the term “a” or “an” as used in the claims does not exclude a plurality.

Unless the context requires otherwise, throughout the specification and claims that follow, the term “comprise” and variations thereof, such as, “comprises” and “comprising” are to be construed in an open, inclusive sense—as in “including, but not limited to.”

The appended claims are not to be interpreted as including means-plus-function limitations, unless a given claim explicitly evokes the means-plus-function clause of 35 USC § 112 para (f) by using the phrase “means for” followed by a verb in gerund form.

A “method” as disclosed herein refers to one or more steps or actions for achieving the described end. Unless a specific order of steps or actions is required for proper operation of the embodiment, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the present invention.

DETAILED DESCRIPTION

Poly-radio XCB devices function as portable radiotags, hubs, sensors, and radio contact reporters with power management synergy. In a first embodiment, the devices include a cellular radio modem with cellular radio antenna and cellular radio unit identifier (aka SIM), the cellular radio modem having a wake mode and at least one power-savings mode, a Bluetooth (BT) radio modem with antenna and BT unique radio unit identifier; a processor and processor support circuitry (the BT radio transceiver having a wake mode, at least one sleep mode, and optionally an “always listening” standby mode). The processor support circuitry may include an alarm apparatus under control of the processor. The processor also has at least one wake mode and at least one sleep mode. Also included is a power supply with portable source of DC current. In a first embodiment, the device is further characterized in that following initialization, the cellular modem, the BT radio transceiver, and the processor are operably linked to the support circuitry and power supply, and are all configured to cycle to a sleep mode or standby mode as a low-power default condition with the exception that the BT radio receiver executes a listening mode at a duty cycle with such low latency that it appears as an “always listening radio”. Generally, the processor is configured to cycle to a wake mode when one of (i) the cellular radio or (ii) the BT radio detects a radio signal that carries one or more symbols or frames that satisfy the characteristics of a “qualified wake signal”. A digital correlator, numerical co-processor, or a DSP (digital signal processor) may be used to match the pattern of the incoming digital radio signal to a repertoire of qualified wake signals or to calculate a probability that an ensemble of radio signals in a current location match the pattern of a known familiar location.

More generally, “intercepted” or “captured” “qualified” BT radio signals may optionally be actionable or may be responded to because they are intended for or are addressed to the particular receiver that is listening, whereas “non-qualified” radio traffic is not generally addressed to the particular receiver but is intercepted incidentally by an “always listening” BT radio on one of the advertising or data channels during the course of regular passive scanning, essentially at random out of the spread spectrum. We distinguish signals that conform to a model in which each signal has a sender and an intended receiver, from signals that may be broadcast in the open, and have both intended and unintended receivers.

In other embodiments, the “qualified wake signal” may be a pattern that emerges by meta-analysis of patterns in a fragment or fragments of BT radio traffic in the BT spread spectrum without a requirement for a coherent message structure or a specific addressee. ARQ, FEC, CRC, parity bit summation and other methods for ensuring reception fidelity are not actively enforced when capturing signals at essentially a random with little or no requirement for frequency hopping instructions or synchronization. Bit error rate BER may be assessed indirectly, but errors do not require correction. This non-stringent, connectionless meta-analysis is conducted in a passive listening mode and results in logging

of radio contact data with timestamps. A chronology of “radio contact log data” is built up in a series of snapshots of radio traffic received by a radiotag. The interpretation of the patterns in the log data may include identification of segments of BT digital signals that are characteristic of a location. In some instances, the characteristic segments such as MAC addresses and access codes may be correlated with a whitelist of transceivers associated with a neighborhood or suite of rooms, but in other cases the characteristic segments may be significant by their alien content and structure, and while the signature-like characteristics are significant, they may be of unknown origin, identity and meaning. User private message content is not recognized or logged. Meta-analysis of the BT radio envelope in a neighborhood or other local area may be classified on a “scale of familiarity” (SOF) or “scale of strangeness” (SOS) from a pattern associated with a home safe zone to a pattern associated with an alien and unsafe environment, and the device and system may use the SOF familiarity or SOS strangeness of the pattern in effecting security measures, in controlling pairing with new devices, and in flagging items that have been left behind or lost before the owner of the item knows that they are lost, for example. Radiotagged items that attach themselves to a user (even at a distance) may also be flagged for inspection.

FIG. 1 depicts a system 100 with XCB radiotag 10 (XCB1), cloud host(s) 1111,2400, reference hub 20, and smart device 30. Also shown is a generic BT radiotag 12 (TD1) of the kind described in U.S. Pat. No. 9,392,404 to Daoura, U.S. Pat. No. 9,357,348 to Evans or U.S. Pat. No. 9,525,971 to Pons Vega, all of which are incorporated here in full by reference. XCB device 10, illustrated here schematically, may be part of system 100 for locating and tracking lost items, for collecting sensor data from networks of radiotags 10,12 and for enforcing safe zones independently or in conjunction with reference hub 20, for example.

Radiotag 10 and smart device 30 both have a BT radio and a cellular modem. Radiotag 12 and reference hub 20 have a BT radio. Reference hub 20 differs from radiotag 12, however, in that the BT radio of the hub is plugged in to the grid and is not power constrained. It may transmit and receive at +4 dBm or higher, whereas radiotag 12 is frequently limited to +0 or −4 dBm maximum power to conserve battery, for example. In addition to a BT radio, smart device 30 and reference hub 20 may have a WiFi connection or serial USB connection that permits data sharing with a network host 1111 or 2400.

BT radiotag 12 transmits BT radio signals that may be received by any of the devices 10, 20, and 30. Device 12 may join a piconet with BT devices 10,20,30 or with other BT devices, but is not able to form a radio link directly to cloud server 1111 or cloud portal 2400.

Device 10 differs from device 12 in several respects. Device 10 includes both cellular and BT radios, whereas device 12 has only a BT radio. Both devices include microcontrollers or processors, but because device 10 has a cellular modem, it can connect 1 via cellular radio to cloud host 1111 (shown here with a virtual private gateway 2400). This is defined as a “CALL HOME” 1, but in the interests of saving power, the cellular modem is used sparingly in tracking, locating, and up-linking data.

Data exchanges with the cloud from device 10 generally occur over a cellular connection 1 but may also be routed through reference hub 20 via a BT 4 or other WLAN radio link 5. Similarly, data exchanges with the cloud from device 10 may be routed through smart device 30. In other instances, data received 9 from a BT piconet or other IoT sensor device may be pooled in memory of the XCB device

10 or in the reference hub 20. Data may be shared over BT radio links 2,4,9 and then uplinked to cloud host 1111 via (i) a wired connection 7 from hub 20, (ii) via a BT connection 6 to a smart device 30 and then to cloud host 1111, or, (iii) via a direct cellular radio connection between device 10 and a cloud server 2400 for example. As used here, cloud server 2400 is representative of a class of cloud hosts termed “virtual private gateways”.

Devices in network 100 that can uplink the BT radio layer to the cloud layer include smart device 30, XCB device 10 and reference hub 20. With the exception of a recent BLE 5.0+ standard, BT data is generally transmitted by frequency-shift keying and hence cannot be directly injected into the IP packet data environment by which most cloud and smart device traffic is routed.

A CALL HOME 1 need not be a voice call—the call may serve to refresh a network connection, to get a location fix from the network, or to update the system with current location and status of the XCB device 10. And, if needed, a CALL HOME can generate a notification to a user/subscriber 11 or to a system administrator. Notifications are generally programmable by rules-based logic resident in device 10 or a system host and are conditional on some aspect of the current status or location of device 10, as will be described below. Thus FIG. 1 provides a general overview of the emergent properties of hybrid lost-and-found networks 100 combining BT and cellular devices in a 5G or LTE packet network environment. A challenge for these devices is to define logic controls that limit when the cellular modem is used and how power savings states eDRX, DRX and PSM for cellular radio use are implemented to reduce battery drain while enabling “on demand” cellular network connectivity. Advantageously, whereas BT connectivity is hit or miss in many areas, cellular connectivity, once authenticated to a network, is much more reliable and structured over large areas of the planet, a key consideration in designing a global lost-and-found system 100.

Links 2,4,6 and 9 are BT signals (connected or extended advertising mode); links 1,3,5 are cellular connections. Link 7 is a link between reference hub 20 through a packet data network environment to cloud host 1111, as may be wired, wireless or a combination of both. Cloud links 1,7,8 are routed through the packet data environment of a 5G or LTE cellular network, through Ethernet connections, through WiFi, or through other wireless or digital networks. Link 8 is a link between smartphone 30 through a packet data network to cloud host 1111, as may be wired, wireless or a combination of both. Device 10 may connect 3 via a cellular link to smart device 30 in one embodiment and connect 1 to cloud host 2400 via cellular or WiFi wireless links, for example. Cellular links generally involve one or more cellular towers, base stations and other elements of a cellular telephonic infrastructure (not shown). WiFi links generally involve local and base station routers using IEEE 802.11a/b/g/n/ac/ah/ax and/or 802.15.4a/g WPAN radio technologies, for example.

BT links to device 10 may be optimized using dynamic gain. In making a response to a qualified incoming BT signal, device 10 will assess the RSSI, (or other index of apparent “path loss” such as RCPI or PER) of the incoming BT signal and can boost transmit power (broadcast power) if the incoming signal is weak or intermittent. Conversely, device 10 can reduce transmit power to save battery if the incoming BT signal from a linked transmitter is strong. A BT transducer operating in dynamic mode at a nominal 0 dBm can increase its transmit power to +4 dBm or +8 dBm if a received signal from a linked transmitter is weak, for

example, and can decrease its transmit power to −4 dBm, or even −12 dBm if a received signal is strong, for example. In field use, a BT receiver may experience intermittent signal loss as the RSSI drops to a threshold of about −100 dBm or lower, but in order to restore a BT link or overcome edge effects, the BT transceiver can increase transmit power temporarily, and send a message that includes TX POWER as a field in the packet payload. The receiving device can calculate path loss from the transmit power minus the received signal power, and can increase its transmission power to compensate if needed. The Apple iBeacon, Eddystone, and other beacon formats include a native field with 8 bits for sending TX POWER, where TX POWER is defined as the nominal received power at 0 meters, in dBm, and the value ranges from −100 dBm to +20 dBm at a resolution of 1 dBm. The value is a signed 8-bit integer as specified by the TX POWER LEVEL characteristic in the BT SEC Specifications. As a rule of thumb, the experimentally determined output as measured at 1-meter distance corresponds to the transmitted power minus 41 dBm. A software development kit (SDK) can be used to implement dynamic gain in BT beacons so as to reduce energy consumption, for example. The kit may include a library or table of path loss-versus-distance calculations based on known factors such as type of phone and environment (such as indoor versus outdoor) by which dynamic assignment of gain can be implemented to improve connection quality while minimizing unnecessary power consumption.

Cloud services are provided by cloud host 1111, optionally in cooperation with virtual private gateway 2400. Cloud services can be accessed via cellular radio from device 10, via smart device 30, represented here as a smartphone, or via wired services. The cloud host may serve as a repository for sensor data and user profiles, for example, and may have much greater resources for analytics than the portable devices.

Reference hub 20 may be in cloud communication via WiFi for example, and smart device 30 (here represented as a smartphone) has conventional connectivity by cellular and WiFi to the IP packet data environment of the World Wide Web, termed here a global area network (GAN).

Smartphone connectivity through GSM, LTE-M and 5G networks is ubiquitous and of interest here is the novel capacity to deploy personal and community radiotag inter-nodes 10 having both cellular and BT network links.

Signals received on the BT radio of an XCB device may cause the cellular modem to be activated when context dictates the need for a CALL HOME 1, for example to establish a location fix, to communicate status to a cloud host 1111,2400, or to generate a notification to user equipment 30. In other instances, an eDRX cycle at 2.5 min intervals or a PSM cycle at 10 min intervals may be feasible within the energy budget of the XCB device’ battery, but in most instances, further limits are needed to achieve a useful field life as defined by a battery size and discharge.

The cellular radio is packaged as a modem that stores the cellular network connectivity and synchronization data including IMEI and ISMI data. By combining the two radios in one device, the main disadvantage of cellular power saving mode (that the radio is unresponsive in sleep mode) is overcome because BT radios have a “flickering” standby mode that is essentially “always listening” (FIG. 8A) for radio traffic at low power even when the rest of the device 10 is in deep sleep. The latency of the system is adjustable, but can be satisfactorily balanced by making millisecond switches from passive listening to standby and back in a continuous repeating loop (FIG. 8A), and increasing active

25

power to one or more device components if and only if relevant radio traffic is intercepted.

The hybrid radio networks enabled by community deployment of devices **10** and **20** result in other emergent properties of system **100**. For example, the virtual geofence **21** (dashed box, FIG. 1) formed around reference hub **20** may be stationary and may be a radio tether: a repeating broadcast that defines a stationary radio geofence. The signal quality of the repeating broadcast, as received by the XCB radiotag in need of location monitoring, is an indication of the integrity of the radio tether: i.e., poor signal quality may indicate a deterioration or breakage of the radio tether as for example if the XCB radiotag leaves the safe zone area within the radio geofence. This enables an owner to implement location services such as monitoring a pet in a fenced yard, and also provides a community resource for creating a BT radio map of neighborhood (the BT radiobeacon “lighthouse” effect).

As a basic tracking system **100** for finding lost objects and radiotagged items, aspects of the system that are relevant include: (i) BT radio proximity sensing functions and BT proximity locator services toolkit; (ii) radio contact data collection, data entry and mapping functions, (iii) network servers with relational database functions and some level of machine intelligence, (iv) open access to global IP packet data networks, and (v) a cellular remote locator services toolkit. The systems may also include one or more private IP networks for providing virtual private gateway (VPN) functions. Synergy is manifested in the merging of the very particulate local network(s) of BT radios capable of ad hoc piconets, micronets and local nets combined with the cellular network(s) that can span 2 miles or 20000 miles over transoceanic cables and orbital satellite relays, for example.

In more detail, when BT signals from reference hub **20** are used to define a radio safe zone **21**, the loss or decay of the BT radio envelope (such as measured by RSSI) around a mobile radiotag **10** is an indication of increasing distance, and if the BT radio signal is lost, the device **10** may be configured to initiate a CALL HOME **1**. Data reported to the cloud **1111** or to a virtual private gateway **2400** is used to assess the location of radiotag **10** and to issue notifications to an owner **11** of the tagged asset or to make other interventions if the location relative to the geofence is not in compliance with a rule programmed by the owner/subscriber. BT radio alone is not sufficient for establishing a location fix where local area BT radio traffic is absent. GPS radio is not conveniently installed in BT radiotags because the process of calculating GPS location is very energy intensive. Thus the capacity to make a network assisted cellular location fix is a compelling value of XCB radiotags.

As illustrated here, reference hub **20** may function as a mobile or stationary radio geofence. The hub **20** may include means for monitoring radio signals from radiotag **10**, or vice versa, and may also monitor signals from radiotag **12** and from smartphone **30**, for example. Hub **20**, radiotag **10**, and smartphone **30** may convey notifications, commands and data to and from compatible radio devices **10,12,20** and **30**. In embodiments, the system **100** is configured so that any detection of a signal from any of radiotag devices **10,12** outside a designated “geofenced” area **21** will result in an alarm or a notification to a responsible party **11**. If device **10** loses the signal from reference hub **20**, it may be triggered to CALL HOME and to obtain its location independently using its cellular radio. BT radiotag **12**, on the other hand, is dependent on system **100** to track its location. The relative proximity of device **10** and **12** can be estimated from the strength of the BT radio signals between the devices, and a

26

rough position can be refined by triangulating BT signals between smartphone **30**, the hub **20** and device **10**, for example. Yet surprisingly, BT device **20** plays an important role in networks of this kind because its spread spectrum signal and those of other BT devices define a BT radio topology or envelope that enables power management of the cellular radio in device **10**.

Referring again to FIG. 1, in other embodiments, reference hub **20** may be a conversational hub, such as the smart home hubs sold as Google Assistant, Echo Plus, Bixby, Siri or Alexa. The computing resources of the cloud have been interfaced with an XCB reference hub **20,22** having a speaker and microphone and a voice-cloud interface for asking simple questions. These plug-in devices **20** have BT radios and are useful to monitor radio proximity and to interact with radiotags **10, 22 1510a**.

Radiotags **10** may be used in conjunction with BT radiotags **12** to keep track of things. Of themselves, autonomous ad hoc BLE networks are unique for several reasons, i) because BLE devices are small and are readily embedded in wearables, in things, or even organisms, ii) because they are digital radios capable of energy-efficient radio communication at 0 dBm or less, and iii) because the radio devices in the network are true peer-to-peer (P2P) networking tools in which one device can act as master in one or more networks while simultaneously acting as slave in multiple other networks, and the roles are interchangeable. While initially developed for pairing accessories to cellphones (Ericsson Mobile, Stockholm), BT pico- and micronets have been discovered to have surprisingly useful emergent properties because they dispense with base stations and access points and spontaneously form autonomous FH-CDMA peer-to-peer (P2P) mesh networks. There is no single point at which transmission in a network can be disrupted. By hybridizing BT networks with XCB networks, yet another level of emergent properties is achieved.

The BT radios are entirely self-sufficient digital radios and can pick up BLE transmissions from up to 1500 ft away. The cellular radios are native to cellular networks, and can quickly be located anywhere around the globe. Having both together in an XCB device **10,22,1510a** provides a dramatic increase in search granularity, the cellular radio providing a general location and the BT radio permitting the owner of a lost article to activate a Bluetooth Proximity Locator Services Toolkit so as to locate the lost article by sight, sound, or feel, as will be described below, or even to display a detailed map with the location of the lost radiotag on a companion smartphone.

To enable user programmability, the system **100** may include an application installable on smartphone **30**. Lost-and-found services are achieved with one or more of radio devices **10,12,20** when used in combination with software installed by user **11** on a smart device **30**, for example. Related tracking functions are enhanced by the participation of a cloud host **1111**, but in many cases, only radiotag **10,12** and active hub **20,22** are needed to monitor asset location within a radio perimeter. In some instances, smartphone **30** can act as a hub. The software supplied to user **11**, when installed on smartphone **30**, functions to relay sensor data and radio contact reports to the cloud host, for example, and may also function to receive notifications sent to user equipment **30** and provide a user interface for setup and customization of features of the lost-and-found network **100**. Webpages are accessible using a smart device **30** that may include administrative tools for navigating through, managing, and customizing programmable features of a Bluetooth or dual radio device **10,12**, and for selecting notification

27

instructions and preferences, entering user information, updating or upgrading subscriptions to cloud services, and so forth. The smartphone need not be in a user's possession to be accessible by radio and many functions of a smartphone can be accessed in background while it remains in the user's pocket.

The application generally supports a graphical user interface (GUI) configured to monitor, track or help locate one or more radiotagged assets such as keys, TV remotes, briefcases, wallets and other valuables. Devices **10** may also function independently to keep track of children and pets, and one can be placed in a car so that the car can be found if the user **11** is having a "senior moment" or is simply unsettled after a long day and doesn't remember where the car is parked. Synergy is immediately apparent in that the devices **10** may include sensor functions such as for detecting and alerting if a child is in hot car or has had a fall, for example. Device **10** may also establish a mobile safe zone when placed in a vehicle where access to WiFi is not available.

The cloud is capable, with permission, of controlling radiotags **10,12**, for example by causing one to go into an alarm state as a convenience in locating a missing asset. But the opposite is also true, the radiotags can control functions of the smartphone **30**. Functions such as taking a picture, responding to an email, sending a hug to a loved one, and indirect or direct control of remote machines, such as opening a garage door on the way home, starting the coffee pot from bed, turning off the alarm clock without getting up, and checking that all the doors in the house are closed and locked can be programmed into the system **100** with the radiotag(s) **10** functioning as both a sensor and an actuator. In some instances the XCB radiotag will be embedded in the effector or remote machine.

FIG. **2A** is a sketch of an exemplary dual-radio XCB device **10** containing a combination Bluetooth radio (B) and a cellular radio (C). The device includes a radiolucent hard case **14**, shown here with clamshell construction with upper case member **14a** and lower case member **14b** joined at seam **14c**. Optionally a battery access port may be provided on an undersurface of the case **14**, or in other embodiments the devices may be sealed and may be inductively rechargeable. A USB port **17** for recharging and data transfer is shown at the back lower end of the device.

XCB devices include a battery or mobile power supply and supporting circuitry as will be described below. The case includes an annulet or slot **16** (FIG. **2B**) for receiving a lanyard or chain. One skilled in the art will readily appreciate that there are various ways of associating a radiotag with an asset in need of tracking or likely to get lost. Also shown is an actuator or switch **15** formed on an upper surface of the case. The switch **15** may function as a "homing button" to cause the device to CALL HOME when the switch is depressed once, as will be described below. These features are representative of radiotags that embody aspects of the invention but are not to be construed as limitations of the inventions as claimed. Optionally the switch **15** can be used as a FIND PHONE utility when pressed twice—the device can be programmed to cause an annunciation display by a companion smartphone; for example the phone will sound off with a repeating chime tone or bell while the user looks for the phone. This is used when the user has the XCB device in hand but is looking for a misplaced phone.

FIG. **2B** is a perspective view of first XCB device **10** with keychain or lanyard through annulet **16**. In these drawings, signals from the radios are indicated by concentric arcs that

28

connote electromagnetic waves, wider for cellular signals (C) and narrower for BT radio signals (B).

FIG. **2C** is a CAD view of a second XCB device **10a** with annulet **16a** for keychain or other attachment. The device includes cellular and BT radio modems with multiband antenna in a sealed package. Inductive charging is achieved with a Qi or NFC antenna on the base of the device. In addition to the center activation switch **15a** on the top of the case, there is an RGB-LED inlaid on the wall of the case that extends in a band along about a 180 degree arc opposite the tabbed annulet. This may be reinforced with a steel frame that extends into the annulet **16**. The RGB-LED may be segmented and may operate in a pulsatile or function specific mode to provide feedback during user setup and interactions. Also enclosed is a speaker with resonant voice-box and microphone. Battery life ranges from weeks to months depending on the frequency of cellular network uplinks and the latency in the BT scanning mode.

FIG. **3** is a view of another XCB device **300** with alternate form factor and user interface **301**. The user interface can include button switches, LEDs and a buzzer or speaker, for example. The piezo element may be suitable as a hypersonic whistle for pet obedience training, if desired, and the radiotag may be attachable to a pet collar. In other use cases, the device slips into a pocket of a jacket and keeps tabs on other BT radiotags under control of a user.

FIG. **4** is a front view of an XCB finder device **310**. The device is operable on battery power and includes a battery access port (reverse side, not shown). Optionally the device may include a USB recharging port or an inductive recharging circuit, for example. The device may include user interface elements selected from button, LED, speaker, or even a microphone. In this instance body form factor card-like and may be inserted in a wallet or daybook, for example.

FIG. **5A** is a view of an alternate discoid device **101** operable on battery power. The device includes a sealed shell **52** with annulus **54** and potted internal components for weatherproofing. Optionally, the battery can be rechargeable, such as by inductive recharging. The device may include a multifunction capacitive or diaphragm-type button switch (**53**, center surface), for example.

In one embodiment, the button switch **53** functions to trigger a CALL HOME when pressed. For example, a passerby, who is able to approach a lost dog or who has found a radiotagged asset, may press the button to activate a notification that goes out to the owner, and the notification may include an updated location. In some instances, a system that monitors radio signals from the device may offer other asset management services. Similar applications are readily apparent in managing lost children and assets generally, and will be described in more detail below.

FIG. **5B** is a puck-shaped XCB radiotag **501** and includes edge-mounted touch capacitive switches and a translucent body for viewing the activity of an RGB LED assembly within the sealed case. An NFC antenna or "Qi" charging antenna is provided along with a 2400 mAh LiPo battery, for example. The case functions as a resonant diaphragm with internal speakers and microphones in a noise-cancelling array. Dual-band antennas are configured for BT and WiFi at 2.4 and 5 GHz and a multi-band LTE antenna are connected to BT, WLAN and cellular radio modems under common control of a BT modem and microprocessor. A battery, SIM card and micro-SSD card slot are accessible in a threaded door on the underside of the device. The device connects to a user's smartphone for access to a more detailed user interface, and tracks user radiotags that have been registered with the device. The device will also respond to voice

inquiries and make voice notifications, for example if one of the user's radiotags is slipping off its radio tether.

FIGS. 6A and 6B are views of a contoured device **601**, pocket sized, that includes an XCB radio pair, a speaker, microphone and multifunction switch accessible on front case. The device may include a mounting port for receiving a strap or lanyard and is intended as a minder for tracking children and pets. The BT radio (B) monitors local radio traffic and reports to a system host periodically. The device learns familiar radio environments and issued notifications if the radio environment manifests characteristics of an unfamiliar or alien location or unexpected presence of strangers. Voice SMS messaging is enabled by pressing the switch once to receive a voice message and twice to record and send a message. Each voice SMS message is recorded as a digital file, transmitted, and the message is unpacked for replay at the receiving device. The location of the device can be monitored remotely using periodic actuation of cellular network-assisted AGPS and will report location and heading on a map to a parent smartphone by subscription.

FIG. 6C is a network schematic showing XCB radiotag **10** and cloud host(s) in a combined BT and cellular network **1110**. Global area network (GAN, **1110**) is built from a cellular network, a Bluetooth network, and a network of cloud hosts **1111**. A single radiotag **10** and a single smartphone **30** are shown for simplicity but each layer of the network can include many radio units and computing machines.

The cloud host **1111**, broadly, is a virtual network and may also include one or more virtual private gateways (VPG, **2400**) with private IP addresses. In a preferred embodiment, CALL HOME traffic **1** is addressed to a dedicated IP address of a VPG. Use of private IP addresses with a VPG **2400** reduces any security concerns with remote location tracking of a child, for example, but the rate of battery power loss due to of inadvertent, unauthorized, and network-incidental messaging is also reduced. We have published our findings that unsolicited cellular radio traffic on a typical commercial network consumes an enormous amount of power, and use of a VPG is an effective solution (Yasukawa et al. 2019. Waking and Interacting with an IoT Device in eDRX Mode on Demand), said non-patent document being incorporated here in full by reference for all it teaches.

In one embodiment, the cloud administrative host **1111** uses an IP address to access the device **10** by the BT radio modem **680** or by the cellular radio modem **682**, depending on which radio(s) are active. For security, the radiotag **10** may be operated as a cellular device accessible by an IP address on VPG **2400** to find and track the whereabouts of the device via a dedicated and secure 5G private network or gateway VPG.

Data uplink and downlink occurs in a packet data network, and may conform to TCP/IP or UDP protocols. Data transfer by SMS messaging is also enabled. Data may include embedded AT commands to the cellular modem **682**, for example, or a qualified BT or cellular signal may cause the processor **670** to generate an AT command to the cellular modem. Packets include a header and payload as known in the art.

Location data is of particular interest in a lost-and-found application. The device **10** may include one or more logic triggers that causes a cellular network connection request, direct or indirect, and the upload of data. We have termed this a CALL HOME **1**. The trigger can be sensor data such as accelerometry or button data, a timer, or may be a trigger inherent in the topology of the physical web, for example. A direct upload of location data to the cloud host **1111** or VPG

2400 can be requested by the network or by the device **10**, either when the cellular modem **682** executes or responds to a paging window call or when the BT radio **680** receives a cellular connection request via BT radio signal **2**. At any time there is a BT connection with a piconet, in which there is a cellular-competent device **10** in the piconet, indirect uplink of data to the network may be executed over BT radio links **2**. Generally smartphone **30** will forward the data to the cloud host via network connection **8**.

In one embodiment, the SIM module **684** may serve to establish an exclusive private IP address for device **10**. VPG network **2400** may collect location information periodically from cellular modem **682** to create a "trail of waypoints" that track locations of device **10**.

The cloud host also adds a layer of artificial intelligence. By supplying and aggregating data from sensor networks composed of devices **10,12,20,30**, dynamic control of cellular radio activity in the individual devices **10** is enabled. As internodes between cellular and BT mesh networks, devices **10** serve an important role in bridging connectivity over much larger areas, while also preserving the proximity and open intimacy that characterizes BT radio. The cloud host may be useful in storing data and preferences, in looking up device identifiers, in making notifications across the Internet, in making long distance connections, and in aggregating large amounts of sensor data. With increasing use of machine intelligence, aggregated sensor data in combination with geostamps and timestamps can be the source of valuable warnings and notifications addressed to particular user/subscribers or the community as a whole. The deployment of XCB devices **10** by a community results in emergent properties of a system that cannot be predicted from the sum of the individual parts. Synergy is immediately apparent in that location and sensor information can be uplinked directly to a cloud host via the XCB cellular radio (when powered on) and at all times shared with local networks of Bluetooth devices via the BT radio set. Location data kept current in this way provides lost-and-found systems with the capacity to downregulate or upregulate location collection as a dynamic process that responds to variable levels of uncertainty in the system and consumer needs. Data may also include logs of radio contacts intercepted by device **10**, and when aggregated with timestamps and geostamps, the radio contact log data is useful to establish situational awareness by which a user in possession of the data can determine whether the radiotag is where is supposed to be or not, and if not, then to find and track it as needed.

The cloud host server **1111** may include a REST API **613**, for example. Once authenticated, the cellular modem **682** with radio **683** can uplink data to the cloud host **1111** and received commands and data. Using an API **613**, the cloud host parses sensor data, radio contact records, extracts relevant information, and combines that information to generate executable commands that may take the form of a notification, a warning, or an intervention. User programmable commands that are conditional on sensor data, location, time or other inputs may be stored in user profiles in database(s) **616** and accessed at administrative engine **612**. Any notification or executable command is handled by the network engine **614** and may involve one or more smart devices **30** or other remote machines as intermediaries, or may be delivered directly to the device **10** during a paging opportunity when the cellular modem **682** is receiving or directly to the device **10** in a BT signal via BT radio **680**.

Cellular modem, **682** includes cellular radio **683**, which is connected to antenna **683a**. Modem **682** may be for example

a Monarch LTE GM01Q (LTE-M/NB-IoT such as the SQN66430 SiP) or NB01Q (NB-IoT) LGA module with integrated SIM platform (Sequans, Paris FR) for machine data exchange. Monarch SOCs such as the SQN3330 generally includes an integrated cellular RF front end, but not BT radio. Sequans modules typically support a variety of LTE bands for worldwide connectivity and consume less than 1 μ A of power with PSM and eDRX modes and providing for batch data transmission in a centimeter-sized combination. An example of the power requirements of a Sequans cellular modem is shown in FIG. 41.

Generally, the information needed to authenticate to the cellular network is stored in a SIM unit 684 that is part of cellular modem 682 and can also be used for higher quality encryption of data. Cellular networks are closed networks and connections are subject to higher authentication security administered by the network. All cellular radio devices are authenticated by IMEI and IMSI information contained in a SIM module, as known in the art. A dedicated control frequency is used for coordinating the connection of user equipment (UE) to the network.

GPS chip 688 is shown with a separate antenna 688a. Antenna 680a is tuned for BT spread spectrum transmission and reception. Notifications may be received via either the BT radio 680 or the cellular radio 683, and may result in a display such as activation of speaker 621 via acoustic driver 622. Optionally, a microphone 620 is included so that responses to notifications can be sent. Both the BT radio and the cellular radio are capable of transmitting and receiving voice signals.

The processor 670 can be programmed, or otherwise configured, using software resident in non-volatile memory (such as Flash memory or EEPROM 650) or as firmware, or a combination of both software and firmware. MCU 670 includes a BT radio die as an SOC 680 configured to transfer data and commands to and from the processor. The BT radio can control the power mode and sleep cycle of the processor and the cellular modem. Exemplary BT chipsets for BT radio include the Nordic nRF52840 (Nordic Semiconductor, Portland OR) with ARM® Cortex M-4 processor, the Dialog DA1468X family, Dialog Semiconductor, Reading UK) or the Texas Instruments CC1640r2F (Texas Instruments, Dallas TX) with low power sensor controller for IoT applications. Other BT chipset manufacturers include STMicroelectronics, On Semiconductor, U-Blox, Silicon Labs, Toshiba, Ankyra, RDA, and Cypress (Infineon). CSR (Qualcomm), Broadcom (Belkin) and MediaTek dominate the supply of BT chips used in smartphones.

In other embodiments, multiband antennas are used. Fractal and diversity antennas are becoming familiar technology and are both inexpensive and compact. Ceramic antennas are also reasonably priced. Because the antenna design includes the circuit board as a whole, the case, the battery, and any NFC or Qi antenna as well, mobile edge radio can benefit from software-defined radio supplemented with software-defined antenna. In a preferred embodiment, FET gate arrays are used for impedance matching over multiple bands. These packages are included in a small radiotag that fits in a hand, in a pocket, on a keychain, a wrist, or is embedded in an asset. Wearable XCB devices find increasing use in monitoring latchkey children in this work-all-day parenting society.

RAM or flash memory 640 is provided for storage of volatile data, such as for data logging of sensor data. Sensor package 660 may include a single sensor or various combinations of sensors as a package. In some instances, one or

more of the sensors are incorporated into the processor. The sensors may include an accelerometer or heading sensor 623.

The size of the RAM or flash memory 640 is dependent on the size of the memory requirement for data. Stored data may include data from sensors 660 and from switches 633. Data from throw- and button-press switches is considered data. Stored data may also include radio contact records (see below). The memory may be supplied as cache memory in the processor, or may include external RAM if data logging functions requires it.

Working memory may also include dedicated registers for handling packet composition and decomposition, for encryption keys, and so forth. BT and cellular radio signal buffers may be gated by the processor and may include registers for parsing commands and command parameters from datastreams. This memory is generally distinct from non-volatile read-only memory 650 for storing processor instructions. Flash or DRAM memory registers may be supplied, or in some instances firmware or combinations of flash memory and firmware are used.

To save power, the cellular modem 682 and the processor 670 may default to a power savings mode and it may be a BT radio signal (received on antenna 680a and conveyed to the processor by BT radio 680) containing a qualified wake signal that tasks the processor to initiate some routine that wakes up various higher functionalities of the circuitry of device 10.

These higher functionalities may include initiating an uplink or a tracking area update (TAU) via the cellular modem 682. In one illustrative embodiment, the cloud host sends a signal to the BT radio 680 via the BT radio of an intermediary device such as smartphone 30, and that signal will cause the cellular modem 682 to initiate a CALL HOME 1, for example, optionally bypassing smartphone 30. In this way, the cellular modem can kept in a dormant or semi-dormant state most of the time but retains the capacity to report to the network and to execute network commands with reduced latency in response to an override command.

Surprisingly, the XCB radiotag is enabled to receive a cellular power management mode override signal (or related power management parameters) in a BT radio signal sent over a piconet or via link 8 to a compatible smart device 30 and therethrough 2 in a connected or connectionless data transfer to the BT radio 680. Alternatively, during a CALL HOME 1, the cellular network may make modifications to the default cellular power savings mode.

Sleep management can include a restricted schedule of cellular activity, for example in a DRX or eDRX mode (extended discontinuous reception cycle) in which the network management node and the user equipment pre-arrange discrete time intervals in which pages will be delivered. The receiving device wakes up to monitor for a paging event (physical downlink control channel) at discrete intervals (along the lines of what is described in U.S. patent Ser. No. 10/313,085 to Nambodiri and PCT Pat. Publ. No. WO2017065671 to Siomina).

During an eDRX event, the cellular receiver is active and linked to the network so as to receive a page. Reception is an active process and may involve transmission of signal quality responses or command acknowledgements. Configurable parameters of eDRX include Paging Time Window (PTW), HSFN (system hyperframe numbering) and eDRX cycle length duration. The eDRX updates the clock synchronization. In a paging window, new commands can be received as a downlink, but generally an eDRX goes by without the need for an uplink of data in a paging opportu-

nity. In modified eDRX, the initial paging window becomes a connection for data uplink so that a cellular location fix by the network PoLTE service or an equivalent can be completed. The device then returns to sleep or idle mode unless other paging instructions are sent. eDRX parameters are established during at ATTACH and TAU data transfer in the initial connection request or in subsequent updates to TAU. By these adaptations, eDRX can become a routine process of acquiring and storing a series of locations or waypoints, each with a timestamp.

Wake up in eDRX can be modified in response to a PDCCH page with a RCC-compatible request for location assistance. The network location assistance request (LAR) involves sending a snippet of signals captured from a plurality of cellular base stations back to the network, generally about 30 Kbytes in length, and receiving in return, a position fix with latitude and longitude from the network. In this way, network location fixes may be obtained every 5 min or 10 min, and there may be a TAU once an hour or three times an hour as required to maintain network synchronization and to balance network loading, for example.

During a tracking area update (TAU), if for example the device 10 has shifted out of a cellular tower coverage area, the cellular modem will lock on to a new tower with stronger signal to authenticate itself and renew its network connection at the new tower. Location data is updated during this “handover” process and will be stored in the memory of the host device or in a network database. The cloud host 1111 can be notified if device 10 is reallocated from one cell to another as it moves. Because this can occur when cell traffic is being levelled (i.e. by shifting users from a crowded cell base station onto an adjacent base station having lighter traffic) the cloud host can monitor the base station carrier channels in the network path to differentiate location changes that are traffic load driven versus changes initiated because the cellular radio 683 and antenna 683a detected a stronger signal from an adjacent base station and elected to initiate a handover to the new system transmitter because it had been moved.

Once the cellular radio is on and authenticated to the network, then network-assisted location fixes on its transmissions may be performed automatically. When requested by the network, the device 10 may supply GPS coordinates or data to assist in AGPS, for example. The device may optionally include a Satnav radio 688 and antenna 688a with specialized processing module for calculating position from the timed signals of satellites in low earth orbit. Some cellular radio chips 683 are provided with the accessory GPS radio integrated into the die. If needed, a network location assistance request (LAR) can seed a Satnav positioning calculation by the onboard GPS chip so as to reduce time and energy for making the calculations. A device energy budget may be used to balance the relative need for Satnav positioning calculations versus network-assisted LAR position data and may be configured according to user specifications or modified on the fly by commands sent from the network. The network can use PoLTE or AGPS to assist in device location calculation, measurement and reporting.

BT radiobeacons or hubs having known fixed locations can also be used to refine location, particularly in indoor environments. Google supplies Eddystone and a Proximity Beacon REST API that allows users to register a beacon with location (Lat/Long) and indoor floor level, for example, which is used to geotag commercial establishments and places of interest as a physical web. Reference hubs 20 may function as “lighthouse radiobeacons” by broadcasting stationary position coordinates at regular intervals.

Uploads of location data stored in memory 640 may be executed from time to time. A direct upload can also be requested by the network or by the device, either when the cellular modem 682 executes a paging window call or when the BT radio 680 receives a cellular connection request. During a BT connection with a piconet, in which there is a cellular-competent device in the piconet, indirect uplink of data may be executed over BT channels. A powerful set of tools for location-directed network services emerges by combining a Cellular Remote Locator Services Toolkit and a Bluetooth Proximity Locator Services Toolkit, as will be described below.

The device may be rechargeable from an optional recharging source 694. Battery 699 may be disposable or rechargeable via circuit 690. Other energy harvesting means known in the art may be used to extend the operating lifetime of the device beyond that offered by one full battery charge and a switching regulator may be used to manage power to the processor and radios.

FIGS. 7A, 7B, and 7C are schematics of alternate dual-radio device embodiments. Radiotags of this kind can be attached to various assets or to pets, for example, and may also be worn or carried by children in need of cloud-based locating and tracking services. The devices are characterized by a functional combination of two radios: (A) a Bluetooth radio operable with low power for extended battery life in portable use; (B) a cellular radio in a modem that is connectable to LTE and 5G cellular network. The two radios work cooperatively to connect to ad hoc local mesh networks characteristic of Bluetooth and to the cellular base-stations (eNodeB) of cellular telephone networks.

FIG. 7A is a schematic of a first device 70a having separate cellular modem 68 and BT radio unit 63 that share control and are linked to a single processor 72 and supporting circuitry. The processor assembly 71 includes ROM 73 and RAM 74. In some instances the processor instructions are not supplied by software but instead are hardcoded into an array of logic gates within the processor.

The unit is battery powered. Digital radio may be by frequency (GFSK) or phase (DPSK) modulation of a carrier baseband. The device includes two antennae, one 62 for the BT radio at about 2.45 GHz and one 66 for the cellular radio at LTE bands. Dual band BT radio may also be supported. The antennae may be printed on the circuit board in some instances, but may extend from the board as by bayonet mounts and are secured to or embedded in faces of the housing members, which are generally made of a radiolucent material. PCB attachment of ceramic antenna is known in the art. Fractal and PIFA antennae are also considered in the design of a working device. In some instances micro-coaxial extensions are used to position the antenna within the peripheral edge faces of the device and away from any battery or display.

BT core 63 is a low energy processor and includes basic computing functionality for executing programming as well as the capacity to compose, broadcast, receive and decompose digital packets. The BT core has a unique feature that allows an external “wake up” command in the form of a “qualified wake signal” can bring it up to full power. By activating the BT core, the entire device 70a, or selected parts of the device, can be selectively activated. Even if the cellular radio is turned off to conserve energy, it can be activated by signaling the BT core 63. The “always listening” mode is awake (on a “flickering” duty cycle), but other parts of the BT core and processor are in standby or are asleep and internal BT radio power management circuitry wakes up accessory functions only when needed and only

35

long enough to complete an assigned function before being put back to sleep mode. This aggressive power saving mode is responsible for the very long battery life of these tiny devices.

The BT radio component **64** is generally a transceiver. As per the BT Specification, the transceiver can be operated in PASSIVE mode, listening only, or in ACTIVE mode, in which the device is discoverable and will respond to INQUIRIES and PAGES so as to make connections or pair with other like BLE radios. The BT radio can also operate in an advertising beacon mode with repetitive transmission while remaining not connectable. Both BT radios **63** and cellular modem **67** have more than one wake or sleep level. Each wake state has an inherent energy draw as will be discussed with reference to FIG. **8A** below.

Each radio includes at least one radio unit identifier (RUI). The BT radio **63** includes a BT core **64** and memory **65** containing a EUID assigned by the manufacturer, for example. A derivative of the EUID or BD_ADDR unique radio identifier may be transmitted with broadcasts in INQUIRY mode. The cellular modem **67** includes radio **68** closely linked to a SIM (**69**, subscription identifier module). The SIM is a microprocessor-based chip that generates an IMSI (international mobile subscription identifier) that is required to register the device on any cellular network. In some instances the SIM is a card inserted into a Molex connector for example. The SIM may be a nano-SIM, but in other embodiments the SIM is an eSIM (embedded SIM) that is integrated into the circuit board and is not removable. Future SIM units may be software based, but all rely on the use of 128-bit keys to authenticate the device. Skyroam is an example of a virtual SIM. The manufacturer's IMEI (international mobile equipment identifier) may also be used for security. The cellular radio generally includes a modem.

The processor **72** may be a microprocessor or microcontroller and may include a co-processor or graphics processors. The processor is a solid-state digital device that can be programmed, or otherwise configured, using software resident in ROM **73** or as firmware, or a combination of both software and firmware.

The master processor **72** can delegate BT radio control to the BT unit **63** to save energy. The BT radio **64** can cycle to a standby "passive listening" mode, a unique feature that allows an external "wake up" signal to bring it up to full power almost instantly. By activating the BT core **64**, the entire device can be activated. Parts of BT radio that are not in use are in standby or are asleep, and accessory functions are activated only when needed and only long enough to complete an assigned function before being put back to sleep mode. "Always listening" mode is like a flickering candle—with very low reception latency. This aggressive power saving mode is responsible for the very long battery life of these tiny devices.

Flash memory **74** is provided to store data, including sensor outputs and history of radio contacts, including any timestamps and location stamps. An alarm apparatus **75** with LED **76** is used for locating the radiotag when at close range as part of the Bluetooth Proximity Locator Services Toolkit. The alarm apparatus may include a speaker that can be actuated to attract attention and/or a buzzer that vibrates.

Switch **77** and sensor module **78** are peripherals that attach to or are surface mounted on circuit boards carrying the chips. The button switch **77** has multiple functions such as in actuating processor commands. A sensor or package of sensors **78** may be included. A motion sensor is typically

36

provided because the information that a radiotag is moving or stationary is often relevant to whether it is lost or needs to initiate a CALL HOME.

FIG. **7B** is a view of an alternate device **70b**. The device schematic is similar to that of FIG. **7A**, but includes a higher level of circuit integration. Microcontroller assembly **71a** is an ASIC or SOC with controller **72**, BT radio **64** and cache memory **65** for storing the radio RUI. Integration of a particular processor and radio components is partly a matter of convenience but is also useful in miniaturizing the tracking device. As shown here, the processor assembly includes ROM **73**, RAM **74**, and an alarm apparatus **75** with surface-mounted LED **76**. The cellular modem **67** with radio **68** and SIM **69** is packaged separately. The interface components, including sensors **78** and button switch **77** for manual actuation of some commands, are separate units so as to be mounted through or on a housing instead of on a circuit board.

FIG. **7C** is a schematic of an embodiment of a device **70c**, this one having an ASIC **71b** that includes a fully integrated combination of both radios **64,68**, at least one general purpose processor **72**, including supporting circuitry **73,74,75**. The cellular radio includes a SIM module **69**. The BT radio includes a radio unit ID cache **65**. A sensor package **78** is also integrated into the ASIC. Again this is useful in miniaturization. Two antennae **62,66** are provided separately as shown, but may also be integrated into the chip. A button switch **77** is wired separately to the processor.

The integration shown in FIG. **7C** is a higher level of integration than currently practiced. According to current practice, the cellular radio is a fully integrated modem **67** with basic processor included. AT commands may be sent to the modem to control cellular activity, but a great deal of the cellular activity is controlled as part of the network connection **1**. FIG. **7C** may also be interpreted as a synthetic radio pair within an ASIC **71b**, in which each integrated processor **64,68** functions as a modem. Optionally the XCB radiotag **10** may include multiple WLAN radios, including WiFi. As synthetic radio is implemented with software or firmware defined radios, the technical barriers are eliminated for putting multiband radio devices in radiotag-sized packages.

FIG. **8A** is a view of power states of a BT radio showing multiple sleep and standby states with modularization of power consumption. The states correspond to the BT protocol stack. The lowest level of functionality of a BT radio is a SLEEP mode running with low power clock only. To approximate an "always listening" radio mode, a minimal STANDBY state alternates in rapid succession with a "PASSIVE LISTENING ONLY" mode, in which the radio will receive Bluetooth radio signal traffic and may forward received signals to the processor, but does not transmit responses or solicit inquiries. In this PASSIVE mode, the device is unresponsive while listening for and recording non-specific radio contacts and may not be discoverable. By adjusting the STANDBY/LISTENING ONLY duty cycle, a very low energy, low latency system can be achieved that meets performance goals in most respects for a close approximation of an "always listening radio" (ALR) and "wake up radio" (WUR). The latency is almost not perceptible to the user, and the radios are readily able to resume the connected state in a familiar piconet without user attention or intervention, for example, by transitioning from STANDBY to CONNECTED in a few quick steps.

A discoverable Bluetooth radio may be configured to listen for (i.e., SCAN) and to respond to PAGE and INQUIRY signals from other units. These energy states correspond to "INQUIRY SCAN" and "PAGE SCAN". In

ACTIVE listening, the device will recognize and respond to inquiries and pages that include a recognized access code. A PAGE is a radio signal that initiates a connection. The radio unit that receives the page responds in a way (by sending an FSH packet) that leads to a formal CONNECTED mode between the receiving and transmitting radio units. In CONNECTED mode there are two substates: MASTER and SLAVE, which for any two devices are interchangeable.

Bluetooth radio is notable for its robust resistance to interference and dropped connections, and has been widely adopted. The BLE radio protocol standard is attractive than BT Classic and BTDM protocols because of its low energy consumption. Advantageously, controllers with integrated BT radio cores operating at 1.8V are readily available. A BT radio in standby “always listening” mode may burn less than 30 μ Ah power while retaining the capacity to wake up the processor and accessory circuitry from deep sleep in response to (i) a qualified radio command from a smartphone or a reference hub (ii) in response to sensor data, (iii) or in response to ambient radio traffic having selected characteristics, and thus supports portable applications for IoT use. Channel listening without response participation consumes only 0.3 mA at parts of the duty cycle when the receiver is on. In standby between listening periods, power consumption drops to less than 60 μ A (Karjalainen O et al. A Comparison of Bluetooth Low Power Modes, 7th Intl Conf Telecomm 2003. IEEE DOI: 10.1109/CONTEL.2003.176900). By controlling latency in a reasonable range, overall power consumption can average out as a sub-milliwatt load (while offering ALR continuously during extended remote deployment).

In the newer BLE standard, inquiry is limited to three out of forty channels in the BT spread spectrum. Selection of the advertising channels was made to avoid most competition with WiFi channels that overlap the BT spectrum. When BT devices elect to exchange data, they can do so in an extended advertising state that uses some of the data channels not reserved for advertising. Devices can also exchange FHS packets and coordinate frequency hopping regime across the data channels to avoid interference. However, what this schema gains in reduced latency for discovery also has the effect of reducing traffic density in the unused data channels. In one application of radio scanning of the data channels, unused channels can be populated in extended advertising mode according to the vacancy rates of the channels so as to better distribute bandwidth usage. What emerges from the scans is a distribution of channel usage across the spread spectrum (data and advertising channels), and an index can be derived from this distribution pattern provides insight into the familiarity or strangeness of the locale surrounding the XCB device. Devices that encounter primarily traffic on the advertising channels are clearly in a public space that is readily differentiable from the more balanced data traffic that characterizes a home or office. With increased availability of synthetic radio, the capacity to “sniff” all the BT channels for traffic is realized and offers the users a simple method for scaling their encryption efforts in direct proportion to the local radio topology or concentration of advertising channel use.

BT radios may also be operated in a BEACON TRANSMISSION mode, which is not connectable. The radio broadcasts a canned message at a regular interval and is unresponsive in this state to any radio responses or inquiry traffic. A baseline energy budget for a BT radio in not-connectable advertising mode may consume about 30 μ Ah assuming an intermittent transmit period of 20 msec, a transmit cycle of 2.5 sec, (i.e., 1440 transmits per hour), and a transmit power

of 3.5 mA (0.3-30 mA depending on packet type and radio hardware). In some BT radiobeacons, the transmission duty cycle is adjustable. Transmit power and frequency may be configured according to the application, and with increasing miniaturization of chip architecture to 14, 10 or even 7 nm gate structures, total energy consumption continues to fall sharply, enabling increasingly longlasting IoT devices in packages using either disposable or rechargeable batteries.

Connected units establish a “pairing” relationship that anticipates the frequency hopping regime and any HOLD, SNIFF, or PARK timing. The BT baseband/link manager configures low power sleep and standby modes that separate active transmission and reception sessions.

Access codes define the specificity of the relationship between the units. These formalities are native to the Bluetooth specification, which has received an essentially global adoption as the BT standard for wireless devices ranging from headsets to keyboards to printers to thermostats, smoke alarms, coffee pots, and smart doorlocks to smartphones.

Power states associated with 5G cellular network management are drawn in FIG. 8B. The energy states correspond to a Cellular radio packet environment protocol stack. ACTIVE/CONNECTED mode can include reduced power states. Power savings is achieved by reducing the duty cycle for the radio and associated processor (DRX, eDRX, PSM). These energy states are established by protocols set forth by network operators and by hardware features native to the devices; the UE and the network agree on a wake up schedule, i.e., a “duty cycle” with designated down time.

Following power up, the radio can exist in a low energy DISCONNECTED IDLE state until authenticated to a network. Once connected to a network, reduced power modes that do not include regular updates at frequent intervals are problematic in cellular radio. If the connection is lost and the device goes into low power disconnected idle, the user equipment (UE), without an assist, cannot be aroused from sleep or contacted by the network except during a designated “wake” period. In connected mode, the system implements only PSM as a standby condition. DRX and eDRX states operate on a reduced duty cycle in which the interval between network refresh windows cannot be extended indefinitely. If the network connection fails, the unit defaults back to DISCONNECTED IDLE and must initiate a new connection in order to be tracked by the network.

A Radio Resource Control (RRC) protocol controls the air Interface. The major functions of the RRC protocol include connection establishment and release functions, broadcast of system information, radio bearer establishment, reconfiguration and release, RRC connection mobility procedures, paging notification and release and power control. By means of signaling functions, the RRC configures the user and control planes according to the network status and allows for Radio Resource Management strategies to be implemented. The operation of the RRC is guided by a state machine which defines specific states of the UE relative to the network.

LTE power states have evolved with the introduction of 5G. In LTE, the EPS (evolved packet system) Connection Manager is in IDLE unless there is an active connection, whereas in 5G, the Connection Manager connects when the user equipment is attaching, and stays connected whether the connection is active or “suspended”. In other words, the RRC is in DISCONNECTED IDLE or CONNECTED for LTE, and is INACTIVE or CONNECTED for 5G.

Standard cellular radios of smartphones may have signal strengths of 0.6 watts or 3 watts (for comparison, most

Citizen Band radios transmit at 4 watts). Battery voltage is typically 3.8V. As a result, a typical transmission consumes 150 to 800 mA, not a trivial amount. Total current load on a smartphone in use ranges from 0.6 to 1.9 A, of which only 50 mA is related to WLAN radio draw. Other processes such as GPS calculations, processor boot and instruction set execution, camera imaging, capacitive screen sensors, and LED display lighting account for the remaining draw. LiPo batteries in Android smartphones may have a battery capacity of 3000 to 4000 mAh. Not surprisingly, most smartphone users find that the battery requires daily recharging, as would not be acceptable for a micro-sized device that is to be remotely deployed without access to a recharging dock for extended periods of time. As a consequence, a radiotag device for IoT use must find energy management solutions that surpass conventional cellular performance benchmarks.

In one embodiment, BT radio power states can be used to manage cellular low power states so as to implement a more robust standby condition. The BT radio core can actuate and adapt the cellular modem (and device processor(s)) according to more flexible rules that override the inflexible duty cycle that governs cellular extended sleep modes in conventional cellular modems. The BT radio can also control a CALL HOME event, which includes the initiation of a cellular Connection Request with network attachment if absent. In other words, the BT radio can assist in recovering from a cellular connection failure.

In our experience, in the XCB devices **10, 22, 1510, 70a, 70b, 70c** and applications described here, only about 4% (or less) of the energy a typical cellular modem expends is actually needed. More than 90% of cellular modem power can be throttled off-on the condition that the cellular radio activity is throttled by the BT radio modem as described here. With stringent application of integrated power management, XCB device field life readily exceeds one week in richly functional use, and can approach 1 year in remote monitoring applications. Power savings are achieved by selectively powering the cellular modem and controlling power to the processor according to the state of the BT radio, and by establishing qualifying radio signals and radio traffic patterns that, when intercepted, wake the processor. By using the BT STANDBY “always listening” mode to control the cellular modem, a “Wake up!” command may be executed so that the cellular modem of the device is activated to initiate a cellular network connection with a cloud administrative center on any available cellular network. The initiation of a cellular network connection is termed here as a “CALL HOME” **1**.

FIGS. **9** and **10** are views comparing and contrasting cellular and BT localization strategies. Cellular location tools are termed a “Cellular Remote Locator Services Toolkit” and may include triangulation and network-assisted location services. The toolkit includes components of the radiotag with cooperative functional connections to a network or smart device that expand the user interface, display and analytical capabilities of the toolkit. The figure defines a CALL HOME **1**. A CALL HOME generally includes a current location update and may require a TAU. In one instance, the cellular modem of radiotag **10** is awakened on a scheduled paging window or by a trigger and a network connection with a cloud host **1111** is refreshed or re-established. The device **10** may call home to report location, or may call home to obtain a location fix by POLTE, for example. Alternatively, location can be triangulated (L1, L2, L3) from cell towers **901a, 901b, 901c**, or by GPS, for example. The location may be stored in a database on cloud host **1111** or may be stored in an internal memory of device

10, or both, for example. The current location fix will inform any notification **91** to a user **11** via smartphone **30**. Once a network connection is established, the user/subscriber **11** has the option of adjusting the cellular radio power management and sleep settings, such as to continuous or more frequent updates so as assist in tracking and recovery of the radiotagged asset. Typically the software application or a cloud host can log and plot aggregated data to generate a map of where the radiotag is located. For example, if device **10** is outside a safe zone and is in motion without being accompanied in proximity to user **11**, then the notification may include the current location with timestamp on a map displayable on the user’s smartphone **30**, and may even show a trail of waypoints over time if the device continues to move. The notification **91** may include more options for the user to respond or offers of further assistance. By switching the cellular radio to continuous operation, battery life will be shortened, but the option may be loss of the radiotagged asset, so having cellular tracking capacity is clearly an improvement over Bluetooth tracking alone, especially over longer distances of separation.

The Cellular Remote Locator Services Toolkit may also include a service for initiating a CALL HOME in response to output from an accelerometer or an electronic heading sensor, for example if there is an impact or the tagged asset is moving in a direction away from where it should be going, circumstances that would warrant getting a current location update reported to the cloud host and perhaps a notification to the owner/subscriber. The same Toolkit can be configured to report body temperature sensor output or temperature sensor output as can be useful in fever monitoring or cold chain tracking, for example. The radiotags can include memory for storing sensor data output so as to be useful for a wide range of specialized applications as will be described further below.

Cellular location fixes are less useful over very short distances, but the combination with BT in radiotag **10** offers an elegant solution that uses the best characteristics of both radios, combining the Cellular Remote Locator Services Toolkit and the Bluetooth Proximity Locator Services Toolkits.

FIG. **10** depicts Bluetooth radio locating capability. This is a short-range function and uses radio proximity determinations to zero in on the location of a lost asset. We term this functionality the Bluetooth Proximity Locator Services Toolkit. In this instance, the BT radio of user handset **30** listens for a signal identifiable as the signal of radiotag **10** and identifies a received signal strength (RSSI) of the BT radio signal. The signal strength is an approximation of proximity within a range of a few feet to a few hundred feet. Bluetooth radio antennae are not generally directional, so RSSI is indicative of a radius around a BT radio signal source, shown here centered on radiotag **10**. As the user **11** gets closer, the signal strength increases and the location of the source of the signal is more obvious. The signals of BT radiotags are distinguishable by their RUIs, and in one embodiment, a user **11** who is searching for a lost item can open a user interface on smartphone **30**, and by pressing a device-specific icon on the interface, cause a particular radiotag to emit a beep or flash so that its location is revealed to the senses, even if partially concealed. The command is transmitted in a BT signal to the radiotag. Alternatively, the handset **30** can display a detailed map of an area, as may be refined by UWB device installations, and pinpoint the lost device emissions in the map space. The Nordic nRF52840 BT chipsets have offered directional antenna hookup for several years, and with increasing sophistication in the

41

radiotag finder market, directional antennas are realized in the miniature radiotag body using diversity and fractal antenna types. By electronically switching the baluns of paired antenna from a balanced to unbalanced receiver characteristic impedance, a simple “hot/cold” directional antenna pair is obtained.

Bluetooth range finding may be initiated at distances of 300 to 500 ft in some conditions, but proximity tracking is strongest within 20-40 yards, and for longer distances, the cellular tracking techniques discussed with respect to FIG. 9 may be more effective. For comparison, commercially available GPS, by itself, is accurate only to 20-30 yards. Thus, the light, speaker and buzzer of an alarm apparatus that can be activated when the user is within sight or hearing of the radiotag is more effective than GPS alone. The cellular and BT tracking systems of FIGS. 9 and 10 are complementary in geolocating a missing asset, child or pet; a missing radiotag can be located via cellular triangulation at a long distance, and then when the user is in BT radio proximity, the user can actuate a beeper or LED on the radiotag 10.

As described in U.S. Pat. Nos. 9,392,404, 9,892,626, and 1,050,281 to Daoura, any happenstance discovery of a BT radio signals by a smartphone belonging to a community of users can also result in recovery of a lost item. Briefly, looking ahead to FIG. 13A smartphones may be configured to report received BT radio contacts and RUIs to a cloud host, and by matching the RUI to a user profile, the cloud host can generate a notification to the owner 11. A map can be displayed in which the map coordinates are obtained from the system and an icon displaying an approximate position of the radiotag.

Advantageously, the transient BT radio contact between a third party 13 community smartphone 31 and a wayward radiotag 10 can be used to leverage actuation of the cellular modem of the radiotag. A report 1301,1302 transmitted to the cloud host via cellular tower 901a brings the cloud resources of the system to bear on recovering the lost asset. When the cloud host receives a report of a signal matching the identifier of a lost radiotag, the system can generate a BT radio signal command to turn on the radiotag's cellular modem, and can send that command 1303 (FIG. 13A) to the radiotag via the community smartphone 31, as described in below.

The Bluetooth Proximity Locator Services toolkit includes components of the radiotag and cooperative functional connections to a network or smart device that expand the user interface, display and analytical capabilities of the toolkit.

The Bluetooth Proximity Locator Services Toolkit may also include a service for initiating a CALL HOME in response to sensor data output. For example, an impact as sensed by an accelerometer can cause the Bluetooth radio core to broadcast an undirected Bluetooth advertisement for help, and to activate the cellular modem. A Bluetooth proximity monitoring system can cause the cellular modem to uplink data if another Bluetooth device breeches a proximity limit threshold mandated for communicable disease mitigation. The same toolbox can cause a CALL HOME if a Bluetooth radio tether is lost or fluctuates. The BT toolbox can include a dynamic gain adjust and can transmit a TX POWER index in its messages. By assessing received signal and transmitted signal power, a path loss can be calculated. The radiotag can increase transmit power if needed to recover a lost or intermittent radio tether, for example. The BT toolbox can also include memory for storing and white-listing radio contacts, such as members of a familiar piconet, and can prepare radio contact records of familiar and unfam-

42

iliar radio contacts as part of a radio snapshot of the surrounding BT radio environment to assist in location-related services. The radiotags can include memory for storing sensor data output so as to be useful for a wide range of specialized applications as will be described further below.

And with implementation of synthetic radio capable of UWB radio at 6-11 GHz operation, proximity monitoring can be significantly improved.

FIG. 11 is a general flow chart of a method 1100 for operating location management services with an economy of power consumption. In the analysis, a radiotag 10 is operated with a companion handset 30 and an application for executing the algorithm 1100 is assumed to be installed in computer-readable media on the handset. The analysis is directed at correctly identifying scenarios in which a location fix is needed and the cellular modem should CALL HOME.

Generally, any monitoring of location begins with a memory that associates an initial “location fix” in memory with a timestamp at time $T=0$. This memory may not be in the device 10, but may instead be stored at a higher network level, for example in a smartphone 30 or in a cloud server 1111. Generally, the initial location of the radiotag and the smartphone are assumed to be in close proximity, as during setup, or in the morning of a workday when both items are where they are supposed to be. At time $T=0$, the location of the smartphone can be taken as a proxy for the location of the radiotag and no further location fix by the radiotag is needed. Updating the location of the smartphone is a routine matter and is not an issue. The more pertinent question 1101 is whether to get a location fix by activating the cellular modem of the radiotag 10. While it may be desirable to have a current location for the radiotag at all times, from a power management standpoint in a portable device, this is not practical. Getting a location fix consumes power.

The decision tree for whether or not to get an updated cellular location fix for the radiotag at a future time $T=T+t$, where t is an interval selected based on predictive accuracy, can be made so that unnecessary location fixes are avoided by attention to a) initial location, b) the presence or absence of defined safe zones, c) data related to accelerometry of the radiotag and the smartphone, and d) any recent change in relative proximity of the radiotag and smartphone.

In a first approach, the location fix at time $T=0$ can be classified 1102 as whether the location is inside a “Safe Zone”, where a safe zone is user defined as a space in which order is maintained and things are where they are supposed to be. By using radio-delimited safe zones, a YES or NO answer can be given to the question of whether a radiotag is inside or outside the safe zone. If the radiotag device is “tethered” to a reference hub or a smartphone by a radio signal that defines and anchors the safe zone, then the need to actually get a new location fix may not be urgent unless the tether is broken and lost or is intermittently broken. If the radiotag is not in a safe zone, then it may be necessary to get an updated location fix, but only if conditions are met that merit the energy consumption. Outside a safe zone, there is no simple assurance that something will stay where it is supposed to be.

A device that has moved cannot be assumed to have remained in its established location. If device 10 includes an accelerometer 623, then the simplest sensor output can be a MOTION truth value, TRUE or FALSE. Because the smartphone also has an accelerometer, a Truth Table can be constructed 1103 comparing motion truth values for the radiotag versus the smartphone. The quality of motion (e.g.,

hard vs soft acceleration), the speed, duration, and the direction are also useful, but the simplest and most economical bit of information from the sensor is whether motion has occurred or not.

If neither the radiotag nor the smartphone have moved (FIG. 11, 1104), then the radiotag can continue to SLEEP. If motion of the radiotag has not occurred 1105, but the smartphone has moved, and the radiotag is not in a safe zone, then it may be appropriate to generate, by the smartphone, a LEFT BEHIND alert, and to actuate the cellular modem of the radiotag for a CALL HOME so that its current location can be tracked and the owner/subscriber can be notified. In some instances, however, smartphone motion may occur that is incidental to normal activity, and the LEFT BEHIND alert is reserved for situations in which the radio tether link is lost or about to be lost. A radiotag that has not moved, but loses its radio tether to a smartphone, may CALL HOME to cause a LEFT BEHIND alert to be sent to the smartphone if the radiotag is not in a safe zone 1105. If motion of the radiotag has occurred 1103 at time $T=T+t$ (i.e., after an elapsed AT, where the time interval is programmable), then it may be useful to look (1106) for a change in proximity. Over a range of several hundred feet, RSSI is a first approximation of distance between the radiotag and the smartphone. The proximity may be increasing or decreasing. Proximity is measure by Bluetooth radios as part of core competencies of BT radio and the smartphone will continuously monitor the RSSI of the radiotag signal as part of routine operations. And if there is no change in proximity, then any motion signal 1103 may be spurious and would not necessitate a need for a new location fix. Proximity will continue to be monitored. Similarly, an increase in proximity (strengthening RSSI 1109) is intuitively not likely to indicate a risk of loss of signal, and the algorithm 1100 can be looped to continue to monitor for motion and proximity.

But a fading or intermittent RSSI 1108, as indicates decreasing proximity and increasing separation, could be followed by a break in the radio tether, and for a tracked asset, may necessitate an immediate CALL HOME to get a new location fix and to generate a LOST or LEFT BEHIND alert notification to an interested party. The LOST alert generally occurs if a motion mismatch was detected between the motion of the smartphone and the radiotag, and the LEFT BEHIND alert occurs if there was not motion of the radiotag but the smartphone was in motion.

The logic of the method helps to distinguish between conditions that necessitate expenditure of energy versus conditions that permit resumption or continuation of a resting sleep or standby state. Under ordinary conditions, a sleep state would not be interrupted to CALL HOME unless there was a pattern of motion and/or a change in radio proximity.

Exceptions could be made if the motion data is more granular, for example a hard impact could merit a CALL HOME with status report even if proximity data is unchanged. And it may be appropriate to maintain an awake cellular network connection if Bluetooth radio proximity measurement is no longer possible because the signal has been lost, even if there is no motion.

The time interval AT for iterations of the method 1100 may be adjusted according to conditions. For example, in a safe zone, infrequent execution of the loop may be unnecessary. An interrupt flag on the processor can be set against the accelerometer output. Iterations can be more frequent if the motion activity is greater. Outside a safe zone, the loop may be executed more frequently, and the timing can be dependent on the nature of the motion input, on temperature

or on a change in temperature, on changes in acoustic patterns, or on changes in Bluetooth radio traffic patterns, for example.

Similar information can be obtained from other sensors. For example, a temperature sensor may provide evidence that a device has moved from a room temperature environment to an outdoor environment. A compass, G-sensor, or gyroscopic sensor output can suggest a change in attitude (or altitude). Acoustic information collected by a microphone may also suggest a transition from one environment to another. Motion sensor data can be collected and transmitted as described in U.S. Pat. Nos. 9,961,523 and 10,638,401 without added length of the BT radio message format, for example, and similar packaging can be used to transmit other sensor outputs. Recently, heading sensors have reached chip scale, and combine gyroscope and compass with 3D accelerometry to vector out fluctuations of momentum around a consensus heading and velocity. These chips are finding increased use in improving the logic needed for efficient utilization of cellular connectivity in XCB devices.

FIG. 12 demonstrates sequential use of Bluetooth and cellular radios as a proximity tracker in a hybrid Bluetooth/cellular wireless system 1200 for monitoring radiotagged assets, here a “lost keys” scenario. Here radiotag 10 is shown attached to a keychain 110, and the scenario is one in which the keys are dropped by an owner 11 while enroute to an appointment. In the sequence of snapshots of FIG. 12, as the owner 11 walks, companion device 30 monitors signals from the radiotag 10. This data may be shared with a cloud host 1111 in sequential transmissions 1231, 1232, 1233. Initially, all is well, but the owner drops the keychain at 1234, and does not realize the keys have been lost.

System 1200 can detect the drop of the keychain in several ways. There may be a loss of the BT signal, but before that, from accelerometer and motion or heading sensor data, the system can infer that the owner is walking ahead without the keychain because motion or heading of the radiotag has stopped even though the owner has kept walking. The system flags the spot 1234 as the last known location of radiotag 10 and recognizing the illogic of the motion sensor data or other indicia of disorder, issues a notification 1235 and causes an alarm such as a bell or a vibrator on the owner’s handset 30.

Devices 10 may use a multi-axis accelerometer 623 to detect movement and velocity of the device, a useful bit of information in understanding what is happening to the device. Is it being moved? Has it fallen still? Answers to these questions can be of great help in knowing where to look for a lost device and systems for generating accompanying alerts and notifications that provide motion sensor data are described for example in U.S. Pat. Nos. 9,564,774, 9,774,410, 9,900,119, 9,961,523, U.S. patent application Ser. No. 15/959,250, and US Pat. Pubs. 2015/0356862, 20150356858, and 20180190103, wherein all said patents and applications are co-assigned at this filing and are incorporated herein in full by reference.

In past versions, the logic for detecting “lost” or “left behind” scenarios is resident in a software package installed on the user’s smart device 30. However, in an advance in the art, the XCB device can be provided with firmware that monitors BT radio traffic from the owner’s smartphone and other BT local sources and if there is an abrupt change, can CALL HOME 1236, uplinking current motion, heading and radio topology data. Because the network is intimately monitoring the smartphone’s position, the mismatch can be almost instantly recognized. This eliminates a nagging latency that has been observed in practice because the

45

smartphone OS forces the finder app to operate in background with a lag of up to 10 minutes between updates. Transferring some of the smarts to the XCB radiotag, and getting a quick uplink from the XCB radiotag to the system host, the problem of late recognition of “lost” and “left behind” scenarios is eliminated.

The owner may realize that there is a “lost” or “left behind” problem within a few seconds and go back to retrieve the keys, but if not the system 1110 continues to intervene. If the owner becomes so far separated that Bluetooth connectivity is about to be lost, as evidenced by a weakening RSSI, then the device 10, lying on the ground, will use its cellular radio to broadcast another alarm 1236 at cell tower 901d that is relayed 1237 to the cloud host 1111. When the owner finally realizes he has lost his keys, he may activate a screen on his handset 30 and is guided by a map display back to the spot 1234 where the keys were lost. The system pushes the map onto the foreground screen when the user pulls out the smartphone. As radio proximity is restored and the owner doubles back toward spot 1234, the owner can use the BT Proximity Locator Toolkit on smartphone 30 to activate a beeper or light on the lost radiotag 10 and then search using hearing and sight. Navigation using a smartphone has increased dramatically in speed and accuracy and is driven by the emerging market for personal location data, and specialty markets such as self-driving cars, for example.

Even if someone else picks up the keys, for example, and turns them in to the cashier at the nearest business from which the owner would have exited, the system 1200 will know that the keys have been moved and via the cellular radio can pinpoint the updated location where the keys are to be found.

The administrative server 1111,612 (FIG. 6C) keeps location data and can help the user/subscriber recover the keys even from across town or across the other side of the world. The system will not alarm unnecessarily, such as when the keys are in a safe place at home, but can be quick enough to catch a user who is about to leave his keys in the car.

Similarly, a user/subscriber can program radiotag 10 attached to a keychain 110 to automatically tag the location where he parks his car. For example, when the car has stopped moving and the owner has exited the car, accelerometer and heading data is recognizable as characteristic of parking a car. Or a button press may be used. The radiotag sends this data to the companion handset 30 or host server, and a map pin is stored in memory that shows the location and the time the car was parked. The map pin will be updated if the car and driver moves again. It is a simple matter to call up this information if the owner has forgotten where his car is parked. And perhaps of equal value, the system can keep track of time on a parking meter, flag a warning when the time is almost up, and even contact an automated metering system operated by a city to add minutes when the owner is unable to tend the meter.

By incorporating cellular radio, the distances at which lost devices are retrievable is increased from tens of meters to tens or hundreds of kilometers. For example, a device that has been pilfered may show an unexpected burst of motion, resulting in an UNAUTHORIZED MOTION alarm, but mere notification is insufficiently speedy to result in its immediate recovery. However, with a cellular connection activated, the device can CALL HOME 1 from wherever it ends up, and will provide a location where it can be recovered. In addition, the owner’s friends and the community of user/subscribers can also watch for it. Any smart-

46

phone that detects the BT radiotag signal or an active cellular network connection becomes a global finder for locating the lost item.

At any time BT connectivity is good and the companion smartphone 30 operated by a user/subscriber 11 is in BT radio proximity, the cellular modem of radiotag 10 may be kept in SLEEP mode to save energy. SLEEP mode will have features of PSM mode, for example (FIG. 8B). Alterations to PSM, DRX and eDRX modes can be triggered remotely using Bluetooth radio signals routed through a smart device or reference hub that is connected to the cloud, or when the cellular radio is AWAKE and receiving direct commands from a cellular radio network in a paging window. The cloud host can then immediately send a notification to the owner’s smartphone 30, or if that opportunity is missed, the hard impact of the keychain when dropped can trigger actuation of the cellular modem of the radiotag, and the radiotag can make a CALL HOME 1, reporting its status and location to the cloud host. The cloud host then can command the radiotag to keep its cellular modem active while simultaneously notifying the owner.

Logic conditions in which the cellular modem is in SLEEP mode are those, for example, in which (a) the radiotag 10 is in a familiar “safe” location, such as at home or at an office, and not moving away from that location; or (b) is receiving a familiar BT radio tether signal from a hub or companion smartphone 30. Also, when accelerometry and motion sensing indicate that the device is stationary, the processor may shut down the cellular module until movement is detected. If movement is detected, the cellular wakes up to CALL HOME if at all, only according to a regular schedule, the periodicity of which is selected by the user or by the requirements of the network to keep the cellular device authenticated on the network.

Some logic rules for activating cellular modem may be based on RSSI. Other contextual information available to the cloud host or to the user’s handset may be used to determine what if any added intervention is appropriate.

FIGS. 13A and 13B form a composite view that illustrates discovery and tracking system features in a “lost dog” scenario over an extended distance that includes several cellular towers (the distance being large enough that cooperative engagement of both the BT and the cellular radios is required to recover of the lost item). FIG. 13A illustrates a series of events that result in a cellular network connection and a location fix. FIG. 13B extends the concept of Bluetooth Proximity Tracking as a complement to cellular tracking.

Initially, at point A, the owner 11 attaches a new radiotag 10 to his dog’s collar, and installs “application” software in a companion smartphone 30 so that the radiotag RUI can be captured and the device set up for use. When setting up a new radiotag 10, a prudent owner will generally select eDRX and PSM settings to limit radio activity of the cellular modem to short periods or intervals. In contrast, the BT radio is typically set to “always listening” and will immediately respond to BT radio commands from smartphone 30.

By way of illustration, the dog jumps the fence and runs off. By point B, the dog is lost. At this point, the cellular modem is asleep and the BT radio is listening but is out of range of the owner’s smartphone 30. At C, the radiotag 10 is out of range of cell tower 901e, and even if it was in range, the cellular modem is in SLEEP mode and not available in a paging window. Because the cellular modem is programmed to be on and to CALL HOME only at certain

47

intervals set up in eDRX or PSM (or if a cell tower is not available at the active time), the call will not go through, such as occurs at point C.

The situation improves at point D. The lost device **10** enters BT radio proximity to a compatible smartphone **31** carried by a community user **13**, who is passing by. A Bluetooth discovery event **1303** occurs that allows the BT radio of the passerby's smartphone **31** to connect to the BT radio of device **10**, and the passerby's smartphone sends transmission **1301** to cloud host **1111**. The message **1301** contains the radio unit identifier (RUI) of the lost radiotag, along with a timestamp and typically a geostamp.

Almost instantly, the cloud host server **1111** sends a command **1302** back to the lost device **10**, commanding the cellular modem to wake up and CALL HOME. A network communications link is established at tower **901f**. During the brief time in which the two devices **10,31** are in BT radio proximity, the cloud host will "borrow" time on the community device to send a BT command **1303** to the lost radiotag from smartphone **31**; essentially using the BT radio of the community device **31** as a proxy for the owner **11**. This can be an AT command sent to the XCB cellular modem over the BT radio, for example. XCB radiotag **10** is preprogrammed to know how to make a CALL HOME and that information is stored with its SIM module. On receipt, radiotag **10** is configured to wake its cellular modem and override any cellular power management protocol.

Typically the network command to CALL HOME is queued for delivery while waiting for any first discovery of a lost radiotag so that it can be automatically and quickly initiated during any brief BT discovery event. In this way any fleeting BT contact can be "leveraged" to power up the cellular radio location tools of the lost device **10**.

Once the cellular modem is ACTIVE and CONNECTED to a network, it can receive direct cellular commands **1304** from the cloud host or from its owner via cell tower **901g**. Almost simultaneously, (point E) the cloud host server **1111** notifies **1304** the owner **11** and displays a map location of the lost dog on the owner's handset **30**. As instructed by the owner, who can now enter direct cell-to-cell radio messages, the lost device at point F is updating location on a more frequent schedule **1305,1306**. The cellular radio of radiotag **10** is now ON and a handoff to a next cell tower **901h** is a routine matter at point G, referring to **13B**.

By points G and H, cell towers **901h** and **901i** have received and forwarded a series of waypoints to the cloud host in signals **1307,1308**; this continues **1309** at cell tower **901j**, so that the owner can log a trail of waypoints tracking the lost device through the series of cell towers and extrapolate where the dog is going. Dashed circle **1310** indicates an approximate cellular location fix at point H.

Each cell tower reports an updated location with timestamp and the cloud host updates a map for the user on handset **30** so that the user can estimate an intercept point to catch the lost dog. By the time the lost dog is at point I, the owner is waiting to intercept it at point J. For final tracking at point J, the owner will switch on the Bluetooth Proximity Tracking Tools on the handset **30**. By doing a sweep, an increase signal strength may be detected. When closing in (dashed circle **1311**), using the Bluetooth Proximity Locator Services Toolkit, the owner can send a command from the handset **30** to the radiotag **10** to launch an alarm state, causing a buzzer to go off. In one embodiment, the radiotag is configured to emit a dog whistle alternating with an audible tone that the dog has learned to obey. Even though the dog is in an alleyway, the owner can readily find it (or the dog can find the owner) using the audible tone and the

48

approximate position of the cellular location fix as shown on a streetmap of the area displayed on smartphone **30**. On command from the Toolkit, an LED on the radiotag may also be illuminated to improve visibility if the alley is dark. Using these tools, the combination of network assisted location fixes and BT proximity radio commands, the dog is quickly re-united with the owner, even if the dog has strayed across town or into another State. Even though cellular coverage is spotty in some rural areas, if the owner can get close enough, BT radio is sufficient to know where to spot the dog.

More generally, in one embodiment, the tracking methods enabled by the system of FIGS. **13A-13B** include receiving a message that identifies a transmission from a lost radiotag at cloud host **1111**, the cloud host having an administrative server configured with an instruction set and an administrative database containing user profiles, such that the instruction set includes instructions for: (i) parsing the message so as to extract the radio unit identifier (RUI), any sensor payload, and any associated timestamp, proximity measurement, or geostamp coded therein; (ii) then, based on the owner identification, sensor payload, and any contextual information associated therewith in a user profile, (iii) formulating a command or a notification, such that the command or notification is based on rules associated with the owner profile in an administrative database and any rules implemented by a system administrator on behalf of a community of members; and finally, (iv) transmitting the command or notification over a network to at least one smart device **30**, to a remote machine **31**, to a radiotag **10** or to any other actuatable device.

In a variant on the lost dog method, the XCB radiotag **10** on the dog's collar can include a homing button, such that if the dog can be approached by a passerby, the passerby can press the button to initiate a CALL HOME. The radiotag has an IP Address and will contact a cloud server over a virtual private gateway to reduce unwanted radio traffic and prevent unauthorized location tracking. The cloud host will receive the dog's current location, and may even be able to patch through a live voice call between the owner and the dog, or the owner and the passerby, by which helpful reassurance and information can be exchanged. Such radiotags may include a speaker and microphone in a weather-resistant package if desired. Any voice call made via a packet data environment with a radiotag **10** necessarily will consume significant battery power, but if it leads to the pet's swift recovery, then the radiotag can be easily recharged or replaced.

While the illustration above relates to a lost pet, the same apparatus can log temperature information in a shipment from point A to point J, for example, or, illustrated in FIG. **40A** and FIG. **40B** by another example, which relates to "cold chain" data logging. The logged data can be periodically reported to cloud host **1111** during a scheduled CALL HOME or if the temperature of the shipment as sensed by the radiotag **10** crosses a threshold. Analogously, a button on the radiotag can elicit a CALL HOME to activate tracking, to alert the shipper that the shipment has been received, or to prepare a report that plots a temperature history **4000** for the shipment, for example as a function of the route and miles travelled on map **4001**. Temperature spikes **4002** may be indicative of handling or stops along the way. Temperature excursion **4003** may indicate a breakdown of the refrigeration system on the truck, for example. A control interface **4004** offers the user different ways to review the cold chain data and make decisions about the quality of the shipment.

In other embodiments, the method may include provision for transmitting a command such that a physical transformation will be achieved, for example opening a garage door, or rolling down a car window, where the owner is not in physical proximity and needs assistance in performing the action. The command to the plurality of remote machines or actuation devices may be a command to execute a machine action or to actuate a device.

The owner typically will preprogram the radiotag **10** with a duty cycle for the cellular radio set, setting wake/sleep duration and frequency, and any eDRX and PSM parameters, along with any geofencing and location suppression so as to reduce energy drain and activation of alarm features where none is needed. Alarms associated with motion patterns are used to further control unwanted activity. These preset features can be reprogrammed remotely; even if the device is lost.

Scheduling cell connectivity conserves power by setting rules that help reduce power loss. The device can be put to sleep for example, when it is safely at home (at a “home location” defined and fixed in memory). Or the sleep mode may kick in only at night, and the clock in the device will turn the device off at 10 PM and back on at 6 AM, for example. Alarms for specific geographical limits may also be preset, as will be described further below.

If the radiotagged asset not found promptly, then the owner can report the lost device **10** to a cloud administrative server, generally on a web page or user interface accessed through a smart device **30**. Because the radiotag device is identified in the database by nickname, radio unit identifier (RUI) and/or IMSI, the server typically can analyze incoming radio traffic reports in real time and flag those reports that include the identifier(s) associated with the lost device. Because a timestamp and location are included with the incoming radio contact reports, the server can notify the owner and show a map locating the most recent contact or plot a series of contacts to extrapolate its location. The system assembles information to construct a preliminary location of the missing device. First contact enables the system to now use cellular communication and more powerful the more wide-ranging location tools of the cellular network(s). These can include AGPS, capture of actual GPS locations from nearby smartphones, triangulation from cell towers, advanced forward link trilateration (AFLT), POLTE, and use of community resources to identify radio landmarks associated with the first contact. A map is constructed and the system may present a detailed GUI to help the owner map and grasp the location information and take action to recover the device. Based on where the initial contact is, for example, the owner can send the information to friends who might be able to assist if the device has been left at a friend’s house.

A strategy for finding the device is quite different if the missing device is stationary versus a device that is on the move. If stationary, the cloud host can assemble aggregate information to finalize a consensus location (the office, the grandparent’s house, the doctors’ office, the restaurant, and so forth). It is then up to the owner to go there and do a sweep using the Bluetooth Proximity Locator Services Toolkit to find the device either using an RSSI sensor as an indicator of distance, or activating a visual or audible alarm. Positions obtained by Bluetooth RSSI mapping are generally about as good as those obtained by commercial GPS, but when in close radio proximity, activation of an alarm display, visual or audible or both, will generally be sufficient to find the lost device, even if it has fallen into a dark corner or is in a coat pocket.

The device is more difficult to locate if it is moving. The assistance of the cloud host is essential. The cloud host will receive periodic reports of cellular contacts, cellular coordinates by GPS, Bluetooth contacts, and so forth, and may be able to use predictive algorithms to extrapolate possible destinations or intercept points were a realistic attempt can be made to recover the device. Receiving location information from a cluster of cell towers by a process of advanced forward link trilateration (AFLT), AGPS or PoLTE as an alternative to the traditional triangulation with directional antennas has proved increasingly accurate in areas covered by cellular service carriers when used in combination with machine intelligence now becoming frequently a part of every online search. Chances are quite good that the location is going to be accurate within a few yards.

As shown in FIGS. **14A**, **14B** and **14C**, XCB trackers are not limited to human use. Many areas of the world have spotty cellular coverage, but by tagging animals with an implantable XCB radiotag **1400**, their location, even in rugged terrain, can be captured by citizen scientists, who can download an application, which when installed on a smartphone **33**, detects any tagged elk in proximity and sends the location data, complete with individual animal identification, and sends the data to a cloud host **1111** operated by the National Wildlife Service for further analysis of the migratory patterns. The application enables crowd-sourced biology and can also be installed in drones for aerial surveys of populations.

Persons who are afraid of the animals can avoid them. Where animals and humans interact, the implantable devices can be monitored, either in cell radio areas around cellular radio towers **1399** or by Bluetooth-to-Bluetooth proximity radio. In this example, a herd of elk **1410** crossing a highway in a rural area are tagged and broadcasting. An approaching truck **1398** can receive a warning to slow down 1200 ft from the animals. In one instance, a smartphone **33** with installed application is alerted to the proximity of the animals and causes a second XCB radiotag **10** on the driver’s keychain to speak in a synthesized voice, (Caption **1420a**: “Slow down, elk crossing!”) XCB-to-XCB radio interactions are implemented by cellular or by BT radio. A voice interface includes a speaker in the device, and optionally a microphone, essentially as a mobile voice-actuated hub.

And in advanced applications, the devices are equipped with piezo buzzers that alert the animals to look up and take caution. The devices are implanted under the skin of anaesthetized animals in the field, and have a service life of more than a year, allowing scientists to gain a rapid understanding of their range and habits. A temperature sensor logs and sends data that indicates the health of each tagged animal. Similar devices can be used to reduce poaching of elephants in the same way.

A cutaway view of an implantable subcutaneous XCB radiotag **1400** is shown in FIG. **14B**. Numbered are a microcontroller **1401** with integral BT radio, a cellular module **1402**, and a stack of coin cell batteries **1403** in a sealed polytetrafluoroethylene shell **1404**, the interior of which is potted to prevent any leakage. Antennae for both radios are surface mounted on the PCB or are bayonet mounted. An RFID chip may also be included.

The systems described above enable generalized “proximity avoidance”, such as useful for “social distancing” required to reduce viral infectivity and transmission rates between humans. The COVID-19 pandemic has led to widespread implementation of a prohibition against close proximity between any two persons, generally defined as a two meter or six-foot radius around any individual. Persons

who approach closer than the prescribed distance are at greater risk of transmitting the virus in an aerosol or by direct contact. Social responsible behavior is to reduce the risk of transmission by maintaining a distance of greater than about 2 meters or 6 feet at all times, but particularly when having a conversation, because the virus can become airborne from an infected host during vigorous breathing or talking, even when the host is asymptomatic. Airborne droplets and aerosols containing virus particles are readily carried in the air over short distances.

Analogously to EXAMPLE I, for proximity contact detection and tracing CB radios are used to assess radio proximity. Cellular radio is used to report radio contact data, including radio proximity data, over a cellular network so that a cloud host **1111** can log close contacts for contact tracing and generate alerts to user/subscribers as needed. Advantageously, the XCB devices are enabled to detect close proximity by measuring RSSI of BT signals emitted from a first radiotag device and received by a second radiotag device. Individuals each wearing a device are automatically monitored for radio proximity. A spike in RSSI value indicates a close approach of the transmitting and receiving radiotags and a threshold in the RSSI value corresponds to a “too close” physical distance. The XCB devices not only can enter a TOO CLOSE alarm state if social distance is too close, with associated buzzer, synthetic voice alert, or blinking light that has a reminder and a training effect, but the devices collect and log an identifier of the proximate transmitter, be it a BT radio unit ID, an IMEI, an IMSI, or just a MAC Address, and transmit a summary of the radio contact record showing the identifier (s) of receiver and the transmitter radio units, the RSSI value as measured, a timestamp and optionally a geostamp. This data is logged by an administrative server and can be used for follow-up if one of the individuals later tests positive for the virus and there is a need to trace all the possible contacts back several weeks in time where a chain of transmission may have begun.

FIG. **15A** illustrates use of a “safe zone” to monitor BT radiotags; shown here describing use of cloud cutting to enforce an exemplary radio geofence **1500**. Radiotag XCB**1** is a dual-radio XCB device **10**; radiotags TD**1**, TD**2**, TD**3**, TDN (**1510**, **1512**, **1514**, **1516**) have only a BT radio. The radiotags are assumed to be associated with selected assets belonging to or of interest to user **11**, where ‘N’ is an integer indicating a plurality of radiotagged assets. While the radiotags may be attached to assets by any convenient mounting or attachment hardware, the radiotags may also be built into or embedded in the body of the assets. Each radiotag is registered in a user profile on a cloud host **1111**. Safe zones may be used with XCB radiotags for enhanced pet finder services or care of children, for example.

Smartphone **30** is operated by user/subscriber **11** and includes software for displaying and operating a user interface designed for tracking the radiotags, entering user commands, receiving notifications, and creating or updating the user profile. The software defines an “application” or “control program”. The user profile includes a programmable definition of geofence **1500** and is stored in computer readable media in the cloud host **1111**.

Geofence **1500** bounds a “safe zone” or “familiar zone” **1501**. The geofence can be defined by GPS coordinates or by reference to a location such as “home” or “office” having fixed coordinates. For example, the geofence can be described in several ways: by (a) coordinates for a northwest corner and a southeast corner of a rectangle (or more properly a “spherical rectangle”) to describe a geographic

area with enclosed topology, (b) a center coordinate and a radius or diameter, (c) a GeoJson-formatted list of points that outline a polygon; (d) the geographic area between any two latitudes and any two longitudes; (e) the area above or below an altitude bounding a hill or a valle; (f) known dimensions of a room or building with reference to a fixed point in space and time; or, (g) a radio contact or contacts known to be associated with the location or zone as will be described in EXAMPLE VII below, without limitation thereto. Once established, the geofence defines a test that a computer can perform: “Given the coordinates and location of a radiotag in space, is the radiotag inside or outside the geofence?” This simple test gives rise to a whole range of rules-based commands that can be programmed into a system and executed by one or more remote machines.

Having received and stored a user-defined geofence **1500** and safe zone **1501**, the cloud host manages a system for enforcing rules related to the safe zone or zones. Any report of radiotags TD**1** through TD**N** or XCB**1** (**10**) at a location that is not within the expected boundary conditions of a geofence will trigger an “exception notification” to the user/owner **11** via smartphone **30** or some system intervention. Effectively, the system becomes a watchdog that monitors location of the assets and issues an alert to the registered user/owner if a detected location is not within the geofence.

The system includes a community of users **13**, represented figuratively, each operating a “community nodal device” (referencing U.S. Pat. Nos. 9,774,410, 9,900,119, 10,063,331, 10,361,800, 10,389,459) such as smartphone **31**, that scans for signals from radiotags and reports them to the cloud host. The reports include a radio unit identifier (RUI) of the transmitter and timestamp and a geostamp that records the time and place the transmission was received.

In practice, smartphones **30,31** each make a location determination when a radio signal is received from radiotags TD**1** through TDN. A record of the radio contact that includes a radio unit identifier (RUI) associated with the radiotag is timestamped and geostamped before being sent to the cloud host **1111**. Signals from radiotag XCB**1** (**10**) may also be reported by smartphones **30,31**, and the cellular modem of the XCB radiotag may also make an independent location fix using internal GPS or network assisted AGPS. Thus the location of each of the radiotags TD**1** through TDN can be captured and reported from devices **10, 30** or **31** whenever a BT radio signal from the tags is detected. These radio contacts are stored in memory or forwarded to the cloud host and are processed according to conditional logic rules established by the owner.

Based on radiotag location information received from smartphones **30,31**, and from any independent contact with radiotag **10**, the cloud host **1111** looks up any geofence definition(s) associated with each radiotag in a user profile by its RUI and generates an intervention such as an “exception notification” if needed. Thus for example if an asset associated with radiotag TD**3** (**1516**) leaves the area demarcated by geofence **1500**, any BT radio contact with community nodal device **31** will result in a radio contact report to the cloud host **1111** and the cloud host will issue a notification to owner’s smart device **30** that indicates the time and location that the errant radiotag **1516** was detected outside safe zone **1501**. By extension, other community nodal devices **31** may provide a continuing trail of waypoints that update movements of any errant radiotag outside the safe zone so that the owner/subscriber’s smartphone can display the trail of waypoints on a map display **30**.

If the asset TD**3** left with some authorized user (such as a friend), the system may be able to note that radiotag TD**3**

has paired via BT radio link with an authorized user's smart device, and that information can be weighted by the system in making reports to the owner/subscriber **11**. In this way, for example, an employer can keep track of assets that are routinely taken to job sites by employees and returned at the end of the day to a shop or central warehouse, for example.

The timestamp may also be helpful in allowing a user to set time restrictions on the geofence and safe zone so that short term borrowing by authorized users is permitted, but overnight absence of a radiotagged asset from the expected location results in an exception notification, for example. If the asset was last linked to an authorized user, a notification can be sent as a query if there is a temporal exception, asking for an explanation to the authorized user, but if the asset left with a stranger, the system can attempt to track the asset and will report any outside radio contacts unless and until the asset is returned to its expected location within the geofence.

The behavior of XCB radiotag XCB1 (**10**) is distinct from radiotags that lack a cellular radio. When radiotag **10** exits the safe zone at **1508**, its BT radio signal is registered by community nodal device **31**. A report will be sent to the cloud and the cloud can immediately respond with a command, sent from smartphone **31** to radiotag **10** to CALL HOME **1** by turning on its cellular modem. Alternatively, the radiotag, recognizing that it is not where it is supposed to be at **10a**, will take a location fix on its own and can CALL HOME to report. On receiving the CALL HOME, the cloud host will generate a notification to owner/subscriber **11** via smartphone **30**. Thus, the devices **10** of the invention enable autonomous location finding and tracking that is not possible with ordinary BT radiotags **12** (FIG. **1**).

A "quiet mode" may be implemented that overrides notifications and alerts if other sensor data is consistent with a "false alarm" condition, such as a bump to a device that could be mistaken for a change in location, even though the device is on the shelf where it should be.

FIG. **15B** extends the concept of radio geofencing to a user who programs a first and a second geofence, shown here as HOME location safe zone **1502** and WORK location safe zone **1503**. Each geofenced area **1502,1503** is defined by GPS coordinates, by a radius around a fixed anchor point, or by a characteristic familiar radio signal ensemble, for example. In this instance, the user **11** may carry an asset tagged with radiotag TD3 **1516** back and forth between the two geofenced areas. Radio contact reports sent to cloud host **1111** from the user/owner's smartphone **30** will show that the radiotag is travelling in uniform proximity to the user. Even though there is motion of both the radiotag and the smartphone, because the radio proximity is unchanged, no system intervention is needed. Radio contact reports received from an anonymous user **13** via community nodal device **31** during transit from one safe zone to another are not actionable if the radiotag is accompanied by the owner's smart device **30**. At any time that radiotag TD3 **1516** is within one of the safe zones **1502** or **1503**, the system will not alarm or initiate a CALL HOME if the signal from smartphone **30** is lost. Thus 'safe zones' provide a valuable tool in reducing the complexity of logic conditions for location management and in reducing power consumption.

By extension, the safe zone logic is adapted to include "radio leash" proximity to the owner/subscriber's smartphone **30**, even while in motion with the smartphone. The geofence may be mobile when so defined. And a geofence can be time-dependent, such that geofences are enforced only in certain time blocks, for example. If any of the radiotagged assets are not in their expected local areas at designated times programmed by the user/owner, and are not

in radio proximity to smartphone **30**, then a "TAKEN AWAY" or "LEFT WITH" alert may be issued to the user/owner via smartphone **30** or to a secondary authorized user. In this way friends and associates can share assets but still ensure that unauthorized removal of an asset will be noted and tracked by the cloud host. These actions are all managed on the cloud host with essentially no burden on the owner, friends or on the community of users.

The owner/user **11** can select the alerts and threshold values on a user interface and the alerts are stored in a user profile. Any radiotag that is stationary when the user's smartphone is mobile could indicate that something has been left behind, and any device that is moving but moving in a direction away from the user's smart device would also trigger a "wayward motion" or LOST alert, for example. For example, if TD3 (**1516**) is moving with user **11** but TDN (**1514**) is not, then a "left behind" alert is caused to be issued even before the owner has left the area defined by work geofence **1503**. Similarly, if heading sensor data is shared, a divergent heading would also indicate a need for an "exception notification" that could prevent loss of the tagged item.

BT radiotags TD1 and TD2 **1510,1512** are dependent on community user device **31** to send a radio contact report to the cloud host so that the system can evaluate and intervene if the radiotag has left the safe zone. In contrast, radiotag XCB1 (**10**) is not dependent on an external monitor, and can CALL HOME at any time to report a current location or to request location assistance from the network, for example if there has been a power failure in an office building, if there has been an earthquake, a shock, a temperature drop, or any of a variety of conditions, including a change in the radio signal environment in its vicinity. TDN **1514** is passive, and reacts to commands from the system but is not able to take action if lost; but XCB1 **10** can report itself lost and act to cause a system notification to the owner or a direct intervention. A wayward XCB radiotag **10** (XCB1) can initiate a cellular network connection to report its own position and if that position is outside the boundaries of a defined safe zone, then the system will intervene, either by notifying an owner/user **11** or by directly causing an alarm display on the radiotag, for example. Radiotag **10** capacity to obtain a location fix can be based on an onboard GPS or AGPS capacity or based on some cellular network serve such as LoLTE or PoLTE, for example.

In one embodiment, power management in radiotag XCB1 **10** follows principles set forth in FIG. **11** (above). While the location of the radiotag is within a safe zone around owner/user **11**, the cellular modem of radiotag **10** is in a SLEEP mode. In response to motion detected by the radiotag, or by loss of radio proximity detected by the BT radio of the radiotag (where proximity to smartphone **30** is assessed), the device can wake up its cellular modem to CALL HOME and get a location fix. Alternatively, the smartphone can monitor RSSI proximity of the BT radio signal and if there is an increased separation (decreased RSSI), for example, the smartphone **30** can send a command to the radiotag **10** (via the BT radio) to wake up the cellular radio and begin more frequent location reports.

The safe zone can also be defined by a boundary condition established by a "radio tether" to a reference smart device or other stationary anchor or hub. The radio tether incorporates two concepts relied on in FIG. **11**: accelerometry and/or heading data and radio proximity. For example, radiotags TD3 (**1516**) and TDN (**1514**) may be tethered in a piconet with radiotag XCB1 **10**. By making radiotag XCB1 the master of the piconet, it can generate a CALL HOME if there is some exceptional circumstances such as motion or

55

heading in radiotag TDN **1514**, which is intended to stay at the owner's desk. The role of master can be switched among radiotags and smart devices, so that, for example, the smartphone **30** can take over the role of master for radiotag slave TD3 when the owner departs for home, but while at work, device XCB1 **10** can be the master of the piconet if the owner carries the phone to lunch but does not take the radiotagged assets with him. Where several radiotags are in use, any discrepancy between one radiotag and the others increases the probability that an alert should be issued. Any discrepancy between the motion of a device that defines a radio tether and a radiotag associated with the tether will also result in an alert.

In FIG. **15C**, a portable XCB device (XCB2, **22**) is centered in a mobile safe zone **1504**. The device is active as a mobile hub with radio tether or mobile geofence. In this embodiment, the device **22** defines a mobile geofence **1504** that travels with owner/subscriber **11**. Radio geofence **1504** is a subdomain of the BT radio envelope and radio tether of BT transceiver **22**. Accessory XCB radiotags like device **1510a** may be radio tethered to mobile hub device **22**. The enforcement of the geofence is further enhanced by the cellular connectivity of device **1510a**, which functions as a radiotag, such as for a pet collar, asset tag or embedded device. Mobile device **1510a** may be operated in cooperation with the cloud server **1111** or with companion smartphone **30** to enforce the safe zone that travels with user **11**. If the device leaves (arrow **1509**) the safe zone, as for device **1510b** (dog collar with phantom lines), a key capacity is the ability to turn on the cellular modem so as to use it to find or track the lost dog in this example. The goal of any logic is to minimize any power loss by eliminating unnecessary cellular radio traffic, but to acquire or reactivate a cellular network connection to device **1510b** if the radiotag outside the safe zone **1504**, for example because the owner has gone in direction **1508**, and dog has gone in direction **1509**.

In an exemplary use case, the mobile safe zone can be used to monitor a companion pet, shown here as a dog with radio collar **1510a**. If there is excess separation **1509** between radiotag **1510a** and the smartphone **30**, such that the length of the radio tether increases to **1510b**, a threshold is crossed before radio contact is completely lost, and that threshold can be a trigger for a CALL HOME **1** or cause an alert to be sent to the owner's smart device **30**. Location data may be routed through the virtual private gateway **2400** using a private IP address that minimizes network traffic and avoids security issues of the public IP networks. The cloud host will cause a notification to the owner/subscriber **11** if the location data violates a conditional logic rule associated with the mobile safe zone.

Loss of signal from a radiotag can result in a system alert, but even before the signal is lost, the system can detect a fading signal or can receive motion data from the radiotag **1510a** if the radiotag includes an accelerometer or an electronic heading sensor, and by comparing that data with motion data from the smartphone **30** for example, can preemptively issue a LEFT BEHIND or LOST alert, for example. If there is a motion mismatch in direction between the smartphone **30** and the radiotag **10**, then the decision to wake the cellular modem is easier. For example, if the smartphone **30** is in motion but the motion sensor in the radiotag **1510a** indicates no motion, then a LEFT BEHIND alert is generated. And if the smartphone **30** is moving **1508** in one direction or at one velocity, and the radiotag **1510a** is moving in another direction **1509** or at another velocity, then a "wayward motion" or LOST alert is triggered and pushed onto the user's smartphone **30**. Before the radio link is

56

broken, the system assesses any drop in radio proximity (e.g., RSSI of radiotag at **1510b**) as sensed by smartphone **30** and causes the smartphone to vibrate or alarm to call attention to the exception. Or alternatively, a buzzer or vibrator in XCB radiotag **1510a** (on black dog collar) may be actuated so that the user takes no more than a few steps before being alerted that the radiotag (and any radiotagged asset) is no longer moving in step. The system, the owner, or the radiotag may actuate a hypersonic whistle built into the radiotag to remind the dog to keep up with the owner on a walk, for example.

In another embodiment, a change in the BT radio environment or topology around an XCB radiotag can be sent to the cloud, an analysis can be made, and if the new radio environment is not copacetic with a "familiar zone" or with the BT radio environment around the owner's smartphone **30**, then an exception notification will be made.

Prompt alerts simplify the process of retracing one's steps to find the lost pet. Radiotags equipped with motion and/or heading detectors can provide a valuable stream of data that can be compared with output from a like motion detector in the user's smart device. With motion or heading data (FIG. **11**, from an on-board sensor package including accelerometer, gyroscope, compass, etc), easily recognized discrepancies between the velocity, acceleration and direction of moving and stationary radiotags are the changes associated with a getting into or riding in a moving vehicle, which would be readily detected even before radio signal was lost. Characteristics of motion may be recognized by machine learning from accelerometry data collected by a radiotag. The cadence of a walking step is distinctive from that taken on a stairway, for example. Direction that is not copacetic between radiotag and smartphone is flagged. Motion can be detected that is characteristic of a lost child or lost pet. Direction that is not copacetic between one radiotag and other radiotags of a group is flagged. In another instance, if the heading sensor data of a first radiotag signal indicates a first direction of motion, and the heading sensor data of a second radiotag signal indicates a non-parallel, diverging direction of motion, the system can generate an alert.

RSSI or other measure of path loss also provides a criterion to test whether the motion of the smartphone and a radiotag are copacetic. Even within the mobile safe zone **1504**, if one radiotag, for example wayward radiotag **1510b** (attached to dog in phantom lines) is moving away from the user **11** (RSSI decreasing), but another radiotag TD1 **1510** is moving with the owner (RSSI unchanged), the relative motion suggests a discrepancy that can trigger an early alert, as would be sent to smartphone **30** and pushed onto the display, or made evident by vibration of the smartphone, for example, to alert the user **11**. Thus the system can signal to the owner that the dog has left the trail even before the owner looks back.

As practiced, the system will include instructions that can be executed by a smartphone or by an XCB radiotag, instructions that will configure the XCB device for receiving a BT radiotag signal (the received BT radio signal having a radio unit identifier and a community identifier); assessing the received radio signal strength (RSSI) of the received radio signal; and determining whether the radio unit identifier in the received radio signal is assigned to a radiotag that is the property of the owner of the smartphone or XCB radiotag. And if the BT radiotag is the property of the owner, and if the RSSI is less than a predefined threshold value, or if the radio signal is lost; then executing a message to a user or a cloud host, the message comprising a notification of a LEFT BEHIND alert. The community identifier in the

received BT radio signal designates an address of the cloud host in an IP packet data network that has stored the needed user profile and instructions for making notifications to the user.

In another use case, radiotags TD3 **1514** and TDN **1516** are stationary and are left within a stationary safe zone **1505** such as a home with fenced yard. Interestingly, XCB radiotag **1510b** (attached to dog in phantom lines) may be in BT radio contact with one or all of the stationary radiotags **1512, 1514, 1516** when leaving home. If wayward radiotag **1510a** is fails to go with the owner and instead goes down the alley behind the house, for example, the radiotag can cause a system-implemented alert to the owner if it senses the continued radio contact with radiotag **1514** as an exception when compared with a fading strength of the radio contact with owner's smartphone **30** headed in direction **1508**. By assessing radio proximity from the standpoint of wayward radiotag **1510b** (as compared to radiotag **1510a**), the level of confidence is achieved that a CALL HOME is needed can be achieved sooner, and for that reason, the BT radio environment around the radiotag offers useful clues to early detection of wayward behaviors. Prompt attention to location management is critical in developing obedience patterns associated with "heel" and "come" commands.

In another embodiment, radiotag **1510a** may remember a radio safe zone. On return to a stationary safe zone such as a fenced yard **1503**, radiotag **10** will expect to encounter the familiar radio signals of the stationary radiotags **1512, 1514** and **1516**. These can be whitelisted so that the signals are recognized. Typically these whitelisted BT radiotags would be restored to a dedicated piconet that includes XCB radiotag **1510a**, but for example if device **1516** is missing on return home, that exception can give rise to a CALL HOME notification to the user **11** via smartphone **30**. In this way, the system can identify missing items before the owner suspects they are missing. Discrepancies in a familiar radio signature ensemble are in some use cases more reliable than RSSI in detecting the need for an exception notification.

FIGS. **16A** and **16B** are views of a reference hub, shown here with a USB-A male power fitting for plugging into a wall adaptor, for example in a home. As shown, reference hub **1600** includes a housing **1602** with face cover and back cover that encloses circuitry. The circuitry includes a green LED visible through a button or window face **1606a** on the front cover and a red LED visible through a second button or window face **1606b**. The green light is an "ALL GOOD" status light that indicates one or more radiotagged assets are in radio proximity to the reference hub **1600**—i.e., where they are is supposed to be. The red light is an alert that means a signal from one of the radiotags has been lost. A blinking LED indicates that one of the signals is fading. The hub **1600** is set up using a GUI provided with installable software on a smartphone or other user equipment that guides the user through set up of a piconet between the reference hub and the one or more radiotags. When the owner/user is home the user's smartphone can be master of the piconet and can monitor the radiotagged items. When the user leaves home, the hub takes over as master of the piconet for any radiotags that remain at home, for example.

Reference hub **1600** may be plugged into an AC adaptor and may be operated indefinitely in a wall plug at a single location. In the event of regional power failure, the hub may have a backup battery power supply, indicated here by a removable battery access panel **1608**.

The hub **1600** may be the default master of a piconet that includes several BT **12** or XCB **10** radiotags of a common owner. As a community service, the hub may broadcasts

location coordinates that a community of users (or the public in general) can use for telemetry-controlled applications, for rangefinding, for wayfinding, for finding lost radiotags, and for creating radio tethers or conditional rules-based actions linked to radio bubble centered on a fixed location. Reference hubs **1600** may be shared. A "Crowd Hub" is a unique service offered by the system. Multiple user/subscribers may establish safe zones for multiple radiotagged assets using a single hub and associated cloud service. In some instances the hub is a client; in other instances the hub is a server in relation to the radiotags and any cloud host. In client role, the hub may forward data such biometrics or voice to a server device. In server role, the hub may perform location management services for the radiotags and for user equipment. The system will manage notifications and alerts for radiotags that are connected to a hub-based piconet. LEFT BEHIND, LEFT WITH, LOST, FOUND and UNAUTHORIZED MOTION services are all supported by the hubs of the invention.

Reference hub **1600** may include an optional speaker and microphone **1609**. The larger housing realizes better acoustics and optionally may be circular or spherical in shape and include a directional speaker and microphone array. U.S. Pat. No. RE47049 to Li teaches a dynamic microphone array for improved voice recognition. US Pat. Nos. U.S. Pat. No. 7,177,798 to Hsu and U.S. Pat. No. 6,766,320 to Wang teach methods for natural language query and response interactions. These patent documents are incorporated in full by reference. Reference hub **1600** may include a natural language interface incorporating cloud-based speech recognition and response.

The radius of the radio bubble around hub **1600** is dependent on power. Generally the transmit power is +0 dBm or +4 dBm for BTLE applications, but can be as high as 20 dBm unless limited by law. A higher power increases the range, but care must be taken so that an impedance matched condition exists between receiver and transmitter antenna and amplifiers for best results. Because signal fade occurs with distance; given signal losses due to refraction of signals around radio opaque structural barriers, and because attenuation due to lossy media such as human bodies can be significant in crowded venues, transmit power may be variable, and a subroutine may be executable by the reference hub so that transmit power can be varied if a radiotag signal is lost. The purpose of increasing radio power is to improve the chances that a lost radiotag can receive a command to turn on its cellular modem.

In other embodiments, the hub may include WiFi so that a WLAN can be formed for reporting BT radio contacts to user equipment or to a cloud server **1111**. A combination power and data connection through a USB port **1604** is provided as another option for establishing a hardwired connection to an internet portal. The device **1600** may be plugged into a household power outlet or into a dashboard of a vehicle, for example, using an appropriate adaptor. In some instances the hub may be solar powered or powered by kinetic energy for use in outdoor venues, for example.

When used portably, XCB radiotags **10** consume significant amounts of energy when maintaining a cellular network connection. By overriding cellular networking whenever a hub **1600** is within BT radio proximity, the battery drain on radiotag **10** can be minimized.

Hub master device **1600** and radiotags **10, 12** have cache memory that can be used to store piconet membership and connection data, so that interrupted connections can be rapidly restored. The hub, as master of the piconet, can also designate PARK, SNIFF, and STANDBY mode for slaved

radiotags, and can manage power consumption of the BT radio, the processor, and any cellular modem in the slaves while in BT radio contact.

The hub can also query qualified devices to determine battery power status and can make appropriate notifications to an owner if particular radiotag needs recharging or a new battery. By extension, where radiotags are embedded in assets such as cameras and other electronics, the radiotag can report asset technical data selected from temperature, battery status, fault status, and so forth to the hub, and that data is forwarded to the cloud host for analysis. Notifications directed at care and maintenance of user assets can be pooled and automated in this way.

In a variant, the piconet around a safe zone may be set up with reference hub **1600** and smartphone **30** as slaves and XCB radiotag **10** as master. The master will minimize its power in scanning for the slaves with a reduced duty cycle and the slaves can be set to broadcast directed advertisements at higher frequency. The XCB radiotag makes the determination if the radio tether has been broken or stretched and can call home as needed. The slaves may be programmed to report radio contacts with the master to a cloud host **1111** when the master is in a safe zone, and the network can follow the master through a companion smartphone **30** when away from home. This eliminates the need to switch master roles from reference hub to smartphone or vice versa.

Each reference hub **1600** may have an IP Address that associates it with the physical web of the IoT, and may be connected by wired, Bluetooth, WiFi or cellular means to a packet data environment via a GAN gateway or portal. The hub will have one or more RUIs and UUIDs that identify it and its services. A cloud host **1111** that receives radiotag data from the hub is able to extract additional identifiers from a user profile associated with a UUID or other radio unit identifier associated with the radiotag. Where WiFi is provided, BT and WiFi may function as complementary radios in overlapping LANs and piconets in which an internet gateway or portal is provided. Where cellular radio is provided, the capacity of the hub to interface directly with a cellular network provides a direct relationship with an administrative server.

The GAN connection may be to a cloud host **1111** or to a virtual private gateway **2400**. The advantage of the VPG is the relative lack of background chatter that can drain battery power and increase latency. The VPG is also much more security friendly for sensitive information such as child location, which may be tracked using the radiotags **10** of the invention.

As a matter of convenience, a user interface can be displayed on user equipment by installing a software application, for example on a smartphone. In other instances, a user interface is accessible at a website with APIs for managing databases containing administrative and user information.

FIG. **17** is a schematic of a system with stationary reference hub **1700** useful for managing radiotags in a network context. Reference hub **1700** includes Bluetooth and WiFi radio sets. In embodiments, the circuit includes a BTLE radio **1702** controlled via a controller circuit **1701**. Power is supplied from a power adaptor **1708** and battery **1705** with power supply circuit **1706**. The device may be insertable into a wall outlet power adaptor **1708** as a USB plug **1604** (as shown in FIG. **16**), or the power adaptor **1708** may be a computer with a USB power and serial data plug **1604**, for example. The power circuit **1706** may be configured to convert the power input into a regulated power signal having a regulated voltage in an approximate range of 1.8

Volts (V) to 3.6 V. For example, the power-supply circuit **1706** can be any suitable type of voltage regulator, such as a linear regulator, a buck converter, a boost converter, a buck-boost converter, or a flyback converter. Logic and analog device power can be supplied via a V_{cc} rail for example, directly from the battery or as regulated by the controller **1701** or power conditioning and management unit of power supply circuit **1706**, for example.

The hub may include a Bluetooth radio **1702**, a LAN radio **1704**, and an optional GPS chip **1714** with associated antennae **1702a**, **1704a**, **1714a**, respectively. GPS chip **1714** is shown as being optional not just because it may not be supplied in inexpensive units, but also because in some instances the GPS functionality will be built into the controller **1701**, into a cellular radio (not shown), or into the LAN radio **1704**. Many cellular radio chips are provided with accessory GPS functionality integrated into the die. The GPS antenna **1714a** may be separate from a cellular radio antenna, but, in some instances, a combination antenna package is used. Based on an autonomous or network-assisted location fix, the reference hub **1700** may function as a “lighthouse radiobeacon”, transmitting Lat/Long or other coordinate information as an open broadcast, and the message may be used to define a radio tether of a geofence or for community uses and applications.

For use in pet location management, reference hub **1700** can define a safe zone in which a pet is free to move around, such as a fenced yard. If the pet jumps the fence and goes for a run, the system will alert the owner. A tethered radiotag **10** may detect a break in the radio tether and wake up its cellular modem and cellular remote locator services toolkit so that a current location update can be sent to the owner at a convenient smartphone **30** and tracking services can be initiated.

The LAN radio **1704** may be a WiFi radio or equivalent. The BT core **1702** of the controller **1701** reports not only BT radio signal data, but also an index of received signal strength such as RSSI, and gives an indication of BT radio proximity within several hundred feet. If a cellular modem is provided, it is generally packaged in a cellular modem SOC and may be controlled by controller **1701** based on input from the Bluetooth or WiFi radios. In other embodiments, data sharing is achieved with a UART **1740** linked via USB port **1604** with the packet data environment of a wired network.

Optionally, the BT radio **1702** may include correlators used for radio signal recognition and “always listening” radio power control. Digital messages include access codes, MAC addresses, and UUIDs, but wake commands and other network commands may also be received in BT signals. Antenna **1702a** is tuned for BT spread spectrum transmission and reception. Notifications may be sent to the device via either the BT radio **1702** or the WiFi radio **1704**, and may result in a display such as activation of speaker **1722** via acoustic driver **1721**. Optionally, a microphone **1723** is included with audio codec **1724** so that responses to notifications can be sent. The hub housing **1602** may be configured for fidelity in reproducing human voice and for capturing voice commands by users. Broadband connection to a cloud server allows relatively low budget smart plug-ins such as these hubs **1700** to have a highly sophisticated voice-actuation and conversation interface.

The circuit diagram of FIG. **17** shows non-volatile memory **1710** for storing program instructions. The circuit may also include flash memory **1711** for data logging. The flash memory may supplement cache memory associated with the controller **1701** for performing edge computing.

61

For example, the non-volatile memory **1710** can store data for configuring one or more sensors **1712** of reference hub **1700**, and a set of software instructions that, when executed by the controller **1701**, cause the controller, or one or more circuits under the control of the controller circuit, to execute routines for transmitting BT and other wireless signals, for receiving BT or other wireless data, and for performing location determination calculations and user-programmed asset management according to conditional rules-based logic.

The controller **1701** may be associated with a packet composer and decomposer that works in concert with buffers and registers of memory. The volatile memory of the memory circuit **1711** can include registers and buffers configured for storing records and data received from sensor package **1712**, from a linked smart device **30**, from WiFi radio **1704**, from BT radio **1702**, from the optional GPS chip **1714**, and for buffering outgoing transmissions, for example. In some embodiments, controller **1701** is an SOC that includes a BT radio core.

RAM **1711** is provided for storage of volatile data, such as for data logging of sensor data from sensor package **1712**, which may contain multiple sensors, for example, such as temperature, humidity, noise, and so forth, and may function as part of a “smart home” or “smart building” platform. Stored data may include data from sensors **1712** and from switches **1707**. Data from throw- and button-press switch array (S1,S2,S3) **1707** is considered data. The size of the RAM memory **1711** is dependent on the size of the memory requirement for data. Large caches of radio contact record logs are not generally stored on board but are uploaded to network when possible.

Stored data may also include radio contact records. The radio contact data may include host-tabulated sensor data and source-tabulated sensor data. The RAM memory may be supplement cache memory in the processor if the data logging function requires it. Memory is generally organized as a rolling stack so that outdated data is dumped from the bottom of the stack and new data is added at the top of the stack if not first uplinked to the network.

The controller circuit **1701** is configured to generate and format output for radio transmission and to select a radio band according to context, radio environment, and power status. When on battery power, BT radio is the preferred radio. When on AC power, WiFi **1704** may supplement BT radio. In circumstances where WiFi is not available, other options may include cellular radio authentication and communication, or a USB connection **1604**, with UART **1740**, for example.

BT radio is used to communicate with BT radiotags and smartphones **30,31** when power savings is important. BT or cellular may be used to communicate with radiotags **10**. WiFi is used for LAN networking where supported, such as by a home computer or smartphone. The above-described configurations of the circuit **1700** allow the hub, smart devices, and radiotags to communication with one another bidirectionally as part of a system for managing asset locations.

The controller circuit **1701** also may command a notification circuit **1730** to call attention to the data. Circuit **1730** can include one or more LEDs **1732a, 1732b**. The hub may include a buzzer driver **1726** and one or more buzzers **1725** configured to provide notification functions. To create a user-friendly experience, in some instances, RGB LEDs are used in combination with an LED configured as a nightlight, for example. The buzzer or LED(s) can serve as an alarm if there is an exception to a rules-based contingency, such as

62

loss of a linked signal, or can indicate an “all clear” if the linked connection(s) are intact and within expected proximity. In some embodiments, an LCD or OLED display screen **1731** may be provided, but generally a fully functional GUI is provided as an installable software application in a smartphone **30** that serves as the master of a piconet formed with the hub device **1700** during setup of features and user profile(s). The companion smartphone, with installed application, also provides remote notification and monitoring that supplements and enhances any user interface directly part of the reference hub housing **1602**.

Still referring to FIG. **17**, alternate embodiments of the hub **1700** are contemplated. For example, the hub **1700** can have an ASIC architecture, with integrated controller circuit **1701** and integrated communications system. The radios **1702,1704** and the controller circuit **1701** can be on separate chips or on a same chip. Radio **1704** may be a WiFi radio in one embodiment, but other suitable LAN radio protocols may include Zigbee and Thread, while not limited thereto.

In one embodiment, using a small solar cell (not shown) associated with a reference hub **1700**, the current needed to maintain the Bluetooth radio for intermittent transmission of sensor data can be met from or supplemented by the solar cell output. In other embodiments, triboelectric structures that harness kinetic movement to generate current sufficient to support an always-listening radio are realized experimentally, demonstrating that the devices of the invention are well positioned to find increasing number of applications for future IoT needs.

FIG. **18** is a flow chart illustrating logic for safe zones established around a stationary reference device such as the reference hub **20,1700** (FIGS. **1** and **17**). By installing a reference “hub” at selected locations, a user may define a virtual radio “geofence” that is anchored to the hub. Radiotags **10,12** are tethered in a piconet to hub **20,1700**. By coupling the reference hub to cloud host **1111** for real-time uplink of data, data from the radiotags can be aggregated to administer a safe zone. The safe zone is useful to prevent radiotagged assets from straying outside the geofence or in reminding the user what to take with them when they go out, for example. “Lost” and “left behind” alerts are readily administered with minimal latency. A smartphone is useful for conveying notifications to a user, but a pocket XCB radiotag **10** with minimal user interface, a wall mounted XCB radiotag **1600** with minimal user interface, or a wearable XCB radiotag **3500** with voice and OLED display, can provide the user with notifications and guidance as part of a cloud system for location monitoring services by incorporating a basic operating system or firmware for administering rules-based conditional logic related to safe zones and radio tracking. The smartphone **30** may provide a more detailed user interface that has utility in setup of customized safe zones, but once established, the system can provide services even in the absence of the smartphone.

A beacon signal from a reference hub **20,1700** can “tether” radiotagged devices to a home or office location, and a radiotagged pet, for example, can trigger an alert if the pet (wearing radiotag **10,12**) leaves a fenced backyard. Children’s activity can be monitored, and so forth. Reference hub **20** is illustrated schematically in FIG. **1** and in more detail in FIG. **17** (**1700**).

Once a radiotag **10,12** is a member of a piconet, it can remember its reference hub “master”, and can resume the piconet even after the initial radio link is broken. Masters can also be switched, so that for example a stationary reference hub **20,1600** can serve as a master of a piconet in a home, while a smartphone **30** can serve as a master of the

63

same piconet when at work. This is achieved by making the reference hub and the smartphone both members of a piconet and then reassigning the role of master according to the location of the smartphone. In another embodiment, the BT piconet can also be defined with a radiotag **10** as master and reference hub **20,1600,1700** and smartphone **30** as slaves, if desired, and can allow the master a prolonged standby duty cycle in listening only mode while the slaves, which have more power, do more frequent directed advertising. The concept of a “lazy master”, in which radio activity is limited to passive listening with active PAGING mode limited to situations in which the radio traffic around the radiotag changes, is an advance in the art, and achieves an improved balance of power utilization and latency for the battery-limited radiotag devices.

To administer a safe zone, in another embodiment, assuming radiotags **10,12** in the role as slaves, the slave devices may transmit their radio identifiers (RUI) and motion or heading sensor data with regular periodicity to a master reference hub **20,1600,1700**. The stationary reference hub is assumed to have a wired power supply and hard data link to the cloud host and can operate independently of a smartphone. The portable radiotags do not need to perform extensive calculations or be provided with complex software. RSSI determinations are inherent functions of the BT radio core, and motion sensor output can be reduced to a single bit for transmission. Heading sensor output can also be simplified for basic functions. Heading is motion, but is motion in a vectored direction, whereas motion per se can sometimes be random, inconsequential or incidental. Algorithms for monitoring radio proximity and vectored motion can be made available as part of software installable on the reference hub **20,1700** or can be implemented by cloud resources. The energy budget for radiotags **10,12** is primarily related to periodic beacon message transmission in which the message includes a RUI and any UUID, plus optional fields for sensor data or a user name, for example, all within 31 to 37 bytes of a typical advertising message. A BT transceiver is generally provided so that the radiotag can respond to a command to enter an alarm state or turn on the cellular radio modem, but a programmable MCU is not required for complex computations. BT radios are provided with firmware that administers the BT radio stack and link manager so that BT radios are capable of advertising and discovery of other BT radios that are close by. CONNECTED and ADVERTISING modes were described earlier with reference to FIG. **8A**.

A local piconet having member smart devices, reference hubs, and radiotags, for example, can be administered to establish stationary safe zones in which slaves in a piconet are set up and remember their master or “reference” device and their CONNECTED state in each safe zone. The master defines the safe zone by a radio tether. The access codes used by the members of the piconet define the relationships among the members and are stored in memory. The master can be a smartphone, but for stationary installations, a reference hub that has a BT radio and a wired power supply is more practical.

Hub devices may include WiFi and cellular, or WiFi instead of cellular, and with either WiFi or a wired connection, the reference hub has the capacity to contact the cloud host independent of the owner’s smartphone. Smart plug-in devices that have a voice interface with speaker and microphone may be adapted as hubs, for example.

The system may include a predictive algorithm, and by machine learning can refine its predictions based on experience. Motion, heading, or radio distance of the radiotags

64

can be compared with position of a stationary reference hub. When averaged, RSSI, relative radio proximity and heading are good indications of the length of the radio tether between the reference hub and the radiotag and can be relied in a decision tree.

Flow chart FIG. **18** illustrates a general method **1800** for operating safe zone services as part of asset location management. Radiotags are attached to or embedded in assets in need of location monitoring. In the analysis, radiotags **10,12** are operated as members of a piconet around reference hub **20,1700**. A radio tether can be unidirectional or bidirectional. An application for executing the algorithm is assumed to be installed in the reference hub **20,1700** or operated remotely on a cloud host **1111**. A user interface may be operated on a smartphone **30** or some other smart device. Remote notifications to an owner/user will be sent to the smartphone or to other user equipment. The initial analysis is directed at correctly identifying scenarios in which the system should issue a notification in response to an exception to one or more rules. The rules relate to the safe zone defined by a radio tether between the radiotag and the reference hub. Generally a notification is not needed if the radiotag is initially in the safe zone and has not moved significantly and the radio proximity is unchanged or increasing in strength. If a radiotag has moved and the radio proximity is fading, then a LOST notification may be appropriate. The relative motion of a smartphone **30** may be a consideration elsewhere (FIG. **11**), but is not considered here so as to focus on what the reference hub can achieve without the smartphone. Radiotag **10** is assumed to include a cellular modem that is in a SLEEP mode by default, but which can receive a location update command during a preset paging window or via a BT radio command to the BT radio. Power to the reference hub is assumed to be unlimited, and an emergency battery can be supplied as part of the hub unit, but careful power management of the radiotag **10** is necessary in order to achieve a satisfactory balance of battery size and portable service life per charge.

Generally, any monitoring of a safe zone **1801** begins with a memory that associates a “location fix” in memory with a timestamp at time $T=0$. This memory may not be in the device **10**, but may instead be stored at a higher network level, for example in a smartphone **30** or in a cloud server **1111**. During setup the radiotag and the radio center are in close range, and the RSSI of a signal from a radiotag, as received by the reference hub, is indicative of radio proximity. At time $T=0$, if the radiotag is in the safe zone **1802**, then no location fix by the radiotag is immediately needed. While it may be desirable to have a current location for the radiotag at all times, from a power management standpoint in a portable device, this is not practical. Getting a location fix consumes power. So the more pertinent question **1803** is the next location fix—when to get it—by activating the cellular remote locator tools of the radiotag **10**?

The decision tree for whether or not to get a next cellular location fix for the radiotag at a future time $T=T+t$, where t is an interval selected based on predictive accuracy, can be made so that unnecessary location fixes are avoided by attention to (a) data related to accelerometry of the radiotag, and (b) any recent change in relative proximity of the radiotag and reference hub.

Assuming radiotag **10** includes an accelerometer **623** and/or heading sensor with digital output, then the simplest sensor output can be a MOTION truth value, TRUE or FALSE **1804** or HEADING vector and velocity for example. The quality of motion (e.g., hard vs soft acceleration), the speed, duration, and the direction are more useful, but the

65

simplest and most economical bit of information from the sensor is whether motion has occurred. If motion has occurred, then the device may have been moved, and the location stored in memory may no longer be valid. If motion of the radiotag has not occurred **1804**, then the radiotag can continue to SLEEP.

If motion of the radiotag has occurred at time $T=T+t$ (i.e., after an elapsed AT, where the time interval is programmable), then it may be useful to look for a change in proximity **1805**. Over a range of several hundred feet, RSSI is a first approximation of distance between the radiotag **10** and the reference hub **20**. The proximity may be increasing, decreasing or stay the same depending on whether the RSSI is strengthening, fading or about the same. Proximity is measure by Bluetooth radios as part of core competencies of BT radio and the reference hub will continuously monitor the RSSI of the radiotag signal as part of routine operations. The radiotag can also measure RSSI of the reference hub signal and report that as part of a smoothing operation to reduce noise in the determination. Binning and averaging are also useful to smooth RSSI data over small intervals.

If there is no change in proximity **1806**, then any motion signal **1803** may be spurious and would not necessitate a need for a new location fix. Proximity will continue to be monitored. Similarly, an increase in proximity (strengthening RSSI, **1808**) is intuitively not likely to indicate a risk of loss of signal, and the algorithm can be looped to continue to monitor for motion and proximity. Heading sensor data, which is output as vectored momentum, is more indicative of a significant change in location, and may be stronger in predicting the need for a location update than any fluctuations in RSSI, which can give false positive indications of significant location change.

A fading RSSI **1807** that is more than a fluctuation in signal strength, as indicates decreasing proximity and increasing separation, could be followed by a loss of signal, and for a tracked asset, a lost radio tether signal may necessitate an immediate CALL HOME by radiotag **10** to get a new location fix and to generate a LOST ALERT notification to any interested party.

Exceptions could be made if the motion data is more granular, for example a hard impact could merit a CALL HOME with status report even if proximity data is unchanged. A sustained motion in a direction is unequivocal in signaling a change in location that could merit a CALL HOME. And it may be appropriate to activate the cellular remote locator tools, even if there is no apparent motion, if the proximity signal has been lost.

The elapsed time interval AT for iterations of the method **1800** may be adjusted according to conditions. For example, in a safe zone, infrequent execution of the loop may be sufficient. An interrupt flag on the processor can be set against the accelerometer output. But if the radiotag **10** is outside a safe zone, the loop may be executed more frequently, and the timing can be dependent on the nature of the motion input, on temperature, or on changes in acoustic patterns, or on changes in Bluetooth radio traffic patterns, for example. The more motion activity, more frequent proximity monitoring may be useful.

If heading sensor data is available and can be analyzed and compared to a companion device with its own heading sensor, the heading sensor data of a smartphone for example, may not match the heading sensor data received in a BT radio signal from a paired radiotag. If the two headings are not parallel, or differ in velocity, then the anomaly shall trigger a notification that can be a LOST or a LEFT BEHIND alert, depending on the details. Where a group of

66

BT radiotags are in communication via radio signals that include motion sensor and/or heading sensor data, then any discrepancy among the group members may indicate a need for user (or cloud host) intervention.

In another method for reducing false LEFT BEHIND alarms, a drop in RSSI of a received radio signal is reassessed after a time delay interval and a notification is generated only if the received radio signal has not recovered above the threshold.

FIG. **19** shows a schematic for power and logic in an alternate embodiment of an XCB radio device. This figure differs from FIGS. **8A** and **8B** in showing active interactions between the BT state controller **1901**, processor **1902**, and the cellular modem **1903**. All three units have SLEEP modes **1911**, **1912**, **1913**. The processor core and cellular modem are generally defaulted to their deep sleep modes. The BT modem **1901** is structured so that different state functions may be separately powered, and the default mode is STANDBY **1914**. STANDBY power consumption is further reduced by alternating with SLEEP **1911**. Other BT functional states are independently powerable, and include INQUIRY SCAN, PAGE SCAN, TRANSMIT DATA, EXTENDED SCAN that make up the link layer **1921**. Also enabled is a PASSIVE, LISTENING ONLY State **1922** that intercepts BT traffic without error correction or ARQ and a TX BEACON MESSAGE State **1923** in which the device emits a periodic repeating radiobeacon message and is not connectable.

In PASSIVE, LISTENING ONLY State **1922**, the device can toggle power and data to the core processor at an I/O pin **1931** in order to generate log entries of received radio traffic in flash memory and can send a command **1932** to wake unit **1933** of the cellular modem if a qualifying signal is received. The cellular modem can also be waked by a command **1934** from the core processor.

BT active standby **1914** is convertible to INQUIRY SCAN state (with extended response in BLE), PAGE STATE, connected state, sleep state, and has two passive modes, one for passive "always listening" and the other for non-connectable beacon transmission. In one embodiment, the passive always-listening state **1922** controls the processor and cellular modem independently.

Power management is hierarchical, with the BT core processor having an override input to the cellular modem 'wake-sleep-idle' cycle. The cellular modem **1903** is shown with a CONNECTED state **1940** alternating with DISCONNECTED IDLE **1941**. RRC CONNECTED (INACTIVE) state may be sustained as a network-connected state, and permits DRX, eDRX and PSM power savings modes. The power of cellular to transmit data is lost after a network DETACH, but re-authentication to the network can be avoided by toggling the CONNECTED state from INACTIVE to ACTIVE if the cellular modem has not been put into SLEEP mode. For this reason, the MODEM WAKE pin can be used to initiate a NETWORK CONNECTION REQUEST if the modem is in RRC CONNECTED but INACTIVE mode or in DISCONNECTED IDLE, and can be used to refresh a CONNECTED link if the cellular modem is in RRC CONNECTED ACTIVE state. By configuring MODEM WAKE **1933** to interpret the current state of the cellular modem **1903** and to direct it with most parsimony to network-connected, active transmission of data, reduction of latency is achieved without excess power consumption. All advertising radiobeacon messages are limited to a maximum of 48 Bytes and may be transmitted with a periodicity that is selected for battery life and minimum acceptable latency in the field.

67

Smartphones also have power management capability. Both iOS and Android will limit the capacity of “Apps” to be “always on”. While many App developers would like to see their App be continuously hot in the phone’s foreground, unregulated power usage would increase the already infamous need for frequent recharging of the phone battery. Many users recharge their phones at least once per day. App developers have come up with “wakelock” subroutines that defeat the power saving features of the phone operating system, and phone manufacturer’s continue to supply updated software that defeats the wakelock workarounds. In contrast, the radiotags described here are not encumbered by unwanted commercial intrusions, and are able to more efficiently manage battery usage so as extend periods between recharging to weeks or even months. In the embodiments of FIGS. 6C, 19, 42, 44, 45, the highly efficient power management of the BT radio modem is extended to put it in charge of power management to both the cellular radioset and the CPU in some use instances. This unconventional approach maximizes battery life of the radiotag and builds a solid approach to power management that reserves constitutive power supply only for defined functions of the CPU such as in memory management and defined functions of the cellular radio such as in adhering to a network connection schedule (eDRX, PSM, SUSPEND). Because these latter network functions, however, are clocked and not spontaneously activated in real time, significant device and network latency exists. In contrast, the capacity to respond immediately to changes in the BT radio environment without the need to decode the BT traffic or turn on an App is an advance in the art that permits a much smaller device with a much smaller battery size than a modern smartphone to outperform the smartphone in tracking, finding and location-dependent notification uses. The XCB radiotag becomes a specialized platform that is capable of taking on additional functions as described for example in U.S. Ser. No. 16/983,965, titled “Private Wireless Network Communications Systems, Methods and Devices”, U.S. Ser. No. 16/950,666, titled “Hybrid Cellular Bluetooth Tracking Devices, Methods and Systems”, which are co-assigned and are incorporated in full for all that they teach.

FIGS. 20A, 20B, 20C, 20D, and 20E are views of digital signal format of a sampling of advertising packets from various radiobeacon types. BT transmissions are tightly structured by the baseband and link manager protocols, but several competing standards have arisen for advertising beacon transmissions. These include the iBeacon®, Eddystone, Altbeacon, Estimote, and Geobeacon standards, for example, all competing for IoT placements in a growing sales market. Eddystone and Altbeacon formats are open source but most devices are not vendor agnostic. Typical messages are 48 octets in length in compliance with the original BT SIG standard and have a maximum payload of 31 bits or less.

In some cases a MAC Address is broadcast with the advertising message, but in other instances, device addresses are given dynamic pseudonyms or are broadcast behind non-discoverable access codes. Devices may instead be identifiable, at least in part, by their “services”. For example, a UUID frame may be broadcast that corresponds to a generic attribute profile (GATT): which codes a general instruction set for how a device works in a particular application that is called by an attribute profile, for example a device could contain a heart rate monitor and a battery level detector, or a speaker and microphone, or a link pointing to a webpage. Each attribute is uniquely identified by a Universally Unique Identifier (UUID) or similar UID,

68

which in the iBeacon standard is a 128-bit string used to uniquely identify information (data) specific to each type of sensor output. The attributes are formatted as characteristics (classes) and services (collections of classes). A “Service UUID” in an extended inquiry response may include shortened 16-bit, 32-bit and global 128-bit service UUIDs. A manufacturer’s UUID constitutes a service and may contain overloaded bits as described in U.S. Pat. No. 9,961,523, which is incorporated herein in full by reference.

Open and proprietary beacon standards are compared. The earliest BT radiobeacon standard was the iBeacon format **2001** introduced by Apple Computer (FIG. 20A, Cupertino, CA) for the iPhone 6 in 2014, which contains as a payload a 128-bit universally unique identifier (**2003**, UUID), along with major **2004** and minor **2005** frames for advertising location. All beacon messages include a preamble **2002** that is used for clock synchronization and for adjusting power gain of the receiver. The preamble results in a calculation of the RSSI of the message. The iBeacon prefix **2003** is proprietary and includes Apple-defined flags for classifying messages plus a company identifier. iBeacon messages include a suffix **2007** that includes an index of the signal power of the transmitting device.

Eddystone is a competing beacon standard introduced by Google in 2015. The format uses an open standard (Apache) that is compatible with both Android and iOS but requires a Google Proximity API installed on a smartphone. Android does not have native iBeacon support. Due to this, to use iBeacon on Android, a developer either has to use an existing library or create code that parses BLE packets to find iBeacon advertisements. As an added service, Google supplies a lookup web server whereby shortened URLs or user identifiers are used to call background resources termed “attachments” from a cloud host server. The attachments can be anything from a subdomain link to a video clip. The Google beacon transmission interval is programmable, unlike the iBeacon standard, permitting operation with a reduced battery consumption rate.

Eddystone consists of a family of standards built around common use cases, each having a different flavor of payload. One flavor (FIG. 20B, **2010**) transmits a URL (as may be compressed when used with a cloud-based look-up table). The beacons can switch to another flavor at intervals or on demand. Cloud services are more extensive with the Google beacon platforms and include the capacity to update messaging on the fly as well as control over transmission frequency and capacity to omit some fields where a shorter message length is preferred.

In the UID beacon format **2010**, radiobeacon message length is transmitted in a prefix, along with Type and Flags that can be used to signal deep intent to a smartphone receiver that includes a developer’s App. The Eddystone UUID **2012** includes a MAC Address, but can be encrypted, and Namespace ID or frame **2013** that may include one or more client services offered by the beacon that again tie into code supplied by the developer using the beacons. The payload frames may also include one or more instances **2014** that specify sub-services. A TX POWER field **2015** that closes the message is used by a receiver to improve the ranging accuracy. TX POWER (dBm) minus RSSI (dBm) is a better indicator of signal distance than RSSI alone.

In the URL beacon format (FIG. 20C, **2020**, the payload includes a webpage reference, often as an encoded URL, that signals “deep intent” to a browser. The developer’s App that opens when the beacon message is encountered takes the smart device to a webpage, and can even fill in fields in

the webpage for an immediate interactive experience that merges the physical world of the IoT with the cloud world of supporting webpages.

Another Google beacon message type is the Telemetry message (TLM), which can include sensor information and a beacon diagnostics suite. TLM message may be interleaved with other Eddystone beacon messages to provide monitoring and maintenance of beacons in the field. Also available is an "EID" payload, which is dynamic and is used with a cloud resolving service to provide timely or context specific messaging by a "fetch" process. The EID message is limited to licensee-specific content, and has stronger privacy settings. Google EID messages contain signal characteristics that are difficult to spoof and can be associated with secure location services. The EID case is included here because it indicates how a radio topology can be shaped by dedicated beacons operating in concert with a dedicated cloud service in joining physical web radios with dynamic Internet content on demand.

The Altbeacon advertising format (FIG. 20D) is an open source, non-connectable beacon signal **2030**. While iBeacons have 20 of 27 bytes available for user data (UUID+Major+Minor) the AltBeacons have 25 of 28 bytes available (MFG ID, BeaconCode, BeaconID, MFG RSVD) and thus appeal to advertisers in need of a larger payload **2033**. The access address **2032** may be used to key the message to specific users or a general audience and conforms to SIG standards, not proprietary standards. A cyclic redundancy check **2035** may be included to ensure message fidelity as received.

Estimote (founded 2012 in Krakow PL) beacon signals (FIG. 20E, **2040**) include formatting that can be generated using an SDK compatible with both iBeacon and Eddystone formats, but the UUID frame structure **2043,2044,2045** that more closely resemble the iBeacon frames. Telemetry packets may be broadcast. Transmit power may be altered to improved location tracking in indoor spaces. UWB and NFC radio support has also been added as an option. The TX Power field **2046** appears to be evolving toward more sophistication in gaining access to location services.

FIG. 21 is a generic view of a BT digital packet structure **2100**. A general structure of a BT message as proposed by BT SIT is shown for comparison with the commercial deployments depicted in FIGS. 20A, 20B, 20C, 20D and 20E. As can be seen, significant modifications are being made to implement proprietary markets.

Starting with the least significant bit (LSB) the message structure specifies a standard preamble **2002** followed by the SYNCH WORD and trailer of the ACCESS CODE **2101**, and then a PDU Header followed by the PAYLOAD and a CRC. The synch word is designed to identify the relationship of the sender and receiver and to specify a basic synchronization sequence and clock offset for exchanging messages in the spread spectrum. The payload can be an advertising data payload of 0-37 Bytes or a data payload of up to 255 Bytes. The trailer is generally a message integrity check. Messages in the advertising channels will include the preamble 10101010 (1 octet); and messages in the data channels may have preambles of 01010101 or 10101010. The PDU header will specify the message type, which may be selected from i) connectable undirected; ii) connectable directed, (iii) non-connectable, non-directed, (iv) scan request, (v) scan response, (vi) connect request, and (vii) scannable undirected. A stealth access code may also be used that will not be accepted by the correlator of BT radios set

up for routine interactions. In this way, the access code confers a significant level of structure on BT radio interactions.

The standard data message structure between transmitting and receiving BT radiosets as designated by the Bluetooth Special Interest Group (SIG) is published as part of the BT Specification. BT radio signals are formatted as packets. BT devices include packet composers and decomposers. BT radios also include correlators with registers for sorting and identifying received signals based on access code or service, for example as described in US Pat. Publ. Nos. 2002/048330 and 2009/0086711, which are incorporated herein for all they teach and reference.

"Access codes" are part of a header that addresses BT radio traffic. For example, a general inquiry access code (GIAC) identifies traffic broadcast to any listening device, and indicates a discoverable device. Other inquiry access codes may be directed to individual devices, such as particular members of a piconet. The access code may be derived from, but is not the radio unit identifier of the transmitting device or the intended receiving device.

BT unique radio identifiers (RUI) or "radio signatures", as used here, may be a MAC address of a BT radio device or may be a universal unique identifier (UUID) or a part of a UUID, and may include a serial identifier assigned by the Bluetooth Special Interest Group (Bluetooth SIG) as administered through the IEEE Standardization group (accessible via a WHOIS-style lookup). The RUI may also include a part number given by the manufacturer. The SIG standard also permits developers to encode a "group identifier" or "community identifier" inside an extended unique identifier (EUID) issued by the manufacturer, inside the BD_ADDR, or inside a Service UUID. Proxy identifiers such as service UUIDs link to services associated with a discovered BT device. Identifiers may include payload URLs and payload unique identifiers (UIDs) that identify proprietary services. The payload may include frames or "values" containing more information. Payload information may include subtype or location, advertising data, sensor output in digital form, and records of Bluetooth radio contacts, for example.

Service identifiers inform the radio of protocols to be followed in sending or receiving data, and allow developers to create tools that incorporate elements of the payload as "deep intent" triggers for software applications. Advertising messages may include one or more identifiers and service UUIDs, for example. Other messages may not provide sufficient information to identify all services associated with a device, but a qualified BT receiver can respond to obtain more information without actually connecting.

For example, identifiers in a message actuate protocols in receiving BT radios that can wake smart devices, direct a smart device to a URL, push a notification to a remote device, or pull attachments from cloud library resources, for example. A smart device can receive Bluetooth radio traffic from any Bluetooth device in radio proximity, and forward that traffic to an IP address associated with a Bluetooth group or community, after adding a timestamp or a location stamp. By doing so, the smart device serves as a "hub" to transfer Bluetooth traffic radio contact records to the broader cellular network (or vice versa), enabling a host of location-driven services that can be modified according to sensor data.

Many devices broadcast their RUI or MAC address in the open, or in response to a SCAN REQUEST. A class of "BT Sniffers" may detect these addresses and compile stacks of addresses and device names as BT traffic metrics. Devices may also be recognized by the services they advertise. For dedicated peripheral devices, a client application can scan

for devices offering services or features associated with a UUID that specifies the GATT services the BT device supports, and in full CONNECTION, data specific to a service or feature can be transmitted across the connection. A BT printer can advertise itself as a ‘printer’, for example.

The RUI address can be an advertising address, a device address, a dedicated address of a piconet device, a virtual address, or a subscriber address, as is useful in mesh networks and for creating whitelists. Some address standards are open, others are proprietary or are obfuscated to prevent BT snooping.

In recent trends (Eddystone, above, for example), BT signal payloads may include URLs that link the device to the physical web. Alternatively a community identifier is transmitted in a message as part of a header, routing address, or payload that when recognized by packet decomposer in a receiving device, causes the message to be forwarded to an IP Address and associated cloud host. This approach has enabled community lost-and-found services (such as described in US Pat. Appl. Publ. No. 2016/0294493, which is incorporated in full by reference).

The radio header and payload may also include resource identifiers that direct communications protocols in the link layer and activate software applications keyed to the resource identifiers. This approach is seen frequently with smartphones-installed Apps react in real time to BT transmissions. For example, a received BT transmission can wake up a sleeping device (US Pat. Appl. Publ. No. 2020/0242549), which is incorporated here by reference. More recently, data supplied in the fields or payload of a BT transmission can cause an App to be installed, or if the App is installed and the appropriate permissions are in place, the App can be run at a particular instance in the program as most relevant to contextual clues in the received BT signal. This is termed “deep intent” to indicate that the App anticipates the user’s thought process and causes the client smartphone to display the most relevant materials from a resource or takes an appropriate action in anticipation of the need. This process has been extended to wall screens, so that “walk up” computing is increasingly automated by invisible BT radio transmissions that identify the user and guess the user’s intent from radio proximity or accelerometry data. For example, if a user picks up a shoe in a shoe store, a BT radiotag attached to the shoe will send a sensor output and a wall monitor will display more information about the shoe, or push that information onto the user’s smartphone.

In connected links, BT signals transmit data. Newer BT 5.0, 5.1 and 5.2 standards permit multi-slot messages for sharing larger amounts of information, even encoding of speech. Connectionless data sharing is also supported in the newer protocols.

In a typical BT interaction, a first BT device will send an INQUIRY packet 128 times in 1.28 seconds, each inquiry packet is sent in 16 time slots (10 msec, 625 usec each) over two alternating sets of frequencies. The INQUIRY packet is short, just an inquiry access code. A second BT device, operating in an unsynchronized listening mode, intercepts one of these transmissions by coincidence (there are 79 possible frequencies [or 40 depending on the standard], three of which are reserved as “advertising” frequencies in BLE). The Baseband protocol causes each radio to use pseudorandom “frequency hops” to jump from frequency to frequency over the spread spectrum (U.S. Pat. No. 2,292, 387). A device that is in INQUIRY SCAN at some crossover hop will intercept a packet with an inquiry access code that it recognizes, or that it chooses to accept. The frequency hop protocol is inherent in the access code, and a device that

accepts an access code can then join the hop sequence with the first device and can send an FHS response packet containing its hardware address and its clock so that the first device can specifically address it with further instructions, if permitted. The interaction may then rapidly be escalated to a PAGE and PAGE SCAN interaction, resulting in a CONNECTION that formally makes a piconet link in which the RUIs of the radios are stored in device memory. The piconet relationship defines one of the devices as a “center” device (“master”) and the other device as a “peripheral” (“slave”) for purposes of organizing the transmission and receive sequences. At the hardware level, these roles are interchangeable and are controllable by a master-slave switch.

A BT device can participate in two or more piconets as separated by time division multiplexing with millisecond separation. While more limited in the newer BLE standards, in one embodiment, any BT device may belong to a hierarchy of piconets, in which its participation in a second piconet is alternated with its active participation in a first piconet. The device in the central role scans for BT radio sources, looking for advertisements and inquiry responses. The device in the peripheral role advertises itself and offers a service. GATT server vs. GATT client determines how two devices talk to each other once they’ve established the connection. GATT metadata is transferred from server sensor node to client center node, for example.

To inquire about other radio units in a receiving area, BT radios may also promiscuously announce their presence to other BT devices by sending a general INQUIRY access code (0x9E8B33, GIAC). An ID Packet may be exchanged in response to a FHS packet. Access codes are classed as DAC, IAC and CAC, indicating Device Access Code, Inquiry Access Code, and Channel Access Code, respectively, the details of which relate to link management. All packets begin with the CAC, a DAC or IAC, and a clock number segment. A correlator identifies relevant packets for processing. BT devices acquire information about other local BT radios in this way.

In a piconet, using link management, devices that are parked or lose a pairing connection can ignore public traffic but will “wake up” (almost instantly) in response to a beacon signal from a familiar or “whitelisted” partner-so as to restore or recover a piconet connection. The listening device can also partially wake up its MCU so as to log any radio contacts, while not responding further. Not all radio interchanges result in a CONNECTION, but the listening radio can record information about the transmission, and by escalating to INQUIRY SCAN without wasting time or energy, will receive more detailed information about the transmitting device.

Bluetooth Core Specification, Version 5.2 and Supplement, (2019, incorporated herein by reference), includes an “Extended Inquiry Response”. Data types may be defined for such things as local name and supported services, information that otherwise would have to be obtained by establishing a CONNECTION. A device that receives a RUI and a list of supported services in an extended inquiry response does not have to connect to do a remote name request and service search, thereby shortening the time to useful information reception. Backchannel communications facilitate the connectionless mode.

FIGS. 22A, 22B and 22C are sample packet structures that relate to the BT advertising and link layers. These advertising beacon message structures that have standardized lengths in the range of 31 to 47 octets. These tend to be proprietary formats having defined fields in the header and two to three open fields or frames in the body of the message

for non-format contents such as an identifier associated with the device or an advertising service. In the case of Eddystone packets, packet types include those for broadcasting URL (or shortened URL lookup) or sensor data. The shortened names tie to a look-up service so that a device that receives the message can be directed to further information or attachments.

The shortest packet is an INQUIRY packet **2110** as shown in FIG. **22A**, with a length of 68 bits. This is an access code **2111** of an inquiry or a device ID_PACKET of a scan response. POLL **2021** and NULL packets used in messaging have a length of 122 or 126 bits (FIG. **22B**) and include an access code **2122** and a header **2123** that defines the packet. The FHS packet (**2141**, FIG. **22C**) is important in exchanging identifiers and clock hopping schema as part of the connection process prior to further data exchange by pairing and is 270 bits in length. The FHS packet includes an access code **2142** for addressing the transmission, a header **2143**, and a payload **2144**. In extended connection mode and connectionless transmissions, packet lengths may span up to 5 slots. For fidelity in voice transmissions, higher numbers of repeats of short packets are used. But packets are typically limited to a maximum payload of 258 octets (about 2 Kbits) with a total length of about 2854 bits in a 5 slot data payload, optionally with enhanced rate packet transmission with DPSK at 2 Mbps or 3 Mbps. These details point to the utility of cellular radio as an enhancement of BT radio for transmission of larger amounts of data at greater speed. 5G tops out at a theoretical 5 Gbps, permitting greater use of broadband services.

FIG. **23** is a more detailed view of an advertising packet and demonstrates how location and proximity data can be extracted from an advertising packet to build a radio contact record. FIG. **23** describes a decomposition schema for an advertising packet **2300** that is harvested to generate a radio contact record **2320** with time stamp **2351** and host ID **2352** (referring to the host device that is capturing the radio contact record). Because advertising packets vary in structure, the correlator is a smart correlator that will categorize incoming radio signals to efficiently pull out some or all of the useful fields and to assemble those data into a radio contact record **2320**, using null fields where data cannot be recovered. The preamble **2002** of the received message will be used to obtain an RSSI for the signal, and any TX POWER **2306** information in the advertising data payload **2305** can be used to calculate proximity **2356**. If heading sensor data is transmitted, that also can be captured. The PDU Header **2303** of the transmitted signal typically includes length L (**2354**) and type T (**2355**) of the packet. The MAC Address TD1 (**2353**), if transmitted, may be in the ADV ADDRESS at the tail of the PDU header or may be part of a UUID transmitted by an iBeacon. The receiving device also has a MAC Address, and will insert that in field **2352** of the radio contact record **2320**. Service UUIDs are also included in the advertising data payload, and can be taken as proxies of device identifiers. Other parts of the advertising data may be useful as sensor data and stored in a separate field DATA2 (**2357**), such as for example telemetry including temperature, battery voltage, biometric readings, and so forth.

Location, field **2358** may be the location of the transmitting device (if transmitted), or may be the current location of an XCB device that receives the transmission, for example, at about the time the transmission is received. XCB devices are fitted with means for capturing location from surrounding networks. The radio contact record may also include the device access code in field **2359** as is received in the

ACCESS CODE or synch work in field **2302**. This information may be useful in establishing relationships between BT devices so that the intercepted signals can be viewed as a family structure of piconets and scatternets with master and slave relationships outlined by the use of GIAC, DIAC (DAC), LIAC access codes, for example.

Each radio contact record **2320** can be packaged in a BT data transmission **2360**, that includes a preamble **2002**, and standard BT data packet structure. Such packets would be sent to a master of a piconet or to a smartphone for example, or to a reference hub, for analysis of the meta-data in the logs of radio contact reports over an interval of time.

“Record making”-refers to creating and logging or storing a record of at least one datum and an associated timestamp in a memory module of a radioset. Record making also refers to a process of making radio contact records. Records of a received radio contact may include an identifier associated with the transmitter and the radio signal strength, for example. Records may include an EUID, a cellular telephone number, or an IMSI for example. Records may also include a UUID. Records may include a geostamp or other contextual information. Records in storage are generally retrievable, such as by accessing or searching a table or a database, or other computer-enabled data retrieval systems known in the art. Records may also be uploaded to a higher layer in a network, such as to a server or other cloud-based service.

In one embodiment, the listening device will make a record of a radio contact in a log, and will add a timestamp to its record, and if available, may also add a geostamp. While the BT core radioset has a limited set of instructions, it controls power to a processor (microcontroller or “MCU”) and actuates the logic for managing computing and memory resources with executable instructions needed to harvest the BT radio traffic in its local environment. Onboard sensor data outputs can also be coupled to actuation of designated logic systems and executables.

“Bluetooth radio contacts” are logged as an array of records in a stack, each record having multiple fields accessible as a database of data entries. The records are stored in a cache register or flash memory of a device and may be transmitted as packet data in response to an INQUIRY or PAGE from a qualified center unit.

In one embodiment, each record in the radio contact array is time stamped and records the radio unit identifier (RUI) of the receiving “host” device where the radio contact record is stored and the radio unit identifier of the transmitting device, the received radio signal strength measured by the receiving device, and optionally a geostamp. A MAC address may also be included. In response to a trigger, the radio contact log is uplinked to a smart device over BT radio or more efficiently to a cloud host by making a cellular radio link to the network.

In logging radio contacts, a convention is used for convenience. The “Source” device refers to a transmitting unit that is the source of a service or data. The “Host” device refers to the unit that enters and “hosts” the radio contact record. Both devices may generate radio contact records that include sensor data and location. For connection request packets, the terminology “INITIATOR” and “ADVERTISER” are used to indicate that the INITIATOR sends a SCAN RESPONSE to the ADVERTISER in response to an advertisement packet.

The trigger to uplink the data may be a correlator dedicated to recognizing particular BT radio signatures, or to a particular digital sensor output from an onboard A/D converter linked to a sensor package, to an IO pin such as a flag

75

from a HOMING button, or to a pattern recognizer, such as a DSP, that can recognize voice fragments, radio signal patterns generated by a software-defined radio (SDR), or unique sound sequences. The DSP may also recognize QR Codes or fingerprints. The trigger may be a value presented to a specific I/O port in a specific logic state of the device. In some instances BT radiosets can be used to transmit larger datasets such as radio contact logs. But in practice, the trigger may instead actuate a packet composer to assemble a table containing radio contact log entries as a message for uplink directly on a cellular network via cellular radio to a cloud host, or indirectly via some other wireless radio technology such as WiFi.

Timestamping may also drive a field termed “Time to Live” that specifies a duration in which a memory is kept or a number of transmissions a memory is kept before it is dumped and fresh context replaces it in a Last in/Last out stack. If the TTL field is an integer, the field can count down the number of transmission that include a record before the record is dumped, for example.

FIG. 24 illustrates a similar decomposition of a BT radio transmission **2400**, here a connection request, but the resultant radio contact report **2410** is packaged in a WiFi IP packet **2420** with timestamp **2411** and geostamp **2412** tailored for transmission through the IP packet data environment to a cloud host and may be analyzed locally or forwarded to a cloud host for analysis of the meta-data in the logs of radio contact reports over an interval of time. The device needed to transmit BT data to a WLAN or WPAN access portal as IP packet data includes a packet decomposer and composer configured to decompose BT packets and assemble IP packets from the radio contact information. By doing the repackaging at the local XCB device, the timestamp and geostamp are valid indications that can be accurately aggregated with other data received at higher network levels from other concurrent reporter nodes. A report from an XCB device that receives an emergency access request from a BT radiotag will be received as a set of reports from several XCB devices that forward the same emergency access request so that the origin of the signal can be accurately triangulated, for example.

IPv6 support over BLE with the adaptation of 6LoWPAN and Thread protocols can result in a crossover between BLE and WLAN comms. IPv6 can be emulated over Bluetooth Low Energy (BLE) as defined in RFC 7668 according to the Internet Engineering Task Force, for example. But this standard does not allow native BLE transmissions to be routed onto IP packet data environment of Internet-compatible/cellular networks. In one embodiment, the devices described here map BLE radio packets to a standardized database entry format or “record”, and these records are shared as a sort of “radio snapshot” of the BT world around the originating device over a WiFi IP packet data environment. This snapshot can be used for authentication, but also finds a myriad of uses in provisioning new devices, in erecting geofences, in wayfinding and locating, and in maximizing throughput in the ISM spread spectrum. Analogous to the 6LoWPAN standard, the snapshot may include 1280 octets or larger when transmitted over WiFi or using the 6LoWPAN standard, but is assembled from smaller snippets extracted from the advertising packets and PDUs in the intercepted BT radio traffic, those snippets going into log entries of BT radio contacts.

In FIG. 25, an IP data packet **2500** containing a BT radio contact record is decomposed to its element parts. The IP data packet includes a preamble **2522**, a header **2523**, an IP packet address **2524** of a host server that acts as a recipient

76

or clearing house for the records, and a payload data packet **2525** and CRC **2526**. The payload (expanded, **2550**) is a radio contact record, and includes a timestamp **2551**, an ID of the host device **2552** that captured the radio contact record, an ID of the transmitting device **2553** that initiated the radio signal that was intercepted, message type **2554** and length **2555** statistics, a proximity indication **2556** that correlates the distance of the receiver and transmitter, any sensor data, UID, service characteristics or URL **2557** in the payload, and a location **2558** of the host device at the time the signal was intercepted.

FIG. 26 makes clear that the IP data packet **2600** can be logged and transmitted across a WiFi or cellular signal in an efficient and data intensive way by concatenating the radio contact reports. A multi-record radio contact payload **2601** is a log of or a “snapshot” of the radio traffic around the host XCB device for a slice of time. Shown here are ten log records **2601a**, **2601b**, **2601c**, **2601d**, **2601e**, **2601f**, **2601g**, **2601h**, **2601i**, **2601j**, of individual BT radio signals that were captured. The snapshot provides a real time look at what kind of radio neighborhood or topology surrounds the receiving device. If the device is in a familiar home office, there will be the BT printers, the smartphone, the user’s radiotags, perhaps a headset or a BT mouse, all expected members of the “radio ensemble” that is characteristic of that home location. The unique radio identifiers of each device are combined to form a “radio signature” that indicates familiar surroundings. When the user leaves the front door, some of those radio identifiers accompany him, and are picked up on the next radio snapshot. Over time, a consensus “portable memory record” of the expected BT topology of a place is stored in memory of the XCB device or host server. The portable memory record **2700** may look like that shown in FIG. 27, where useful elements include the Radio ID **2701** of the XCB device that generated the record, a timestamp **2702**, and a geostamp **2703** that relates to the snapshot as a whole. Within the snapshot the records **2721**, **2722** and so forth take the form of an array of fields in a relational database. Data may be containerized to limit access and to open the data for access by a variety of user Apps.

The relational database includes fields for SOURCE ID **2720** (the ID of the transmitter of the signal that was intercepted), the type of BT message, the length of the message, the proximity **2706** of the host device to the transmitter, and a field for data **2708** that can be anything from sensor readings to service characteristics, biometric readings, temperature, or even URLs indicating deep intent pushed by the transmitters. Field **2370** may cross-reference a whitelist of BT devices that are owned by the owner of the XCB radiotag. Field **2732** may indicate battery life residual in the XCB radiotag device, or some other telemetry useful in maintenance.

Thus it may be useful to look at the local BT radio topology from the point of view of a meta-analysis at a first level, and then based on correlator fits to known message types, to set up permission structure for controlling the “deep intent” features of smartphones and installed apps. So at the first level, defensive smartphone control may be possible without specificity and without identification of the incoming signals. Particularly, by calculating a running “strangeness index” of the surrounding BT chatter or radio topology, the level of trust extended to recognizable BT messages may have to be adjusted, additional authentications may be needed, and further sandboxing or “drying” of wet code may have to be engineered to suit the danger represented by the environmental milieu of BT traffic. On the positive side, the BT traffic can also have a “familiarity

index”, and this can be used to recognize particular devices having or in need of a special relationship of sharing, and for autoprovisioning a whitelist of those devices for immediate access based on history of pairing and history of DIAC inquiries, or other local chatter that is characteristic of a safe, familiar place, such as home or a secure office.

Let’s take some examples. De-identification of data is doable. De-identified data is data that has not been associated with a user identity. It is possible to know, for example, how many people in a county have been vaccinated without out knowing who has been vaccinated. It is possible to know how many dogs within whistling distance have collars that are paired by a BT radio to their owner’s smartphone without knowing where the owner’s live. It is possible to know how many persons in a city have firearms in a house with or without having a BT triggerlock if BT transceivers are embedded in firearms. It is possible to know if your camera has a link to your laptop, or needs one. And it is possible to walk up to an ATM machine, tap a smartphone on a pad, and have your encrypted balance information downloaded to your finance advisor in your BT device along with link for login set up to be biometrically authenticated and to cause a bootstrapping setup of a proprietary WiFi, 6LoPAN or UWB connection. Higher data transfer speeds are possible within BT, but the ubiquity of the BT radio interface as a number of installed devices clearly supports increased effort at making the BT radio interface into a flexible hand within the glove of a synthetic radio and antenna capable of M2M communication over a range of authenticated connections.

In another simple example of a user case study, consider a user who notices something is lost—that something has an attached XCB radiotag. The user consults the cloud to look up the last known snapshot of BT radio traffic captured by the lost radiotag and sent to the cloud. The cloud shows the time and place and reports that other similar radio traffic “snapshots” are identified as associated with the user’s garage, so the user goes there and can use other BT proximity tools to turn on a beeper or a lamp of the radiotag so that the lost item can easily be found under the front seat when the car door is open. Those tools are described in U.S. patent Ser. No. 16/575,315 (as an FIGS. 15A and 15B therein), titled “XCB Devices and Methods”, which is co-assigned and incorporated herein in full by reference for all that it teaches, and in FIGS. 9 and 10. GPS with latitude/longitude coordinates are not needed for making location-based interventions. It is not unusual that BT radio signals would be associable with a user’s garage. Some garage door openers use BT radio; some cars have BT speakers or user interfaces, some user’s place a BT radio-beacon in their car so that they can find the car if they forget where they parked; and in some instances, the next-door neighbor has BT devices in close proximity to the user’s garage, so a reasonably good BT “radio snapshot”, “radio fingerprint”, or “radio signature” of the garage location is commonly achieved. And even if the car is not in the garage, but is in the driveway, a signature characteristic of the car in the driveway is also known to the cloud or to the XCB device or to the user’s smartphone, for example. If necessary, the XCB device can obtain a network-assisted location fix using the cellular radio, or can obtain a GPS location fix using a GPS chip and radio, or in many cases, the GPS capacity can be resident in the cellular radio, so that triangulation of cellular signals or PoLTE are not needed. Multiple features and pathways allow the system to quickly converge on a consensus location for the lost radiotag by accessing the last BT radio snapshot uploaded to the system.

FIG. 28 identifies an XCB device 2800 (XCB1) surrounded by a local cluster of BT signals and network components of a poly-radio system. Shown are BT radiotags TD1 (2801), TD2 (2802), TD3 (2803), TDN (2814), a smartphone RO1 (33) and having access to a VPG 2400 and a general cloud IP environment 1111. Devices TD1, TD2, TD3 belong to the owner’s personal piconet 2810 and generally travel together with smartphone 33. The XCB device (2800) and the phone RO1 (33) are capable of interfacing with both the cloud nodes and each other, but also with the BT radiotags. If radiotag TD3, for example an umbrella, is left behind as the owner 11 leaves home, the XCB radiotag and the smartphone receive an indication of decreasing RSSI and possibly heading sensor data that indicates the disconnect, but the smartphone has other priorities. However the XCB device has a primary logic level dedicated to monitoring the radiotag envelope and immediately pushes a notification to the smartphone that alerts the user. The XCB device may also have its own speaker and display, but either way, the owner is instantly notified that the TD3 radiotag is at risk of being left behind. This occurs at 8 AM while stepping out the door if there is a prediction of rain in the afternoon. RSSI data can be jumpy; multi-axis heading sensor data less so, but it is the snapshot of radio signals assembled by the XCB radiotag that shows the TD3 signal and sensor output as being an anomaly by several parameters and results in the exception notification and intervention following conditional logic rules or derived by machine-learning.

Similarly, TDN (2814), by example, is lying on the sidewalk ahead of the user 13, and is connected to a headset or radio apparently dropped by a passerby. User 13 can tap his smartphone 33 to the TDN radiotag, and the shock of the impact is digitized and sent to the XCB device in a radio contact report among with the UUID and MAC address of radiotag TDN. Alternatively, very close radio proximity of the smartphone (within a meter or two) is sufficient to trigger a radio contact report. The XCB radiotag sends this data to cloud 1111, where a look-up table is used to identify the true owner of the headset or radio, and a notification is sent to the owner that the lost item has been found. The XCB device also uses a voice interface to tell user 13 that the true owner (user 11, not shown here) can be contacted by text and invites the user to enter a text describing what was found. The cloud service sends the text message and the current location of the true owner and offers to convey a reply that sets up a phone conversation that could lead to recovery of the item, or at least to an exchange of instructions so that the lost item can be returned.

Alternatively, the cloud host can assist in putting any passerby in communication with the police if the radiotag is attached to a lost child. Any BT device carried by the passerby can cause the XCB radiotag to activate its cellular modem and CALL HOME if the BT device comes into unusually close proximity, for example. The cloud host can then assist in setting up a 911 call via Voice-over-BT (VoBT) with the passerby. The activation of the cellular modem need not involve a switch or tap, but instead may occur when any unfamiliar BT signal is detected within a one or two meter proximity of the radiotag. The RSSI of a BT signal at a range of less than one meter, even if the transmitting device is transmitting at low power (0 dBm), will be in the range of -30 to -50 dBm and hence can trigger a predefined protocol in the receiving XCB radiotag such as a command to call in a radio contact record update.

This is a simple means by which shared community resources can be implemented on a cloud host for operating

a lost-and-found service that protects privacy but maximizes the small efforts of kindness by community members in restoring lost items to their rightful place.

Simple user-directed control functions are readily established and executed. User/subscribers interacting with cloud hosts via XCB devices can manage or entirely automate many remote machine tasks ranging from turning on a coffee pot, opening doors and windows on a sunny day, turning out the lights at night, starting a video on a nearby screen, taking a user to a website, finding parking, initiating a purchase, tracking a lost pet, sharing a file with another device or printer, synchronizing contact lists between devices, and so forth.

In some instances, BT devices and hubs define “safe zones” (also termed here “familiar zones”) and are programmable, for example. Radiotags associated with each asset are radio tethered to the safe zone. In some embodiments, reference hub devices that are not smartphones may also include software or firmware and may include a user interface in the form of one or more buttons or other control surfaces for operating a safe zone. Conditional logic rules are established by a user/subscriber for controlling transit of objects out of or into a safe zone. Alternatively, a smartphone or smartwatch can include programming for setting up and administering safe zones. By integrating the basic XCB package into target assets, “smart objects” are created and can be managed as part of network services by individual users or system administrators.

BT radiotags may also be used to manage or control proximity. By using XCB radiotags, BT proximity-related data may be collected, as may be useful in contact tracing to reduce spread of infections, for example, and any contacts that need follow-up can be uplinked to a cloud host over a cellular network connection. Software viruses can also be monitored. XCB devices that encounter malicious code will report local radio devices to a network administrator. Proximity alerts can then be used to quarantine the infected devices.

In another embodiment, a database of radio contact data is achieved using a software- or firmware-defined BT radio “sniffer” in an inexpensive portable package—i.e., the XCB radiotag. The BT “radio envelope” around an XCB device, including all the exogenous frames and snippets of advertising and data that have been described in the literature as “BT pollution”, is sampled to achieve “situational awareness”. The sampling device or distributed learning process includes a correlator, DSP or numerical co-processor that assesses a bitstring for a match probability corresponding to one or more whitelists or blacklists, with supplemental logic criteria that are built into a gate array as part of a decision process. The device may implement Kalman filters and semantic algorithms in a miniature platform that achieves a measure of situational awareness as may be supplemented to support or draw on machine learning (ML) using cloud resources. “Situational awareness” feeds context-driven features, in which context includes location, familiarity, and discovery, within an overall low energy budget for a consumer device capable of extended use without battery recharge. Situational awareness results in notifications, alerts, alarms, executable function activation, and commands to remote devices, for example.

And by facilitating internodal transfer and porting language for containerized file management with rolling temporary storage, sampling of the radio envelope topology around a user becomes a tool for safe wayfinding for a community of users. Patterns in the google of bits in a radio environment may be anonymously sampled and reported as

“radio contact reports”. Finding, tracking, scanning, locating and proximity monitoring are provided as complementary services supplemented by a Bluetooth Proximity Locator Services Toolkit and a Cellular Remote Locator Services Toolkit (FIGS. 9,10). Data logging of sensor data and radio contacts enables provisioning of the physical web, piconets, tracking minders, geofences, libraries, maps, clusters, human interactive, and “proximity avoidance tools” such as useful for managing social and network interactions.

Looking to the future, it is clear that BT radiobeacon signal formatting (for example iBeacon or Eddystone) has not evolved as an open resource, but instead is tied to the particulars of dominant handsets, the iPhones and Android smartphones that control access to the cloud servers. Proprietary features are built around commercial advertising and the ecosystems of Apps provided by developers. Formatting is so specific that data can be decoded only by a compatible App. There is no open source data for open source software to operate on. But there is another level of data that is less unambiguous in meaning, is openly accessible on the advertising channels of the BT SIG standard, and from which inferences can be made without access to a decoder or to a lookup table provided by the specific developer who released and supports the radiobeacon and its commercial accounts. Google’s initial offering might have been a roadmap to a BT topology for the physical web of the IoT, with embedded deep intent links, but it is increasingly clear that Eddystone as released ties beacon message content to commercial uses and is not an open format. Subscribers must register to have their radiobeacons displayed in a scan result as a marker for their storefront locations. In contrast, the “BT pollution” that can be captured by a user with a BT XCB sniffer in their pocket or on their keychain, can instead be used as an ad hoc source of dynamic information gleaned from the patterns in the pollution, as described here. Hints of meaning include the channel used by the signal (advertising or data), the length and type of the signal, any access code, any identifiable MAC address if any, any UUID or fragment of a UUID, and patterns in the payload or format. Data patterns on the non-advertising channels may be assessed without decoding content. This is a meta-function that can be achieved with suitable correlators or a DSP as part of the XCB radiotag or at a cloud level.

FIG. 29 illustrates a series of radio snapshots 2900a, 2900b, 2900c in a time series, each recording an ensemble of BT radio signals captured from the proximity of the receiver. The snapshots are taken by intercepting radio traffic in BT advertising and data channels at 10 second intervals. Passive listening is used preferentially so as to not consume more battery power by generating back-and-forth radio messages. Each message includes the HOST ID 2801 of the XCB device collecting the snapshot, a timestamp 2802, and a geostamp 2803. In this instance the top ten BT signals intercepted at a moment of time is captured in each snapshot. Five columns correspond to the transmitter identification, the type of message, a length of the message, RSSI, and any data content in the message. The gain on the receiver can be adjusted to construct conditions by which a tabular content of suitable meta-complexity is filled.

From each snapshot, a “familiarity index” (a probability) may be calculated that analyzes the integrity of the BT topology as it accounts for each member of the owner’s BT ensemble (intrinsic signals) and the indigenous endemic and heterochthonous signals (extrinsic signals) that are associated with the location at which the snapshot is taken. Some extrinsic signals may be reliably associated with a particular location; some extrinsic signals may be intermittent or

81

non-recurring in a series of snapshots taken at different times. The intrinsic signals arise from the several radiotags users carry with them, but there will also be endemic signals from BT radios encountered in passing as the user moves from place to place. With familiar locations, these endemic signals will form a pattern known to the system. The familiarity index is highest in familiar locations; the system scores device signatures that are endemic to a familiar zone differently from device signatures that always accompany the owner. A “strangeness index” can also be provided that accounts for anomalous or unexpected BT signatures associated with a new or disrupted location. By quickly recognizing the pattern of familiar and strange or unfamiliar BT transmissions in a location, the XCB device can quickly autoprovision a kit list of what to keep track of, infer actions to be taken if there is an exception, and also note the overall characteristic of the location with its mélange of BT radio traffic so that in the future, when that pattern is again detected, the device can know where it is. In this way, the XCB device can know something is lost before the owner does, can guide the owner to the lost device, and can also know where it is without the need for GPS, AGPS, POLTE, or other advanced, more energy intensive location acquisition means.

As another case study, referring back to FIG. 28, radiotag TD3 (2803) is left behind at time and location point 2822 marked by an X (accelerometer impact). As a result, in FIG. 29, the combination of a sudden decrease in proximity and a vectored impact in the data columns at 2922 in radio log 2900c is manifested in a summary indication (P-INDEX 3, 2916) that radiotag TD3 has been left behind. P-INDEX 3 may be configured as an “anomaly index” and may trigger an exception notification if it crosses a threshold. The logic algorithm to generate an alert is executed by the XCB device and the alert is communicated to the owner directly or via the smartphone. The XCB device performs the analysis autonomously; further confirmations are obtained if radiotag 2803 is itself a smart object having the capacity to intercept surrounding radio traffic and write a log of recent radio contacts that is shared with a master device 2800 or with a system server 1111. By comparing the logs of multiple radio devices, a consensus emerges and the “left behind” or “lost” alert can be issued instantly while the owner is still well positioned to backtrack and recover the lost radiotag. In contrast, other radiotags may supply radiotag log records 2900a, 2900b in which the familiarity indices 2912, 2914 are recognizably close to nominal, and hence require no intervention.

FIG. 30 is a view of a rolling memory stack 3030 containing radio contact records. In this model, a continuous rolling stack of BT radio contact records, each a snapshot, is taken. Anomalies in the data are detected by machine learning and probabilities associated with predictable outcomes are used to design and execute interventions. A ten second lag is built into the memory stack, for example. Rapid identification of issues results in a significant uptick in consumer confidence and successful resolutions of location management issues. Old data is sent to the network for use in building models for machine learning and is then discarded from the local device so that the memory resources inside the XCB radiotag are not overloaded. An XCB radiotag might contain several gigabits of DRAM, or just a few thousand Kb, depending on the computational load that is done locally versus in the cloud. The percentage of memory that is flash memory versus volatile RAM memory is a design decision and depends on the depth of programming in the device and the need for permanent

82

memory that includes whitelists, blacklists, and so forth, versus short term memory used in making calculations and comparisons, for example. For more background on machine learning in pattern recognition, U.S. patent Ser. No. 10/937,156, titled “Saliency mapping of imagery during artificially intelligent image classification” to Bilenko is incorporated in full by reference.

FIG. 31 is a schematic 3100 of another embodiment of an XCB device in a system context. Processor 3170 includes a BLE radioset 3180 electronically coupled to an antenna 3181a, including, if needed, an encoder/decoder for parsing digital radio signals. The processor can be programmed, or otherwise configured, using software resident in non-volatile memory (such as Flash memory or EEPROM 3185) or as firmware, or a combination of both software and firmware. RAM 3184 is also provided for storage of volatile data, such as for data logging of sensor data that is transmissible by cellular or BT radio.

The MCU 3170 defaults to low power mode with clock function as a default, but can be powered up by a signal from the BT radioset or according to a power management schedule that also controls the cell transceiver 3183. The power management schedule includes PSM mode for deep sleep and eDRX mode for intermittent wake at paging windows and for TAU.

Cellular radio modem 3183 with cellular antenna 3183a is configured to provide simplified communication on a private network. In one embodiment, the XCB device is operated as a cellular device accessible by an IP address on a private network to find and track the whereabouts of the device via a dedicated and secure 5G private network or gateway VPG that is administered by a cloud-host administrative server. A SIM module may be installed to establish the exclusive IP address in the XCB device, with network access restricted to authorized parties such as smartphone 30.

In one embodiment, the cloud administrative host implements the VPG or cloud host 1111 network and uses the IP address to access the XCB device. By using the VPG to wake up the XCB device in a paging window as needed, sleep modes can be increased to save power. For example, an eDRX protocol can be overridden, or the parameters of a power saving mode modified. And by waking up the cellular device, then added commands and data can be transmitted to the XCB device and data can be uploaded from the XCB device. Once the cellular radio is on, then network-assisted location fixes on its transmissions may be performed automatically.

In one embodiment, use of private IP addresses with a VPG reduces the incidence of inadvertent, unauthorized, and network-incidental messaging that can drain battery life. The cloud host also adds a layer of artificial intelligence.

XCB devices having a cellular radio may wake up periodically, get a location fix, report the location to the cloud host, and then return to sleep so that the battery life is months or years. In one embodiment, the VPG network may use the location information to create a “trail of waypoints” of locations of the XCB devices over time by periodically generating and logging locations obtained by AGPS in an energy efficient manner. A motion sensor or heading sensor also improves the efficiency of the devices. For any given time period, if accelerometric motion is detected that is characteristic of motion, or a vectored heading, a position fix is requested and fulfilled. For example, the position fix is not repeated unless motion is again detected. In a variant on geofencing, XCB devices in identified “safe” or “familiar locations” are queried less frequently for location updates and not unless motion data is consistent with an excursion

that would take the XCB device outside a designated range of the safe location. The tempo of a walking person is one flag, the higher frequency vibration of an automobile ride is another flag that would trip a location update command inside the XCB device. Thus only the accelerometer or heading sensor needs to be monitored on battery power unless and until a location update command is scheduled in advance or a query is received from the cloud host.

The signal strength of a cellular base station can also be monitored, as is typically the case in cellular networks to monitor connections and when needed transfer connections from one cell to another cell. Typically, the XCB device location is updated by TAU (Tracking Area Update) when a handoff is made between two cells. Depending on rules set by the cloud host that can be linked to the XCB device user's profile, to local events, time of day, and so forth, the cloud host can also be notified if the XCB device is reallocated from one cell to another. Because this can also occur when cell traffic is being levelled (i.e. by moving users from a crowded cell base station onto an adjacent base station having lighter traffic) the cloud host can monitor the base station carrier channels in the network path to differentiate location changes that are traffic load driven versus changes driven by a change in cell initiated because the XCB device detected a stronger signal from an adjacent base station and elected to initiate a handover to the new system transmitter.

The device optionally includes an OLED display **3130**, **3402** and display driver **3131** such as designed per the artist's sketch of FIG. **34A**. The OLED display **3130** may include a touch-sensitive interface with capacitive sensor fields, or may be limited in functions to button controls.

In some instances, the cellular radio chip **3183** will also contain a GPS position locator. In other instances, a GPS chip **3188** and antenna **3188a** will be supplied as a separate component(s) (dashed boxes). Because GPS involves an energy-intensive signal acquisition and calculation, triangulation methods for determining location may instead be implemented using the cellular or BT radiosets and MIMO for example, and such network-assisted methods are satisfactory where multiple basestations having known locations are available, such as in most urban environments. GPS chip **3188** with GPS antenna **3188a** is shown as being optional because in some instances the GPS functionality will be built into the processor or into one of the radiosets, if present at all. Some cellular radio chips are provided with accessory GPS functionality integrated into the die. The GPS antenna **3188a** may be separate from the cellular antenna **3183a** as shown, but, in some instances, a combination package is used. GPS may be actuated infrequently at extended intervals to save power, and may be smart GPS, that is, activation occurs when there is a need, such as when there is motion of the wireless device or there is a situation in proximity to the wireless device (as detected from other data feeds) that necessitates, or can benefit from, closer tracking and monitoring of location. AGPS may be used to reduce power consumption of making a GPS location determination.

USB port **3181** is intended to operate with charger **3198** to recharge battery **3199** but may also be used to download program upgrades, for product qualification and troubleshooting, and for any other purpose for which a USB port may be utilized.

Audio codec **3190** is coupled by a LINE OUT to amplifier **3192**, which drives speaker **3193**. The speaker may be mounted on the housing rather than on a circuit board so as to take advantage of any resonance of the housing shell. The XCB device may include a vibrator driver **3194** and one or more vibrators **3195** configured to provide notification func-

tions and may be combined with one or more buzzers. By selecting a higher dB piezoelectric buzzer (not shown), a FIND function can be realized analogous to the FIND PHONE function taught in U.S. Pat. No. 9,892,626, herein incorporated in full by reference for all it teaches. Using the vibrator, an XCB device may be "nudged." A nudge is useful when a user of a parent device wants to attract the attention of the user of the XCB device, such as when a message is sent that needs a prompt reply.

Sensor package **3196** may include one or more sensors that are not switch sensors and are thus distinct from switches **3186** (S1, S2, S3). Various combinations of sensors may be provided in a sensor package. In some preferred embodiments, a sensor is a combined 9-axis motion sensor and temperature sensor. In one preferred device, a sensor is an integrated package having an accelerometer, gyroscope, and magnetometer for each axis. In some instances, the sensor package is incorporated into the processor.

As illustrated here, accelerometer and/or heading sensor **3197** is associated with processor **3170** and may be used to trigger processor functions as in motion/heading control and left-behind notifications. Generally, an XYZ three-axis accelerometer is included but may also include a 3D gyroscope and magnetic compass with firmware that generates a pre-processed, vectored heading output to the processor. In some instances, the accelerometer may be integrated into the processor and has a number of uses. Input from the accelerometer, such as a double or triple tap, can be used as a wakeup signal as part of a power-savings sleep routine.

The device may also be equipped with a Qi charger antenna (not shown) or an NFC antenna **3133** for receiving taps. Each tap can be associated with a transmission of an RFID identifier from the XCB radiotag to a device such as a smartphone equipped with an NFC reader and suitable App. On sensing a tap impact, the XCB device powers on the NFC circuit to transmit the RFID ID and causes a bootstrap routine in cooperation with the smart device to initiate a BT connection or a WiFi connection, for example.

FIG. **32** is a block diagram view of an XCB radiotag **3200** and a global area network with cellular network, Bluetooth network, and cloud host **1111**. A single radiotag **3200** and a single smartphone **30** are shown for simplicity but each layer of the network can include many radio units. The radiotag device **3200** has some variations from device **3100** described in FIG. **31** and will be described in detail in a systems context.

The processor **3270** can be programmed, or otherwise configured, using software resident in non-volatile memory (such as Flash memory or EEPROM **3285**) or as firmware, or a combination of both software and firmware. The processor may be for example a Monarch LTE GM01Q (LTE-M/NB-IoT such as the SQN66430 SiP) or NB01Q (NB-IoT) LGA module with integrated SIM platform (Sequans, Paris FR) for machine data exchange. Monarch SOCs such as the SQN3330 generally include an integrated cellular RF front end, but not Bluetooth radio, thus the different layout here as compared to FIG. **31**. Sequans modules typically support a variety of LTE bands for worldwide connectivity and consume less than 1 μ A of power with PSM and eDRX modes while providing for batch data transmission in a centimeter-sized combination.

The cloud host **1111** is able to engage directly with smartphone **30** using conventional cellular or WiFi radio technology. The cloud host may also engage directly with cellular radio **3283** of the XCB radiotag via a cellular network such as LTE. But to save power, the cellular radioset **3283** and the processor **3270** may default to a power

savings mode and it is a Bluetooth radio signal from the network or from a companion smart device **30** to the XCB radiotag (received on antenna **3281a** and conveyed to the processor by Bluetooth radioset **3280**) that tasks the processor to execute some routine that wakes up various higher functionalities of the device **3200**. These higher functionalities may include a network connection request, uplink, or a TAU by the cellular radioset as in a CALL HOME. In short, in one embodiment, the cloud host sends a signal to the Bluetooth radioset via an intermediary device such as smartphone **30**, and that signal will cause the cellular radioset **3283** to initiate a direct CALL HOME, optionally bypassing smartphone **30**. In this way, the cellular radioset can kept in a dormant or semi-dormant state most of the time. By engineering the device power management under BT radio control (control according to received BT traffic) the cellular radioset can minimize or at least manage the kinds of energy demands illustrated in FIG. **41**, where the power consumption of a full TAU is plotted.

During a CALL HOME, the network can send commands to the XCB device that include modifications to the default cellular power savings mode. A DRX protocol can be initiated, with provision to extend it in eDRX mode, in which the frequency with which the cellular radioset uplinks for a network location fix or initiates a data uplink is prescribed by network command. The increased call activity will drain battery power, but the network uses the increased location data to assist in recovering or tracking a lost or errant item, for example, only when needed.

Cellular radioset **3283** with cellular antenna **3283a** is configured to provide simplified cellular communication. The XCB device may be operated as a cellular device accessible by an IP address and may be used to obtain and report location fixes to a cloud host. The SIM module may serve to establish an exclusive private IP address in the XCB device. Typically, network access is restricted to authorized parties.

The information needed to authenticate to the cellular network is stored in a SIM unit that is part of cellular radioset **3283** and can also be used for high quality encryption of data exchanged. Once authenticated, the cellular radioset can be used to uplink data to the cloud host **1111**. Using an API, the cloud host parses sensor data, radio contact records, extracts relevant information, and combines that information to generate an executable command that may take the form of a notification, a warning, or an intervention. The executable command is handled by the network engine and may involve one or more smart devices **30** or other remote machines as intermediaries or may be delivered directly to the XCB device **3200** during a paging opportunity when the cellular radioset **3283** is receiving.

In one use case, the XCB device is enabled to receive a cellular power management mode override signal in a Bluetooth radio signal. That is, the cloud host can command the processor and cellular radioset to override a cellular default power management parameter by sending a signal to the Bluetooth radioset. That signal is conveyed to the processor, and the command is executed.

Sleep management can be by a restricted schedule of cellular activity, for example an eDRX mode in which network location fixes are obtained every 5 min or 10 min, and in which there is a TAU once an hour or three times an hour as required to maintain network synchronization and to balance network loading, for example. During a TAU, if the XCB device has shifted out of a tower coverage area, the device will lock on to a new tower to authenticate itself and re-establish an extended DRX mode with the new tower. In

this way, hours of location data can be stored in the XCB memory and uploaded only when an opportunity arises, or can be requested either when the cellular radioset executes a paging window call or when the Bluetooth radioset receives a connection request. Expired location data will be dumped after it is uploaded or in response to an exceeded timeout that necessitates dumping memory to make space for new location data.

During an eDRX event, the cellular radio is fully active and linked to the network, so data can be exchanged and new commands received, but generally an eDRX goes by without the need for an uplink of data in a paging opportunity and the only network interaction is to request a cellular location fix by the network PoLTE service or an equivalent.

The signal strength of a cellular base station can also be monitored, as is typically the case in cellular networks to monitor connections and transfer connections from one cell to another cell in response to signal strength changes. Typically, the XCB device location is updated by TAU when a handoff is made between two cells. Depending on rules set by the cloud host that can be linked to the XCB device user's profile, to local events, time of day, and so forth, the cloud host **1111** can be notified if the XCB device is reallocated from one cell to another. Because this can also occur when cell traffic is being levelled (i.e. by moving users from a crowded cell base station onto an adjacent base station having lighter traffic) the cloud host can monitor the base station carrier channels in the network path to differentiate location changes that are traffic load driven versus changes driven by a change in cell initiated because the XCB device detected a stronger signal from an adjacent base station and elected to initiate a handover to the new system transmitter.

Data is transmitted via multiple channels, one termed a "traffic channel" for example, another a "control channel," and yet another as a data channel containing GPS-related data, although the terminology varies. The control channel carries commands to SIM cards of the parent devices and also carries data packets for SMS text messaging. The traffic channel is organized into slots for carrying symbols and on the uplink is controlled by a dynamic allocation of slots to each end user device. To avoid an imbalance on the backhaul, slot traffic is asymmetrical, and may be greater in the downlink than the uplink. In newer systems, slots can be mini-slots for carrying small payloads, slots can be aggregated as needed if reception is good, and packet data may be transmitted in the aggregated slots to support bit streaming applications. Dynamic allocation of slots is known in the art.

Continuing with FIG. **32**, the XCB device **3200** may include voice and display capability for cellular and Bluetooth voice and data networking. The device here incorporates a voice-quality speaker **3222**, speaker driver **3221**, and display **3230** with display driver **3231**. The circuitry includes a microphone **3225** and voice encoder **3226** for voice transmission. A simple buzzer, bell, or vibrator **3220** and driver **3226** may also be provided. Separate audio and visual codecs are supplied for higher fidelity quality. The chipset can include a Monarch controller (Sequans, Paris FR) with integrated LTE modem. RAM memory **3284** for use in data logging is adapted as a buffer for audio/video transmission or separate memory and buffering is provided. The recharging circuit **3298** may include a connector for a USB power cord or other adaptor **3297**, for example. The processor **3270** may also include a digital signal processor (DSP) or neural engine for recognizing voice patterns. In one embodiment, a CEVA (Mountain View CA) Bluetooth BLE or BTDM RF front end chip is combined with a Monarch (Sequans, Paris FR) cellular front end chip in

which the Bluetooth radio controls the power state of both chips. VoBT (Voice-over-Bluetooth) may be implemented.

By adding a voice interface with microphone and speaker, the user/subscriber can communicate instructions directly to and from the XCB device without the need for a keyboard. This is enabled by VOLTE (voice over LTE) or VoBT (voice over Bluetooth). A speaker and microphone have utility in wearable XCB devices, such as one worn as a wristband or a headset. The scope and concepts of the invention are not limited to particular device forms, but may encompass other form factors as are readily apparent from these teachings. Any embodiments, alternatives, modifications and equivalents may be combined to provide further embodiments of the present invention that include an energy harvesting human interface or other user interface for a XCB device without departing from the true spirit and scope of the disclosure.

Interestingly, voice-over-Bluetooth (VoBT) can be enabled for short range voice interactions between an XCB unit and a smart device without the need for the more complex voice codec used by a cellular radioset. To save power, the data-over-LTE functions are sometimes limited to machine M2M data, not voice. "FIND PHONE" functions can be implemented using BT or cellular comms between an XCB device and a smartphone, with BT being the more power efficient method where radio proximity is established.

Larger BT transmissions such as audio recordings and video recordings can be transmitted using "store-and-share" ("audio SNS" and "video SNS") protocols in which there is an initial processing of data on the portable device that includes compression by a vocoder. The device stores for example the audio in memory temporarily, followed by processing to encapsulate the digital recording into the frame/slot structure of a transmission. Transmissions are completed in one or more bursts. The data then transits the packet data environment and is received by a cloud host at a designated IP Address. The cloud host examines the data, looks up an associated profile of the users, including any permissions, and forwards the message to one or more smart devices **30**. In this way, the digital radio signals are made compatible with an Internet Protocol (IP) format and can be displayed or played back as audio on a smart device or on a XCB device equipped with a speaker and microphone. Any smart device or XCB device can also send audio (or video) messages by the same process of packetization followed by radio transmission in one or more frames.

Frame structure is complex and includes hyperframes, superframes, frames, and slots. In TDD, the frames and slots are transmitted at designated times. Each carrier is part of a transmission protocol managed by the network and includes its own meta-data, including error checking, synchronization, indexing, guard periods, and so forth. For example, a 5G network supports text message traffic from a smart device to an XCB device in larger slots, but can use mini-slots to carry a reply from a XCB device to a smart device if replies are limited to a button press of on the smaller XCB device.

RAM **3284** is provided for storage of volatile data, such as for data logging of sensor data and buffering of messages. The size of the RAM or flash memory **3284** is dependent on the size of the memory requirement for data. Stored data may include data from sensors **3297** and from switches **3286**. Data from throw-switches, capacitive switches, and button-press switches are considered data. Stored data may also include radio contact records as described above in FIGS. **27**, **29** and **30**. The radio contact data may include sensor data received from other radio units. The memory

may be supplied as cache memory in the processor, or may be provided with extra RAM or flash memory chip space if more space is needed for a data logger device. Memory is generally organized as a rolling stack so that outdated data is dumped from the bottom of the stack and new data is added at the top of the stack if not first uplinked to the network. Memory may also include dedicated registers for handling packet composition and decomposition for example, for encryption keys, and so forth. This memory is generally distinct from non-volatile dedicated read-only memory **1185** for storing processor instructions.

Sensor package **3297** may include a single sensor or various combinations of sensors. In some preferred embodiments, a sensor is a combined 9-axis motion sensor and temperature sensor. In one preferred device, a sensor is an integrated package having an accelerometer, gyroscope, and magnetometer for each axis in which the sensor output is a pre-processed heading vector with corrections for motion artifact of the XCB device (such as tumbling around in a pocket). In some instances, the sensor package is incorporated into the processor.

Bluetooth radioset **3280** is electronically coupled to an antenna **3281a**. Notifications may be sent to the XCB device **3200** via either the cellular or the Bluetooth radioset, and may result in a display such as activation of buzzer **3220** via piezo driver **3221**.

The device may be rechargeable from an optional recharging source **3296**. Battery **3299** may be disposable or rechargeable via circuit **3298**. Other energy harvesting means known in the art may be used to extend the operating lifetime of the device beyond that offered by one full battery charge and a switching regulator may be used to manage power to the processor and radiosets. Depending on the kind of power available, flags may be set that configure the switching regulator and processor.

Alternatively, at least some of the devices **3200** may be hardwired to a grid power supply that has a degree of always-on reliability, and these specialized devices may function as "reference hubs" **20** (FIG. **1**). The role of hubs may be to organize low energy slave beacons into piconets and to relay data to a network server, for example. In one embodiment, an XCB device may be master in a piconet of Bluetooth devices. The XCB device may optionally be battery powered, or may be wired to be always on. Combinations of powered and portable XCB devices may also be deployed.

The meta-data can also be used to activate XCB devices to CALL HOME and to uplink relevant data in memory with a cloud host, and to receive new commands and instructions.

FIGS. **33A** and **33B** are views of an XCB radiotag **3300** with LED Array Display configured for "banner" message display and with capacity for simple inquiries and responses using tactile switches on the front panel. The device is adapted for dual BT/cellular communications of data and voice. The device includes a cellular radioset, a BT radioset, a processor, memory for storing radio and sensor data records, and a battery. The device also includes a speaker and microphone and a pair of button press switches for simplified texting of replies to Yes or No queries. The housing encloses BT and cellular radiosets and a processor as described in any of the schematics of FIG. **6C**, **7A**, **7B**, **7C**, **17**, or **31**, as representative schematics of a device of this kind—but including here an LED dot matrix array display and speaker with microphone.

FIGS. **34A** and **34B** show an XCB device **3400** that includes an OLED display screen for video graphics, shown here as a color OLED screen with sufficient pixel resolution

to display QR codes, emojis, icons, faces and simplified text. The body includes a shell surrounding a hollow core for the electronics. The device includes an OLED screen display **3402**, with miniport for USB-A recharging **3403** and an accessory USB-C or HDMI port may also be included. The face includes pressure sensitive pads for button press commands.

The sensor package may include a temperature sensor. The sensor and device may be configured for attachment to an object in need of cold chain tracking with temperature data logging as a function of position or location. The sensor package may be adapted for use in measuring body temperature, and the device may be strapped onto or otherwise contacted with a living body to monitor temperature over time and to periodically report location and temperature, as is useful for fever mapping in a workplace or community. Typically the temperature sensor (not shown) is mounted on an exposed surface of the underside of the device.

For setup, the device can have a peel-off decal with QR Code over the OLED screen. The initial setup is relatively easy. Each XCB radiotag device is provided with a QR code label on the housing. Scanning the code with a companion smartphone intended as a "parent" device (once the needed software is installed) causes a folder to be created on the smartphone display and takes the user to a menu for assigning the XCB "child" device to a particular person (such as a dependent minor, an elderly aunt, travelling companion or friend designated as an addressee and respondent associated with a particular device) or pet, and entering any context or relevant background such as a profile, an appointment list, a calendar, a schedule of regular destinations and time brackets for each, and so forth. Once programmed, the child device is electronically tethered to the smartphone via a VPG. The smartphone can direct text messages to the child device, or can even call the device using VOIP cellular service. Immediately after setup, the smart device can also display a map showing the current location of the child device and any direction of movement (if the circuit in the device includes an accelerometer). This innovation is discussed in more detail in US. Pat. Publ. No. 2021/0035429, which is co-assigned and is incorporated here in full by reference.

The transparent QR decal may include a first QR Code that is static, and when the device is powered up, other details of the QR Code may be added, so that a smartphone that is used to read the QR code will see a first authentication code prior to power-on and a second authentication code after power on as a way of preventing counterfeiting. Devices are provided with SIM cards inside and a QR sticker outside, and the initial pairing is through BT radio of a user's smartphone. The QR code directs the user's smartphone to a cloud host, the cloud host will recognize the new SIM IMSI and will go through an activation setup in which BT commands are sent to the device processor via the BT radio. Subscription cellular network access can be bundled at very low cost, enabling private virtual gateways (VPG) in which network access is gated through a private IP Address and all packeted data is routed to a private server.

The battery is for example a 2400 mAh battery. An NFC antenna is mounted on the underside of the battery, which is a LiPo foil pouch battery. The NFC antenna can be used for tap-to-touch pairing, for example, and for exchanging secret keys using direct contact to ensure lack of eavesdropping.

The shell exterior supports ports for a speaker **3404** and microphone **3406**. Yes and No button switches and a radio display LED are also mounted in the housing. The housing can include an interior speaker that causes the housing to

resonate for voice and buzzer applications. Several antenna are embedded in the housing, one for Bluetooth, another for cellular radio. There may be multiple antenna for cellular radio on different frequency bands, or the antenna may be articulated to allow use at multiple frequencies for receiving and transmission.

As shown in FIG. **34B**, device housing **3401** is perforated with a slot for a lanyard or belt. Other means for attaching a device to an asset are known in the art. The attachment hardware can be adapted to use for attaching the device to a container or package. The device includes buckles for use on a lanyard **3212** and an OLED display **3402** with microphone and speaker for audiovisual communications as well as data logging.

FIG. **35** is a view of an exemplary XCB device **3500** worn as a wrist strap that functions as a location and wayfinding monitor and messaging center. The device has voice and display capability for cellular and BT voice and data networking, and incorporates a speaker **3503** and display **3502**. The circuitry includes a microphone **3504**, voice encoder (internal) for voice transmission, a USB connector **3507**, and tactile buttons **3508** as part of a voice and touch user interface. Separate audio codecs are supplied for higher quality. The chipset can include a Monarch controller (Sequans, Paris FR) with integrated LTE modem. RAM memory for use in data logging is adapted as a buffer for audio/video transmission or separate memory and buffering is provided. Options may include a Satnav positioning module and a recharging circuit with external power supply, which may include for example a connector for a USB power cord. The device may also include a DSP for recognizing voice patterns, but voice recognition interfaces have increasingly been realized as a system service and are implemented on a system level, not as extra hardware that must be supported onboard the device. In one embodiment, a CEVA (Mountain View CA) Bluetooth BLE or BTDM RF front end is combined with a Monarch (Sequans, Paris FR) cellular modem in which the BT radio controls of the power state of both chips via the processor. In another embodiment, an AeroFONE single-chip core from NPX Semiconductor may be combined with a BT radio modem.

The display on a device of this kind enables mixed media communication on a small screen. By also including a camera (not shown), higher functions (such as recognition or display of QR codes, biometric security features, and capturing photographs or videos) can be enabled, but the energy and buffering requirements for broadband capture, storage or transmission of images may preclude many of these uses in the smallest coin-sized devices. Devices without a camera can be simplified to operate on a lower energy budget but to supply a powerful tool for communicating with a user. The kinds of data that can be sent to a device with a display include QR codes that can be read by remote machines, text messages, machine-readable icons, pictures of faces, statistics, biographical information, biometrics, emojis that accompany voice messages, maps, directions, instructions in graphical form, animations, plots, graphs, decorative images, and so forth, any of which can be presented on a display screen without a need for a camera and associated processing power.

An NFC antenna on the walls adds the capacity for Tap-to-Pay and Tap-to-Connect. Qi charging is also envisaged. Devices of this kind are described further in U.S. Prov. Pat. Appl. No. 62/968,105, which is co-owned and is incorporated herein by reference for all it teaches.

Example I: Mobile Voice Hub

FIG. **36** is a view of a compact XCB mobile device for providing location management services with a natural

language voice interface. In this exemplary embodiment, owner/subscriber **11** can communicate using an interactive voice interface through XCB devices **3600**. Devices include a speaker **3603** with resonant voicebox, a microphone **3604** or microphones in a noise-cancelling array, a USB connector **3607**, tactile buttons **3608** (“YES”), **3609** (“NO”), and audio codexes for processing speech via radio. The communication can be conducted by an automated cloud host on one end and a human on the other, or can be a human-to-human interaction. The user can conduct a conversation with an intelligent machine analogous to voice-actuated user interfaces such as Google Assistant, Bixby and Alexa that are becoming more widely implemented in consumer electronics. The response back to the device can be as simple as a beep in acknowledgement of a button press, or can be a decorous “thank you,” a “bien sur”, or a “do itashimashite”, depending on the native language of the user.

The speaker and microphone array may optionally be contained in a circular, geodesic, prolate spheroid, or spherical acoustic housing. U.S. Pat. No. RE 47,049 to Li teaches a dynamic microphone array for improved voice recognition. US Pat. Nos. U.S. Pat. No. 7,177,798 to Hsu and U.S. Pat. No. 6,766,320 to Wang teach methods for natural language query and response interactions. These patent documents are incorporated in full by reference. Reference hub **20** may include a natural language interface incorporating cloud-based speech recognition and response, for example. A DSP (not shown) may be incorporated in the circuitry for recognizing basic wake words, for example. Mobile devices **10** may also include a natural language interface incorporating cloud-based speech recognition and response, for example. Voice codec transmissions over BT are often at about 250 KB, and the conventional technology is well developed. However, modifications in the housing of a radiotag to create a true voicebox are innovative.

Any interactive response can lead to further assistance, or to a two-way conversation between an owner/administrator and for example a community member who found the lost object or pet and pressed the button or somehow generated a CALL HOME **1** (FIG. 1). Typically a message might be, in the case of a child with wrist radiotag, dog or cat wearing a radio collar, or a lost asset that carries an attached radiotag, “Your child/asset/pet has been found . . . and here is the location where the pet is now [. . . see displayed map], for example.” Arrangements can then be made to recover the/ pet, or the owner can simply go to the spot and repeat the process of refining the current location until the animal is within reach. Extended voice interactions may be offered as part of the Cellular Remote Locator Services Toolkit (FIG. 9). At close range, the Bluetooth Proximity Locator Services Toolkit (FIG. 10) is valuable for finding concealed objects if needed. The back and forth of a voice channel allows for direct communication and speeds recovery. Items such as keys, jackets, purses, vehicles, valuables of any kind that can have an attached finder device, are readily tracked if lost.

In an important application, children who have strayed can be re-united with their parents or teacher using this system. Also, using machine learning, devices that are about to be lost and children or pets who are about to stray can also be detected and preventative interventions taken by the system. The object of a smart system that can detect a lost child/asset/pet scenario before the owner knows the child/asset/pet is lost is realized by this system when combined with BT radio topology awareness, radio contact record data aggregation, and machine learning.

The promise of the IoT is a sea of information that empowers people to manage their lives. By incorporating

XCB polyradio capacity in a portable device, we realize a platform for tracking, finding, and sensing that can provision itself with location data. This device can be used as a data logger for collecting all kinds of information—including the surrounding BT radio topology—that a cloud host can then use to steer events to a successful outcome without user intervention, or to notify the user of the need for intervention, such as by flagging a lost item status before the owner knows it is lost. The computing resources onboard an XCB radiotag may be limited, but open source radio contact logs containing (optionally de-identified) location and sensor data, when uplinked, can power the computing resources of the cloud for the benefit of whole communities of users.

Location of a radiotag is readily determined by GPS or by network-assisted location service, for example, and the location is readily acquired by a system administrative host. In some instances, BT radio topology has the characteristics that allow a location prediction to be made by the device or the system. If needed, the owner/subscriber can request a location fix of the radiotag from the system host. If the XCB device is not awake and in a cellular paging window, the device can be contacted via its “always listening” BT radio with minimal latency. If the radiotag is not in a familiar radio environment, then a proxy smartphone or hub operated by an anonymous user of a community of users, in response to a BT advertising signal from the lost radiotag, will alert the system administrative host that the radiotag has been detected, and the system will acquire a location and notify the owner. This community approach is termed the “Community Open Arbors” system, analogous to a living forest in which branches, trunks and roots enable intercommunication between trees and leaves as nodes.

Example II: Device with Remote Found Notification Capability

FIGS. 37A, 37B and 37C are plan and elevation views of an XCB radiotag. The radiotags **10a** are discoid in shape with a diameter of about 4 cm and a thickness of about 8 mm. The upper case include an actuation switch **15a** and the lower wall has an inlaid LED **18** that functions as part of the user interface and can also serve as a flashlight, such as is useful on a keychain in a dark parking lot. A tab on the bezel includes an annulet **16a** for mounting the radiotag to an asset such as a keychain. The devices are inductively rechargeable and sealed. Setup and wireless synchronization and data services are achieved using the cellular, BT and NFC radios inside. These devices also include a SIM card or embedded SIM for authentication to a cellular network. As currently practiced, a virtual private gateway (VPG) with private IP address is used to connect the cellular modem to the network. For setup, a “tap-2-connect” scheme may be used, for example, by which the NFC radio of the radiotag, when in radio proximity to a smartphone having the required software, causes a link to be opened to a web server and the server handles registration of the devices. After initiating the device by pressing the actuation switch **15a**, the case can be tapped on the smartphone, and an NFC carrier wave initiated by the tap engages an NFC transceiver in the smartphone. Authentication data and setup data are exchanged. NFC is used to bootstrap a higher bandwidth radio, such as the BT or the cellular radio of the XCB radiotag. The smartphone may use WiFi to connect to the cloud server, for example, or the XCB radiotag can make a direct cellular connection once it receives the needed control channel packets. User account information may be synchronized with user data already on file and the authentication is made with the smartphone in

order to link the radiotag to the user's account. Once setup is completed, direct cellular network connectivity is established. The setup may also include updating software, firmware, network settings, and any information that the user wants entered to associate the radiotag with a particular smart object, such as a child, a pet, a backpack, camera or other asset.

A cloud host is used to provide subscription lost-and-found services. FIG. 37D indicates some of the subscription services available via a user interface 3700 on a smartphone 30. The services can include biometrics telemetry from a mammal to which the radiotag is secured, real time tracking, establishment of safe zones enforced by radio tethers (as described earlier), tracking history, extrapolation of destination, detection of an escape from a geofenced area, community-supported lost asset recovery, voice interaction services, for example. In other embodiments, a handy FIND PHONE utility lets the actuation switch 15a on the radiotag cause your smartphone to ring so that you can quickly locate it if misplaced in the house.

Several levels of tracking services may be provided, each with a different expected battery life on a single charge. These are estimates based on balanced radio usage and latency for the cellular and/or BT radio modems. The "Dynamic Tracking" mode, which updates location at a central server every 30 to 90 minutes, is recommended for most routine use, but in the event a smart object (with radiotag) is lost, the devices can be reset remotely to uplink data every 4-8 min in an "Emergency Tracking" mode. Lower power tracking provides updates every 4-6 hours, and for maximum battery life (up to 1 year), a BT-only tracking regimen is supplied. Each of the cellular plans is associated with radio parameters for EDRX, PSM and/or SUSPEND, and can be modified anytime the radiotag opens a network connection in a paging window. As described earlier, the BT radio and sensor package, including motion and heading sensors, can trigger a CALL HOME during which the power savings parameters can be reset. Also, the CALL HOME can include biometric data as well as a timestamp and geostamp.

And with learning, the device can recognize when it is lost, before the owner knows it is lost, by monitoring other BT radio traffic topology in its environment and using strangeness and familiarity indices to build situational awareness into its power management routines. The actuator button 15a can also cause a cellular network connection to be initiated, or a tap by an NFC-enabled smartphone will actuate a network connection.

FIG. 37E provides a screenshot 3701 of the tracking application on a smartphone 30, shown here configured to display the location of one or more XCB radiotags (such as "My Wallet") on a map of a city. By using a combination of BT and cellular, very large areas of the planet can be scanned for a missing radiotag and the precise position pinpointed and displayed on a map, with the capacity to zoom in to street level and get directions for recovering the lost smart object.

The XCB radiotag need not be a standalone device, but in some instances may be embedded directly in a smart object and receive power from the smart object.

Example III: Global Finder Management Services

FIG. 38 shows a flow chart for the steps of a found/recovery method 3800 effected by a cloud host in cooperation with an XCB radiotag and the smartphone or other BT device of a passerby. This can be a global finder management service.

In an initial step 3801, a passerby will discover a lost smart object having an attached or embedded XCB radiotag unit. On inspection, the radiotag will be found to provide instructions for making a notification to a lost-and-found recovery service. Generally, this will be an automated system operated as a cloud server. While it is possible to initiate a link to the cloud server by (1) scanning a QR code on the radiotag, for example, other actuation steps for initiating a network link may involve (2) tapping the radiotag 3802 to create a distinctive output at a motion sensor in the radiotag that wakes up the radiotag processor, optionally activating the radiotag by (3) pressing a switch 15, 15a on the front face, or, if the smartphone includes a compatible processor or modem-executable application, (4) capturing a BT radio inquiry from the radiotag and according to the community identifier in the signal, forwarding a message with the BT radio unit identifier to the cloud host. Alternatively, (5) an NFC field of the radiotag is recognized by an NFC reader of the smartphone, and a limited amount of data is exchanged over the NFC connection. In yet another activation method, the passerby (6) activating the XCB radiotag unit by bringing a BT radio device in their possession close to the XCB radiotag so that the XCB radiotag will detect that a "strange" BT signal is within a one-to-two meter range, and that will trigger the radiotag to wake up the cellular modem and initiate a CALL HOME independently of the passerby's smartphone.

Regardless, the initial data from the XCB radiotag can generate a deep intent link sent to the cloud server's webpage and/or can cause the smartphone browser to be taken to the webpage, where an automated assistant, chatbot, or human agent may provide found/recovery services. Also transmitted is identifying data that can be used to look up the radiotag and pull up a user profile. By the smartphone (or the XCB radiotag), the initial link may be escalated cellular connection. When the task of reporting the lost radiotag is handled by the passerby's smartphone, the escalation may be by a bootstrapping process that is managed by an App in the smartphone or is downloaded from the cloud server and installed on demand when the website is accessed. The escalated WLAN connection can bootstrapped via the BLE radio of the XCB radiotag, and then via a WiFi radio of the smartphone, or can be a direct cellular connection through the smartphone of the passerby, optionally using BT to link the radiotag into the call, or can be a direct cellular connection from the XCB radiotag to the cloud, while linking in the passerby's smartphone by pushing a browser link through a BT connection, for example. If the smartphone initiates the cloud connection, the radiotag may receive an instruction from the cloud host to initiate its own cellular connection.

Regardless, the net effect is that the cloud host will list the radiotag as having been found 3803 at a location (the location can be provided by the passerby's smartphone, for example) and can notify the owner. The cloud host can also, for example, offer a reward 3804 to the passerby for assistance in getting the lost smart object back to its owner.

As a way to protect the privacy of the passerby and the owner, the cloud host can set up a chat room so that some anonymous arrangement can be settled on to recover the lost smart object or can provide a chat bot for mediating recovery. Assistance can be as simple as entering into a conversation or exchange of messages with the owner to make arrangements to meet, for example. Or obtaining a mailing address. The cloud administrator can operate a pickup and delivery service if needed. A taxi driver or police officer can serve as an intermediary, as another example. The nature of the services 3805 is dependent on the goodwill of the

95

passerby and the nature of the lost smart object. A variety of telecommunications and network-enabled services may be offered.

If the radiotag is attached to a child, for example, the police can assist. Pets also can receive assistance from authorities, and in some instances, an arrangement can be made to leave the smart object at a pre-arranged shop or office where the owner can go by and pick it up. The net effect is that the community can extend the search network to find a lost radiotag, and the XCB radiotag can facilitate or enable itself to be reunited with its true owner by engaging a cloud host directly or indirectly **3806**. The cloud host can also coordinate a reward to the passerby.

Minor variations in the method, for example, allow the owner to map the location where the smart object is found and to navigate a path to go there. Alternatively, the radiotag can adjust its power management so as to broadcast more frequent location updates to the cloud host, and the owner can catch up with the lost item even if it continues to move.

The method of operating a global lost-and-found service directed to reuniting the asset (for example a child or pet) with its true owner may include cloud services selected from: i) by the cloud server, sharing a chat link by which the true owner is enabled to arrange for return of the smart object, the chat link opening on a browser window of a first smartphone of a passerby who has activated the radio tag; ii) sending a voice message to the radio tag, causing the radio tag to audibly articulate the voice message in a speaker of the radio tag; iii) receiving from the radio tag, a voice reply audibly articulated into a microphone of the radio tag. A voice-over-Bluetooth (VoBT) link can also be established as a relay to carry a voice message to the radiotag or vice versa. Third parties may also be recruited to assist in rejoining the smart asset with its true owner.

Example IV: Charging Pad

FIG. **39** is a view of an ensemble of XCB radiotags with a smartphone on a charging pad **3900** having dimensions of about 22x12 cm and a few millimeters thick. The pad, for example a Qi Pad, allows for charging of three XCB radiotags (one is card-shaped) along with the phone simultaneously and is for home use. The pad can also include provision for autosynchronization of data to an external server via one or more radios built into the pad. In future releases, an NFC pad having the capacity to charge the system may be used so as to also achieve data transfer over the NFC pulse connection. Data transfer is useful for example in synchronizing calendars, contact lists, and network settings. Each NFC radio is configured to enable escalation of a network connection to one of the higher bandwidth radios of the poly-radio devices; for example an NFC interrogatory can trigger setup and/or activation of a BT radio link with the NFC paired device, or a cellular link as a CALL HOME to a cellular base station. NFC interrogatory pulses can also trigger setup or activation of a WiFi connection with a home reference hub, for example. But in other embodiments, the BT radio of an XCB Radiotag is used to initiate data transfers to, or respond to data inquiries from, a home hub or a user's smartphone, for example, while recharging. By incorporating memory for data logging in the radiotag, particularly motion sensor data, heading sensor data, and radio contact record data, surprising new capabilities of the system are achieved.

Example V: Cold Chain Data Logging

As shown in FIGS. **40A** and **40B**, these devices may also be used for data logging. Cold-chain tracking is performed

96

by attaching a radiotag **10** with temperature sensor to an asset in need of temperature monitoring. Here, data logging of temperature **4000** follows a shipment from Seattle to Chicago **4000**. A failure of the coolant system occurs at timepoint **4003**, as is evident from a sudden increase to room temperature in the data.

The radiotag is attached in a way so as to be thermally connected to the asset such that the temperature sensed by the radiotag is meaningful in assessing a corresponding temperature condition of the shipment/asset. The radiotag serves as a "data logger" for recording the sensed temperature as a function of time and optionally as a function of location and is able to report the data wirelessly. The processor of the radiotag is set up to perform routine cyclical temperature measurements and store the data in a table in flash memory. A simple user interface **4004** allows the user to select a variety of display formats for data.

The raw digital data records are simple in structure. Each record includes a TIMESTAMP, a radio unit ID (RUI) for the source radiotag device **10**, the LOCATION where the data was collected, and one or more fields for sensor data DATA1 and DATA2, for example. In each record, sensor data is paired with a timestamp and a geostamp.

Using M2M data sharing protocols, the radiotag can establish a cellular network connection to report the data on a schedule, when a sensor datum is out of range, or when interrogated by a radiotag reader such as a smartphone with an installed software application configured to read the tags. To initiate a query to a radiotag in BT radio proximity, the tag reader transmits an inquiry to the BT radiotag, escalates the inquiry to a PAGE using standard Bluetooth methods, and develops a connection link with the radiotag whereby digital radio data exchange occurs. A button press on the radiotag may also be used to initiate a pairing or bonding of the radiotag and the tag reader. At greater distances, the radiotag may be caused to initialize the cellular modem for transmission of tabulated data across a cellular network, where the data is a chronology of temperature measurements that includes geostamp.

Location can be obtained using any of the cellular tools described here, including AGPS and POLTE, for example, or BT mapping, and uploads of data can be triggered by sensor data, by proximity to a smart device configured as a reader, at pre-provisioned destinations, or on a schedule established by the network, for example. The upload can be transferred from a tag reader to a cloud host. A single upload when a package is delivered may be sufficient if all is well, but the radiotag can request network access at any point en route if there is a deviation from a preset temperature range as configured in software or firmware. By limiting the frequency of cellular network calls, the duration of sensor monitoring is extended without loss of data.

The data can be collected at regular intervals and uploaded discontinuously. For economy of storage and transmission, the data can be pruned by binning time intervals with temperatures that are within expected limits, but itemizing intervals where temperatures were outside expected limits, either too cold or too hot. An excursion of temperature sensor data output outside of threshold limits can trigger an upload on a cellular channel and can result in interventions by the system operator and/or notifications to the customer.

Example VI: Cellular Radio Power Consumption

FIG. **41** reproduces an oscilloscope image **4100** of instantaneous power consumption during a connection event fol-

lowed by a series of paging opportunities in DRX mode of a cellular modem. This picture represents a full TAU cycle with a series of brief paging opportunities by eDRX events on the right of the WAKE, authentication and synchronization routine that appears as a series of steps in the plot on the left, each with higher power consumption. The device is controlled by a Monarch series processor with integrated LTE RF front end (Sequans, Paris FR).

Example VII: Provisioning of a "Familiar Location" Recognition Feature by Use of Radio Contact Scanning

FIG. 42 sketches out an exemplary method 4200 for recognizing a familiar location by the BT radio signals most associated with that location and for programming power management, notification management, and "airplane mode" features (defined generally as a reduction in transmission and reception, with corresponding reduced activity of commerce-related radio traffic and network maintenance, but with exceptions for messages and notifications having a high priority that can be configured by the user) based on that location. Locations that benefit from being recognized by the radiotag include HOME, OFFICE, VEHICLE #1, FRIEND'S HOUSE #2, and so forth. The underlying apparatus is the radiotag that operates a BT scanning program 4201 configured to scan periodically (with or without making connections) for other BT signals.

In one embodiment, the BT radio functions to scan in "LISTEN ONLY" mode (1922, FIG. 19), although a similar process can take place in BT INQUIRY SCAN AND PAGE SCAN modes. Scan results are stored in a table (FIG. 27), typically including MAC Address or other identifier(s) of the transmitting radioset that sent the signal for GIAC signals, radiobeacon broadcasts, and optionally with RSSI and any sensor data captured in the signal, without connecting or responding to an inquiry, as well as scan results for signals from familiar BT devices. Signals that are picked up in the "sniffing" process by the listening device may include connectable signals and non-connectable signals; public signals and private signals, user-owned device signals and user-not-owned device signals.

To initiate the process, scanning begins by collecting traffic from a given location, so that a consensus emerges that identifies the BT signals to be expected at that location. The scan results are stored in a table (FIG. 27), typically including MAC Address or any other RUI identifier(s) of the transmitting radioset and a timestamp or geostamp, optionally with RSSI and any sensor data captured in the signal. Examples of signals that may be captured include the user's wireless modem, smartphone(s), a BT-equipped television, the neighbor's smart coffee machine, a home office printer, other BT radiotags that the user carries in a pocket or leaves by the nightstand, a WiFi hotspot with BT transceiver at the library across the street-any or all such signals could form the signature or part of a signature of BT signals that are a characteristic of the given location, for example a HOME location. Similarly, office printers, scanners, fax machines, computers, and office worker's phones, an advertising radio-beacon in a shop, and other devices may be scanned at an OFFICE location. A log of signals captured at a VEHICLE location would include a few radiotags in a user's jacket, the user's smartphone, plus any vehicle BT radio and BT headset or earpods that the user keeps in the car, albeit the signals at a mobile location such as a vehicle will also sometimes include one-time encounters with radio units passing near the vehicle. There will be a mix of familiar

signals that characterize a familiar place, and some unfamiliar signals that are non-recurring. By a process of iteration, as can be performed automatically, the device can "learn" to recognize the location from the radio signal ensemble. This learning process can be performed in the device (as in "edge computing"), can be performed or assisted by a central server, or can be performed or assisted by a companion smartphone with suitable "app".

Each of these BT signal clusters of a "radio snapshot" are "radio signatures of a place" or "radio snapshots of a place" and can be programmed 4202 to be identified as and associated with an identifier by which the group can be "nicknamed" according to its location (or stored with location coordinates such as GPS coordinates). Parameters such as time and day of the week can also be useful and may be included in a timestamp with each record. Each radio signal ensemble, based on the snapshot(s), becomes a "whitelist" that identifies the location, and the BT device, upon receiving mixed signals and matching them to the most highly correlated whitelist, knows where it is. A "correlator" is linked to a signal logging buffer and operates by assigning a logic value of a "match function" based on the contents of the buffer as compared to a reference list in memory. Once that location recognition is affirmed, user-programmable rules-based logic can conditionally cause the execution of functions by the receiving device, and/or commands to be issued to remote devices, networks and systems. For example, when at a familiar location 4203, power to the radiotag's cellular modem can be reduced to standby or sleep. When HOME, executable commands may include: send a text to wife, open garage door, turn on the air conditioning, and so forth. And in the event that the radiotag is moved, such as when carried to a new location, or carried from the kitchen to the living room, the motion signal clears the logic flags associated with the familiar location recognition event and may cause the device to recheck, using the scanning function, so as to confirm or identify the new location. This process can be compressed if signals highly correlated with a location are still being received. To prevent unnecessary scanning, a heading sensor, which outputs an integration of motion, compass and gyroscopic micromachined multi-axis sensors, can prevent false "motion" activations that might occur if the device is bumped or picked up and put down. Multiple activations that would otherwise drain the battery are avoided.

By extension, any discrepancy in the signal log of the BT radio ensemble (4310, FIG. 43 below) can be detected. Each BT device capable of reporting a radio signal inventory of its surroundings, and knowing where it is, will also be capable of reporting that something is missing or out of place by comparing the BT topology with the expected BT signals on a whitelist or whitelists.

When there is no match, the device may generate an exception notification, may attempt to learn a new location, or if a "strangeness index" is activated, may add protective features such as transmission anonymity (DIAC, LIAC), may increase anti-viral measures, or may increase the frequency of cellular CALL HOME events so that the cloud host and user can track the device as it moves. At the very least, the device can rescan periodically for changes in the BT signal environment 4204, can then activate the cellular radio modem to request a network-assisted location fix, take a GPS location fix, and/or to request instructions if an anomaly is detected. For devices having on-board GPS, in order to save power, a location fix command can be executed only when truly needed.

With cloud intelligence or by refinement of edge computing capability, this “scan to locate” process can be an “autoprovisioning” process, by which the system assigns location coordinates determining where a BT radio signature or radio signal cluster is uniquely found, and the user has only to give each location a nickname, or confirm a system-suggested nickname, for instance. The user can also set up any actions, notifications or commands to be executed when the signals correlate with that place. Similarly, the cloud host can evaluate the radio contact logs of one or more radiotags as transmitted to it (**2500,2600,2700**, FIG. 30) periodically, and can generate a notification **4205** to the user via the XCB radiotag or via a text or push notification to a user’s smartphone (or guardian’s smartphone), allowing the user/guardian to evaluate any warnings or alerts. Programmed actions may be automatically performed when the device or system analyzes a BT signal snapshot and determines that the device is in a place identifiable by those signals.

Avoidance of certain signals and certain kinds of signals is also contemplated, again with possible actions to be taken and notifications to be made; and this can be termed a “blacklist” function that is similar to a “whitelist” function, but is a list of signals which are to be avoided, or for which some defensive action is to be taken. As an example, if a pet owner has a dog that frequently strays to a neighbor’s yard or a park down the street, the pet owner can attach a radiotag to the dog, and can capture a “reference” radio snapshot of the BT radio ensemble that characterizes the neighbor’s yard or park, and program an alert to be displayed if the radiotag reports that signal cluster. The user programs the system by storing the reference snapshot in a library of stored locations, some of which are allowed locations and some of which are restricted locations, but each of which has a characteristic radio signature. Radio snapshots captured in real time are compared by the system with reference snapshots from the library. The system, a smart device or smartphone will notify the user if the dog is in one of the restricted areas. In one embodiment, a warning voice or tone will be played from the dog’s collar so as to train the dog to avoid that location. In another embodiment, the proximity of the dog to the source of the radio signals can be assessed using RSSI technology so as to better assess the dog’s position. But in a further advance in the art, the fine mapping of BT signal complexity in the signature is a much more sensitive indicator of location and proximity, given the low energy character of BT radio. At low gain in the receiver, only the strongest and closest signals will be captured, at higher gain, a broader range of signals will be captured. In one embodiment, the amplification may be dynamically adjusted to fine tune the certainty of the dog’s location. Similarly, a young adult’s whereabouts can be monitored for time spent properly in school and time spent in unauthorized locations. Devices that include poly-radio capabilities for UWB radio can also be used to map proximity-received signal strength decreases with increasing frequency at any fixed distance; a tone scale or “chirp” can be analyzed to estimate distance.

These improvements have the effect of complementing and clarifying the information provided by RSSI, and the same hardware can be applied to finding missing items—using BT signal complexity capture as an adjustable measure, with a matching function of a correlator configured to assess the depth of a match, to calculate a most probable location and even a direction of travel that can be confirmed with a motion sensor and/or a heading sensor. In some instances, proximity to a well-characterized, strong signal, such as the BT printer upstairs (that broadcasts its UUID or

nickname at frequent intervals) will render more complex analysis unnecessary, but between the radiotag, smartphone, and cloud host, the analytical power of the system to generate a probable location notification with high confidence in near real time is realized.

In contrast, RSSI proximity data can have a significant level of variability and be harder to reliably interpret, going high/low/high/low without an obvious change in position, but the radio contact records gain predictive strength with the number of signals in the group cluster, so that locations having a plurality of known signals can be assigned with a high degree of reliability. Conceptually, the population of BT devices on the planet is so large that what has been termed “BT pollution” (an overlapping mélange of signals) can be turned to advantage in locating objects having an attached or embedded BT radioset, particularly in commercial, residential and generally urban areas. And with the increasing size of cellular networks, the synergy of a BT and cellular combined network realizes a global lost-and-found system and supporting services.

The log of radio contact records can be preserved in memory ((**2500,2600,2700**, FIG. 30) as a library entry and if a radiotag is missing, the last known radio contact record in memory may provide the clue to its current location **4206**. System exception notification latency can be reduced, for example, by incorporating motion or heading sensor data into the log entries. In practice, discrepancies in the motion of a group of tags (expected to be travelling together), such that one tag becomes stationary while others are still on the move, or one tag takes one heading while the others of a group take another heading or velocity, result in unambiguous rapid notifications to the user. The notifications can be handled by a local device, either the smartphone or an XCB radiotag, or can be handled by the system’s distributed computing power. Each time a device verifies to the cloud host or local device that it is in an expected, familiar location, all warning flags and counters are deleted. A new iteration **4207** of the scanning program is commenced when motion is detected or when a companion device such as a smartphone or other radiotag sends an inquiry with status update.

Iterations of step **4201** may also be a part of a process for learning familiar sites by collecting model data in multiple scans and rescans over days or weeks and extracting a more robust pattern with ranked data from the variability and complexity of the scans when taken as a whole. The learned pattern(s) are stored in a library of patterns and used to improve location matching.

RSSI proximity data can have a significant level of variability and be harder to reliably interpret, going high/low/high/low without an obvious change in position, but in contrast the radio contact records gain predictive strength in the number of signals in the group cluster, so that locations having a plurality of known signals can be assigned with a high degree of reliability. Because there are six times more BT devices on the planet than people, what has been termed “BT pollution” (an overlapping mélange of signals) can be turned to advantage in locating objects having an attached or embedded BT radioset, particularly in commercial, residential and generally urban areas.

FIG. 43 is a system view of an XCB radiotag (**4300**) at the center of a cluster of BT radio signals (double-headed arrows), and includes major elements of a system operable to leverage radio contact record data as actionable information, even without disturbing the user. The system may include a smartphone **30**, other BT transmitters and transceivers, a cloud with IP packet network **1111**, a virtual

101

private gateway (VPG, **3000**), a cloud host server **1112** with algorithms, administrative files **1113** and databases **1114**, and of course one or more users. The users known to a cloud administrator of a system receive the benefit of all BT radio signals, not just the one's registered with the system. Extensions to multiple database concerted computing, APIs, distributed computing, and databases that grow organically by collection of data and metadata are realized as extensions of a basic system built on the XCB platform.

In operation, the system databases **1114** include a library of BT radio snapshots associated with various locations. Bayesian algorithms, vectored matrices, and probability match functions are provided as OTA upgradable firmware or software. The radiotag devices, smartphones, hubs, smartwatches, or smart devices generally, comprise software and firmware configured to solve probability match functions from matrix data, for example. The administrative files **1113** may include user profiles, programmable executables, and data, for example.

The XCB radiotag **4300**, and related dual radio BT/Cellular devices illustrated here (**2800**, **3100**, **3200**) may lack the large OLED display characteristic of modern smartphones **30**, but has the capacity to request a cellular connection **4301** with a cloud host (e.g., VPG **3000**) and to form BT radio links with a variety of BT devices. In some instances, the BT links may be two-way connections that advance to the level of "bonding" or dedicated "pairing"; in other instances, the XCB radiotag detects a transmission from a radiobeacon that is functional only as a transmitter (a "lighthouse" beacon). While in a scanning mode, the radiotag receiver captures and parses or inspects received signals from both transceiver-capable and transmitter-limited radiosets from any and all manufacturer's and hence is not limited to a small community of users. For energy conservation, in a "listen only" mode, the receiver may be configured not to respond to intercepted radio traffic, and may not advertise by transmitting GIAC signals, for example. The process of capturing, parsing and inspection of intercepted BT radio traffic need not involve analyzing the BT radio signals for content, but simply archiving, curating, and noting patterns in the signal that are consistent with a unique identity of the transmitting radio. The signals that are recognized as unique signatures of a BT transmitter are then assigned to or matched to a location for purposes of establishing the current location of the radiotag **4300** and for activating user-programmable features and services of the global cloud-based system. The system administrator enables the user-programmable features and services provided, and controls access for services provided across the system by third parties. Also enabled, are services provided between members of a community, as part of an "open arbors" community that sponsors user-localized control of private data while openly sharing de-identified data. The current art is built around large private enterprises that commercialize all such data, but by by-passing the handset-controlled infrastructure operated by telecommunications companies and their social media partners, new levels of privacy and personal empowerment are achieved.

Without establishing a radio connection with enough signal structure that the synchronization and frequency hops of a transmitting BT radio can be followed, capture of a full data exchange is limited. However, from snippets of bit-strings of identifiers and data, some patterns can be assembled as a "signature signal" that can subsequently be used to define a location or place that is characterized by the presence of that signature signal. At the level of radio reception in which "BT sniffing" is enabled, capture and

102

recording of UUID or MAC Addresses (the payload of many BT signals includes one or the other) radio place matching is enabled, for example, by listening to INQUIRY SCANS on the advertising channels. However, other parts of a signature such as a "nickname", unique preamble, or service can be informative. The matching process need not occur at the level of the receiving radio, but alternatively the signals may be uploaded to the cloud host for pattern recognition. Any signal patterns that are significant indicators of locations can be downloaded to a correlator in the local radioset and used to more speedily make location determinations. Regardless, a pattern that matches a location with a reasonably specific level of certainty can be used to cause a variety of notifications and commands to be issued. Notifications, alerts and warnings, for example may include LEFT BEHIND, WAYWARD SIGNAL, DISTURBANCE, HOT/COLD (referring to distance from a target area), LOST CONTACT, LEASH BROKEN, and so forth. Commands can include instructions to smart devices at or near the defined location, for example, or to any remote device accessible by radio, more generally. Remote device actuation can be system-implemented, cloud-implemented, or local device implemented.

The XCB radiotag **4300** is not a full-function smartphone of the kind familiar in the 21st century. Generally, the cellular modem of the XCB radiotag is standardized and isolated so that it will CALL HOME to a particular IP Address via the cloud, but is not set up to call other telephonic devices or random phone numbers, and hence needs no dialpad. Thus the link with smartphone **30** is typically a BT link **4302** (with the exception that for XCB devices, the link can be a cellular connection if desired. Radiotag-to-smartphone and smartphone-to-radiotag cellular connections may be made through the IP packet data environment **1111**, with or without a VPG **3000**). BT link connections are made on BT channels, as can be used to operate smoothly with a companion smartphone running an installed "App". Smartphone **30** of course is able to make cellular connection **4303** with a service provider via a cloud portal **1111** and to be connected **4304** with cloud server **1112** when functioning as an intermediary or "bootstrapping device" between the system administrative cloud server and the XCB radiotag **4300** and can also make BT links as a CENTER device with multiple BT devices. In this example, the smartphone can be prioritized as CENTER for the BT radiotags of piconet **4310**, but the XCB radiotag can take over when the smartphone is out of range, for example.

XCB devices **4300** differ in other respects. XCB device can function as a CENTER DEVICE with respect to a BT piconet **4310** or as a PERIPHERAL DEVICE in other BT mesh networks. The BT piconet **4310** includes the XCB radiotag with five satellite BT radiotags shown here **4311**, **4312**, **4313**, **4314**, **4315**. These radiotag devices can belong to multiple piconets (as indicated by the double-ended arrows) to the XCB radiotag CENTER and the smartphone CENTER. But XCB radiotag **4300** can scan for and detect other BT transmitters in a local area, including for example modem **4330**, and printers **4331** and **4332**, which are located in a HOME OFFICE (not necessarily owned by the operator of the XCB device) and are detectable within radio proximity of a few hundred feet or less. Thus the neighbor's modem **4335**, while not connectable, may be detected even in stealth mode and can serve as part of the radio signal cluster or "BT snapshot" that defines the familiar place nicknamed HOME. The characteristics in the signals in the cluster, including MAC Address when available, can be pulled up in a table on a screen on the smartphone **30** and

will be maintained in a database 1114 on the cloud server 1112. Because XCB radiotag 4300 can scan without connecting, and can record radio contacts in a log, the number of radio contacts is not limited to the maximum size of a piconet. Only the most reliable and important radio contacts captured in a series of snapshots can be selected to define a “whitelist” used to characterize HOME as a familiar and safe location.

Radio contact logs (FIGS. 25, 26, 27 and 30) may be consolidated in memory at the local device, smartphone, or cloud host level, and may include other kinds of sensor data. Of particular interest in managing lost and found functions and conserving energy, motion and heading data may be included with each record. Motion and heading data are useful in accelerating generation of exception notifications. If device 10 includes an accelerometer or heading sensor 623 (FIG. 6), then the simplest sensor output can be a MOTION truth value, TRUE or FALSE. Because a companion smartphone also has an accelerometer, a Truth Table can be constructed comparing motion or heading values for the radiotag versus the user’s smartphone. Any discrepancy between sensor outputs may indicate a need for an exception notification to the user. The simplest and most basic bit of information from the sensor(s) is whether motion has occurred or not, but the quality of motion (e.g., hard vs soft acceleration), the speed, duration, and the direction are all useful in detecting discrepancies and resolving false positives.

The use of radio contact chronologies and associations of clusters of signals has benefit of improving the confidence in predicting lost, left behind, and find me events. Unlike RSSI, which can be suddenly lost, the radio contact log preserves the last known location of the missing device if it is in a recognized familiar zone.

Radiotags 4312 and 4313 may be configured to form a radio link 4320 in which data is exchanged. Further details of mixed mesh networks are described in US. Pat. Publ. No. US2021/0274315, titled “Tracking Device Systems” to Daoura, which is incorporated herein in full by reference and is co-assigned. Simple BT tags are also described in U.S. Pat. No. 6,297,737 to Irwin. U.S. Pat. Nos. 9,392,404, 9,564,774, 9,892,626, 10,580,281, 10,937,286, 11,043,086 to Daoura, all of which are co-assigned, are incorporated in full by reference for all that they teach.

Interestingly, a cloud system designed to capture BT “snapshot” topologies can be trained to associate those radio signature envelopes with particular places, even with open streets and parks, until an entire map and “model” of a neighborhood, village or city is developed. The last reported radio contact log entry from a lost or missing item can be matched to the metadata stored in the cloud database 1114, and with conditional logic or artificial intelligence, a prediction can be developed by which the lost item is recoverable at a most likely location, based on the BT signal pattern and Bayesian logic. This process can also support radio tethers, and provide advance warning if a radio tether is broken or weakening. The cloud host is able to detect the change in the scanned radio signal environment (i.e., in a radio contact record anomaly) and make an instant notification with low network latency. The RSSI may be inconclusive, but a definite prediction is generated by dynamic analysis of radio contact records.

Interestingly, the energy cost in doing radio contact scanning (in “LISTEN ONLY” mode) is quite low and can be achieved at 2-3 mA/day with a scan frequency of 5 to 10× per minute. A conservative scan length can be 200 msec, for example. Receiver and transmitter functions of the radio

require a similar amount of power, but the energy use is cut significantly if back-and-forth radio communications are cut or eliminated. Scanning itself may operate at 60 μ A or less for BLE radiosets. Battery size for these radiotags may be on the order of 500 to 2400 mAh, for example. The scan time can be dynamic, as described in US. Pat. Publ. No. US20090221242 to Bergstrom, titled “RLC Window Size Reconfiguration”. Receiver amplification may also be dynamically adjusted if desired to stretch the duty life on a single charge.

By scanning the BT advertising channels, only three channels are captured, but by scanning data channels (without connecting), a more comprehensive snapshot of snippets of BT traffic in a surrounding locale can be captured. Snippets of bitstreams may be cataloged as advertising signals, paging signals, and data response signals. The ratio of advertising inquiries to paging inquiries or the ratio of advertising channel traffic to data channel traffic can be an index of the relative character and “strangeness” or “familiarity” of a surrounding area. Local areas with higher advertising inquiries (GIAC) may be more risky for “fishing attacks” and more “downtown” than private or suburban locations with higher DIAC and LIAC inquiry levels. Areas with extended data transmissions may be indicative of a large concentration of BT headsets, for example. These bits of metadata are linked, by machine learning, with notifications to a user for mapping, finding and tracking, and any threat level that is within BLE radio range.

Also new here is the role of the XCB radiotag in taking local control of the scanning process. IC chips such as the Nordic nRF52810 and nRF52840 (Nordic, Trondheim, Norway) have advanced BT radiosets, as well as a modem that functions as an advanced microprocessor, but have insufficient memory for storing scanning results and logic or for analyzing the metadata for patterns. Because Nordic Semiconductor is a fabless supplier, other manufacturers such as Samsung Semiconductor (Austin, TX), Intel (Hillsboro, OR), Texas Instruments (Dallas, TX), Ericsson Silicon (Lewisville, TX), Toshiba (Minato City, Japan), ST Microelectronics (Grenoble, France) and TSMC (Hsinchu, Taiwan) for example, are able to combine the core features of the BT and cellular radio modems with a custom-featured modem that manages radio contact archiving and pattern analysis in a local device (as will be described in FIGS. 44-45) having a required amount of flash memory and computational logic support. Cloud-based data management is also an option.

Dual-band Bluetooth and MIMO are also becoming available for consumer electronics. Newer poly-synthetic radio devices and radio access networks (RAN) have the capacity for UWB communications as well as BT, and may be built into next generation BT devices. This explosion in the basic BT platform retains reverse compatibility but offers yet greater power for future generations of 5G+ global finder technology. XCB radiotags are expected to develop as a platform that is independent and distinct from smartphones: leading to new and unexpected uses.

In another embodiment, the XCB device functions in classical INQUIRY SCAN and PAGE SCAN mode, and will advertise and respond to inquiries from other BT devices in order to collect MAC addresses and keep track of group movement, for example. The transmit and receive radio functions are relatively similar in their energy requirements, so the significant factor is the duration of the scan inquiry and listen functions. In some instances, where traffic is at high volume from multiple radio units, a 10 msec scan window may be sufficient; in other instances a 200 msec

window scan may provide the basic information needed to establish or confirm a set of BT signatures for a whitelist and a location fix. The BT SIG provides a UUID that can be used in full or in part to establish location when made a part of a whitelist, but because the UUID is an 128-bit identifier and has significant bits that are highly likely to be unique, even one or two local BT devices may be sufficient to verify a location based on their signatures. The scan time is adjustable, and may be dynamically adjustable by logic in the device, so as to optimize certainty of scanning-based location finding, while minimizing battery drain.

As currently practiced, not only the BT or BLE radio receiver and modem, but also an MCU of the XCB device may be required to make a location determination. However, with increased sophistication of the BT modem architecture, a process of matching radio location identity can be performed within the BT chip (for example using a simple DSP/correlator with a few registers for storing whitelists). These enhanced BLE chips are provided with multiple layers of native power management for “edge computing” so that radio functions such as transmit and analyze do not receive power until needed. For that reason, the BT chip can serve as the master power controller for other chips of the device circuitry, including the cellular modem and the MCU or CPU.

As introduced here, a combination in which an MCU or CPU is provided with i) sufficient memory for data logging of a library of radio contact scans, ii) is linked to a BT radioset for reporting radio contacts, and iii) may also include a cellular radio and modem. The MCU can be a multi-threaded processor with a Cortex ARM instruction set, for example, and may be programmed to analyze radio contact data and to develop and administer whitelists and blacklists as location indicators. A DSP or numerical coprocessor configured for matrix algebra may also be included for higher level pattern matching analysis linked to a Bayesian prediction engine with machine learning. Radio snapshots captured in real time by a radiotag can be compared with reference snapshots from the library.

Systems **4350** of the kind described FIG. **43** are configured to build smart communities and cities on the BT platform that already exists, even while preserving user-sensitive anonymity and privacy. Detection of radio linkages or inquiry scans that are malicious and enable unwanted tracking, for example, may be detected without pairing.

Conventional cellular modems **682**, **3183** (FIG. **6**, FIG. **31**) are solid state chips having a high degree of autonomy build around an isolated SIM card that functions in user authentication of the cellular radio to the cellular network. The physical isolation in the architecture of these chips in a device is impressive; they are literally surrounded by a limited number of ports that allow commands and data to enter the cellular modem processor and (via a proprietary network connection) access the network. Problematically, and again driven by the desire to enforce network exclusivity, cellular modems require a great deal of energy to maintain the cellular network connection, with traffic back and forth mostly of a housekeeping and commercial nature, even over a VPG **3000**. [We have shown that use of a VPG dramatically reduces the extraneous traffic found in typical cellular networks.] But with use of a BT radioset to control power to the cellular modem, an effective means is achieved for reducing battery consumption in dual-radio BT/Cellular devices, while augmenting functions such as LOST ALERT and LEFT BEHIND ALERT, for example, and reducing network latency in generating important notifications. New

classes of executable remote commands are also enabled; while improved security is also realized.

In one embodiment, the global BT radiotag franchise is expanded by use of XCB radiotags as companion and standalone devices from a standard use in tracking and sharing data with connected devices to a tracking model in which signals of both known and unknown origin are part of the BT signal signature capture ensemble used to assign familiar locations, to make a “location fix”, and to monitor radiotag movement in and out of the familiar locations. In advanced versions, for example, the device can include firmware of software for activating the cellular radio to alert the cloud administrator (and by extension the end user) if anything is at risk of being lost or left behind, even before the loss actually occurs; an advance in the art that has brought us to the realization of a global finder and tracker management service with power to spare for extended IoT deployment and growth potential for an intermediate level of machine learning in a memory-stick-sized device.

Firmware modifications in support of capture of radio contact information and logging while in LISTEN ONLY mode are available by adapting SDK kits that accompany the newer chips, such as the Nordic nRF52840, which already allows for OTA (over the air) reprogramming in use so that improvements can be retrofitted as they are released.

As a part of systems **4350**, radiotag **4300** may comprise: a processor and logic circuit in electrical connection to a Bluetooth radio modem with a unique radio unit identifier and a community identifier; a cellular radio modem having a wake mode and a sleep mode, said sleep mode defining a default mode; wherein the cellular radio modem, in response to a received Bluetooth radio signal containing the unique radio unit identifier, is caused to transition from the sleep mode to the wake mode, to transmit a cellular modem identifier transmissible with a network connection request, and, over a network connection, to send a real-time determined current location of the radiotag to a remote monitor device or to a cloud administrative host associated with the community identifier. The radiotag may comprise a location-determination apparatus selected from: a) a correlator or digital signal processor apparatus configured to match a snapshot of BT radio signals received by the Bluetooth radio modem in a location with a whitelist or a blacklist of BT radio signal patterns assigned to known locations; b) a network-assisted cellular radio location apparatus; c) a GPS radio chip apparatus; and/or d) a radio proximity apparatus configured to measure signal strength and proximity to an incoming BT signal from a transmitter having a known location. The BT radio modem may be configured to intercept BT radio traffic, or signal fragments thereof, without transmitting INQUIRY or PAGE scan signals as part of the interception of said all BT radio traffic. The Bluetooth radio modem is enabled to: i) intercept at least a part of a Bluetooth radio signal broadcast in a Bluetooth spread spectrum, to demodulate at least one digital segment of the Bluetooth radio signal that is characteristic of the proximate radio environment, and to generate a radio contact record of the digital segment with a timestamp in a writeable memory; ii) to operate in a virtual always-listening state with passive mode and active mode, wherein the always-listening mode is defined by a low latency listening window frequency. The Cellular radio modem is enabled to: i) establish a cellular network connection and make a CALL HOME; ii) establish a present location of the radiotag, add a geostamp to the radio contact record, and transmit the radio contact record with timestamp and geostamp to the cloud administrative host over an IP packet data network. In operation, the

processor and cellular radio modem may default to a sleep mode unless a “wake override” is generated, and the Bluetooth radio modem comprises program instructions to execute a wake override signal to the cellular radio modem or the processor according to rules-based logic executed by the Bluetooth radio modem upon interception of any digital segment of a received signal that has a probability of a predictable outcome, wherein the predictable outcome is learned by machine learning, programmed and stored in a flash memory accessible by the Bluetooth radio modem, or programmed in a user profile. The program instructions may include an instruction by the Bluetooth radio modem to send the wake override to the processor if the predictable outcome is not urgent or to send the wake override directly to the cellular radio modem and thereby initiate a CALL HOME if the predictable outcome is urgent, for example.

Example VIII: Learning from a Model Dataset

In order to make a location determination and take action, the match of a radio contact record need not be an exact match to a reference record. In some instances, a close match involving many radio signals provides more information than a match made with a single signal. The complexity of radio contact data captured by the local device can be varied by increasing the gain of the receiver. More powerful and closer transmissions will be captured at lower gain. A weighting algorithm may be used to rank radio contacts and assign a reliability index for their use in assessing location. This reflects the variety of BT sources available. Some BT radiosets are plugged in to AC power and are in fixed locations. These include printers, televisions, smart appliances and so forth. Many BLE radiosets are portable and may be in different locations depending on time of day or user’s choice. For example, a smartwatch may not transmit a consistent signal because the battery is often drained. Depending on the density of BT signals, movement of a receiver around corners and walls may be detectable, and the signal of a BT ensemble from inside a structure will be different from the signature outside, where other BT transmitters are added to the ensemble. Thus even in a walk across the street, the character of the BT signatures will change. The power of pattern matching is greater when individual signals of the ensemble are ranked by reliability, with weighting also given to the overall number of matches in a cluster. To leverage the power of complexity, the system may take multiple snapshots of a location over time and develop a “model” as used in AI. For example the user may identify a location by a nickname “HOME” during setup, and the system will periodically iterate scans at that location so as to build a deeper “model dataset” of the signal pattern at various times and days. In other words, the system will continue to “learn” the location and how signals in the general area vary in content and complexity. Over time, more reliable predictions and notifications will result. This task may occur at the cloud host level, or in the individual XCB radiotag, or in a companion smartphone, and the consensus pattern data is then shared with multiple devices, including friends and family. During the learning period, the user may occasionally be queried to reduce the incidence of false alarms or to confirm the validity of the learning experience. In this way, libraries of BT ensemble signal technologies will be accumulated that can result in “situational awareness” at the radiotag device level. Because of the intimacy of BT radio sharing, this kind of distributed machine intelligence ultimately allows the user to store and share personal data without use of cloud resources and

without being limited to wired connections or password authentication. Location-specific data is valuable, and can be directed to user control of location-specific action without the handset/smartphone as an all-seeing interlocutor.

In simplified practice, the system predicts a current location of the radiotag by matching one or more digital radio snapshots captured in current time with a snapshot from a library of digital radio snapshots stored in the memory from a past time, each snapshot in the library from a past time defining a reference snapshot assigned to a known location; and, may initiate a CALL HOME to a designated IP Address via the cellular radio or cause other programmable action in a remote machine if the current location meets one or more criteria defined by a user in a user profile, for example.

Example IX: Chip Design and Device Integration

FIG. 44 is a block diagram of a PCB 4401a containing a unique integrated circuit (IC, 4400). IC 4400 is integrated into a device circuit for operating a finder and tracking device 4401. The IC 4400 incorporates two (or more) radiosets, each with a modem that guides the transmit and receive protocols for the digital radios. The BT radioset 4402 includes an antenna and oscillator optimized to the 2.5 and/or 5 GHz ISM bands. The cellular radioset 4403 includes for example a more complex antenna tree capable of broadcasts in multiple bands that may vary with the jurisdiction for which the device is designed. The SIM card (Subscriber Identity Module, 4404) is a sophisticated key that is recognized by a cellular network and is available only to authorized users. The cellular radio modem is protected from intrusion and cloning and supports an almost continuous stream of housekeeping and commercial messages that keep an open connection “hot” even when not in use. This “housekeeping traffic” has expanded of course to include spam, email, text, and data, including robocalls, much of which is unwanted, but consume battery nonetheless. Power saving mechanisms such as EDRX, PSM and SUSPEND reduce the duty cycle by permitting short gaps in the exchange of messages, but these gaps are typically a few seconds in length or minutes at most. When combined with the higher power requirement for transmission and reception, the cellular modem is at par with GPS chips as an “energy hog” in mobile devices. A sample TAU (Tracking Area Update) plot showing power versus time to complete a LOGON and user validation was shown in US Pat. No. U.S. Pat. No. 11,184,858 (see FIG. 13 therein).

A more complete discussion of the required clocks is found in <http://www.thalesgroup.com/en/markets/digital-identity-and-security/iot/resources/developers/lpwan-module-configuration-and-timers-calculation>, for example, and illustrates the AT command structure that is directed to the cellular modem processor and logic stack. During gaps in EDRX paging cycles and the extended deep sleep of SUSPEND or PSM mode, the device cannot be reached by any cellular message. It is dead to the world until the next paging window, and by then the network connection may have been lost or synchronization may have failed. The paradox is that power savings can be achieved, but only by increasing network latency.

A solution offered here is based on building into IC 4400 a “wake pin” (4405, sensu lato) driven by BLE modem 4402 that activates the cellular modem 4403 on demand. The bitwise signal carried on conductor 4407 from the BT modem to the cellular modem may be an AT command, but the firmware of the cellular modem can instead be modified to recognize that when the “wake pin” goes high, it should

activate itself and can self-initiate a network connection. When awakened, the cellular modem can communicate with an off-chip CPU 4408 as needed, and can send and receive information in its active connection with cloud host 1111 or VPG 3000. Data bus 4410 is not always on; the CPU 4408 may be placed in a sleep mode as a default state, and it is the BT modem that can activate both the cellular modem and the CPU via two wake pins 4405, 4406, independently or in concert. In some instances, given the autonomy build into the cellular radio modems, the CPU may not be needed until a network connection is established and questions such as “Where am I?” or “Why was I awakened?” rise up in the logic stack. By separately depositing on wafer or chip 4400 an electrical connection 4407 from the BT modem to wake pin 4405 of the cellular modem and an electrical connection 4409 that connects off-chip to GPIO wake pin 4406 to the CPU, separate power control to the two processors (cellular modem 4403 and CPU 4408) is established and can be coordinated as needed. The BT modem, which is highly engineered for power management, is re-engineered to extend its power control to other functions of the device such as a block of functions (described here loosely as supporting logic circuitry and memory(s) 4412 and power management circuitry 4414). In short, the BLE modem, with its parsimonious energy budget, takes charge of power to the CPU, the cellular radio, and possibly other device functions such as a GPS radio, as is useful for longer term deployment of IoT devices such as radiotags so as to maximize the time between battery recharging cycles.

In advanced chip design, IC 4400 may include supplemental hardware and firmware for operating a matching algorithm in which radio contact signatures captured by the BLE modem 4402 are stored in a radio contact log, and then matched, using correlator(s) 4415 with programmable registers that serve as a whitelist of reference signatures needed to decide location based on the BT radio environment and its “familiarity index”. This can occur on the chip 4400 rather than in CPU 4408 and can involve flash memory and registers built into the chip for full interoperability between devices having various CPUs. In other embodiments, the needed flash memory and correlators 4415 may be built into the BLE modem 4402. The correlators 4415 include AND gates that trigger an output to the CPU or to the cellular modem when a match is found, and can trigger a CALL HOME. The radio contact record package 4416 has direct connections to the primary bus 4410 by which data is shared between the radios and the CPU 4408, and to supporting logic circuitry 4412. In the exemplary embodiment shown here, the BT modem 4402 includes firmware 4417, but the chip 4400 also includes flash memory and registers 4416 for creating radio contact records and for comparing those records to a whitelist when making location determinations according to the BT radio topology around the receiver. In other embodiments, the needed flash memory, field gates and firmware are designed into the BT modem 4402 as a stock item 4400. IC 4400 is mountable on a PCB and is soldered to leads on the board to the needed antennae and logic connections. Flash memory size for registers and logs may vary according to the application, so in some instances expansion of the memory size may be accomplished by designing a series of chips each with larger memory or by including expansion ports that can be wired to larger memory units. The firmware is also expandable as needed, and generally, as coordinated by the CPU, can be upgraded OTA to permit future upgrades.

Supporting logic circuitry 4412 may include one or more clocks as needed to operate the radios and CPU. In some

instances, such as for the base layer of the BT modem and cellular modem, the clock is part of the modem. The clocks of the BT radiotag (not shown) execute synchronization in an early stage of listening, but the earliest convergence between transmitter and receiver is entirely by chance. Spread spectrum bandwidth is distributed so that the preamble to a digital signal is received where two or more BT radiotags are within radio range and operating on a common channel. Programmable scan settings for capturing the sending unit’s preamble, instantly adjusting the gain, and demodulating the access code and remainder of the signal, which includes frequency hop and synchronization routines and also optional channel mapping instructions that allow the two devices to follow the transmission(s) while shifting channels in unison despite interference. BT clock precision is reasonably good; scan intervals may be 10 msec or longer; packets are transmitted in 2.5 to 40 msec bursts, and using 2.5 or 5.0 GHz modulation, achieve bit rates of up to 3 Mb/sec (up to 2 Mb/sec for BLE).

BT radio modems may include multiple clocks. Details of digital device clocking and radio antenna harmonics are established in the art. A low jitter BT radiotag clock can provide a cellular clock signal and radio frequencies of cellular transmissions by clock division. Alternatively, separate clocks may be provided for the BT and cellular modems and transmitters. The oscillators and antenna outputs may be integrated on a single chip with one or more clocks and two or more radios, including Bluetooth and Cellular LTE radio transceivers. Where synthetic radio is implemented, a high precision clock with multiple clock dividers may be used to clock other chip functions.

Dual radio chip 4400 receives power via lead 4420. The power in the chip is carried on conductive leads to the BT modem 4402 and the cellular modem 4403, and may be gated at several levels so that some parts of the chip 4400 are not powered until needed. The correlators of the BT modem may require access to memory 4412 in order to recognize signal patterns, that memory can be accessed via lead 4411 without mediation of the CPU, for example. In other instances, the memory 4415 is provided on-chip with the correlators and supporting firmware. Persistent memory may require low power, or may be flash memory as is widely available.

The term “wake pin” (4405, 4406) is used more broadly than would be connoted by “general purpose I/O pin” (GPIO), and may include any digital signal path or paths by which an instruction is introduced from the BT modem 4402 and executed by the cellular modem 4403 or the CPU 4408, for example, initially to wake the downstream chips from a sleep or idle mode. When a “wake signal” goes high, the receiving element may have all the needed instructions to respond, but in other instances, a digital key may be sent to the “wake pin” that includes one or more needed instructions to activate certain functions. In some instances, the received signal at the “wake pin” is an AT-command. Hierarchical control of power and wake states builds on the established capacity of the BLE chip to manage its power consumption at a subcircuit level, and extends it to manage power to the cellular modem 4403 and the microprocessor 4408. Thus the microprocessor is no longer the sole master of the logic stack, but instead is in a hierarchical relationship in which its tasks may be initiated by a radio contact event, a wake signal, or other clock signal from the BT modem 4402. Most instructions executed by the BT modem are encoded in firmware whereas the microprocessor may access a combination of volatile and non-volatile memory, including flash memory and volatile DRAM useful in making probability

calculations, and may receive OTA upgrades of needed software and firmware. Upon receiving an upgrade, the microprocessor may perform needed updates on the BT modem firmware, for example, as known for firmware over-the-air (FOTA) technologies, in which the cellular

modem is better suited for transmitting secure firmware updates of locked firmware to the CPU for flash installation to the radio circuitry and supporting logic gate arrays. One advantage of a single chip IC **4400** with BT and cellular radiosets and a separate CPU **4408** is borne out by the experience with BT consumer products in the market. Solid state radiosets do not evolve at the same rate as microprocessors and memory devices, so segregating the ICs provides a means for conserving costs when constructing next generation devices with newer CPUs or peripherals and allowing for evolution of security measures.

Other parts of the circuitry may also be slaved to the BT radio modem, and thus may be caused to be asleep or awake depending on the state of BT traffic around the device. This is a unique way of looking at the architecture of a micro-electronic device, but it builds on the notion that the radio signal environment around the device is part of the device, is the glue that connects the device to the real world, is the “community” that drives the need for and value of radio system **4500** (FIG. 45), and as a matter of intelligent design, is the controlling stimulus that drives energy use and device function. Integration of the external radio environment and chipset(s) in IoT devices is a directed evolution in chip architecture illustrated here. As synthetic radio becomes more readily available, the chipset for cellular radio is not expected to change, but the BLE chip may acquire other radio frequencies and capacity, such as for UWB, dual-band capacity, and MIMO where the needed antennae can be built into the radiotag.

In contrast to the innovative design described here, the trend in the art has been to integrate all features. The ESP32-S3 and -C2/3 SOC chip series (Espressif Systems, Shanghai CN) includes a CPU and also BT and WiFi radios, essentially a complete package, but a disadvantage is the inability to update the package design with a newer CPU without also buying the radios. By separating the radios onto a separate chip and attaching a bus for offloading data and memory to a CPU or MCU that is mounted on the PCB separately, the latest CPU or most efficient MCU **4408,4530** can be easily substituted. This is advantageous because CPU technology is evolving very quickly whereas radio chipsets evolve more slowly if at all over generations of products, in part because they must conform to regulatory standards for sharing the public airwaves and are thus constrained in ways that CPUs and MCUs are not.

The integrated circuit (IC, **4400**) comprises a BT radio modem, a cellular radio modem, and a first wake pin on the cellular modem, the first wake pin having an exclusive input from the BT radio modem; wherein the input is configurable as a “wake signal” to the cellular radio modem. The wake signal to the cellular radio modem from the BT radio modem operates independently from the processor. The BT radio modem may also include an output addressable to an off-chip processor, and that output is configurable as a wake signal to the processor. The IC may include at least one bus configured to share data between the BT radio modem, the cellular radio modem, and an off-chip microprocessor. A power supply may be shared by the BT radio modem, the cellular radio modem, and the processor. The IC may also include a GPS radio location device and an accelerometer or a heading sensor. Flash memory may be provided on chip for storing BT radio snapshots. Firmware may be provided on

chip for operating a locating and alert service based on BT radio snapshots, or the BT radio modem may be configured for offloading locating and alert services based on BT radio snapshots to an off-chip processor, DSP, or numerical co-processor. The IC will operate at low power for sustained periods as part of a circuit of a portable radiotag.

As embodied here, the radiotag **4401** includes a printed circuit board with dual radio integrated circuit **4400**. IC **4400** has a Bluetooth (BT) radio modem, a cellular radio modem, flash memory configured for storing an ensemble of received BT radio signals, and a correlator configured to match the ensemble of received BT radio signals to one or more reference signals associable with a defined location, wherein an output of the correlator causes a wake command to be executed in the cellular radio modem. An output of the correlator causes a wake command to be executed in the microprocessor. In one use case, the cellular radio modem is configured to CALL HOME in response to a non-match of the ensemble of received BT radio signals as determined by a correlator with or without computational assistance from the microprocessor, from a smart device, or from cloud system resources.

The printed circuit board also includes a microprocessor **4408** in digital communication with the dual radio integrated circuit via a databus and specialized pin connections. The microprocessor is configured to execute a conditional rule in response to an output from the correlator. Conditional rules are programmable by the user or by a system administrator.

FIG. 45 shows the radio modem IC **4400** as part of a device **4501** in the context of a radio system **4500**. The radio system may have global reach via its cellular radioset **4403** and intimate radio connections with its immediate environment via BLE radioset **4402**, all with minimal power for maximum duration of standalone IoT use, such as in radiotags **4501**. The device includes a BLE antenna **4402a** and a cellular antenna or antennae **4403a**. A GPS chip **4550** and antenna **4550a** may also be provided as an option. The unit is provided with a rechargeable battery **4560a** and power source **4561** for recharging the battery. The battery is protected by a power regulator **4562**. The power regulator may be configured to switch power from the battery to a superconductor **4560b** sized to provide backup power for several days. While not shown, a USB port can serve the functions of both recharging the battery and port for serial data exchange with the microcontroller.

In this embodiment, the MCU **4530** includes a suite of optional peripherals, including a display **4535** and display driver **4535a**, which may be a single LED, multiple RGB LEDs, or an LCD and display screen (**3502**, as was depicted in FIG. 35), for example. Also included is a microphone **4534** and driver **4534a** used for dictating messages, and a speaker **4533** and speaker driver **4533a** for audio. A piezo buzzer, vibrator or bell is included **4532** with driver **4532a** for simple audio notifications.

The MCU also may be provided with one or more sensors **4540**, including motion and heading sensors for example, and one or more switches **4541** that function as sensors, such as for sensing commands from a user. The switches, display, buzzer, microphone and speaker form a user interface. Other more complex setup and programmable tasks may be performed by installing a full user interface on a smart device **30**, linking the smart device to the MCU, and programming features of and instructions to the radiotag remotely that are then stored in non-volatile memory **4570**. Memory (such as Flash memory or EEPROM) is provided for storing an instruction set and can be reprogrammed as needed. Volatile data buffering for the microprocessor or co-processor is

generally provided by RAM 4531. The MCU may also include support for vector matrix instructions, as is useful for acceleration of neural network computing and the signal processing workloads associated with matrix pattern matching calculations.

The system provides for cloud-to-smart device (and smart device-to-cloud) communication, cloud-to-radiotag (and radiotag-to-cloud) communication over a cellular network. While the smart device may also be enabled to communicate with the cellular radioset of the radiotag, BT is the more commonly chosen vehicle for communication from smart device-to-radiotag and radiotag-to-smart device. This specialization allows the cellular radioset to use a simple instruction set by which all network contacts are routed through a common cloud portal such as a VPG 3000. In contrast, BT is a promiscuous radio technology and can communicate with any other BT radioset directly. Thus the BT radioset 4402 will be able to sense and to connect with all BT radiosets in radio proximity, but the cellular radioset 4403 will have a limited but direct pipeline to the cloud host.

The BT radio modem 4402 is specialized in another way. In this embodiment, the BT radio controls a “wake up” pin 4405 (FIG. 44) to the cellular radioset on a single chip 4400, and may also control power levels in the MCU 4530 via a “wake up” pin 4406. Control is not exclusive, during regular operation the MCU may also share commands, data and instructions with the BT radioset via bus 4570, and may share commands, data and instructions with the cellular radioset via bus 4571. But the resting state of a radiotag as useful in IoT applications is necessarily a standby mode at low power, and the design decision disclosed here is to establish that standby state by putting the MCU and the cellular radioset to sleep as much as possible under power management implemented to depend on the BT radio environment around the device and other sensor outputs. Sleep is the default state of the MCU and cellular radio modem—in which power is restricted to clock and essential constitutive chip functions. The MCU may have periodic activity cycles in which it wakes up to perform maintenance of data archives in memory and sample sensor input, for example. The Cellular radioset may have periodic activity cycles in which it wakes up to report to the network and/or update its location (TAU) as needed, while spending most of its time in EDRX, PSM or SUSPEND sleep. But the BT radioset 4402 is able to maintain low energy surveillance of the proximate radio signal environment while controlling power to the entire device. The proximate radio environment includes signals from any proximate smartphone or radiotag, for example. If a signal is detected that requires system attention, the BT chip can wake up the Cellular radioset at 4405 to make a CALL HOME, or can wake up the MCU to get a GPS location fix, or can rely on a library or log of familiar places to identify the location according to the BT signal signature, and with data from the motion or heading sensors, then use a simple logic tree to decide whether to escalate the radio contact or to just log it and await further instructions. Power is supplied to IC chip 4400 via pin 4420 and is shared by the BT radio modem and the cellular radio modem, but beyond basic standby power, the cellular modem is in sleep mode unless either routine wake activity is scheduled with the network or the BT modem wakes it up to perform a needed task.

In some instances, the BT radioset will cause the Cellular radioset to execute a CALL HOME to the cloud administrator, and following establishment of a network connection or TAU, instructions will be relayed to the MCU (and BT radioset) directly from the Cellular radioset at 4571.

The individual devices, BT radioset, Cellular radioset and processor may share a databus and power. The BT modem may receive sensor outputs on the same databus that carries data to the MCU, and in some embodiments, the BT modem may have a logic stack that pre-disposes it to analyze and act on BT radio contact data according to a sensor context more broadly.

In one embodiment, during up time, the MCU will collect BT “radio snapshots” from the BT radioset and log them as radio contacts, optionally with other location and sensor data. A comparator at the level of the MCU (or the BT radio modem) is used to determine whether the signals match a known location. Or in another embodiment, the radio contact data and any location data that is collected is periodically uploaded to the cloud for archiving and analysis. Cloud administrative services provided as part of the global finder and tracker network will include notifications to parties of interest if a familiar pattern or an anomaly is detected in the data. Users can be notified when a partner or pet enters radio proximity, for example, or alerted if the partner or pet has left or is leaving radio proximity.

In conventional devices, services may use RSSI to detect LEFT BEHIND items, to establish safe zones, and to create radio leashes. However, the RSSI signals output by conventional BT devices are quite noisy (for a number of reasons) and the most straightforward solution to the noise has been to implement a time filter in which a persistently rising, falling or lost RSSI over several minutes results in a notification. However, by correlating and analyzing radio contact data in real time at the local level for patterns that include loss prevention indicia, the network latency of a time filter, which can require several minutes, can be eliminated.

Whitelisting may be complemented by blacklisting, which may avert unwanted or dangerous situations, such as a family’s youngsters hanging out at a place of misfortune, just as an example. The cloud host would receive data from a child’s smartphone or XCB device that shows where the child is spending his time, and the cloud host will supply a notification to the parents. Similarly, a dog that persists in returning to an undesirable location can be tagged and tracked so that any future return to that location will result in a realtime alert to the owner. In each case, the user collects a library of radio signatures or snapshots, programs a conditional rule or command with any new radio data that matches a pattern in the library, and the system takes over automated monitoring and execution of the rule.

Example X: Global Lost and Found System

An advantage of including a cellular modem 4403 in device 4501 is that the range of cellular connectivity is as broad as the range at which cellular coverage is available, so the finding of missing devices that have been tagged by one of the cellular-enabled tracking devices or radiotags of the invention allows us to offer a global lost-and-found system 4500 (FIG. 45) by use of these radiotags.

Implementation is straightforward. An “app” installed on smartphone 30 or on a laptop computer may be used to set up the group of tracking devices 4310 (FIG. 43) and also to set up XCB radiotag 4300, which has a limited user interface. Once the group is defined, the XCB radiotag can monitor and report on the group location and the cloud host 1111 can send relevant alerts and notifications to the owner via smartphone 30 as described above (FIGS. 37D and 37E). In some embodiments, the app may be upgraded using OTA (over-the-air) methods. Similarly, an image update of firm-

ware in any of the tracking devices or XCB radiotags can be downloaded into the devices using OTA methods.

Detecting “Left Behind” Assets using “Find and Track Group” Feature: A program with a “find and track group” feature is installed in a smartphone **30**. The feature enables finding and tracking by radio proximity or by using a bell and light built into each tracking device of group **4310**. The program monitors accelerometer output on the smartphone, and if sustained activity is detected that is consistent with movement that would result in a change in a user’s location, the program causes the XCB radio to verify radio contact with each of the items of the larger group and to interrogate the accelerometer of each member of the group. If any items in the group are not in motion, the program causes the screen of the control apparatus to display a menu list (see FIG. 17, US Pat. Publ. No. US20210274315, titled “Tracking Device Systems”, with a highlight on the items not in motion. The highlight indicates to the user that some items may be “left behind”. If indeed left behind, the program will assist in locating the items. So for example, pressing an icon on the display causes the mislaid tracking device to sound a distinctive tone that can be cleared by pressing the button again after the user finds the missing item (such as in the pocket of the jacket worn yesterday).

The system **4350, 4500** may also receive a weather forecast of pending rain, and may generate a reminder to take an umbrella that is tagged with a radiotag of the group **4310**. All the items in the group are managed from a single screen, and the associated displays or actions are fully programmable. User programmable features are included and may be set up or run on a smartphone, for example. The program will receive sensor information and other information that the user can use as a “trigger condition” to cause the system to invoke an executable action or notification in the smartphone, in a tracking device, or in a remote machine. The program can also monitor battery status of tracking devices in the group, and issue an alert if one or more of the items has a low battery. By “tracking device”, both the BT tracking devices and the XCB radiotags may be included as group members, and in some instances, the XCB radiotags can share management responsibility for monitoring the group and communicating any changes to the cloud host **1111** or sending a notification to an owner’s smartphone **30**, for example.

On the smartphone, or on a companion XCB device that is managing the group **4310**, the program or system more generally may be enabled to run a counter that integrates motion sensor or heading sensor output in a rolling window of time, and if there is sufficient motion to cause an expected relocation of the smartphone or XCB device, then the program will compare motion sensor output received from the members of the radiobeacon group **4310**, and flag any anomalies and inconsistencies. The managing device can beep or vibrate as a notification, and a screen is displayed on a companion smartphone that lists the members of the group that have been flagged or even lists their relative motion(s) and current locations. This monitoring is automatic. An override of power savings features occurs if motion of the smartphone and associated radiotags is not copacetic, and the counter is reset if there is a period in which the smartphone is stationary or all devices are moving in unison. The smartphone may include a menu for monitoring the relative motion sensor outputs of a group of radiotags on a single display screen so as to reduce loss of assets, children, and pets that have been radiotagged.

Similarly, any of the XCB radiotags **10, 4300, 4501** can monitor a group of BT radiotags. For example, an increased

motion sensor or heading sensor output from a member of the group of radiotags may be inconsistent with a more stationary motion sensor output of an internal motion sensor in an XCB radiotag. If that sensor output is consistent with a change in location of the radiotag, then the system will generate an alert to a user. The alert can be a vibration or a tone, and a display may include a listing of the radiotags and their motion status. A companion smartphone may be caused to display a menu of actions for user consideration.

These left behind protocols are improved by providing radio contact information. Essentially the group of radiotags and the XCB radiotag look out for each other and report to a smartphone, XCB radiotag, smartwatch, or a cloud host, if there is a lost signal or if the motion and heading of one of the tags is leading it into a local area where the BT signal environment is not familiar or unexpected. Often this can occur just by turning a corner or going out a door, so the latency, the time delay between a left behind event and a notification of the event, can be dramatically reduced.

Example XI: “Familiar Zones” for Pets

Alternatively, the cloud host can analyze BT traffic using records of radio call logs transmitted from the XCB radiotag to the cloud host and can identify radio signals reliably associated with familiar locations such as home or office, each of which may define a “familiar zone” (aka “safe zone”). The cloud host can then autoprovision the XCB radiotag to toggle power to its cellular modem, activating more frequent location updates when the radiotag is not in a safe zone and reducing or limiting power to the cellular modem when the radiotag is in the familiar zone. The radio correlator output is coupled to a flag in the instruction stack so that if a fixed location signal is detected, the cellular modem will negotiate a power-savings protocol with the cellular network provider, for example an extended eDRX or PSM power savings mode. If an alien signal pattern is detected that crosses a threshold “strangeness index”, the device may be programmed to attempt a CALL HOME.

Pet Location And Monitoring Services: In one use case, a dog is provided with a collar or chip that includes an XCB radiotag **4401** or **4300**. The cloud host auto-provisions the radiotag with one or more BT radio signatures that are reliably associated with a particular location, for example a backyard of a home. The instruction stack in the radiotag is configured so that if a correlator of the BT radio modem detects a familiar radio signature or cluster that defines a BACKYARD location, then the cellular modem will be deactivated to operate in a power savings mode. But if upon scanning its radio environment (after the dog has jumped the fence, for example), the radiotag scanner no longer detects the familiar radio signature, then the instruction stack will execute commands to activate the transmission of more frequent location updates to a cloud host. This is a radio leash, but is operated by sampling the BT radio signal environment rather than by monitoring RSSI values of specific companion BT radiosets, and hence can be operated without the user’s presence or the need for a radio hub with “lighthouse” signal. The programmable radio leash can also be reset to a new location just by capturing a BT radio signature of the new place.

A similar regimen can be operated using a WiFi hub to define a virtual leash around a familiar zone or location. But the advantage of a BT virtual safe zone or “familiar radio signature location” is that it can be defined using BT radio signatures that are “native” to a particular location and require no point-to-point connection to serve as a trigger

condition for activating a conditional rule by which an action or a notification is executed. The BT signals used for flagging an escaped dog need not be associated with owner-provided hardware, and can be anonymous. Any “unfamiliar” BT signal from a passerby that the dog encounters can activate a notification and cause execution of a CALL HOME, for example. If the dog approaches a stranger who has a BT transmitter, the radio contact may be sufficient to activate the CALL HOME protocol. In short, signals from strange beacons may provision and even autoprovision notifications. An escaped dog notification will be presented to the owner if there is a transition in the BT signal environment from familiar to strange or alien, and if the radio collar detects a persistent alien signal environment for more than a minute or so, then it will begin generating cellular location data that can be sent to the owner via the system host. The owner can select the notification sensitivity and frequency based on experience and in some instances, the alert can be sent instantly if the dog collar detects a signal that is not expected and loses a signal that is expected, for example, so as to reduce the incidence of false positives. This regimen can be implemented with a few flash memory registers for storing BT reference signals, a correlator that compares the BT radio traffic with the reference signals, and a program that periodically executes a scan of local BT radio traffic. Motion and heading sensor data can also be reported if a threshold of probability is crossed that the dog jumped the fence.

In addition, other locations can be readily defined as “familiar and safe” or “dangerous and to be avoided” without the need to install new hardware. These are the “whitelist”/“blacklist” snapshots that have been discussed before. By limiting the scans of the BT radio environment to 200 msec at a time or less, and selecting a latency of 1 or 10 minutes or less between scans, for example, a compromise can be achieved between battery service life and system latency in which the dog’s owner will receive a timely notification from the cloud host if the dog has jumped the fence. The system will decrease the response latency as the radio contact strangeness index increases. Once the cellular radio modem of the radiotag has been activated, the notification to the owner’s smartphone 30 may include a map with a moving map pin that shows the dog’s location in real time, for example. The cloud host 1111 causes the owner’s smartphone to display the notification and map and provides other assistance in tracking the dog on a frequent basis. A passerby who sees the dog may also receive a notification that the dog is lost if the passerby has a smartphone that operates a copacetic lost-and-found software package (US Pat. Publ. No. US2021/0274315, titled “Tracking Device Systems”, which is incorporated in full by reference and co-assigned), if the passerby presses a multi-function button (4541, FIG. 45) on the collar, or if the passerby merely approaches the dog within a few feet so as to inadvertently activate the BT radio modem of the radiotag, which senses the approach of the unrecognized BT signal from the passerby’s smartphone (as mediated by a strong BT INQUIRY SCAN or REPLY transmitted routinely by the smartphone). More details of this system are provided in co-pending and co-owned US. Pat. Appl. Publ. No. US2021/0152976, which is co-owned and incorporated in full by reference. In one approach, a community of users is supplied with smartphone software that assists in lost-and-found recovery, in the radio contacts record-enabled system, anyone with smartphone may assist 65 without conscious effort and without a special “App” installed on the phone.

The improved radiotag and system disclosed here enables owners to expect and to realize an extraordinary level of power management sophistication while enabling both long distance and short range location management tools, including global finding, wayfinding, and tracking. The system is versatile and provides Lost, Found, Left Behind, Familiar Location, Geofencing, Save Zone, Radio Leash, and a variety of other services managed locally or by a cloud host—with privacy and anonymity. Power management is stringent and is modelled on BLE (which typically can operate at less than 60 μ A with a subsecond duty cycle), so that an owner can attach a radio device to a pet or other asset, and be able to track the signal over long distances if the asset goes missing weeks or months later—while having adequate broadcast range and no unacceptable latency or dead time.

In another case study for pet finding and tracking, an owner clips or attaches a BT radiobeacon to a belt or keychain and carries it along, while placing an XCB radiotag on a collar attached to the pet. The XCB radiotag plays the role of master in a BT piconet linking the radiobeacon to the XCB radiotag. The BT radiobeacon worn by the owner acts as a mobile “familiar zone” and is recognized by its periodic signal to the XCB radiotag. The radiobeacon is instructed to report in at selected intervals and if a report is missed, then the XCB radiotag will take the initiative to begin tracking the pet and reporting its location to the owner, for example by displaying the pet’s location on a map pushed to the owner’s smartphone. Alternatively, the XCB radiotag will initiate a network connection request and begin supplying location information to a cloud host. The cloud host is programmable to generate an exception alert to the owner and begin other lost-and-found services. Also, the XCB radiotag is enabled to react to the close approach of any passerby with an active BT radio unit, and to attempt to elicit a connection with the cloud host or owner through a BT-to-BT link that can be bootstrapped to a cellular connection, such as through a VPG (3000, FIG. 43; 2400, FIG. 6C). In one embodiment, the cloud host may mediate a chat-room style interaction between the owner of the pet and the passerby so that the owner can quickly retrieve the pet without an issue of the passerby’s privacy. Similar capabilities may be employed to look out for children who have lost their way or become separated from their guardians. This system is strengthened by the XCB radiotag’s capacity to determine its own location and to override alerts if the pet or child is in a familiar zone, as determined from the BT radio signal topography.

Example XII: Lost Luggage

Airport congestion has lead to increasing claims for lost luggage by upset passengers. But for baggage that is fitted with an XCB tracking device, the device can be tracked by the user. In some instances, the device itself discovers that it is lost if it is isolated in an alien radio environment for hours. If lost, the device may activate its cellular phone to CALL HOME. A network assistant or user receives the call and an approximate current location is displayed on a viewscreen. Devices used in international travel can include a GPS locator that is subject to the same power management rules of the BT radio modem. The missing bag may be found using the BT Proximity search tool described in FIG. 10. By building the device with a quality piezo speaker, the device can be commanded to bleep loudly on command to signal its location to a nearby user at a baggage terminal, or in the “lost luggage” office.

A heading sensor provides the needed signal to detect that the device is airborne and can override the CALL HOME protocol until the airplane is back on the ground. Extended use without power loss is essential. Lithium ion batteries may be used in combination with a supercapacitor as shown in FIG. 24 as a backup. Power management circuitry 4562 may be enabled to maintain power from a 3 Farad supercapacitor 4560b for up to 5 days, assuming a BLE advertising frequency of once per minute for a duration of 10 msec per INQUIRY SCAN, for example. The superconductor is enclosed in the radiotag housing, and may be inserted inside the luggage, inside a luggage pocket, or attached by an adhesive patch around the handle, for instance. Superconductors as small as a pen cell battery are obtained for example from Eaton Electronics, Cleveland OH.

Rugged XCB units can be built into the traveller's bag, inserted in the bag, or attached to a strap, and do not require recharging for weeks or months. A community of users may detect the BLE advertising signal and forward contact and location information to a cloud host as described in U.S. patent Ser. No. 10/389,459, for example. In other instances, the XCB radiotag will turn on its cellular radio to receive instructions from the cloud administrator if its location is stationary and no familiar BT radio activity is detected in the location.

INCORPORATION BY REFERENCE

All of the U.S. Patents, U.S. Patent application publications, U.S. Patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and related filings are incorporated herein by reference in their entirety for all purposes.

SCOPE OF THE CLAIMS

The disclosure set forth herein of certain exemplary embodiments, including all text, drawings, annotations, and graphs, is sufficient to enable one of ordinary skill in the art to practice the invention. Various alternatives, modifications and equivalents are possible, as will readily occur to those skilled in the art in practice of the invention. The inventions, examples, and embodiments described herein are not limited to particularly exemplified materials, methods, and/or structures and various changes may be made in the size, shape, type, number and arrangement of parts described herein. All embodiments, alternatives, modifications and equivalents may be combined to provide further embodiments of the present invention without departing from the true spirit and scope of the invention.

In general, in the following claims, the terms used in the written description should not be construed to limit the claims to specific embodiments described herein for illustration, but should be construed to include all possible embodiments, both specific and generic, along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited in haec verba by the disclosure.

We claim:

1. A method for operating a radio-enabled tracking and locating system, the method comprising:
 scanning for and capturing, by a radiotag, an ensemble of Bluetooth (BT) radio signals from a BT spread spectrum signal topology in radio proximity thereto, said capture defining a digital radio snapshot, wherein the radiotag includes a processor, a BT radio modem, a unique radio unit identifier, a community identifier, a

cellular radio modem, and a flash memory unit configured to store one or more digital radio snapshots of intercepted BT radio signals each with a timestamp;
 predicting, by the radiotag, a current location of the radiotag in near real time by matching the one or more digital radio snapshots of intercepted BT radio signals with a snapshot from a library of digital radio snapshots stored in the library from a past time, each snapshot in the library from a past time defining a reference snapshot assignable to a known location; and
 controlling, by the radiotag, a CALL HOME event in response to determining that the current location meets one or more criteria defined by a user in a user profile, wherein controlling the CALL HOME event includes initiating a cellular Connection Request to a designated IP Address via the cellular radio modem.

2. The method of claim 1, further comprising collecting a library of reference snapshots in memory and, by a user's preference, assigning reference snapshots to a whitelist or a blacklist according to locations at which the snapshots were captured.

3. The method of claim 2, further comprising providing a user interface for managing at least one of the whitelist or the blacklist.

4. The method of claim 1, further comprising providing a user interface for mapping the current location of the radiotag.

5. The method of claim 1, further comprising providing a user interface for programming a command that is executable according to the current location.

6. The method of claim 1, further comprising storing a chronology of current locations in a temporal sequence, and displaying the temporal sequence on a user interface.

7. The method of claim 1, further comprising providing a cloud administrative host for storing user profiles and for managing radiotag CALL HOME events.

8. The method of claim 1, wherein determining that the current location meets one or more criteria defined by the user in the user profile includes determining that the BT spread spectrum signal topology in radio proximity to the radiotag meets an index value of strangeness or familiarity.

9. The method of claim 1, further comprising attaching or embedding the radiotag on or in an object in need of tracking or locating.

10. The method of claim 1, wherein the BT radio modem comprises an amplifier having an adjustable gain, and wherein the method further comprises dynamically adjusting the gain of the amplifier of the BT radio modem.

11. A system for Bluetooth (BT)-radiotag enabled tracking and locating, the system comprising:
 a radiotag having:

a BT radio transceiver modem with a unique radio unit identifier, a community identifier, and executable instructions that, in response to execution by the BT radio transceiver modem, cause the BT radio transceiver modem to scan for and capture an ensemble of BT radio signals from a BT spread spectrum signal topology in radio proximity thereto, said capture defining a digital radio snapshot;

a cellular radio transceiver modem; and

a flash memory unit configured to store one or more digital radio snapshots of intercepted BT radio signals, each with a timestamp;

wherein the system is configured to predict a current location of the radiotag by comparing the one or more digital radio snapshots with a snapshot or snapshots from a library of reference snapshots having each a

121

known location, each said reference snapshot or snapshots defining a reference location;
 wherein the system is further configured to execute an executable instruction in response to determining that the current location meets one or more criteria defined by a user in a user profile;
 wherein executing the executable instruction includes controlling a CALL HOME event, and
 wherein controlling the CALL HOME event includes initiating a cellular Connection Request to a designated IP Address via the cellular radio transceiver modem.

12. The system of claim 11, further comprising a library of reference snapshots in memory and a user interface configured to assign said reference snapshots to a whitelist or blacklist of familiar locations.

13. The system of claim 11, further comprising a user interface configured to map and display the current location of the radiotag.

14. The system of claim 11, further comprising a user interface configured to program a command that is triggered by a match of the snapshot at the current location with a reference location.

15. The system of claim 11, further comprising a cloud administrative host configured to store user profiles, to manage radiotag CALL HOME events, and to generate notifications and alerts according to data received from the radiotag.

16. The system of claim 11, wherein determining that the current location meets one or more criteria defined by the user in the user profile includes determining that the BT spread spectrum signal topology in radio proximity to the radiotag matches a location on a whitelist or a blacklist.

17. The system of claim 11, wherein determining that the current location meets one or more criteria defined by the user in the user profile includes determining that the BT spread spectrum signal topology in radio proximity to the radiotag meets an index value of strangeness or familiarity.

18. The system of claim 11, wherein the radiotag is configured to enter a sleep or standby state in response to determining that the BT radio is actively paired or bonded in

122

a radio link to a companion smartphone or hub, and wherein the BT radio is configured to wake up the cellular radio modem in response to determining that the radio link is broken.

19. The system of claim 11, further comprising a user device having a display configurable as a user interface for managing whitelists or blacklists and for managing a user profile on a cloud administrative host.

20. The system of claim 11, wherein the BT radio modem is configured to control a second CALL HOME event in response to detecting a BT signal in a proximity of less than two meters, and the signal identifier or identifiers are not registered on any whitelist, wherein controlling the second CALL HOME event includes initiating a second cellular Connection Request to the designated IP Address via the cellular radio transceiver modem.

21. The system of claim 11, wherein the BT radio modem is configured to control a second CALL HOME event in response to determining that a motion or heading sensor data is consistent with a lost pet or child, wherein controlling the second CALL HOME event includes initiating a second cellular Connection Request to the designated IP Address via the cellular radio transceiver modem.

22. The system of claim 11, wherein the radiotag comprises a multi-function switch that when touched, pressed or toggled, activates a radio link with the cloud administrative host; or, the radiotag comprises a QR code that when scanned by a compatible smartphone of a passerby, activates a radio link with the cloud administrative host.

23. The system of claim 22, wherein the radio link is a chat link operable on a compatible smartphone of a passerby.

24. The system of claim 22, wherein the radio link is a cellular connection with an agent or chatbot prepared to provide tracking and locating assistance for an owner of the radiotag.

25. The system of claim 11, wherein the radiotag is attachable to or embeddable in an object in need of tracking or locating.

* * * * *